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ASYMPTOTIC q,t-FUSS–CATALAN NUMBERS FOR TYPE B

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Abstract. Let $W = W(B_n)$ act diagonally on $\mathfrak{h} \oplus \mathfrak{h}^*$, let $S = \mathbb{C}[\mathfrak{h} \oplus \mathfrak{h}^*]$, let $J \subseteq S$ be the ideal generated by the W-alternating polynomials and $\mathfrak{m}_S$ is the maximal ideal of the origin. For sufficiently large m we compute q,t-Fuss-Catalan polynomial $\mathrm{Cat}^{(m)}(B_n;q,t) := \mathrm{Hilb}_{q,t}\left(\frac{J^m}{\mathfrak{m}_S J^m}\right)_{\det\text{-part}}$ and imply $\mathrm{Cat}^{(m)}(B_n;1,1) = \binom{n(m+1)}{n}$. For proofs, we work with the Γ-equivariant Hilbert scheme $Y_n = n\Gamma\text{-Hilb}(\mathbb{A}^2)$, $\Gamma = \mu_2$ and Haiman-type Koszul complex that defines the punctual locus of Y_n. Our formula for $\mathrm{Cat}^{(m)}(B_n;q,t)$ is derived from a localization computaion for the Haiman-type Koszul complex.

1. Introduction

1.1. **The type A vs B.** Let W be a finite real reflection group acting on its reflection representation $\mathfrak{h}$ and diagonally on $\mathfrak{h} \oplus \mathfrak{h}^*$, and let

$$S = \mathbb{C}[\mathfrak{h} \oplus \mathfrak{h}^*], \qquad A = S^W, \qquad \mathfrak{m} = A_+.$$

Let $S^\varepsilon = \{f \in S :\ w(f) = \det(w)f\}$ be the space of alternating polynomials and $J = S \cdot S^\varepsilon$ the ideal it generates. Every alternating polynomial vanishes on every reflection diagonal $\{\alpha_s = \alpha_s^\vee = 0\}$, so

$$J \ \subseteq\ I := \bigcap_{s \in \mathrm{Ref}(W)} (\alpha_s, \alpha_s^\vee), \qquad \text{and} \qquad J^m \subseteq I^m \subseteq I^{(m)} := \bigcap_{s \in \mathrm{Ref}(W)} (\alpha_s, \alpha_s^\vee)^m. \tag{1}$$

For $W = S_n$ Haiman's polygraph theorem [41, 42] implies the remarkable equalities $J^m = I^m = I^{(m)}$ for all m as well as

$$\mathrm{Cat}^{(m)}(S_n;q,t) = \chi_{\mathbb{C}_q^* \times \mathbb{C}_t^*}(\mathcal{O}_{Z_n}(m)),$$

where $Z_n \subset \mathrm{Hilb}^n(\mathbb{C}^2)$ is the punctual Hilbert scheme [40, 42]. Stump [74] defined, for every finite reflection group,

$$\mathrm{Cat}^{(m)}(W;q,t) := \mathrm{Hilb}_{q,t}\Big(\, e_\varepsilon\, DR^{(m)}(W)\Big) = \mathrm{Hilb}_{q,t}\Big(\frac{J^m}{\mathfrak{m}_S J^m}\Big)_{\det\text{-part}}, \tag{2}$$

the bigraded Hilbert series of the determinantal component of the generalized diagonal coinvariants, equivalently of the minimal generating space of J^m; and he conjectured that its dimension is the Fuss–Catalan number of W and that it matches the graded character of the corresponding finite-dimensional spherical Cherednik module (see details below).

The Fuss–Catalan numbers in Stump's conjecture are the numbers $\mathrm{Cat}^{(m)}(W) = \prod_{i=1}^{\ell} \frac{d_i + mh}{d_i}$, where ℓ is the rank of W, $d_1, \dots, d_\ell$ are its degrees and h is its Coxeter number, and they are the subject of an extensive combinatorial theory, the Coxeter–Catalan combinatorics. For $m = 1$ they count the noncrossing partitions of Reiner [69] and Bessis [12], the clusters of Fomin and Zelevinsky [26], the Coxeter-sortable elements of Reading [68], and the nonnesting partitions and the dominant regions of the Catalan arrangement [4]. For general m they count the m-divisible noncrossing partitions of Armstrong [2], the facets of the generalized cluster complexes of Fomin and Reading [25], and the dominant regions of the extended Catalan arrangements and the m-nonnesting partitions of Athanasiadis [4, 5]. The recent memoir of Stump, Thomas and Williams [75] gives a uniform framework for the noncrossing side of this theory, and the rational Catalan combinatorics, in which the parameter $m + \frac{1}{h}$ is replaced by an arbitrary rational number, is developed in [3, 30]. The q,t-Catalan numbers of type A were introduced by Haiman [39] and by Garsia and Haiman [27],

and the book of Haglund [35] is a reference for their combinatorics and for the space of diagonal harmonics. For $W = B_n$ one has $\mathrm{Cat}^{(m)}(B_n) = \binom{n(m+1)}{n}$, the number of lattice paths with unit north and east steps from $(0,0)$ to (mn, n), and we discuss the type B lattice path models in Sections 9.3, 10 and 11.

For groups other than S_n the inclusions (1) are strict in general: Kivinen [51, Ex. B.2] computes for $W(B_3)$ that $J \subsetneq I$ while $eJ = eI$ for the symmetrizing idempotent e. This suggests that the correct general statements are *spherical*. Kivinen [51] attaches to these data three "Hilbert-scheme" models,

$$X_{\mathfrak{g},\mathrm{sgn}} = \mathrm{Proj} \bigoplus_{m\geq 0} (\Delta S^{\varepsilon})^m, \quad X_{\mathfrak{g},\mathrm{diag}} = \mathrm{Proj} \bigoplus_{m\geq 0} \big(e(\Delta I)\big)^m, \quad X_{\mathfrak{g},\mathrm{symb}} = \mathrm{Proj} \bigoplus_{m\geq 0} e(\Delta^m I^{(m)}), \tag{3}$$

where $\Delta = \prod_{\alpha\in\Phi^+} \alpha$ ([51, Defs. 3.1, 3.4, 3.6]); the symbolic model is normal [51, Thm. 3.8] and is the commutative degeneration of a $\mathbb{Z}$-algebra built from shift bimodules of rational Cherednik algebras [51, Thm. 4.3, Cor. 4.5]. The three models are connected by canonical birational morphisms $X_{\mathrm{symb}} \to X_{\mathrm{diag}} \to X_{\mathrm{sgn}}$, and it is not clear in general whether these are isomorphisms.

This paper solves the type-B problem asymptotically in the Fuss parameter m, with explicit q,t-characters.

1.2. **Main results.** Fix $W = W(B_n) = \mu_2 \wr S_n$, with $\mathrm{Ref}(W)$ consisting of the n coordinate reflections and the $2\binom{n}{2}$ signed transpositions; $h = 2n$ is the Coxeter number, $\deg\Delta = n^2$. Write $e_m = \frac{1}{|W|}\sum_w \det(w)^m w$, so $e_0 = e$ and $e_1 = e_-$. The phrase "$m \gg 0$" always means: for each fixed n there is a (non-effective) $m_0(n)$ such that the statement holds for all $m \geq m_0(n)$.

Our first result settles the comparison of the three models.

Theorem A (Asymptotic spherical collapse; Theorem 9.2)**.** *For every fixed n and all $m \gg 0$,*

$$e_m J^m \;=\; e_m I^m \;=\; e_m I^{(m)}.$$

Consequently the three spherical models (3) coincide for $\mathfrak{g}$ of type B_n/C_n: $X_{\mathfrak{g},\mathrm{symb}} \cong X_{\mathfrak{g},\mathrm{diag}} \cong X_{\mathfrak{g},\mathrm{sgn}}$.

The q,t-characters are expressed by an explicit localization polynomial. Let $\mathrm{Bal}(2n)$ denote the partitions $\lambda \vdash 2n$ with empty 2-core (equivalently, with n boxes of each checkerboard parity), and for $\lambda \in \mathrm{Bal}(2n)$ define, with boxes $(a,b) \in \mathbb{N}^2$ in English notation and arm and leg lengths $a_\lambda(s), l_\lambda(s)$,

$$\begin{aligned} B_\lambda &= \sum_{(a,b)\in\lambda} q^a t^b, \qquad \ell_\lambda = \prod_{\substack{(a,b)\in\lambda \\ a+b \text{ odd}}} q^a t^b, \qquad \Pi_\lambda^{(0)} = \prod_{\substack{(a,b)\in\lambda\setminus\{(0,0)\} \\ a+b \text{ even}}} (1-q^a t^b), \\ D_\lambda &= \prod_{\substack{s\in\lambda \\ a_\lambda(s)+l_\lambda(s) \text{ odd}}} \big(1-q^{a_\lambda(s)+1}t^{-l_\lambda(s)}\big)\big(1-q^{-a_\lambda(s)}t^{l_\lambda(s)+1}\big), \end{aligned} \tag{4}$$

and set

$$\frac{F_{n,m}(q,t)}{(1-q)(1-t)} = \sum_{\lambda\in\mathrm{Bal}(2n)} \frac{\ell_\lambda^m\, B_\lambda \Pi_\lambda^{(0)}}{D_\lambda}, \quad P_{n,m}(q,t) = \tfrac{1}{2}\Big(F_{n,m}(q,t) + (-1)^{nm} F_{n,m}(-q,-t)\Big). \tag{5}$$

Theorem B (q,t-Fuss–Catalan character; Theorem 9.4)**.** *For every fixed n and all $m \gg 0$,*

$$\mathrm{Cat}^{(m)}(B_n; q, t) \;=\; P_{n,m}(q,t).$$

Theorem C (Principal specialization; Theorem 7.2)**.** *For all $n \geq 1$ and $m \geq 0$, with $M = n(m+1)$ and $Q = q^2$,*

$$F_{n,m}(q,q^{-1}) = q^{-mn^2} \begin{bmatrix} M \\ n \end{bmatrix}_Q - q^{1-mn^2}\,[m]_{Q^n} \begin{bmatrix} M \\ n-1 \end{bmatrix}_Q, \qquad [m]_{Q^n} = \tfrac{1-Q^{nm}}{1-Q^n},$$

the two summands having opposite exponent parities. Hence

$$q^{mn^2}\,P_{n,m}(q,q^{-1}) = \begin{bmatrix} n(m+1) \\ n \end{bmatrix}_{q^2}, \qquad P_{n,m}(1,1) = \binom{n(m+1)}{n},$$

and the opposite-parity part $N_{n,m} := P_{n,m} - F_{n,m}$ *satisfies* $N_{n,m}(1,1) = m\binom{n(m+1)}{n-1}$ *and* $F_{n,m}(1,1) = \frac{1}{nm+1}\binom{n(m+1)}{n}$. *In particular* $\mathrm{Cat}^{(m)}(B_n;q,q^{-1}) = q^{-mn^2}\begin{bmatrix} n(m+1) \\ n \end{bmatrix}_{q^2}$ *and* $\mathrm{Cat}^{(m)}(B_n;1,1) = \binom{n(m+1)}{n}$ *for* $m \gg 0$.

Let H_c be the equal-parameter rational Cherednik algebra of $W(B_n)$, $U_c = eH_ce$, and let $L_c(\mathrm{triv})$ be the finite-dimensional simple at $c = \frac{1}{2n} + m$. Set $\mathfrak{S}^{\mathrm{symb}}_{n,m} = e_m I^{(m)}/\mathfrak{m}\, e_m I^{(m)}$.

Theorem D (Cherednik identification; Theorem 8.12)**.** *For every fixed* n *and all* $m \gg 0$ *the canonical filtered shift-bimodule surjection is an isomorphism of Gordon-bigraded vector spaces*

$$\mathfrak{S}^{\mathrm{symb}}_{n,m} \xrightarrow{\sim} \mathrm{gr}_F\, eL_{\frac{1}{2n}+m}(\mathrm{triv}), \qquad \text{and} \qquad P_{n,m}(q,t) = \mathrm{ch}^G_{q,t}\ \mathrm{gr}_F\, eL_{\frac{1}{2n}+m}(\mathrm{triv}),$$

where F *is the iterated tensor-product filtration of Theorem 8.1.*

All of the above rests on the following geometric theorem, which occupies the bulk of the paper. Let $\Gamma = \mu_2 \subset SL_2(\mathbb{C})$ act on $\mathbb{A}^2$ by $(x,y) \mapsto (-x,-y)$, let $Y_n = n\Gamma\text{-}\mathrm{Hilb}(\mathbb{A}^2)$ be the corresponding smooth $2n$-dimensional Nakajima quiver variety, $\eta : \mathcal{U}_n \to Y_n$ the universal Γ-cluster, $\mathcal{P}_n = \eta_*\mathcal{O}_{\mathcal{U}_n} = \mathcal{R}_0 \oplus \mathcal{R}_1$ the rank-$2n$ cluster algebra bundle, $\mathcal{T}_n = \ker(\mathrm{Tr}_{\mathcal{R}_0})$ the trace-zero even part, and

$$\mathcal{E}_n = \mathcal{T}_n \oplus \mathcal{O}_q \oplus \mathcal{O}_t, \qquad \mathcal{K}_n^{-a} = \mathcal{P}_n \otimes \textstyle\bigwedge^a \mathcal{E}_n, \qquad \mathcal{I}_n = \mathcal{P}_n/J_n,$$

the type-B analogue of the Haiman–Garsia Koszul complex (Section 3); here J_n is the ideal generated by $\mathcal{T}_n$ and the universal sections x, y. Let $\rho : Y_n \to Q_n$ be the projective symplectic resolution of the wall quiver variety $Q_n \cong \mathrm{Quot}^n_X(R_1)_{\mathrm{red}} \cong X_{\mathfrak{g},\mathrm{symb}}$ (Section 2), $\mathcal{L}_n = \det \mathcal{R}_1 = \rho^*\mathcal{O}_{Q_n}(1)$, and $\mathcal{Z}^{\mathrm{symb}}_n = Q_n \times_{S_n} \{0\}$ the scheme-theoretic symbolic punctual fiber, $S_n = (\mathfrak{h} \oplus \mathfrak{h}^*)/W$.

Theorem E (Geometric theorem; Theorems 3.13, 3.14, 5.1, 6.22)**.**

(1) *The complex* $\mathcal{K}_n^\bullet$ *has exactly two nonzero homology sheaves,* $\mathcal{H}^0(\mathcal{K}_n^\bullet) = \mathcal{I}_n$ *and* $\mathcal{M}_n := \mathcal{H}^{-1}(\mathcal{K}_n^\bullet)$*; and* $\mathcal{I}_n \cong \mathcal{O}_{F_n}$*, the structure sheaf of the scheme-theoretic punctual fiber.*

(2) *There is a canonical* T*-equivariant isomorphism* $\mathcal{M}_n \cong \mathcal{E}xt^n_{\mathcal{O}_{Y_n}}(\mathcal{S}_n, \mathcal{O}_{Y_n}) \otimes \det \mathcal{E}_n$*, where* $\mathcal{S}_n = \mathcal{P}_n^\vee/J_n\mathcal{P}_n^\vee$ *is the dual-socle sheaf.*

(3) $R\rho_*\mathcal{M}_n^+ = 0$*, where* $\mathcal{M}_n^\pm$ *are the central-parity summands.*

(4) (*Symbolic descent*) *For every* $d \in \mathbb{Z}$*,*

$$R\rho_*(\mathcal{L}_n^{\otimes d} \otimes \mathcal{I}_n) \ \simeq\ \mathcal{O}_{\mathcal{Z}^{\mathrm{symb}}_n}(d).$$

A feature with no type-A counterpart is that $\mathcal{K}_n^\bullet$ is *not* a resolution: it has $n+1$ generators against expected codimension n, and the excess sheaf $\mathcal{M}_n$, in particular its even part $\mathcal{M}_n^+$, is nonzero in every rank $n \geq 2$ (Theorem 5.26). Part (3) says that this excess is invisible to the wall. We prove it in Section 5 through a rank-one Hecke correspondence $Y_n \xleftarrow{p} \mathscr{H} \xrightarrow{q} \mathfrak{M}_\theta((n-1,n),(1,0))$; the Grassmannian wall fibers $F_J \cong \mathrm{Gr}(k, 2k-1)$ and their Borel–Weil–Bott vanishing (Section 4) reappear as the fixed-point shadow of this proof, but are not needed for it. Part (4) is proved in Section 6 on the formal neighbourhood of each wall fiber, where, by the Bellamy–Craw normal form, the even cluster bundle is a trivial summand plus the tautological quotient bundle of a cotangent Grassmannian: Borel–Weil–Bott makes the positive Haiman complex termwise acyclic for the pushforward and its first differential computable, an elementary normalized-trace lemma identifies the cokernel with the punctual quotient, and part (3) forces concentration in degree zero.

The paper is organized as follows. Section 2 fixes notation and identifies the wall variety and its polarization. Section 3 introduces the Koszul complex and proves parts (1) and (2) of Theorem E. Section 4 describes the wall fibers, the local model at a wall point and the Brill–Noether thresholds.

Section 5 proves part (3) and Section 6 part (4). Sections 7 to 9 deduce Theorems C, B, D and A, in that order. Section 10 formulates a lattice path model for the polynomials $P_{n,m}$, a type-B analogue of the shuffle conjecture. Section 11 lifts this model to the level of symmetric functions. The paths of the rectangle acquire labels, each labelled path contributes a fundamental quasisymmetric function, and we prove that the resulting series $\mathcal{H}^+_{n,m}$ is symmetric and Schur positive, by identifying its path-by-path pieces with LLT polynomials, and that its s_n-coefficient is the polynomial $\widehat{P}_{n,m}$ of Section 10. We conjecture that $\mathcal{H}^+_{n,m}$ is the bigraded Frobenius character of the G_n-invariant part of Gordon's module $L_{\frac{1}{2n}+m}(\mathrm{triv})$, whose spherical part is the subject of Theorem D. At $m = 1$ we prove a schedule formula in the sense of Haglund and Loehr, a sum over the permutations of $\{0, 1, \ldots, n\}$ of a monomial in q, t times a product of qt-integers, whose two extreme sectors are the type-A formula, and we conjecture that the corresponding $(n + 1)^n$ monomials form a basis of the module.

1.3. **Techniques.** Let us describe the techniques which enter the proofs of the main results. The geometric part of the paper rests on the theory of Nakajima quiver varieties for the framed affine A_1 McKay quiver [64, 65]. The space Y_n is the quiver variety at a generic stability, the wall variety Q_n is the quiver variety at the wall stability, and ρ is the variation of GIT morphism. The identification of Q_n with the reduced Quot scheme of the odd McKay module is due to Craw, Gammelgaard, Gyenge and Szendrői [21, 22], and its identification with the symbolic model is due to Kivinen [51]. We add to this the identification of the polarization of Q_n with the sign component of the Procesi bundle of Losev [60] and of Boixeda Alvarez and Losev [16] (Theorem 2.7), which is proved by intersecting with the two extremal curves of Y_n.

The proof of Theorem E combines several tools. Parts (1) and (2) follow from a local cohomology argument, which shows that a perfect complex of Tor-amplitude $[-n - 1, 0]$ whose cohomology is supported in codimension n has only two homology sheaves, and from Koszul duality, which converts the second homology sheaf into an Ext sheaf of the dual-socle sheaf. The codimension bound is the semismallness of symplectic resolutions [48, 49]. Part (3), the even-excess vanishing, is proved with the rank-one Hecke correspondence of González, Gorsky and Simental [29], which parametrizes a cluster together with an even socle line. Its two projections are cut out by regular sections in projective bundles, and the proof uses Bott's formula for twisted differential forms on projective space, the Hilbert–Burch and Buchsbaum–Rim resolutions of the even socle locus, and Grothendieck duality for finite morphisms. Part (4), symbolic descent, uses the étale-local normal form of Bellamy and Craw [8] at a wall point, which exhibits the formal neighbourhood of a wall fiber as a cotangent Grassmannian times a smooth core, and the Borel–Weil–Bott theorem on the Grassmannian, applied to positive tensor constructions of the tautological quotient bundle. An elementary normalized-trace lemma then computes the pushforward of the punctual quotient, and the theorem on formal functions passes from the formal model to Q_n.

The characters of Theorems B and C are computed with equivariant localization on Y_n, the principal specialization reduces to the finite q-binomial theorem, and Serre vanishing with cohomology and base change turns the Euler characteristic into a Hilbert series for $m \gg 0$. On the representation-theoretic side we use the shift bimodules of the rational Cherednik algebra of type B_n. Kivinen's associated-graded theorem [51] identifies the associated graded of the iterated shift bimodule with $e(\Delta^m I^{(m)})$, the Procesi bundles of Losev and of Boixeda Alvarez and Losev give the surjection from the symbolic model onto the finite-dimensional module $eL_{\frac{1}{2n}+m}(\mathrm{triv})$, and the theorem of Berest, Etingof and Ginzburg [6] gives its dimension. Theorem D is then a dimension count, and Theorem A follows from Theorem D, from Stump's generalized Gordon surjection [32, 74] and from graded Nakayama.

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2. Notation and the spaces

In this section we fix the notation which is used throughout the paper, and we collect the facts about the spaces Y_n and Q_n which we need. The two spaces are the Nakajima quiver varieties of the framed affine A_1 McKay quiver at a generic stability and at the wall stability, and the morphism $\rho : Y_n \to Q_n$ between them is the variation of GIT morphism. The results of Craw, Gammelgaard, Gyenge and Szendrői [21, 22] and of Kivinen [51] identify Q_n with the reduced Quot scheme of the odd McKay module and with the symbolic model $X_{\mathfrak{g},\mathrm{symb}}$, and this is what connects the geometry of the paper with the ideals $I^{(m)}$ of the introduction.

The main new statement of the section is the identification of the polarization, Theorem 2.7. Boixeda Alvarez and Losev construct a partial resolution of $(\mathfrak{h}\oplus\mathfrak{h}^*)/W$ on which a power of the sign component of the Procesi sheaf is ample, and they remark that in type B_n it is the wall quiver variety. We prove that the sign component is the determinant $\det\mathcal{R}_1$ of the odd tautological bundle, by pairing both line bundles with the two extremal curves of Y_n. This is what allows us to transport the Procesi package of Boixeda Alvarez and Losev to Q_n in Section 8. In Section 2.3 we also compare the two finite-dimensional quotients of e_mJ^m which occur in the literature, Stump's quotient and the quotient by the invariants, since Theorem B is stated for the former while the geometry of the paper computes the latter.

2.1. **McKay data.** Throughout, $\Gamma=\mu_2=\{\pm1\}\subset SL_2(\mathbb{C})$ acts on $\mathbb{A}^2$ by $(-1)\cdot(x,y)=(-x,-y)$, and

$$R=\mathbb{C}[x,y]=R_0\oplus R_1,\qquad R_0=\mathbb{C}[x,y]^\Gamma=\mathbb{C}[u,v,w]/(uv-w^2),\qquad R_1=\mathbb{C}[x,y]^{\mathrm{odd}}=R_0x+R_0y,$$

with $u=x^2$, $v=y^2$, $w=xy$. The quotient $X=\operatorname{Spec}R_0=\mathbb{A}^2/\Gamma$ is the A_1 singularity and $\mathfrak{m}_X=(u,v,w)=R_1R_1$ is its maximal ideal at the origin.

We identify $\mathbb{C}[x,y]=\mathrm{Sym}(\mathfrak{h}\oplus\mathfrak{h}^*)$ for the rank-one situation and, for rank n, $S=\mathbb{C}[\mathfrak{h}\oplus\mathfrak{h}^*]=\mathbb{C}[x_1,\dots,x_n,y_1,\dots,y_n]$ with $W=W(B_n)=\mu_2\wr S_n$ acting by signed permutations, diagonally in x's and y's. The torus $T=\mathbb{C}_q^*\times\mathbb{C}_t^*$ scales (x,y) with weights (q,t). The central element $\gamma=(-1,-1)\in T$ generates Γ, and for any T-equivariant sheaf $\mathcal{F}$ on a space with trivial γ-action we write $\mathcal{F}=\mathcal{F}^+\oplus\mathcal{F}^-$ for the γ-parity decomposition. On local classes, parity is the parity of the total (q,t)-degree. We write $\Lambda=\mathbb{C}[x,y]\rtimes\Gamma$ for the skew group algebra, and S_0, S_1 for its even and odd one-dimensional simple modules at the origin, hence x and y act by zero on S_0 and S_1.

For type B_n we normalize the reflection data such that, up to nonzero scalars,

$$I^{(m)}=\bigcap_{i=1}^n(x_i,y_i)^m\cap\bigcap_{i<j}(x_i-x_j,\,y_i-y_j)^m\cap\bigcap_{i<j}(x_i+x_j,\,y_i+y_j)^m,\qquad \Delta=\prod_{i=1}^n x_i\prod_{i<j}(x_i^2-x_j^2),\tag{6}$$

hence $\deg_q\Delta=n^2$. We write $A=S^W$, $\mathfrak{m}=A_+$, $e_m=\frac{1}{|W|}\sum_w\det(w)^mw$, $e=e_0$, $e_-=e_1$. Since W is real, $e(\Delta^mf)=\Delta^m\,e_m(f)$, hence the multiplication by Δ^m identifies

$$\mathfrak{S}^{\mathrm{symb}}_{n,m}:=\frac{e_mI^{(m)}}{\mathfrak{m}\,e_mI^{(m)}}\xrightarrow[\cdot\Delta^m]{\sim}\widetilde{\mathfrak{S}}^{\mathrm{symb}}_{n,m}:=\frac{e(\Delta^mI^{(m)})}{\mathfrak{m}\,e(\Delta^mI^{(m)})},\tag{7}$$

a bigrading shift by q^{mn^2}.

2.2. **The generic and wall quiver varieties.** Let $Y_n=n\Gamma\text{-}\operatorname{Hilb}(\mathbb{A}^2)$ be the Γ-equivariant Hilbert scheme of clusters $Z\subset\mathbb{A}^2$ with $\mathbb{C}[x,y]/I_Z\cong\mathbb{C}[\Gamma]^{\oplus n}$ as Γ-modules. Equivalently, Y_n is the Nakajima quiver variety $\mathfrak{M}_\theta((n,n),(1,0))$ for the framed affine A_1 McKay quiver at the cyclic-chamber stability θ [53, 22]. It is smooth, symplectic, of dimension $2n$ [65], with universal cluster $\eta:\mathcal{U}_n\to Y_n$ finite flat of degree $2n$ and cluster algebra bundle

$$\mathcal{P}_n=\eta_*\mathcal{O}_{\mathcal{U}_n}=\mathcal{R}_0\oplus\mathcal{R}_1,\qquad \operatorname{rk}\mathcal{R}_0=\operatorname{rk}\mathcal{R}_1=n,\qquad \mathcal{R}_0=\mathcal{O}_{Y_n}\cdot1\oplus\mathcal{T}_n,\quad \mathcal{T}_n=\ker(\mathrm{Tr}_{\mathcal{R}_0}).$$

The splitting uses $\mathrm{Tr}_{\mathcal{R}_0}(1)=n$, and $\mathcal{T}_n$ is a vector bundle of rank $n-1$.

Let $\theta_0 = (0,1)$ be the wall character and $Q_n := \mathfrak{M}_{\theta_0}((n,n),(1,0))$ (with its reduced structure), and let

$$\rho : Y_n \longrightarrow Q_n$$

be the VGIT morphism. Both spaces map to $\mathrm{Sym}^n(X) = (\mathbb{A}^2)^n/(\mu_2 \wr S_n)$. We write $\pi_Q : Q_n \to S_n := \operatorname{Spec} A$ for the support-cycle morphism, $\mathfrak{m}_0$ for the ideal of the origin, and

$$\mathcal{Z}_n^{\mathrm{symb}} := Q_n \times_{S_n} \operatorname{Spec}(A/\mathfrak{m}_0), \qquad F_n := Y_n \times_{S_n} \operatorname{Spec}(A/\mathfrak{m}_0)$$

for the scheme-theoretic central fibers downstairs and upstairs. Below we collect known facts about the spaces from above.

Proposition 2.1.

(a) [21, Thm. 1.1] *The reduced scheme Q_n is isomorphic to the reduced Quot scheme $\mathrm{Quot}_X^n(R_1)_{\mathrm{red}}$ of colength-n quotients of the odd McKay module. It is irreducible and normal with symplectic, hence rational Gorenstein, singularities, and it admits a projective symplectic resolution.*

(b) [21, Prop. 5.3] *The VGIT morphism ρ is such a resolution, that is, it is projective, symplectic and birational.*

(c) [51, Prop. A.1, Def. 3.6, Thm. 3.8] *There is a torus-equivariant isomorphism $Q_n \cong X_{\mathfrak{g},\mathrm{symb}}$ for $\mathfrak{g}$ of type B_n/C_n, identifying $\mathcal{O}_{Q_n}(m)$ with the sheaf whose module of global sections is $e(\Delta^m I^{(m)})$, Q_n is normal and*

$$\Gamma(Q_n, \mathcal{O}_{Q_n}(m)) \;=\; e(\Delta^m I^{(m)}), \qquad \pi_{Q*}\mathcal{O}_{Q_n}(m) \text{ is the sheaf associated to } e(\Delta^m I^{(m)}). \tag{8}$$

(d) *The polarization pulls back to the determinant of the odd tautological bundle, namely*

$$\mathcal{L}_n := \rho^* \mathcal{O}_{Q_n}(1) = \det \mathcal{R}_1.$$

Proof. (a)–(c) are the cited results. Let us give more details for (c). The two-vertex cyclic quiver of [51, Prop. A.1] with dimension vector (n,n) at the wall stability is the framed affine A_1 McKay quiver at θ_0. Kivinen's parameter computation gives Gordon's stability $(0,-2c)$ in the equal-parameter case, and quiver varieties are isomorphic under $\theta \mapsto -\theta$. Hence we can use the identification of the section rings in [51, Thm. 4.3(2)] and the normality statement [51, Thm. 3.8]. Let us also notice that the symbolic Rees algebra need not be known to be finitely generated for (8), since $e(\Delta^m I^{(m)})$ is the space of sections of a unique line bundle on the normal Q_n, as observed in [51].

For (d), note that on the Quot side $\mathcal{O}(1)$ is the determinant line bundle of the tautological quotient, whose pullback is $\det(\mathcal{R}_1)$. Equivalently, on the GIT side, $\theta_0 = (0,1)$ is the determinant character of the odd vertex. □

2.3. **The two punctual quotients.** Two finite-dimensional quotients of $e_m J^m$ occur in the literature and we compare them below. Recall the notation $\mathfrak{m}_S = S_+ \subset S$ for the maximal ideal of the origin. For a W-representation χ let e_χ be the corresponding idempotent, for example $e_m = e_{\det^m}$.

Definition 2.2. For a W-stable ideal $\mathcal{J} \subseteq S$ set

$$\mathsf{Q}_m^{\mathrm{inv}}(\mathcal{J}) := \frac{e_m \mathcal{J}}{\mathfrak{m} \cdot e_m \mathcal{J}}, \qquad \mathsf{Q}_m^{\mathrm{full}}(\mathcal{J}) := e_m\left(\frac{\mathcal{J}}{\mathfrak{m}_S \mathcal{J}}\right) = \frac{e_m \mathcal{J}}{e_m(\mathfrak{m}_S \mathcal{J})}.$$

Stump's $\mathrm{Cat}^{(m)}$ is $\dim_{q,t} \mathsf{Q}_m^{\mathrm{full}}(J^m)$ [74, Def. 3.8, Thm. 3.5], while the geometric arguments of the paper compute $\mathsf{Q}_m^{\mathrm{inv}}$. Since $\mathfrak{m} \cdot e_m \mathcal{J} \subseteq e_m(\mathfrak{m}_S \mathcal{J})$, there is a canonical bigraded surjection

$$\pi_m : \; \mathsf{Q}_m^{\mathrm{inv}}(\mathcal{J}) \twoheadrightarrow \mathsf{Q}_m^{\mathrm{full}}(\mathcal{J}). \tag{9}$$

Lemma 2.3 (Sandwich)**.** *Fix n, and let $m \gg 0$ be in the range of Theorem 9.2 and Theorem 7.5. Then π_m is an isomorphism for $\mathcal{J} = J^m$, and*

$$\mathrm{Cat}^{(m)}(B_n; q, t) = \dim_{q,t} \mathsf{Q}_m^{\mathrm{full}}(J^m) = \dim_{q,t} \mathsf{Q}_m^{\mathrm{inv}}(J^m) = P_{n,m}(q,t).$$

Proof. By Theorem 9.2, $e_m J^m = e_m I^{(m)}$, and by Theorem 7.5, we have $\dim \mathsf{Q}_m^{\mathrm{inv}}(I^{(m)}) = P_{n,m}(1,1) = \binom{n(m+1)}{n}$. By (9), we have $\dim \mathsf{Q}_m^{\mathrm{full}}(J^m) \le \binom{n(m+1)}{n}$. On the other hand Stump's surjection [74, Thm. 4.5 and §4.2.1] (reproduced as Theorem 9.1 below) maps $\mathsf{Q}_m^{\mathrm{full}}(J^m)$, up to the determinant twist, *onto* $\mathrm{gr}_F\, eL_{\frac{1}{2n}+m}(\mathrm{triv})$, whose dimension is $\binom{n(m+1)}{n}$ by Theorem 8.11. Hence $\dim \mathsf{Q}_m^{\mathrm{full}}(J^m) \ge \binom{n(m+1)}{n}$, and therefore both dimensions equal $\binom{n(m+1)}{n}$ and π_m is an isomorphism of bigraded spaces. The common bigraded character is $P_{n,m}$ by Theorem 7.5. □

2.4. **The polarization and the sign component of the Procesi bundle.** Boixeda Alvarez–Losev [16, §2] construct, for any finite Coxeter group W, a partial resolution X of $(\mathfrak{h}\oplus\mathfrak{h}^*)/W$, which is dominated by a $\mathbb{Q}$-factorial terminalization $\widetilde{X}$, together with Procesi sheaves on both spaces. From their construction we see that on X a power of the line bundle $\mathcal{O}^{\mathrm{reg}}(1) := \widetilde{\mathscr{P}}^{\mathrm{reg}}{}_{\varepsilon_-}$ becomes ample [16, §3.3], where $\widetilde{\mathscr{P}}^{\mathrm{reg}}{}_{\varepsilon_-}$ is the sign-isotypic component of the Procesi sheaf $\widetilde{\mathscr{P}}$ on $\widetilde{X}$, normalized to the unit of the Namikawa–Weyl group. In type B_n they remark that X is the wall quiver variety at $\theta_0 = (0,1)$ [16, Rem. 2.2].

We expand on their remark below. Here $\widetilde{X} = Y_n$ and $\nu := c_1(\widetilde{\mathscr{P}}_n{}_{\varepsilon_-})$, and the statement to be proved is

$$\nu = c_1(\det \mathcal{R}_1) \qquad \text{in } N^1(Y_n/S_n). \tag{10}$$

Definition 2.4 (The two extremal curves)**.** Let s be a reflection of $W(B_n)$ and $y \in S_n = \mathfrak{h}\oplus\mathfrak{h}^*/W$ a point of the codimension-two symplectic leaf attached to its conjugacy class. All formal slices to these leaves are of type A_1 [16, §2.1], hence the fibre of $Y_n \to S_n$ over y is the exceptional curve of the minimal resolution of an A_1 surface germ, a $\mathbb{P}^1$. Write

- C_{sh} for the fibre at the *short* class $s_i : e_i \mapsto -e_i$, along which $n-1$ distinct free Γ-orbits are frozen and one Γ-cluster supported at the origin moves.
- C_{lg} for the fibre at the *long* class (ij) (needs $n \ge 2$), along which $n-2$ free orbits are frozen and two free orbits collide.

Lemma 2.5 (The tautological degrees)**.**

$$\det \mathcal{R}_0 \cdot C_{\mathrm{sh}} = 0, \quad \det \mathcal{R}_1 \cdot C_{\mathrm{sh}} = 1, \quad \det \mathcal{R}_0 \cdot C_{\mathrm{lg}} = \det \mathcal{R}_1 \cdot C_{\mathrm{lg}} = 1.$$

Proof. The fibre of $Y_n \to S_n$ over a point of either leaf is the exceptional curve of the minimal resolution of an A_1 surface germ, hence a *reduced* $\mathbb{P}^1$, since the minimal resolution of a surface rational double point has reduced exceptional divisor, and for A_1 it is a single (-2)-curve. Along either curve the supports stay disjoint throughout. Indeed, on C_{sh} the $n-1$ frozen orbits sit at fixed distinct points of $\mathbb{C}^2/\Gamma$ while the moving cluster stays over the origin. Respectively, on C_{lg} the $n-2$ frozen orbits stay away from the colliding pair.

Hence along either curve the Γ-cluster Z is a disjoint union of a frozen part and a moving part, the sheaf $\mathcal{O}_Z$ splits accordingly, and the frozen summands are constant, of degree 0.

On C_{sh} the moving part is a Γ-cluster Z_0 at the origin, hence $\mathcal{O}_{Z_0} \cong \mathbb{C}[\Gamma] = \rho_0 \oplus \rho_1$ with each isotypic piece one-dimensional. Its even part is $\mathbb{C}\cdot 1$, constant, of degree 0, and its odd part is $\langle x,y\rangle/(I \cap \langle x,y\rangle)$. Writing $V = \langle x,y\rangle$, the curve is $\mathbb{P}(V)$ with $\mathcal{S} := I \cap V$ the tautological sub line bundle, hence the odd part is the tautological quotient $V/\mathcal{S}$ and $\det(V/\mathcal{S}) = \mathcal{O}_{\mathbb{P}^1}(1)$, of degree $+1$.

On C_{lg} the two colliding orbits are free, hence on a small ball $U \subset \mathbb{C}^2/\Gamma$ around their common image the covering is trivial, $\pi^{-1}(U) \cong U \times \Gamma$, and for the length-two subscheme $W_0 \subset U$ determined by the cluster we get $\mathcal{O}_{\widetilde{W}_0} \cong \mathcal{O}_{W_0} \otimes \mathbb{C}[\Gamma]$. Hence $\mathcal{R}_0$ and $\mathcal{R}_1$ both restrict to the tautological bundle $\mathcal{O}^{[2]}$ of $\mathrm{Hilb}^2(U)$. Its determinant has degree $+1$ on the fibre. Indeed, over a point of the diagonal the fibre is $\mathbb{P}(V)$, $V = \mathfrak{m}/\mathfrak{m}^2$, and $\mathcal{O}^{[2]}|_{[I]} = \mathbb{C}[x,y]/I = \mathcal{O} \oplus (\mathfrak{m}/I) = \mathcal{O} \oplus (V/\mathcal{S})$, hence again $\det = \mathcal{O}_{\mathbb{P}^1}(1)$. Different trivializations of $\pi^{-1}(U)$ swap $\rho_0 \leftrightarrow \rho_1$, but both give $+1$, hence nothing depends on the choice. □

Lemma 2.6 (The sign degrees, from [16]). $\nu \cdot C_{\mathrm{sh}} = \nu \cdot C_{\mathrm{lg}} = 1$.

Proof. This is contained in the proof of [16, Lemma 2.5], run there for an *arbitrary* reflection s and hence covering both classes at once. At a point y of the leaf of s they show, by the direct analogue of [60, Prop. 4.1], that $\widetilde{\mathscr{P}}^{\wedge y} = \mathrm{Hom}_{\mathbb{C}\{1,s\}}(\mathbb{C}W, \mathscr{P}^{\wedge 0})$ with $\mathscr{P}$ the Procesi bundle over $(\mathfrak{h} \oplus \mathfrak{h}^*)^s \times T^*\mathbb{P}^1$. Then they argue that the two possibilities for $\mathscr{P}$ are $\mathcal{O} \oplus \mathcal{O}(1)$ and $\mathcal{O} \oplus \mathcal{O}(-1)$. Using results from [60, Section 4.2] they conclude that the restriction of the line bundle $\mathscr{P}\varepsilon_-$ to $X^{\wedge y}$ is $\mathcal{O}(1)$. That is degree $+1$ on the slice $\mathbb{P}^1$, which is C_{sh} or C_{lg} according to the class of s. □

Theorem 2.7 (The polarization). *For every $n \geq 2$,*

$$c_1(\widetilde{\mathscr{P}}_n\varepsilon_-) = c_1(\det \mathcal{R}_1) \qquad \text{in } N^1(Y_n/S_n),$$

an equality of classes, not merely of rays, hence $\widetilde{\mathscr{P}}_n\varepsilon_- \cong \det \mathcal{R}_1 = \rho^*\mathcal{O}_{Q_n}(1)$. Consequently the partial resolution X of [16] is $Q_n = \mathfrak{M}_{\theta_0}$, $\bar{\rho} = \rho$, and $\mathcal{O}_{Q_n}(1)$ is its VGIT ample generator.

Proof. By Theorem 2.5 the matrix of $\{\det \mathcal{R}_0, \det \mathcal{R}_1\}$ against $\{C_{\mathrm{sh}}, C_{\mathrm{lg}}\}$ is $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, of determinant -1. Hence the two curve classes are independent, and they span $N_1(Y_n/S_n)_{\mathbb{Q}}$, which is two-dimensional because $b_2(Y_n) = 2$. Respectively, $\det \mathcal{R}_0, \det \mathcal{R}_1$ form a basis of N^1, hence $\nu = a\, c_1(\det \mathcal{R}_0) + b\, c_1(\det \mathcal{R}_1)$ and pairing with the two curves, Theorem 2.6 gives $b = 1$ and $a + b = 1$, i.e. $(a, b) = (0, 1)$. Since numerical and linear equivalence agree on Y_n ($H^1(\mathcal{O}) = 0$ and Pic is torsion-free), $\widetilde{\mathscr{P}}_n\varepsilon_- \cong \det \mathcal{R}_1$.

The last sentence follows because X is by construction the partial resolution on which a power of $\widetilde{\mathscr{P}}_n\varepsilon_-$ is ample, and $\det \mathcal{R}_1 = \rho^*\mathcal{O}_{Q_n}(1)$ with $\mathcal{O}_{Q_n}(1)$ relatively ample on Q_n (Theorem 2.1(d)). To complete the argument, we use that partial resolutions between Y_n and S_n are classified by the faces of the ample cone of Y_n [16, proof of Prop. 2.1], and the face is determined by the class. □

Remark 2.8. Let us mention some special cases and discuss the situation in type A. At $n = 1$ only C_{sh} exists, $\det \mathcal{R}_0 = \mathcal{O}$, and $Y_1 = T^*\mathbb{P}^1$ with Procesi bundle the McKay tautological bundle $\mathcal{R}_0 \oplus \mathcal{R}_1$. Then $\widetilde{\mathscr{P}}_1\varepsilon_- = \mathcal{R}_1$ and $\nu = c_1(\det \mathcal{R}_1)$ directly. For Γ trivial we have $W = S_n$ and $Y_n = \mathrm{Hilb}^n(\mathbb{C}^2)$, only C_{lg} exists, and the same argument gives $\nu = c_1(\det \mathcal{R})$. Thus we recover Haiman's $\mathscr{P}\varepsilon_- = \bigwedge^n \mathcal{B}$.

3. The Haiman complex and the dual-socle identity

In this section we introduce the type B analogue $\mathcal{K}_n$ of the Koszul complex which Haiman used to resolve the punctual Hilbert scheme, and we prove parts (1) and (2) of Theorem E. The complex is built from the trace-zero part $\mathcal{T}_n$ of the even cluster bundle together with the two universal coordinates x, y. Its zeroth homology is the structure sheaf of the scheme-theoretic punctual fiber F_n (Theorem 3.4). The new feature, compared with type A, is that $\mathcal{K}_n$ has $n + 1$ generators against the expected codimension n, hence it is not a resolution.

The main idea of the section is that this failure is very controlled. The homology of $\mathcal{K}_n$ is supported on F_n, which has codimension at least n by semismallness, and a general local-cohomology argument (Theorem 3.8) shows that only two homology sheaves survive, namely $\mathcal{I}_n = \mathcal{O}_{F_n}$ and the excess sheaf $\mathcal{M}_n$ in degree -1. Koszul duality then identifies $\mathcal{M}_n$ with an Ext sheaf of the dual-socle sheaf $\mathcal{S}_n$, whose fibers are the duals of the socles of the cluster algebras. This dual-socle formula is what makes the excess computable in Section 5.

3.1. **Conventions.** We use cohomological indexing, $(A[r])^i = A^{i+r}$ and $\mathcal{H}^i(A[r]) = \mathcal{H}^{i+r}(A)$, hence a sheaf $F[1]$ sits in degree -1. For a vector bundle V on a scheme X and a map $s : V \to \mathcal{O}_X$ the Koszul complex is

$$K_X(V, s)^{-j} = \Lambda^j V, \qquad d = \iota_s,$$

in degrees $[-\operatorname{rk} V, 0]$. If the operators act on a locally free module P rather than on $\mathcal{O}_X$ we write $K_P(V \to P)$ for the complex with terms $P \otimes \Lambda^j V$. $R\mathcal{H}om_X$ denotes *internal* derived Hom.

For a perfect complex, Tor-amplitude $[u,v]$ means that its derived tensor product with every quasicoherent module has cohomology only in $[u,v]$. Equivalently, locally it admits a finite locally free representative in those degrees. This is stronger than having its own cohomology in $[u,v]$.

3.2. **The complex.** Recall from Section 2.2 the splitting $\mathcal{R}_0 = \mathcal{O}_{Y_n}\cdot 1 \oplus \mathcal{T}_n$, $\mathcal{T}_n = \ker(\mathrm{Tr}_{\mathcal{R}_0})$, $\operatorname{rk}\mathcal{T}_n = n-1$. Set

$$\mathcal{E}_n = \mathcal{T}_n \oplus \mathcal{O}_q \oplus \mathcal{O}_t, \qquad \operatorname{rk}\mathcal{E}_n = n+1,$$

where $\mathcal{O}_q$, $\mathcal{O}_t$ are the trivial line bundles with the T-characters of the coordinates x, y (as ordinary bundles, $\mathcal{E}_n = \mathcal{T}_n \oplus \mathbb{C}x \oplus \mathbb{C}y$). Let $\nu : \mathcal{E}_n \to \mathcal{P}_n$ be the inclusion on $\mathcal{T}_n$ together with the universal sections $x, y \in \mathcal{P}_n$, let $J_n \subseteq \mathcal{P}_n$ be the ideal sheaf generated by $\operatorname{im}\nu$, and let

$$\mathcal{K}_n = K_{\mathcal{P}_n}(\mathcal{E}_n \to \mathcal{P}_n): \qquad \mathcal{K}_n^{-a} = \mathcal{P}_n \otimes \textstyle\bigwedge^a \mathcal{E}_n, \quad 0 \le a \le n+1,$$
$$d(p \otimes e_1 \wedge \cdots \wedge e_a) = \sum_i (-1)^{i-1} p\,\nu(e_i) \otimes e_1 \wedge \cdots \widehat{e_i} \cdots \wedge e_a.$$

This is the direct A_1-McKay analogue of the complex Haiman used to resolve the punctual fiber in type A [42, Prop. 2.4, eq. (34)]. The main difference is that here $\operatorname{rk}\mathcal{E}_n = n+1$ exceeds the expected codimension n by one, and exactness fails. Define

$$\mathcal{I}_n = \mathcal{H}^0(\mathcal{K}_n) = \mathcal{P}_n/J_n, \qquad \mathcal{M}_n = \mathcal{H}^{-1}(\mathcal{K}_n), \qquad \mathcal{M}_n = \mathcal{M}_n^+ \oplus \mathcal{M}_n^-, \qquad \mathcal{M}_n^+ = \mathcal{M}_n^\Gamma.$$

We shall also use the *transposed* complex. The dual $\mathcal{P}_n^\vee$ is a $\mathcal{P}_n$-module through $(a\phi)(b) = \phi(ab)$, and

$$\mathcal{K}_n^T = K_{\mathcal{P}_n^\vee}(\mathcal{E}_n \to \mathcal{P}_n^\vee), \qquad \mathcal{S}_n = \mathcal{H}^0(\mathcal{K}_n^T) = \mathcal{P}_n^\vee/J_n\mathcal{P}_n^\vee, \qquad \mathcal{N}_n = \mathcal{H}^{-1}(\mathcal{K}_n^T).$$

The sheaf $\mathcal{S}_n$ is the *dual-socle sheaf*. When n is fixed we drop the subscript n from all of these symbols.

Remark 3.1 (Equivariance). The parity summands are selected in the T-equivariant Koszul complex. In particular, the generator γ of Γ acts trivially on Y_n, hence the parity decomposition is a decomposition of complexes of $\mathcal{O}_{Y_n}$-modules. Where it is convenient we suppress the torus linearizations and compute with ordinary line bundles. In particular, $\det\mathcal{E}_n = \det\mathcal{T}_n$ as ordinary line bundles. The two coordinate lines contribute the constant character qt, whose central Γ-character is trivial. The vanishing $R\rho_*\mathcal{M}_n^+ = 0$ for ordinary coherent sheaves implies the equivariant statement, since forgetting equivariance is conservative.

3.3. **Punctual support and the zeroth homology.** In this subsection we collect the most basic facts about the Haiman complex $\mathcal{K}_\bullet$.

Lemma 3.2. *Let $z \in Y$ be a closed point and $E = \mathcal{P}_z$ the corresponding cluster algebra. Then $E/J_zE \neq 0$ if and only if E is supported at the origin. Consequently all cohomology sheaves of $\mathcal{K}$ and of $\mathcal{K}^T$ are supported on the set $|F_n|$.*

Proof. A point in the support of E/J_zE is killed by x, y, hence it is the origin. For every even polynomial f vanishing at the origin, the element $f - \frac{1}{n}\mathrm{Tr}_{E_0}(f)\,1$ belongs to the trace-zero space $\mathcal{T}_z$, and its value in the residue field at the origin is $-\frac{1}{n}\mathrm{Tr}_{E_0}(f)$. Thus a nonzero quotient forces all such traces to vanish. If E has a nonzero support orbit, choose an invariant polynomial f which is zero at the origin and at the other support orbits and equals one at the chosen orbit. Multiplication by f on each local factor has trace equal to its residue value times the dimension, the nilpotent part contributing zero. Hence the chosen free orbit contributes a strictly positive integer to $\mathrm{Tr}_{E_0}(f)$, a contradiction. Conversely, for a punctual algebra the trace of multiplication by $a \in E_0$ is n times its residue, hence $\mathcal{T}_z$, x and y lie in the maximal ideal and the quotient is nonzero.

Away from this locus, $J = \mathcal{P}$ locally, and an expression $1 = \sum a_i f_i$ gives the standard contracting homotopy of the Koszul complex. Transposing the commuting multiplication operators gives the same contraction for $\mathcal{K}^T$. □

Lemma 3.3 (Power sums generate the symmetric tensors)**.** *Let C be a commutative $\mathbb{C}$-algebra with an augmentation $a : C \to \mathbb{C}$, and for $h \in C$ let $p(h) = \sum_{j=1}^{n} h^{(j)} \in (C^{\otimes n})^{S_n}$, where $h^{(j)}$ is h in the j-th factor and 1 elsewhere. Then $(C^{\otimes n})^{S_n}$ is generated as an algebra by the $p(h)$, and the kernel of the augmentation $a^{\otimes n}$ is generated as an ideal by the $p(h)$ with $a(h) = 0$.*

Proof. For $1 \le s \le n$ put $D_s(h_1, \dots, h_s) = \sum h_1^{(j_1)} \cdots h_s^{(j_s)}$, the sum over pairwise distinct $j_1, \dots, j_s$. In the product $p(h_1)D_{s-1}(h_2, \dots, h_s)$ the position assigned to h_1 is either distinct from all others or agrees with exactly one of them, hence

$$p(h_1)D_{s-1}(h_2, \dots, h_s) = D_s(h_1, \dots, h_s) + \sum_{b=2}^{s} D_{s-1}(h_2, \dots, h_1 h_b, \dots, h_s),$$

and induction on s expresses every D_s in power sums. Every invariant tensor is a linear combination of symmetrizations of pure tensors (average over S_n), and the symmetrization of $h_1 \otimes \cdots \otimes h_n$ is $D_n(h_1, \dots, h_n)/n!$. This proves generation. Writing $h = a(h)1 + h_+$ with $a(h_+) = 0$ gives $p(h) = na(h) + p(h_+)$. Hence modulo the ideal generated by the $p(h_+)$ every generator is a scalar, the quotient is a quotient of $\mathbb{C}$, and the augmentation shows it is $\mathbb{C}$. □

With $C = R_0$ and $a(f) = f(0)$ this describes $\mathcal{O}(\mathrm{Sym}^n X) = (R_0^{\otimes n})^{S_n}$ and its maximal ideal $\mathfrak{m}_0 = (p(f) : \ f \in R_0, \ f(0) = 0)$ at $n[0]$.

Proposition 3.4. *There is a canonical T-equivariant isomorphism of $\mathcal{O}_{Y_n}$-algebras $\mathcal{I}_n \cong \mathcal{O}_{F_n}$, where $F_n = Y_n \times_{S_n} \{0\}$ is the scheme-theoretic punctual fiber. In particular $\mathfrak{m}_0 \mathcal{I}_n = 0$.*

Proof. Step 0. Since $\mathcal{U}_n \subseteq Y_n \times \mathbb{A}^2$, we have $\mathcal{P}_n/(x,y)\mathcal{P}_n = \eta_* \mathcal{O}_{\mathcal{U}_n^0}$ with $\mathcal{U}_n^0 := \mathcal{U}_n \cap (Y_n \times \{0\})$. This sheaf is $\mathcal{O}_{Y_n}$-cyclic, generated by the image of 1, hence equals $\mathcal{O}_{Y_n}/\mathfrak{b}$ for the ideal $\mathfrak{b} = \mathrm{Ann}(\eta_* \mathcal{O}_{\mathcal{U}_n^0})$. Note that $\mathfrak{b} \neq 0$, since fiberwise $E/(x,y)E \neq 0$ iff $0 \in \mathrm{Supp}\, E$, which is a codimension-two condition.

Step 1: $\mathfrak{b} \subseteq \mathfrak{m}_0 \mathcal{O}_{Y_n}$. Equivalently $\mathfrak{b} \cdot \mathcal{O}_{F_n} = 0$, i.e. the natural surjection $\mathcal{O}_{F_n} \twoheadrightarrow (\mathcal{P}_n/(x,y)\mathcal{P}_n)|_{F_n}$ is injective. For $y \in F_n$ the fiber E is the local Artinian algebra of a *punctual* cluster, with maximal ideal $(x,y)E$, hence $(\mathcal{P}_n|_{F_n}/(x,y)) \otimes k(y) = E/(x,y)E = \mathbb{C}$. Thus the displayed surjection of coherent sheaves has kernel with vanishing fibers, and by Nakayama the kernel is zero.

Step 2: the trace ideal. Let $\sigma : \mathcal{T}_n \to \mathcal{P}_n/(x,y)\mathcal{P}_n$ be the composite of the inclusion with the projection, and therefore $\mathcal{I}_n = \mathcal{O}_{Y_n}/(\mathfrak{b} + \widetilde{\mathrm{im}}\, \sigma)$ for any lift $\widetilde{\mathrm{im}}\, \sigma \subseteq \mathcal{O}_{Y_n}$ of its image. For a positive-degree invariant $f \in (R_0)_+$ let $\bar{f} \in \mathcal{R}_0$ be its universal image and $\tau_f := \mathrm{Tr}_{\mathcal{R}_0}(m_{\bar{f}}) \in \mathcal{O}_{Y_n}$. Then $\bar{f}^{\,0} := \bar{f} - \frac{1}{n}\tau_f \cdot 1 \in \mathcal{T}_n \subseteq J_n$, while $\bar{f} \in (x,y)\mathcal{P}_n \subseteq J_n$ (positive degree). Subtracting, σ takes the value $-\frac{1}{n}\tau_f$ on $\bar{f}^{\,0}$, and every element of $\mathcal{T}_n$ is an $\mathcal{O}_{Y_n}$-combination of such. Hence $\mathfrak{b} + \widetilde{\mathrm{im}}\, \sigma = \mathfrak{b} + (\tau_f : \ f \in (R_0)_+)$. Write $\mathfrak{a}$ for the latter ideal.

By the equivariant form of Haiman's trace identity [42, eq. (28)] (take Γ-invariants along the fixed locus $Y_n \subset \mathrm{Hilb}^{2n}(\mathbb{A}^2)$), the function τ_f is the pullback, under the support-cycle morphism $Y_n \to \mathrm{Sym}^n(X)$, of the polarized power sum attached to f. (The identity can also be checked directly. On the dense open locus of n disjoint free Γ-orbits, $\mathcal{R}_0$ is a sum of n one-dimensional algebras and $\mathrm{Tr}(m_{\bar{f}})$ is the sum of the values of f at the orbits, and both sides are regular functions on the reduced irreducible Y_n.) The positive-degree polarized power sums generate the maximal ideal $\mathfrak{m}_0$ of $n[0] \in \mathrm{Sym}^n(X)$ (Theorem 3.3). Hence $(\tau_f) = \mathfrak{m}_0 \mathcal{O}_{Y_n}$, and with Step 1, $\mathfrak{a} = \mathfrak{b} + (\tau_f) = \mathfrak{m}_0 \mathcal{O}_{Y_n}$, hence $\mathcal{I}_n = \mathcal{O}_{F_n}$. □

Remark 3.5 (Generator count). Step 1 is not a formality. The map σ has locally free source of rank $n-1$, hence $\mathrm{im}\, \sigma$ is locally $(n-1)$-generated, whereas $\mathrm{codim}\, F_n = n$. The ideal of F_n can never be generated by $n-1$ elements, and the missing generators come from $\mathfrak{b}$, the ideal of the divisor of clusters through the origin. In the Koszul complex $\mathcal{K}_n$ the ideal $\mathfrak{b}$ is accounted for by the two sections x, y of $\mathcal{P}_n$, hence J_n has $n+1$ generators for a height-n ideal. This deviation by one is the source of $\mathcal{M}_n \neq 0$. In type A the same count gives $n+1$ generators of height $n+1$, a complete intersection.

The following fiberwise form of Theorem 3.4 will be used in the unit normalization of Section 6.

Lemma 3.6 (The incidence quotient is the augmentation line)**.** *Let $\xi \in F_n$ be a closed point, $A_\xi = \mathcal{P}_n \otimes k(\xi)$ the corresponding punctual cluster algebra, $\varepsilon_\xi : A_\xi \twoheadrightarrow \mathbb{C}$ its residue-field augmentation, and $J_{n,\xi} \subset A_\xi$ the specialization of J_n. Then*

$$J_{n,\xi} = \ker \varepsilon_\xi = (x,y)A_\xi, \qquad \textit{hence} \qquad \mathcal{I}_n \otimes k(\xi) = A_\xi / J_{n,\xi} \cong \mathbb{C} \cdot [1],$$

and the unit map $\mathbb{C} \to A_\xi/J_{n,\xi}$, $1 \mapsto [1]$, is the identity under this identification.

Proof. The inclusion $(x,y)A_\xi \subseteq J_{n,\xi}$ holds because x, y are among the generators of J_n. Conversely, let g be a specialized trace-zero even generator and write $g = \varepsilon_\xi(g)1 + g_{\mathrm{nil}}$ with $g_{\mathrm{nil}} \in \ker \varepsilon_\xi$, which is nilpotent. Being even, g_{nil} preserves the even summand $(A_\xi)_0$ and is nilpotent there, hence has trace zero, while the scalar term acts on the n-dimensional even summand as $\varepsilon_\xi(g)$ times the identity. Therefore $\mathrm{Tr}_{(A_\xi)_0}(m_g) = n\,\varepsilon_\xi(g)$. The left side is zero because g is trace-free, hence $\varepsilon_\xi(g) = 0$. Thus every trace-zero even generator lies in $\ker \varepsilon_\xi = (x,y)A_\xi$. □

3.4. **The dimension of the punctual fiber.**

Lemma 3.7. $\dim F_n \le n$.

Proof. The affinization map $Y_n \to \mathrm{Sym}^n(\mathbb{A}^2/\Gamma)$ is a projective symplectic resolution, since Y_n is smooth symplectic and the map is birational onto the symmetric power [53, 22]. Symplectic resolutions are semismall [49, Lemma 2.11], [48, Thm. 1.9], and a proper map $f : Y \to X$ is semismall exactly when $\dim(Y \times_X Y) \le \dim Y$. Since $F_n \times F_n \subset Y_n \times_{\mathrm{Sym}^n(\mathbb{A}^2/\Gamma)} Y_n$, we get $2\dim F_n \le \dim Y_n = 2n$. (Alternatively, the central fiber of a quiver variety over its affinization is Lagrangian by Nakajima [64, Thm. 5.8], see also [31, §§3–4].) Only this inequality is needed, not reducedness or Cohen–Macaulayness of the scheme F_n. □

3.5. **Homological lemmas.**

Lemma 3.8 (Support versus Tor-amplitude)**.** *Let X be a regular Noetherian scheme, let $Z \subset X$ be closed of codimension at least c, and let $A \in \mathrm{Perf}(X)$ have Tor-amplitude $[-m, 0]$. If every $\mathcal{H}^i(A)$ is supported on Z, then $A \in D^{[c-m,0]}(X)$. In particular, if $m = c$, the complex is a sheaf in degree zero.*

Proof. The assertion is local on X. Write $X = \operatorname{Spec} R$, choose an ideal I defining Z, and represent A by finite free modules in degrees $-m, \dots, 0$. Since R is regular, it is Cohen–Macaulay, and the codimension assumption gives $H^r_I(R) = 0$ for $r < c$, and the same for each finite free term. This is Grothendieck's local-cohomology characterization of depth, $\mathrm{depth}_I(R) = \min\{r : H^r_I(R) \neq 0\}$, together with $\mathrm{depth}_I(R) = \mathrm{grade}(I) = \mathrm{ht}(I)$ in a Cohen–Macaulay ring [18, Thm. 3.5.7, Cor. 2.1.4], [73, Tag 0AVZ]. The hypercohomology spectral sequence of $R\Gamma_I$ applied to the bounded complex of free modules A has $E_1^{i,r} = H^r_I(A^i)$, and every nonzero entry has $i \ge -m$ and $r \ge c$, hence the total degree is at least $c - m$. As the cohomology of A is supported on $V(I)$, the natural map $R\Gamma_I(A) \to A$ is an isomorphism (A restricts to zero on $X \setminus Z$). Thus $A \in D^{\ge c-m}$, and the locally free representative gives $A \in D^{\le 0}$. □

Remark 3.9. The hypothesis is Tor-amplitude, not ordinary cohomological amplitude. A sheaf supported in positive codimension has amplitude $[0,0]$ and arbitrary projective dimension. In our applications the amplitude comes from pushing down a regular-section Koszul resolution.

Lemma 3.10 (Low Ext vanishing)**.** *If X is regular and a coherent sheaf N is supported in codimension at least c, then $\mathcal{E}xt^i_X(N, \mathcal{O}_X) = 0$ for $i < c$.*

Proof. Locally N is annihilated by an ideal of grade at least c. The grade $\mathrm{grade}(\mathrm{Ann}\, N, R)$ is the first nonzero index of $\mathrm{Ext}^i_R(N, R)$ [18, §1.2], and over a Cohen–Macaulay ring grade equals height [18, Cor. 2.1.4], i.e. the codimension of the support. □

Lemma 3.11 (Koszul duality, including shifts)**.** *Let V have rank r, and let $K_X(V,s)$ be in degrees $[-r,0]$. Then*

$$R\mathcal{H}om_X(K_X(V,s),\mathcal{O}_X) \simeq K_X(V,s)[-r]\otimes(\det V)^{-1}. \tag{11}$$

If the Koszul operators act on a locally free module P rather than on $\mathcal{O}_X$, the right-hand complex has coefficient module $P^\vee$ and the transposed operators.

Proof. In dual cohomological degree j the term is $(\Lambda^j V)^\vee$, which the perfect exterior pairing identifies with $\Lambda^{r-j}V\otimes(\det V)^{-1}$, the degree-$j$ term of the right-hand side. Under this pairing the transpose of contraction becomes contraction on the complementary exterior power up to sign. Rescaling the degree-j identification by $(-1)^{\binom{j}{2}}$, together with the shift sign, makes the differentials agree. With coefficients, the same calculation transposes the multiplication operators. □

Lemma 3.12 (Canonical modules are semidualizing)**.** *Let B be Cohen–Macaulay with a dualizing module ω_B, and let L be invertible. For $\Sigma=\omega_B\otimes L$, $R\mathcal{H}om_B(\Sigma,\Sigma)\simeq\mathcal{O}_B$.*

Proof. A defining property of a dualizing complex is $\mathcal{O}_B\simeq R\mathcal{H}om_B(\omega_B^\bullet,\omega_B^\bullet)$, and on a Cohen–Macaulay scheme the complex is locally a single canonical module with a shift, which cancels in its self-Hom. Concretely, for a Cohen–Macaulay local ring with canonical module ω one has $\mathrm{Hom}(\omega,\omega)=R$ and $\mathrm{Ext}^i(\omega,\omega)=0$ for $i>0$ [18, Thm. 3.3.10], and tensoring both entries with L does not change internal derived Hom. See also [73, Tags 0A7A, 0DW6]. □

3.6. **Only two homology sheaves.**

Theorem 3.13. *$\mathcal{K}_n,\mathcal{K}_n^T\in D^{[-1,0]}(Y_n)$. In other words, the only homology sheaves of $\mathcal{K}_n$ are $\mathcal{H}^0=\mathcal{I}_n$ and $\mathcal{H}^{-1}=\mathcal{M}_n$, and those of $\mathcal{K}_n^T$ are $\mathcal{S}_n$ and $\mathcal{N}_n$.*

Proof. Both complexes have locally free terms in $[-n-1,0]$ on the smooth $2n$-dimensional Y_n, and by Theorem 3.2 their cohomology is supported on F_n, of codimension at least n by Theorem 3.7. Apply Theorem 3.8 with $(m,c)=(n+1,n)$. □

3.7. **The dual-socle formula.**

Theorem 3.14 (Dual-socle formula)**.** *There is a canonical T-equivariant isomorphism*

$$\mathcal{M}_n \;\simeq\; \mathcal{E}xt^n_{Y_n}(\mathcal{S}_n,\mathcal{O}_{Y_n})\otimes\det\mathcal{E}_n. \tag{12}$$

Its even part is

$$\mathcal{M}_n^+ \;\simeq\; \mathcal{E}xt^n_{Y_n}(\mathcal{S}_n^+,\mathcal{O}_{Y_n})\otimes\det\mathcal{T}_n, \tag{13}$$

where the last equality suppresses the constant coordinate character qt.

Proof. Dualize the truncation triangle $\mathcal{N}_n[1]\to\mathcal{K}_n^T\to\mathcal{S}_n\to\mathcal{N}_n[2]$ of Theorem 3.13. By Theorem 3.10 (supports have codimension $\geq n$), the two outer groups in

$$\mathcal{E}xt_Y^{n-2}(\mathcal{N}_n,\mathcal{O}_Y)\longrightarrow\mathcal{E}xt_Y^n(\mathcal{S}_n,\mathcal{O}_Y)\longrightarrow\mathcal{H}^nR\mathcal{H}om_Y(\mathcal{K}_n^T,\mathcal{O}_Y)\longrightarrow\mathcal{E}xt_Y^{n-1}(\mathcal{N}_n,\mathcal{O}_Y)$$

vanish, hence the middle arrow is an isomorphism. By Theorem 3.11, $R\mathcal{H}om_Y(\mathcal{K}_n^T,\mathcal{O}_Y)\simeq\mathcal{K}_n[-n-1]\otimes(\det\mathcal{E}_n)^{-1}$, whose degree-$n$ cohomology is $\mathcal{M}_n\otimes(\det\mathcal{E}_n)^{-1}$. This proves (12). The determinant of the two coordinate lines is Γ-even, as is $\det\mathcal{T}_n$, and taking the even summand proves (13). □

Remark 3.15 (The fiber interpretation). At a punctual closed point z, the dual of $\mathcal{S}_z\otimes k(z)$ is the socle of $E=\mathcal{P}_z$, since $(E^\vee/\mathfrak{m}_EE^\vee)^\vee=\{a\in E:\mathfrak{m}_Ea=0\}$. This is a statement about the cokernel of transposed multiplication, and the sheaf kernel of multiplication on the ambient Y is not a substitute for it.

Since $\mathcal{I}_n$ is γ-even, the parity split of the truncation triangle $\mathcal{M}_n[1] \to \mathcal{K}_n \to \mathcal{I}_n \to$ gives the canonical triangle

$$\mathcal{M}_n^+[1] \longrightarrow \mathcal{K}_n^+ \longrightarrow \mathcal{I}_n \xrightarrow{+1}, \tag{14}$$

and $\mathcal{K}_n^- \simeq \mathcal{M}_n^-[1]$.

Remark 3.16. The bound of Theorem 3.13 is sharp, that is, $\mathcal{M}_n^+ \neq 0$ for every $n \geq 2$. The balanced complete intersection $S/(xy,\ x^2+y^{2n-2})$ has basis $1, x, y, y^2, \dots, y^{2n-2}$, with n even and n odd basis elements, and its socle is the even line spanned by y^{2n-2}. Theorem 5.26 shows that $\mathcal{M}_n^+$ is nonzero at this point and is Cohen–Macaulay of pure dimension n. Thus, unlike type A, the incidence is never a complete intersection compatible with the natural generators, and Sections 4 and 5 are the management of the excess $\mathcal{M}_n^+$. For $n = 1$, $\mathcal{M}_1^+ = 0$ (Theorem 5.1).

Parts (1) and (2) of Theorem E are Theorems 3.4, 3.13 and 3.14.

4. Wall fibers, length duality, and the local model

In this section we describe the geometry of the contraction $\rho : Y_n \to Q_n$ over the torus-fixed points of the wall. The fiber over a fixed point z_J is a Grassmannian $\mathrm{Gr}(k, V_J)$ (Theorem 4.1), and the length duality $\dim V_J = 2k-1$ is the combinatorial fact which makes these Grassmannians special. We then compute the cluster bundle on such a fiber, the polystable representative and the Ext-quiver of a wall point, and we deduce from the theorem of Bellamy and Craw a local model of ρ at a wall point, namely the Springer resolution $T^*\,\mathrm{Gr}(r+1, 2r+1) \to \mathfrak{N}_{r,2r+1}$ times a smooth core. The last subsection recalls the Brill–Noether theory of González, Gorsky and Simental for the socle strata and derives the threshold $n \geq r(r+1)$ for the even socle rank r.

The main idea is that everything is governed by the even socle of the cluster algebras. Its rank jumps along the wall fibers, the tautological quotient bundle of the Grassmannian is the varying socle, and the local model is determined by the dimension $2r+1$ of the extension space. The results of this section are used in Section 6, where the positivity of the tautological quotient bundle drives the symbolic descent argument. The Grassmannian acyclicity of Theorem 4.5 is the fixed-point shadow of the vanishing theorem of Section 5, and it explains at a single wall fiber why the excess sheaf is invisible to the wall.

4.1. Hearts and Grassmannian fibers.

A T-fixed point $z_J \in Q_n^T$ corresponds to a monomial submodule $J \subset R_1$ with $\dim_{\mathbb{C}} R_1/J = n$. Its *heart* is the order ideal $H = H(J) = \{x^a y^b \notin J\}$ of n odd lattice points. Define

$$J' = R_1 J \subseteq R_0, \qquad J'' = (J : R_1) = \{f \in R_0 :\ xf, yf \in J\} \subseteq R_0, \qquad V_J = J''/J'.$$

Theorem 4.1 (Grassmannian fibers and length duality)**.**

(a) $\mathfrak{m}_X V_J = 0$, *and consequently every $\mathbb{C}$-subspace of V_J is an R_0-submodule.*

(b) *With $k = k_J := \mathrm{length}(R_0/J') - n$, the reduced fiber is*

$$F_J := \rho^{-1}(z_J) \ \cong\ \mathrm{Gr}(k, V_J),$$

where a point of the fiber is a balanced ideal $I = I^+ \oplus J$ with even part $J' \subseteq I^+ \subseteq J''$, and $W = I^+/J' \subset V_J$ is an arbitrary k-plane.

(c) (*Length duality*) $\mathrm{length}(R_0/J') + \mathrm{length}(R_0/J'') = 2n+1$, *hence*

$$N := \dim V_J = 2k-1,$$

and every positive-dimensional fixed fiber has $k \geq 2$.

Proof. (a) $R_1 J'' \subseteq J$, hence $\mathfrak{m}_X J'' = R_1(R_1 J'') \subseteq R_1 J = J'$.

(b) A cluster in the fiber has odd part exactly J, because the wall parameter $\theta_0 = (0,1)$ fixes the odd quotient $R_1 \twoheadrightarrow R_1/J$, and ρ remembers exactly this datum (Theorem 2.1(a)). The even

part I^+ must satisfy $R_1 I^+ \subseteq J$ (ideal condition), i.e. $I^+ \subseteq J''$, and $I^+ \supseteq R_1 J = J'$. Conversely, by (a) any intermediate subspace is an ideal. The colength condition $\operatorname{length}(R_0/I^+) = n$ becomes $\dim(I^+/J') = \operatorname{length}(R_0/J') - n = k$.

(c) Let C and D be the monomial bases of R_0/J' and R_0/J''. Directly from the definitions,

$$C = \Big\{ e \text{ even}: \ (a{=}0 \text{ or } e{-}(1,0) \in H) \text{ and } (b{=}0 \text{ or } e{-}(0,1) \in H) \Big\},$$

$$D = \Big\{ f \text{ even}: \ f{+}(1,0) \in H \text{ or } f{+}(0,1) \in H \Big\}.$$

Let $h_x = |H \cap \{b = 0\}|$, $h_y = |H \cap \{a = 0\}|$, let P be the number of even points (a, b), $a, b \geq 1$, whose two odd predecessors $(a{-}1, b)$, $(a, b{-}1)$ both lie in H, and let Q be the number of even points whose two odd successors both lie in H. Every element of C is the origin, or a boundary point $(a, 0)$ with $(a{-}1, 0) \in H$ (there are h_x of these), or $(0, b)$ with $(0, b{-}1) \in H$ (h_y of them), or an interior point counted by P, hence $|C| = 1 + h_x + h_y + P$. For D, count the even points with odd successor in H in the x-direction. These biject with $H \setminus \{a = 0\}$, giving $n - h_y$, and similarly there are $n - h_x$ in the y-direction. The double-counted points are exactly those counted by Q, hence $|D| = 2n - h_x - h_y - Q$. Translation by $(-1, -1)$ is a bijection between P-configurations and Q-configurations, hence $P = Q$ and $|C| + |D| = 2n + 1$. Then $|D| = n - k + 1$ and $N = |C| - |D| = 2k - 1$. If $\dim F_J > 0$ then $0 < k < N$, i.e. $k \geq 2$. □

Remark 4.2. Exact enumeration through rank 16 shows that the positive-dimensional fixed fibers occurring in Q_n are precisely of the form $\operatorname{Gr}(m, 2m-1)$, $m \geq 2$, with $\operatorname{Gr}(m, 2m-1)$ appearing first at $n = m(m-1)$, where $\dim \operatorname{Gr}(m, 2m-1) = m(m-1) = n$. Thus the first occurrence is an entire Lagrangian irreducible component of F_n contracted to a point. See Section A for the census through $n = 12$. The thresholds $n \geq m(m-1) = r(r+1)$, $r = m - 1$, are those of the Brill–Noether theory of Section 4.6.

4.2. **The cluster bundle, the socle, and the universal extension.** Fix a heart J and write $F_J = \operatorname{Gr}(k, V)$, $V = V_J$, with tautological sequence $0 \to \mathcal{S} \to V \otimes \mathcal{O} \to \mathcal{Q}_J \to 0$ and $\mathcal{O}_{\mathrm{Pl}}(1) = \det \mathcal{Q}_J$ (quotient Plücker convention). Let

$$\mathsf{M}_J := \mathbb{C}[x, y]/(J'' \oplus J), \qquad A_W := \mathbb{C}[x, y]/(I_W^+ \oplus J) \quad (W \subset V,\ I_W^+ = \text{preimage of } W),$$

regarded as modules over $\Lambda = \mathbb{C}[x, y] \rtimes \Gamma$.

Proposition 4.3.

(a) $\mathcal{P}_n|_{F_J} \cong ((R_0/J'') \oplus (R_1/J)) \otimes \mathcal{O}_{F_J} \oplus \mathcal{Q}_J$. *In particular* $\mathcal{L}_n|_{F_J}$ *is the constant character* $\ell_J = \prod_{s \in H} w(s)$, *and* $\det \mathcal{E}_n|_{F_J} \cong \det \mathcal{T}_n|_{F_J} \otimes qt \cong \mathcal{O}_{\mathrm{Pl}}(1) \otimes \chi_J$ *for a fixed character* χ_J.

(b) $\operatorname{soc}_0(A_W) = J''/I_W^+ = V/W$, *and globally* $\operatorname{soc}_0(\mathcal{P}_n|_{F_J}) \cong \mathcal{Q}_J$. (*This holds with no Gorenstein hypothesis. On the non-Gorenstein strata the even socle simply has rank* > 1.)

(c) *There is a universal extension of families of* Λ*-modules*

$$0 \longrightarrow S_0 \otimes \mathcal{Q}_J \longrightarrow \mathcal{P}_n|_{F_J} \longrightarrow \mathsf{M}_J \otimes \mathcal{O}_{F_J} \longrightarrow 0. \tag{15}$$

Proof. At W, the even part splits T-equivariantly as $R_0/I_W^+ \cong R_0/J'' \oplus J''/I_W^+$ (choose the monomial complement, the splitting is canonical as a T-module and the extension is by the $\mathfrak{m}_X$-torsion piece), and $J''/I_W^+ = V/W$, while the odd part is the constant R_1/J. This gives (a), and the determinant statements follow since $\det \mathcal{E}_n = qt \cdot \det \mathcal{T}_n$ and the only varying summand of $\mathcal{T}_n|_{F_J}$ is $\mathcal{Q}_J$. For (b), note that $f \in R_0/I_W^+$ is killed by x and y iff $f \in J''/I_W^+$. For (c), the surjection $A_W \twoheadrightarrow \mathsf{M}_J$ has kernel J''/I_W^+ concentrated in even degree and killed by x, y (as $xJ'', yJ'' \subseteq J$), i.e. $S_0 \otimes (V/W)$, and this globalizes. □

4.3. **The wall polystable and the Ext-quiver of a heart.**

Proposition 4.4.

(a) M_J *is generated by* 1 *and has no* Λ*-submodule isomorphic to* S_0*. The* θ_0*-polystable representative of every point of* F_J *is*

$$\mathsf{M}_J \ \oplus \ S_0^{\oplus m}, \qquad m = k-1,$$

and no S_1*-summand ever splits off.*

(b) *There are natural* T*-equivariant isomorphisms*

$$\mathrm{Ext}^1_\Lambda(S_0, \mathsf{M}_J) \cong V_J, \qquad \mathrm{Ext}^1_\Lambda(S_0, S_0) = 0, \qquad \mathrm{Ext}^1_\Lambda(\mathsf{M}_J, S_0) \cong V_J^\vee \otimes (qt).$$

(c) *Under* (b)*, the class of the extension* (15) *at* W *is the quotient map* $V_J \to V_J/W$*, and globally the classifying map of* (15) *is the tautological quotient* $V_J \otimes \mathcal{O}_{F_J} \twoheadrightarrow \mathcal{Q}_J$*.*

Proof. (a) If $\bar f \in (R_0/J'')$ satisfies $x\bar f = y\bar f = 0$ then $f \in (J : R_1) = J''$, hence $\bar f = 0$, and there is no S_0-submodule. For the polystable representative, we use King stability at $\theta_0 = (0,1)$ (framing normalized such that the framed module has slope 0). A subrepresentation not containing the framing vector has θ_0-pairing equal to its odd dimension, which is ≥ 0 with equality iff the subrepresentation is a sum of copies of S_0. Dually, S_1 has θ_0-degree $1 \neq 0$ and can neither destabilize nor split off at the wall. The largest S_0-isotypic submodule of A_W is $\mathrm{soc}_0(A_W) = V/W$ of dimension $k-1$ (Theorem 4.3(b)), and the quotient is the stable framed factor M_J.

(b) Apply $\mathrm{Hom}_\Lambda(-, \mathsf{M}_J)$ to the Γ-equivariant Koszul resolution of S_0,

$$0 \to P_0 \otimes (qt) \xrightarrow{\binom{-y}{x}} (P_1 \otimes q) \oplus (P_1 \otimes t) \xrightarrow{(x \quad y)} P_0 \to S_0 \to 0, \tag{16}$$

$P_i = \Lambda e_i$. The resulting complex is $(\mathsf{M}_J)_0 \to (\mathsf{M}_J)_1^{\oplus 2} \to (\mathsf{M}_J)_0$ with maps $m \mapsto (xm, ym)$ and $(b,c) \mapsto -yb + xc$, whose middle cohomology is the even part of $(J'' \oplus J)/(x,y)(J'' \oplus J)$, namely $J''/(xJ + yJ) = J''/J' = V_J$. The first map is injective by (a), hence Ext^0 contributes nothing in the middle. Taking $\mathsf{M}_J = S_0$ gives zero middle term. The last isomorphism is the 2-Calabi–Yau duality of Λ ($\Gamma \subset SL_2$), whose Serre twist is the symplectic character qt.

(c) The extension of M_J by $S_0 \otimes (V/W)$ determined by A_W corresponds, under the Koszul identification in (b), to the cocycle $J'' \ni g \mapsto g \bmod I_W^+ \in V/W$, i.e. to the quotient map with kernel W, and these globalize to the tautological quotient. □

4.4. **Grassmannian acyclicity.** The following vanishing is the fixed-point shadow of the excess vanishing of Section 5. It is not used in the proof of the vanishing, but it explains the mechanism at the level of a single wall fiber.

Theorem 4.5. *Let* $F = \mathrm{Gr}(k,N)$ *with quotient bundle* $\mathcal{Q}$*,* $\mathcal{O}_{\mathrm{Pl}}(1) = \det \mathcal{Q}$*, and suppose* $N = 2k-1$*. Then*

$$R\Gamma\big(F, \ \mathcal{Q} \otimes \mathcal{O}_{\mathrm{Pl}}(1-N)\big) = 0 \quad \Longleftrightarrow \quad k \geq 2.$$

Proof. In the quotient-block convention, $\mathcal{Q} \otimes \mathcal{O}_{\mathrm{Pl}}(1-N)$ corresponds to the GL_N-weight $\mu = (2-N, \underbrace{1-N, \dots, 1-N}_{N-k-1}, \underbrace{0, \dots, 0}_{k})$. Adding $\varrho = (N-1, \dots, 1, 0)$,

$$\mu + \varrho = \big(1, \ -1, -2, \dots, 1-(N-k), \ k-1, k-2, \dots, 0\big).$$

Since $N - k = k-1$, the final block is $(k-1, \dots, 1, 0)$ and contains the entry 1 exactly when $k \geq 2$, repeating the first entry. A ϱ-shifted weight with a repetition is singular, and Bott's theorem [17] gives vanishing in all degrees. For $k = 1$ the entries are pairwise distinct and exactly one cohomology group (of dimension one) survives. □

4.5. **The local model at a wall point.**

Theorem 4.6 (Bellamy–Craw [8, Thm. 1.4; Thms. 3.2, 3.7; Prop. 3.8, §3.4])**.** *Let $f : \mathfrak{M}_\theta(\mathbf{v},\mathbf{w}) \to \mathfrak{M}_{\theta_0}(\mathbf{v},\mathbf{w})$ be the VGIT morphism of framed McKay quiver varieties for $\Gamma \subset SL_2(\mathbb{C})$, from a generic chamber to a wall. Fix a closed point z with polystable representative y and local Ext-quiver datum $(\mathbf{m},\mathbf{n})$. Then the following hold.*

(a) *étale locally over z, f is the product of the morphism $\mathfrak{M}_\varrho(\mathbf{m},\mathbf{n}) \to \mathfrak{M}_0(\mathbf{m},\mathbf{n})$ of local quiver varieties with the identity of a smooth symplectic factor;*

(b) *the equivalence is realized before GIT, on a common affine H-variety U with étale maps to both moment-map fibers and Cartesian squares before and after quotient. At representation-space level the map is $c \mapsto y + c$, hence the pullback of the global universal representation is the family $y + c$ of representations of the* original *framed preprojective algebra;*

(c) *$p_\theta^*\mathcal{R}_i(\mathbf{v},\mathbf{w}) \cong \bigoplus_j q_\varrho^*\mathcal{R}_j(\mathbf{m},\mathbf{n})^{\oplus\beta_i^{(j)}}$, where $\beta^{(j)}$ run over the dimension vectors of the stable summands of y. The framing summand $\mathcal{R}_\infty$ is trivialized, removing the common line-bundle ambiguity of the universal family.*

Fix a closed point $z \in \mathcal{Z}_n^{\mathrm{symb}}$ of exact even-socle rank r. By Theorem 4.4(a) (the odd socle never splits off at this wall) the polystable representative is

$$E_z = E_{\mathrm{fr}} \oplus S_0^{\oplus r}, \tag{17}$$

where E_{fr} is the stable summand containing the framing. The framing rigidification kills its scalar automorphisms, while the repeated simple factor has stabilizer $H = GL(U)$, $U = \mathbb{C}^r$. Put $V := \mathrm{Ext}^1(S_0, E_{\mathrm{fr}})$. At a heart, $E_{\mathrm{fr}} = \mathsf{M}_J$, $V = V_J$ and $\dim V = 2r+1$ by Theorem 4.1(c) (the dimension is constant along the stratum by the étale-local triviality of the stratification [11]). The 2-Calabi–Yau pairing identifies $\mathrm{Ext}^1(E_{\mathrm{fr}}, S_0) \cong V^\vee$ up to the symplectic character, and $\mathrm{Ext}^1(S_0,S_0) = 0$, that is, there are no loop coordinates at the multiplicity vertex. Applying Theorem 4.6 at z, the local model is

$$\{(A,B,u) :\ AB = 0\}/\!\!/GL(U), \quad A \in \mathrm{Hom}(V,U),\ B \in \mathrm{Hom}(U,V),\ u \in U_z := \mathrm{Ext}^1(E_{\mathrm{fr}}, E_{\mathrm{fr}}), \tag{18}$$

with the invariant-theoretic wall map $(A,B,u) \mapsto (X,u)$, $X := BA \in \mathrm{End}(V)$. The moment-map equation gives $X^2 = B(AB)A = 0$ and $\mathrm{rk}\, X \le r$, hence the transverse wall singularity is the orbit closure

$$\mathfrak{N}_{r,2r+1} := \{X \in \mathrm{End}(V) :\ X^2 = 0,\ \mathrm{rk}\, X \le r\}. \tag{19}$$

The A-surjective chamber, the source of ρ. In the chamber defining Y_n, stability requires A to be surjective. Let $K := \ker A$, of dimension $(2r+1) - r = r+1$. Then A identifies $U \cong V/K$, and $AB = 0$ says $\mathrm{im}\, B \subset K$, hence B becomes a map $V/K \to K$, i.e. a cotangent vector $T^*_K \mathrm{Gr}(r+1,V) = \mathrm{Hom}(V/K, K)$. Hence

$$Y_+^{\mathrm{loc}} \cong T^*\,\mathrm{Gr}(r+1,V) \times (\mathrm{core}), \qquad (K,B) \longmapsto BA. \tag{20}$$

The fiber over $X = 0$ is the zero section, because if $BA = 0$ and A is surjective then $B = 0$. Hence $\rho^{-1}(z) \cong \mathrm{Gr}(r+1,V)$, which is the wall fiber F_J with $k = r+1$, its tautological quotient $V \otimes \mathcal{O} \twoheadrightarrow \mathcal{Q}$ of rank r being the varying even socle (Theorem 4.3). On the standard big cell $V = W_0 \oplus Q_0$, $\dim W_0 = k$, $\dim Q_0 = r$,

$$A_C = (C \ \ I_{Q_0}), \qquad B_{C,D} = \begin{pmatrix} D \\ -CD \end{pmatrix}, \qquad B_{C,D}A_C = \begin{pmatrix} DC & D \\ -CDC & -CD \end{pmatrix}, \tag{21}$$

hence $BA = 0$ if and only if $D = 0$. Thus the zero fiber of the local wall contraction is cut out by the cotangent coordinates, independently of the Grassmannian coordinates C. Moreover the center of H acts by $\zeta \cdot (A,B,u) = (\zeta A, \zeta^{-1}B, u)$. Hence every H-equivariant formal coordinate change preserves $\{B = 0\}$, and on $\{B = 0\}$ transforms $A \mapsto A\Psi(u)$ with $\Psi(u)$ invertible. Therefore the induced point of the Grassmannian and its tautological quotient bundle are canonically unchanged.

The B-injective chamber. For the opposite sign of the stability character, stability requires B to be injective. With $L := \operatorname{im} B$, of dimension r, A factors through $V/L \to L$, and the opposite quotient is $T^*\operatorname{Gr}(r, V)$ with zero fiber $\operatorname{Gr}(r, V)$. We obtain the two-sided local diagram

$$T^*\operatorname{Gr}(r,V) \longrightarrow \mathfrak{N}_{r,2r+1} \longleftarrow T^*\operatorname{Gr}(r+1,V). \tag{22}$$

Only the right-hand map is the local model of the given ρ. For X of maximal rank r, $L = \operatorname{im} X \subset K = \ker X$ because $X^2 = 0$. The two resolutions remember the two halves of the same flag $L \subset K \subset V$, and are canonically dual through the 2-Calabi–Yau pairing. The left-hand Grassmannian $\operatorname{Gr}(r, 2r+1)$ is the one of the Brill–Noether recursion below.

4.6. **Brill–Noether loci and the thresholds.** For a stable representation $D \in Y_n$ let $\sigma_i(D) = \dim \operatorname{Hom}(S_i, D)$ be the even and odd socle dimensions, and

$$\mathrm{BN}_n(a,b) := \{D \in Y_n :\ \sigma_0(D) \geq a,\ \sigma_1(D) \geq b\}, \qquad \mathrm{BN}^+_{n,r} := \mathrm{BN}_n(r,0).$$

For the framed affine A_1 quiver with dimension vector (n,n) and framing $(1,0)$ the theory of González–Gorsky–Simental [29, Thms. 1.3, 1.11] gives, whenever the locus is nonempty,

$$\dim \mathrm{BN}_n(a,b) = 2n - a(a+1) - b^2, \tag{23}$$

that $\mathrm{BN}_n(a,b)$ is irreducible and Cohen–Macaulay, that the exact-socle stratum $\{\sigma_0 = a, \sigma_1 = b\}$ is smooth and dense in it, and that it is resolved by a split-parabolic correspondence. In particular $\dim \mathrm{BN}^+_{n,r} = 2n - r(r+1)$, and the boundary on which the even socle jumps from r to at least $r+1$ has codimension

$$\dim \mathrm{BN}^+_{n,r} - \dim \mathrm{BN}^+_{n,r+1} = 2r+2 \tag{24}$$

in $\mathrm{BN}^+_{n,r}$. We use (23) with $(a,b) = (1,0)$ and $(2,0)$ in Section 5.

The split-parabolic resolution of [29] is built from a recursion, which we make explicit. For a dimension vector $d = (d_0, d_1)$ define the defect vector

$$k^0[d] = (2d_0 - 2d_1 - 1,\ 2d_1 - 2d_0), \tag{25}$$

start with $d^{(0)} = (n,n)$, $k^{(0)} = (r,0)$, and set recursively

$$d^{(j+1)} = d^{(j)} - k^{(j)}, \qquad k^{(j+1)} = \max\bigl(k^{(j)} + k^0[d^{(j+1)}],\, 0\bigr) \tag{26}$$

(componentwise maximum). In words, one removes the prescribed socle, computes the forced socle of the quotient, and repeats.

Proposition 4.7 (Explicit recursive sequence)**.** *The sequence* (26) *is*

$$(r,0),\ (0,2r),\ (2r-1,0),\ (0,2r-2),\ \ldots,\ (0,2),\ (1,0),\ (0,0),$$

that is, $k^{(2a+1)} = (0, 2r-2a)$ for $0 \leq a \leq r-1$ and $k^{(2a)} = (2r-2a+1, 0)$ for $1 \leq a \leq r$, with the exceptional initial value $k^{(0)} = (r,0)$. The total amount removed at each vertex is $r(r+1)$, hence the terminal dimension vector is $d^{(2r+1)} = (n - r(r+1),\, n - r(r+1))$.

Proof. From $d^{(1)} = (n-r, n)$ we get $k^0[d^{(1)}] = (-2r-1, 2r)$, hence $k^{(1)} = \max((r,0) + (-2r-1, 2r), 0) = (0, 2r)$. Suppose $k^{(2a)} = (2r-2a+1, 0)$ with $1 \leq a \leq r$. One checks $d_0^{(2a+1)} - d_1^{(2a+1)} = a - r$, hence $k^0[d^{(2a+1)}] = (-2(r-a)-1, 2(r-a))$ and $k^{(2a+1)} = (0, 2(r-a))$. Subtracting reverses the sign of the difference, $d_0^{(2a+2)} - d_1^{(2a+2)} = r - a$, hence $k^0[d^{(2a+2)}] = (2(r-a)-1, -2(r-a))$ and $k^{(2a+2)} = (2(r-a)-1, 0)$, the even-stage formula with $a+1$. The sums are $r + (2r-1) + \cdots + 1 = r(r+1)$ and $2r + (2r-2) + \cdots + 2 = r(r+1)$. □

Theorem 4.8 (Even-socle Brill–Noether recursion)**.** $\mathrm{BN}^+_{n,r} \neq \emptyset$ *if and only if* $n \geq r(r+1)$. *When nonempty,* $\mathrm{BN}^+_{n,r}$ *is birational to a* $\operatorname{Gr}(r, 2r+1)$*-bundle over a dense open subset of* $Y_{n-r(r+1)}$.

Proof. The split-parabolic variety resolving $\mathrm{BN}^+_{n,r}$ [29, Thm. 1.11] is nonempty precisely when the recursion reaches a nonempty terminal quiver variety, which by Theorem 4.7 is $Y_{n-r(r+1)}$. The generic fiber of the first split-parabolic projection is $\mathrm{Gr}(k_0^{(0)}, -k_0^0[d^{(1)}]) = \mathrm{Gr}(r, 2r+1)$. At every later step the recurrence forces $-k_v^0[d^{(j+1)}] = k_v^{(j)}$ at the active vertex, hence the generic Grassmannian is a point. □

Corollary 4.9 (Triangular threshold)**.** *An exact even-socle rank r occurs on Y_n only when $r(r+1) \le n$. The first possible values are $r = 1, 2, 3, 4$ for $n \ge 2, 6, 12, 20$. In particular every positive-dimensional wall fiber for $n \le 5$ is a $\mathbb{P}^2 = \mathrm{Gr}(2, 3)$, and $\mathrm{Gr}(3, 5)$ first appears at $n = 6$.*

Lemma 4.10 (Trace preservation along the recursion)**.** *Let $0 \to K \to D \to \widehat{D} \to 0$ be one of the universal split-parabolic exact sequences, where the successive subquotients of K are vertex simples. Then for every positive-degree invariant polynomial f, $\mathrm{Tr}(f|D) = \mathrm{Tr}(f|\widehat{D})$. Consequently the recursion restricts to the punctual fibers, and reversing it over an irreducible component of the punctual fiber of $Y_{n-r(r+1)}$ produces an n-dimensional subvariety of $F_n \cap \mathrm{BN}^+_{n,r}$ birational to a $\mathrm{Gr}(r, 2r+1)$-bundle over that component.*

Proof. Every positive-length path acts by zero on a vertex simple, hence relative to a filtration adapted to K the action of f on D is block upper triangular with zero diagonal block on K. □

Remark 4.11 (What the recursion does and does not prove). The recursion identifies the complementary Grassmannian $\mathrm{Gr}(r, 2r+1)$ of (22), gives the threshold $r(r+1) \le n$, proves that the exact rank stratum is smooth and dense with boundary of codimension $2r+2$, and shows that the scalar trace functions are unchanged along it. It is not an induction on n proving symbolic descent, and it will not be used as one. The descent argument of Section 6 is local at each wall point and uses only the Grassmannian structure of the fiber.

5. The rank-one Hecke correspondence and the excess vanishing

In this section we prove part (3) of Theorem E, the vanishing $R\rho_*\mathcal{M}_n^+ = 0$ of the even part of the excess sheaf. This is the central geometric result of the paper. The main idea is to realize $\mathcal{M}_n^+$ as a pushforward from the rank-one Hecke correspondence $\mathscr{H}$, which parametrizes a cluster together with an even socle line. The correspondence has two projections, to Y_n and to the smaller quiver variety $Y' = \mathfrak{M}_\theta((n-1, n), (1, 0))$, and both projections are cut out by regular sections inside projective bundles. Bott's formula on the first projective bundle bounds the Tor-amplitude of the pushforward of every term by $[-2, 0]$, and since the excess is supported in codimension n, the pushed-forward complex is concentrated in a single degree. On the other hand the inverse $\mathcal{L}^{-1}$ of the universal socle line lies in the negative acyclic window of the second projection, hence the pushforward to the wall vanishes.

The identification of $\mathcal{M}_n^+$ with the object on the Hecke correspondence is the technical heart of the section. It passes through a determinantal model of the even socle locus, the canonical module of that locus, and the dual-socle formula of Section 3. We conclude with the Cohen–Macaulay consequences and with the two lowest ranks in which the argument can be followed by hand.

Throughout the section n is fixed and we write $Y = Y_n$, $\mathcal{P} = \mathcal{P}_n = \mathcal{R}_0 \oplus \mathcal{R}_1$, $\mathcal{T} = \mathcal{T}_n$, $\mathcal{E} = \mathcal{E}_n$, $\mathcal{K} = \mathcal{K}_n$, $J = J_n$, $\mathcal{I} = \mathcal{I}_n$, $\mathcal{M} = \mathcal{M}_n$, $\mathcal{S} = \mathcal{S}_n$. The main result of the section is the following theorem.

Theorem 5.1 (Even-excess vanishing)**.** *For every $n \ge 1$,*

$$R\rho_*\mathcal{M}_n^+ = 0, \tag{27}$$

i.e. $R^i\rho_*\mathcal{M}_n^+ = 0$ for every $i \ge 0$, and the same holds in the T-equivariant derived category.

For $n \geq 2$ the proof produces a more precise statement. Put

$$Y' = \mathfrak{M}_\theta((n-1,n),(1,0)), \qquad Y \xleftarrow{p} \mathscr{H} \xrightarrow{q} Y',$$

where $\mathscr{H}$ classifies exact sequences

$$0 \longrightarrow S_0 \otimes \mathcal{L} \longrightarrow p^*\mathcal{P} \longrightarrow q^*\mathcal{P}' \longrightarrow 0 \tag{28}$$

of framed Λ-modules, and $\mathcal{L}$ is the universal even socle line. Define

$$\mathscr{D} = (\det \mathcal{R}_1)^2 (\det \mathcal{R}_0)^{-2}, \qquad \mathscr{D}' = (\det \mathcal{R}_1')^2 (\det \mathcal{R}_0')^{-2}. \tag{29}$$

There is a universal augmentation $\varepsilon' : \mathcal{P}' \to \mathcal{O}_{Y'}$ at the origin (Theorem 5.6), and we use the modified trace and its kernel

$$\lambda' = \mathrm{Tr}_{\mathcal{R}_0'} + \varepsilon', \qquad \mathcal{T}' = \ker \lambda', \qquad C' = K_{Y'}(\mathcal{T}', \varepsilon'|_{\mathcal{T}'}). \tag{30}$$

Note that $\lambda'(1) = (n-1)+1 = n$ and $\mathrm{rk}\, \mathcal{T}' = n-2$.

Theorem 5.2 (Hecke realization)**.** *Let $n \geq 2$. After forgetting the auxiliary torus linearizations, there is an isomorphism*

$$\mathcal{M}_n^+ \simeq Rp_*(\mathcal{L}^{-1} \otimes q^*(\mathscr{D}' \otimes C')), \tag{31}$$

and the object on the right is concentrated in degree zero. Moreover

$$Rq_*\mathcal{L}^{-1} = Rq_*\mathcal{L}^{-2} = 0. \tag{32}$$

The order of the argument matters. The statement that a complex is annihilated by $R\rho_*$ does not imply that its zeroth cohomology is annihilated, and we prove that the right-hand side of (31) is a sheaf *before* identifying it with $\mathcal{M}_n^+$. The argument proceeds in six steps. We construct a codimension-two determinantal model of the even socle locus, we identify the canonical module of that locus by means of the dual-socle formula, we express the excess as the first homology of a trace Koszul complex, we use the rank-one Hecke filtration to obtain a uniform Tor-amplitude $[-2,0]$, we deduce concentration from the punctual support in codimension n, and finally we use the negative projective window for the second projection.

5.1. **Geometric inputs.**

Geometric input 5.3. *In the cyclic chamber, the quiver varieties with dimension vectors (n,n) and $(n-1,n)$ are smooth schemes of cyclic commutative Γ-cluster algebras, of dimensions $2n$ and $2n-4$. The rank-one even Hecke scheme $\mathscr{H}$ is smooth of pure dimension $2n-2$. Its first image is the even socle locus of codimension two, which has a dense exact socle-rank-one stratum. If nonempty, the locus of even socle rank at least two has codimension six in Y. Finally,* $\dim F_n \leq n$.

Smoothness of $\mathfrak{M}_\theta(v,w)$ for generic θ and the dimension formula $\dim \mathfrak{M}_\theta(v,w) = 2w^Tv - v^TCv$ are Nakajima's [65]. That the cyclic chamber produces $n\Gamma$-Hilb, with the universal object a commutative cluster algebra rather than merely a framed representation, is [53, 22] (Section B gives the algebraic content of this identification over an arbitrary base). For the correspondence and the Brill–Noether assertions we use [29, Thms. 1.3, 1.11, 2.23; Lemmas 3.7, 3.14], in the version arXiv:2601.22287v1 of 29 January 2026. The stability convention there is generation by the framing, which is ours, and there are no loops at the modified vertex, hence their split-parabolic correspondence is the usual rank-one Nakajima correspondence. The last assertion is Theorem 3.7.

Let us record the numerical checks. With $C = \left(\begin{smallmatrix} 2 & -2 \\ -2 & 2 \end{smallmatrix}\right)$ and $w = (1,0)$ we get $\dim Y = 2n$ and $\dim Y' = 2(n-1) - 2 = 2n-4$. For $v = (n,n)$ the vector $k^0 = Cv - w$ is $(-1,0)$, and for the modification $k = (1,0)$ the codimension is $\sum_i k_i(k_i - k_i^0) = 1 \cdot (1+1) = 2$, hence the correspondence has dimension $2n-2$. For $v' = (n-1,n)$, $k^0 = (-3,2)$, and the second projection has fiber $\mathbb{P}^{s_0(E')+2}$ where $s_0(E') = \dim \mathrm{soc}_0(E')$, which also follows from the bundle presentation of Section 5.2.5.

Remark 5.4 (Nonemptiness). For $n = 2$ the smaller cluster is $S/(x, y)^2$, of dimension vector $(1, 2)$. For $n > 2$, adjoining $n-2$ distinct free Γ-orbits gives a point of Y', and a balanced length-four cluster with even socle together with free orbits gives a point of $\mathscr{H}$. Thus the nonemptiness qualifications in the cited results hold.

Convention 5.5 (Projectivizations)**.** For a vector bundle V, $\mathbb{P}_{\text{lines}}(V)$ carries $\mathcal{O}(-1) \hookrightarrow V$ and $\mathbb{P}_{\text{quot}}(V)$ carries $V \twoheadrightarrow \mathcal{O}(1)$. On $\mathscr{H}$ the same line satisfies

$$\mathcal{L} = \mathcal{O}_{\mathbb{P}_{\text{lines}}(\mathcal{T})}(-1)|_{\mathscr{H}} = \mathcal{O}_{\mathbb{P}_{\text{quot}}(\mathcal{V}')}(1)|_{\mathscr{H}}, \tag{33}$$

and the two projective bundles have different bases and different moduli meanings.

5.2. **The Hecke scheme and its two incidence presentations.**

5.2.1. *The moduli functor.* For a test scheme T_0, a T_0-point of $\mathscr{H}$ consists of a family of balanced cyclic S-algebras E (a family of points of Y) and an even line subbundle $L \subset E_0$ annihilated by x and y. The quotient $E' = E/L$ is locally free with ranks $(n-1, n)$ and cyclic with the induced framing, hence a family of points of Y'. Conversely, a family of exact sequences of this form gives the same datum. The two projections send the sequence to its larger and smaller algebra, the universal sequence is (28), and in particular

$$0 \to \mathcal{L} \to p^*\mathcal{R}_0 \to q^*\mathcal{R}'_0 \to 0, \qquad p^*\mathcal{R}_1 \simeq q^*\mathcal{R}'_1. \tag{34}$$

The descriptions below are descriptions of this functor, not only of its closed points.

5.2.2. *The smaller cluster has a universal augmentation.*

Lemma 5.6. *Every closed point of Y' contains the origin in its support. Evaluation at the origin descends to a morphism of universal algebras $\varepsilon' : \mathcal{P}' \to \mathcal{O}_{Y'}$, compatible with arbitrary base change.*

Proof. A Γ-stable finite subscheme supported away from the origin is a union of free Γ-orbits with multiplicities, each of which has equal even and odd dimensions. The dimension vector $(n-1, n)$ is unbalanced, hence there is support at the origin. To pass from points to the universal family, work on an affine open $\operatorname{Spec} A \subset Y'$ with universal ideal $I' \subset A[x, y]$. For $f \in I'$ the regular function $f(0, 0) \in A$ vanishes at every closed point of $\operatorname{Spec} A$. Since Y' is smooth, A is reduced, and therefore $f(0, 0) = 0$. Hence evaluation $A[x, y] \to A$ kills I' and factors through the universal quotient. These local maps glue, and the resulting universal map pulls back to any, including nonreduced, test scheme. □

Warning 5.7. Fiberwise containment of the origin is not by itself enough to produce such an augmentation over an arbitrary nonreduced base. The reducedness of the universal moduli scheme is used first, and the universal map is then pulled back.

5.2.3. *Trace additivity and the universal trace-bundle sequence.* Pull back ε' along q and compose with the quotient in (28). This gives an augmentation $\varepsilon_{\mathscr{H}} : p^*\mathcal{P} \to \mathcal{O}_{\mathscr{H}}$. Polynomial multiplication on $\mathcal{L}$ is through evaluation at the origin, because $x\mathcal{L} = y\mathcal{L} = 0$. Hence for a section a of $p^*\mathcal{R}_0$ with image a' in $q^*\mathcal{R}'_0$, multiplication on the line $\mathcal{L}$ is the scalar $\varepsilon'(a')$.

Proposition 5.8. *There is an identity of linear functionals on $p^*\mathcal{R}_0$,*

$$\operatorname{Tr}_{p^*\mathcal{R}_0}(a) = \operatorname{Tr}_{q^*\mathcal{R}'_0}(a') + \varepsilon'(a'). \tag{35}$$

In particular $\mathcal{L} \subset p^*\mathcal{T}$, and there is a canonical exact sequence of vector bundles

$$0 \longrightarrow \mathcal{L} \longrightarrow p^*\mathcal{T} \longrightarrow q^*\mathcal{T}' \longrightarrow 0, \qquad \mathcal{T}' = \ker(\operatorname{Tr}_{\mathcal{R}'_0} + \varepsilon'). \tag{36}$$

Proof. Multiplication by a preserves the subbundle $\mathcal{L}$ in (34), and in a local splitting its matrix is block triangular with diagonal blocks $\varepsilon'(a')$ and multiplication by a' on $q^*\mathcal{R}_0'$. Trace is additive on these blocks over any base ring, proving (35). For $a \in \mathcal{L}$ the image a' is zero, hence $\mathrm{Tr}(a) = 0$ and $\mathcal{L} \subset p^*\mathcal{T}$. Since $\mathrm{Tr}_{\mathcal{R}_0}(1) = n$ and $(\mathrm{Tr}_{\mathcal{R}_0'} + \varepsilon')(1) = (n-1) + 1 = n$, both functionals are split surjections, and taking kernels in (34) gives (36). □

Remark 5.9 (Why the unmodified trace is wrong). The lost even line contributes its scalar action to the trace, hence the trace inherited by the quotient is $\mathrm{Tr}_{\mathcal{R}_0'} + \varepsilon'$, not $\mathrm{Tr}_{\mathcal{R}_0'}$, and the ranks alone do not repair the discrepancy. For example, let the smaller algebra be the disjoint union of $S/(x,y)^2$ and a free two-point orbit. Its even algebra has two idempotent factors, on which the trace is $(1,1)$ whereas the augmentation is $(1,0)$, and adding an even socle line at the origin changes the trace to $(2,1)$, exactly the modified trace.

Taking determinants in (34) gives

$$p^*\mathscr{D} = \mathcal{L}^{-2} \otimes q^*\mathscr{D}'. \tag{37}$$

5.2.4. *The presentation over the larger cluster.* On Y put

$$\mathcal{X} = \mathcal{R}_1 \oplus \mathcal{R}_1 \oplus \mathcal{O}_Y, \qquad M_2 = (x, y, i) : \mathcal{X} \to \mathcal{R}_0, \qquad M_1 = (-y, x, 0) : \mathcal{R}_0 \to \mathcal{X},$$

i the framing. The relation $M_2M_1 = 0$ is commutativity. Cyclicity implies that M_2 is surjective, because every nonconstant even monomial is x or y times an odd monomial, and the constant monomial is the framing. Hence $\mathcal{V} = \ker M_2$ is locally free of rank $n+1$. The section $u = M_1(1)$ is nowhere zero, because if it vanished in a fiber, both x and y would vanish in a cyclic algebra of dimension $2n > 1$. Thus $\mathcal{O}_Y u$ is a line subbundle of $\mathcal{V}$, and we set

$$\mathcal{G} = \mathcal{V}/\mathcal{O}_Y u, \qquad \psi : \mathcal{T} \longrightarrow \mathcal{G}, \qquad \mathrm{rk}\,\mathcal{T} = a := n-1, \quad \mathrm{rk}\,\mathcal{G} = a+1 = n, \tag{38}$$

ψ being induced by $M_1|_{\mathcal{T}}$.

Proposition 5.10 (Unit cancellation on the incidence functor)**.** *The Hecke scheme is the zero scheme*

$$\mathscr{H} = Z\Big(\mathcal{O}_{\mathbb{P}_{\mathrm{lines}}(\mathcal{T})}(-1) \longrightarrow \pi^*\mathcal{G}\Big) \subset \mathbb{P}_{\mathrm{lines},Y}(\mathcal{T}), \tag{39}$$

with tautological line $\mathcal{L}$. This is an equality of moduli schemes, compatible with arbitrary base change.

Proof. In the uncancelled presentation, choosing a line in $\ker M_1 \subset \mathcal{R}_0$ is precisely choosing an even line killed by the arrows. Such a line is an ideal in the cyclic algebra, and its quotient is the smaller cluster, hence this kernel incidence is the Hecke functor. Use the splitting $\mathcal{R}_0 = \mathcal{O}\cdot 1 \oplus \mathcal{T}$. Let L_0 be a line in $\ker M_1$ over a test scheme. Its projection to $\mathcal{T}$ is a line subbundle (it cannot meet the unit line in any geometric fiber because $M_1(1)$ is nowhere zero) and is annihilated by ψ. Conversely, if $\widetilde{L} \subset \mathcal{T}$ is a line subbundle with $\psi|_{\widetilde{L}} = 0$, then $M_1|_{\widetilde{L}}$ factors uniquely through $\mathcal{O}u$, say by $c : \widetilde{L} \to \mathcal{O}$, and the graph $\ell \mapsto \ell - c(\ell)1$ is a line subbundle of $\mathcal{R}_0$ annihilated by M_1. These constructions are inverse over arbitrary rings. Under this isomorphism the graph line is the universal kernel $\mathcal{L}$. By Theorem 5.8 it has trace zero, and since $\mathrm{Tr}(\ell - c(\ell)1) = -nc(\ell)$ for $\ell \in \mathcal{T}$, one has $c = 0$ on the incidence. Thus the actual socle line is the tautological line of (39). □

The ambient projective bundle has dimension $2n + (n-2) = 3n-2$, whereas $\mathscr{H}$ is smooth of dimension $2n-2$, and the defining section has rank n. Therefore it is a regular section (Section B.1), the zero scheme carries its ordinary scheme structure, and its Koszul resolution is exact. Let $\overline{\mathcal{W}}$ be the universal quotient on $\mathbb{P}_{\mathrm{lines}}(\mathcal{T})$ and $\mathcal{W} = \overline{\mathcal{W}}|_{\mathscr{H}}$, thus

$$0 \to \mathcal{L} \to p^*\mathcal{T} \to \mathcal{W} \to 0, \qquad \mathcal{W} \simeq q^*\mathcal{T}' \tag{40}$$

by Theorem 5.8.

5.2.5. *The presentation over the smaller cluster.* On Y' define $\mathcal{X}' = \mathcal{R}'_1 \oplus \mathcal{R}'_1 \oplus \mathcal{O}_{Y'}$, $M'_2 = (x', y', i') : \mathcal{X}' \to \mathcal{R}'_0$ and $M'_1 = (-y', x', 0) : \mathcal{R}'_0 \to \mathcal{X}'$. Again M'_2 is surjective, its kernel $\mathcal{V}'$ has rank $n+2$, and M'_1 factors through a map $\phi' : \mathcal{R}'_0 \to \mathcal{V}'$.

Proposition 5.11 (Quotient incidence, with reconstruction)**.** *There is a functorial isomorphism*

$$\mathscr{H} \simeq \mathbb{P}_{\mathrm{quot},Y'}(\operatorname{coker}\phi') \subset \mathbb{P}_{\mathrm{quot},Y'}(\mathcal{V}'), \tag{41}$$

with universal quotient line $\mathcal{L}$, and the closed immersion is cut out by the section $\mathcal{R}'_0 \to \mathcal{V}' \to \mathcal{O}(1)$.

Proof. A test-scheme point on the right is a line quotient of $\mathcal{V}'$ killing $\operatorname{im} M'_1$. If $U \subset \mathcal{V}'$ is its kernel, then $\operatorname{im} M'_1 \subset U \subset \ker M'_2 \subset \mathcal{X}'$ and $\mathcal{V}'/U = L$. Define $E_0 = \mathcal{X}'/U$ and $E_1 = E'_1$. The map M'_2 induces $\pi_0 : E_0 \twoheadrightarrow E'_0$ with kernel L. The incoming maps $x_1, y_1 : E'_1 \to E_0$ are induced by the first and second inclusions into $\mathcal{X}'$, the framing by the third, and the outgoing maps are $x_0 = x'_0\pi_0$, $y_0 = y'_0\pi_0$. The even commutativity relation is exactly $\operatorname{im} M'_1 \subset U$, the odd one is inherited from E', and the line L is killed by x_0, y_0. As for cyclicity, the submodule generated by the new framing surjects onto the cyclic module E'. Because the kernel is even, this submodule contains all of $E_1 = E'_1$, and applying x_1, y_1 and adjoining the framing then generates all of $E_0 = \mathcal{X}'/U$. This works over every base, using surjectivity between the corresponding locally free modules. For the inverse, start with a Hecke sequence. The map $M_2 : E'_1 \oplus E'_1 \oplus \mathcal{O} \to E_0$ is surjective by cyclicity, its kernel U lies in $\ker M'_2$ since $M'_2 = \pi_0 M_2$, the outgoing maps kill the kernel line and therefore factor through E'_0, the even moment-map equation gives $M_2 M'_1 = 0$ hence $\operatorname{im} M'_1 \subset U$, and $(\ker M'_2)/U$ is canonically L. The constructions are inverse and commute with base change. □

The ambient projective bundle has dimension $(2n-4)+(n+1) = 3n-3$, the defining section has rank $n-1$, and its zero scheme is the smooth $(2n-2)$-dimensional $\mathscr{H}$, hence the section is regular. The fiber over E' is a projective space of dimension $s_0(E')+2$, because $\ker\phi'(E') = \operatorname{soc}_0(E')$ and therefore $\dim\operatorname{coker}\phi'(E') = (n+2) - (n-1-s_0(E')) = 3 + s_0(E')$. Thus q is not a projective *bundle*, it is an incidence inside one.

5.3. **The determinantal scheme and its canonical module.** Let $B \subset Y$ have the scheme structure defined by the maximal minors of $\psi : \mathcal{T} \to \mathcal{G}$, and $i : B \hookrightarrow Y$. By Theorem 5.10 its underlying set is the positive even-socle locus $\mathrm{BN}^+_{n,1}$, of codimension two by (23).

5.3.1. *Determinants and Hilbert–Burch.* The two exact sequences defining $\mathcal{V}$ and $\mathcal{G}$ give $\det\mathcal{G} = \det\mathcal{V} = (\det\mathcal{R}_1)^2(\det\mathcal{R}_0)^{-1}$ and $\det\mathcal{T} = \det\mathcal{R}_0$, hence

$$\det\mathcal{G} \otimes (\det\mathcal{T})^{-1} = \mathscr{D}. \tag{42}$$

The maximal-minor map is naturally $\mathcal{G} \otimes \mathscr{D}^{-1} \to \mathcal{O}_Y$, $g \otimes (\det\mathcal{T} \otimes \det\mathcal{G}^\vee) \mapsto \Lambda^a\psi \wedge g$, with image the ideal of B.

Proposition 5.12. *The sequence*

$$0 \longrightarrow \mathcal{T} \otimes \mathscr{D}^{-1} \xrightarrow{\psi} \mathcal{G} \otimes \mathscr{D}^{-1} \longrightarrow \mathcal{O}_Y \longrightarrow i_*\mathcal{O}_B \longrightarrow 0 \tag{43}$$

is exact. In particular B is Cohen–Macaulay, pure of codimension two, and reduced.

Proof. The grade of the maximal-minor ideal is two, since Y is regular and its zero set has codimension two, and Hilbert–Burch gives (43) [24, Thm. 20.15], [18, Thm. 1.4.17]. An ideal of grade two with a length-two free resolution is perfect, hence B is Cohen–Macaulay of pure codimension two [18, Thm. 2.1.5, Cor. 2.1.4]. On the dense exact-corank-one locus the kernel-incidence map $\mathscr{H} \to B$ is an isomorphism of schemes. Indeed, locally choose an invertible $(a-1)$-minor of ψ and reduce the matrix to an identity block and a remaining 2×1 column, then both the determinantal ideal and the incidence are cut out by the two entries of that column. This locus is smooth because $\mathscr{H}$ is, hence B is generically reduced. A Cohen–Macaulay scheme has no embedded primes [18, §2.1], hence B is reduced. □

5.3.2. *Identifying the transpose cokernel.* The image of M_1 lies in $\mathcal{V}$, and since $\mathcal{X}^\vee \twoheadrightarrow \mathcal{V}^\vee$ is surjective, passing from $\mathcal{X}$ to $\mathcal{V}$ does not change the cokernel of the transpose. Cancelling the common unit line in domain and image, in local compatible splittings the map has block form $\left(\begin{smallmatrix} 1 & c \\ 0 & \psi \end{smallmatrix}\right)$, and elementary invertible row and column operations on its transpose identify its cokernel with $\operatorname{coker}\psi^\vee$, and the zero framing component of M_1 does not affect that cokernel either. Therefore

$$\operatorname{coker}((-y,x)^\vee : \mathcal{R}_1^\vee \oplus \mathcal{R}_1^\vee \to \mathcal{R}_0^\vee) \simeq \operatorname{coker}\psi^\vee, \tag{44}$$

and the sheaf on the left is the even part of $\mathcal{P}^\vee/(x,y)\mathcal{P}^\vee$, the even ambient dual-socle sheaf. Dualizing (43),

$$\mathcal{E}xt^2_Y(i_*\mathcal{O}_B, \mathcal{O}_Y) \simeq \operatorname{coker}\psi^\vee \otimes \mathscr{D}, \tag{45}$$

and the right side is annihilated by the ideal of B. We write

$$i_*\Sigma = \operatorname{coker}\psi^\vee, \tag{46}$$

hence $\omega_{B/Y} := \mathcal{E}xt^2_Y(i_*\mathcal{O}_B, \mathcal{O}_Y) = i_*(\Sigma \otimes \mathscr{D}|_B)$. Since the absolute canonical module differs from this by the invertible $\omega_Y|_B$, Σ is a canonical module of B up to an invertible twist. In particular it is maximal Cohen–Macaulay on B and

$$R\mathcal{H}om_B(\Sigma, \Sigma) \simeq \mathcal{O}_B \tag{47}$$

by Theorem 3.12.

Warning 5.13 (Not a regular-embedding argument). For larger even socle rank the scheme B is singular and need not be a local complete intersection. We use the canonical module supplied by (43) and never write $i^!\mathcal{O}_Y$ as a determinant of a normal bundle. Instead, $i^!\mathcal{O}_Y \simeq (\Sigma \otimes \mathscr{D}|_B)[-2]$, which follows from the explicit resolution and is valid on all of B.

Remark 5.14 (The uncancelled form). Before cancelling the unit, $\operatorname{coker}\psi^\vee$ is the cokernel of $\sigma^\vee : \mathcal{V}^\vee \to \mathcal{R}_0^\vee$ with $\operatorname{rk}\mathcal{V} = n+1$, $\operatorname{rk}\mathcal{R}_0 = n$, and (43) is the Buchsbaum–Rim complex of $\sigma^\vee$, $0 \to (\det\mathcal{V})^{-1} \otimes \det\mathcal{R}_0 \to \mathcal{V}^\vee \to \mathcal{R}_0^\vee \to \operatorname{coker}\sigma^\vee \to 0$, whose acyclicity criterion is again grade two of the maximal minors. The two presentations agree because $\det\mathcal{G} = \det\mathcal{V}$ and the unit column is split.

5.4. **Trace complexes and the expression for the excess.**

5.4.1. *The augmentation on the determinantal scheme.* Every point of B contains the origin, since it has a nonzero element annihilated by x and y. Since B is reduced by Theorem 5.12, the argument of Theorem 5.6 gives a universal augmentation $\varepsilon_B : \mathcal{P}|_B \to \mathcal{O}_B$. Define

$$s_B = \varepsilon_B|_{\mathcal{T}|_B}, \qquad C_B = K_B(\mathcal{T}|_B, s_B), \qquad Z = V(s_B) \subset B. \tag{48}$$

For a positive-degree even polynomial f, $\varepsilon_B(f - \frac{\operatorname{Tr}(f)}{n}1) = -\frac{\operatorname{Tr}(f)}{n}$, and the projections of even polynomials to $\mathcal{T}$ generate it. Hence the ideal of Z is the ideal of all positive-degree even traces restricted to B, and

$$|Z| = |B| \cap |F_n|, \qquad \dim Z \le n. \tag{49}$$

This concerns support and the trace ideal on B, and it does not assert that the punctual scheme F_n is Cohen–Macaulay.

5.4.2. *The dual-socle quotient is a trace quotient.* Set $\mathbf{\Sigma} = \mathcal{P}^\vee/(x,y)\mathcal{P}^\vee$. The S-action on this sheaf factors through evaluation at the origin, and its even summand is $i_*\Sigma$ by (44). Hence multiplication by $t \in \mathcal{T}$ on $i_*\Sigma$ is the scalar $\varepsilon_B(t)$, and

$$\mathcal{S}^+ = i_*(\Sigma \otimes \mathcal{O}_Z) = i_*\mathcal{H}^0(\Sigma \otimes C_B). \tag{50}$$

This is an ordinary quotient statement, and the whole complex $\Sigma \otimes C_B$ is retained when applying duality. The scheme B and the module Σ are Cohen–Macaulay of dimension $2n-2$. For a maximal Cohen–Macaulay module M over a Cohen–Macaulay ring and an ideal I we have $\operatorname{grade}(I,M) = \dim M - \dim(M/IM)$ [18, Thm. 2.1.2], hence by (49) the trace ideal has grade at least $(2n-2)-n = n-2$ on each of $\mathcal{O}_B$ and Σ. Since C_B has $n-1$ generators, Koszul grade-sensitivity [18, Thm. 1.6.17] gives

$$C_B,\ \Sigma \otimes C_B \in D^{[-1,0]}(B), \tag{51}$$

with all cohomology sheaves supported on Z.

Proposition 5.15 (The excess as first trace homology)**.** *There is an isomorphism of ordinary coherent sheaves*

$$\mathcal{M}_n^+ \simeq i_*(\mathscr{D}|_B \otimes \mathcal{H}^{-1}(C_B)). \tag{52}$$

Proof. Let $N_B = \mathcal{H}^{-1}(\Sigma \otimes C_B)$ and $S_B = \mathcal{H}^0(\Sigma \otimes C_B)$. The triangle $i_*N_B[1] \to i_*(\Sigma \otimes C_B) \to i_*S_B \to i_*N_B[2]$ has outer sheaves supported in codimension $\geq n$ in Y, hence Theorem 3.10, exactly as in the proof of Theorem 3.14, yields

$$\mathcal{E}xt_Y^n(\mathcal{S}^+, \mathcal{O}_Y) \simeq \mathcal{H}^n R\mathcal{H}om_Y(i_*(\Sigma \otimes C_B), \mathcal{O}_Y), \tag{53}$$

the obstruction terms $\mathcal{E}xt_Y^{n-2}(i_*N_B, \mathcal{O}_Y)$ and $\mathcal{E}xt_Y^{n-1}(i_*N_B, \mathcal{O}_Y)$ both vanishing. Duality for the finite morphism i, in the form $R\mathcal{H}om_Y(i_*F, \mathcal{O}_Y) \simeq i_*R\mathcal{H}om_B(F, i^!\mathcal{O}_Y)$ with $i^!\mathcal{O}_Y = R\mathcal{H}om_Y(i_*\mathcal{O}_B, \mathcal{O}_Y)$ regarded on B [45, Ch. III, §6], [73, Tag 0A7A], together with (45) and (47), gives

$$R\mathcal{H}om_Y(i_*(\Sigma \otimes C_B), \mathcal{O}_Y) \simeq i_*R\mathcal{H}om_B(\Sigma \otimes C_B, \Sigma \otimes \mathscr{D}|_B)[-2] \simeq i_*(\mathscr{D}|_B \otimes C_B^\vee)[-2], \tag{54}$$

the second step because C_B is a bounded complex of vector bundles, hence perfect, and $R\mathcal{H}om(C \otimes X, W) \simeq C^\vee \otimes R\mathcal{H}om(X,W)$ for perfect C. By Koszul self-duality in rank $n-1$ (Theorem 3.11), $C_B^\vee \simeq C_B[-n+1] \otimes (\det \mathcal{T}|_B)^{-1}$, hence the degree-$n$ cohomology of (54) is $i_*(\mathscr{D}|_B \otimes \mathcal{H}^{-1}(C_B) \otimes (\det \mathcal{T}|_B)^{-1})$. Multiplication by $\det \mathcal{T}$ in (13) cancels the last factor. □

5.4.3. *The smaller trace complex and its support.* Recall $\mathcal{T}'$ and C' from (30), and set $Z' = V(\varepsilon'|_{\mathcal{T}'}) \subset Y'$. If f is positive-degree and even then $\varepsilon'(f) = 0$, hence $f - \frac{\operatorname{Tr}_{\mathcal{R}_0'}(f)}{n}1 \in \mathcal{T}'$ with ε'-value $-\frac{\operatorname{Tr}_{\mathcal{R}_0'}(f)}{n}$. Thus Z' is defined by the positive-degree even traces on Y', and at a closed point vanishing of the section is equivalent to $\operatorname{Tr}_{E_0'} = (n-1)\varepsilon'$, which by the separating-polynomial argument of Theorem 3.2 holds exactly when the smaller cluster is punctual. We use only the resulting support statement, namely

$$\operatorname{Supp}(q^*C') \subset q^{-1}(|Z'|), \qquad p(q^{-1}(|Z'|)) \subset |F_n|, \tag{55}$$

the second inclusion because an extension of a module supported at the origin by S_0 is again supported at the origin. Here q^*C' denotes the termwise pullback of the locally free complex C', and no flatness and no assertion that q^*C' has only zeroth cohomology is used.

5.4.4. *The exact Koszul filtration.* The two augmentations on $\mathscr{H}$ agree, both evaluating the original polynomial algebra at the origin. By Theorem 5.8 the section of $p^*(\mathcal{T}|_B)$ vanishes on $\mathcal{L}$ and descends to the section of $q^*\mathcal{T}'$. Let $\bar{p}:\mathscr{H}\to B$ be the factorization of p.

Proposition 5.16. *There is a short exact sequence of complexes of vector bundles on $\mathscr{H}$,*

$$0 \longrightarrow \mathcal{L}\otimes q^*C'[1] \longrightarrow \bar{p}^*C_B \longrightarrow q^*C' \longrightarrow 0. \tag{56}$$

Proof. For an exact sequence $0\to L\to V\to W\to 0$ with L a line there are canonical exact sequences $0\to L\otimes\Lambda^{j-1}W\to\Lambda^jV\to\Lambda^jW\to 0$. If the Koszul section kills L these are compatible with the Koszul differentials, and locally $\iota_s(\ell\wedge w)=-\ell\wedge\iota_s(w)$, hence the differential on the kernel has the minus sign of the shift [1]. Apply this to (36), where local splittings are used only to check the sign. □

5.5. The uniform projective-space calculation.

5.5.1. *Cohomology of twisted differential forms.* Let V be an a-dimensional vector space, $\pi:\mathbb{P}_{\text{lines}}(V)\to\operatorname{Spec}\mathbb{C}$, $m=a-1$, with universal sequence $0\to L=\mathcal{O}(-1)\to V\otimes\mathcal{O}\to W\to 0$. Dualizing, $W^\vee\simeq\Omega^1(1)$.

Lemma 5.17 (Projective Bott calculation)**.** *For $0\le b\le m$ the only possible nonzero cohomology of $\Omega^b(k)$ on $\mathbb{P}^m$ is given by the following table.*

condition	*cohomological degree*
$k>b$	0
$k=0$	b
$k<b-m$	m

(57)

All other cases are acyclic, and in the middle case the cohomology is one-dimensional.

Proof. This is Bott's formula [17], [66, Ch. I, §1.1], and here is the elementary derivation. The exterior Euler sequence is

$$0\to\Omega^b(k)\to\Lambda^bV^\vee\otimes\mathcal{O}(k-b)\to\Lambda^{b-1}V^\vee\otimes\mathcal{O}(k-b+1)\to\cdots\to\mathcal{O}(k)\to 0 \tag{58}$$

[73, Tags 0FUK, 01XS]. If $k>0$, every twist is $>-m-1$, hence no term has higher cohomology, and the complex of global sections is the indicated truncation of the degree-k strand of the polynomial Koszul complex on $V^\vee$, which is exact for $k>0$ ($dh+hd=k\,\mathrm{id}$ on total degree k for $d=\sum z_i\iota_i$ and $h=\sum e_i\wedge\partial_{z_i}$). If $0<k\le b$ the full strand occurs and $\Omega^b(k)$ is acyclic, and if $k>b$ only the leftmost kernel, $H^0(\Omega^b(k))$, remains. If $k=0$, all negative twists in (58) lie in $[-m,-1]$ and are acyclic. Only $H^0(\mathcal{O})$ remains, and the long exact sequences place it in $H^b(\Omega^b)$. If $k<0$, Serre duality with $\omega=\Omega^m=\mathcal{O}(-m-1)$ and the perfect pairing $\Omega^b\otimes\Omega^{m-b}\to\Omega^m$ give $H^r(\Omega^b(k))^\vee\simeq H^{m-r}(\Omega^{m-b}(-k))$, and the positive-twist case gives the third row. The case $m=0$ is included. □

The same degree statements hold relatively for a projective bundle over any base, by checking on trivializing opens, and every nonzero relative cohomology sheaf occurring below is locally free and commutes with arbitrary base change.

5.5.2. *The two rows of Hecke bundles.* Retain $a=n-1$ and the presentation $\mathscr{H}\subset\mathbb{P}_{\text{lines},Y}(\mathcal{T})$ of Theorem 5.10, with ambient projection π. Its exact Koszul resolution has terms $\Lambda^\ell\pi^*\mathcal{G}^\vee\otimes\mathcal{O}(-\ell)$, $0\le\ell\le a+1$, in degrees $-\ell$.

Proposition 5.18 (Uniform Tor-amplitude)**.** *For every $n\ge 2$, $e\in\{0,1\}$ and $0\le j\le n-2$,*

$$Rp_*(\mathcal{L}^e\otimes\Lambda^j\mathcal{W})\in\operatorname{Perf}^{[-2,0]}(Y). \tag{59}$$

This is a Tor-amplitude statement, not only an ordinary cohomology bound.

Proof. The line and quotient bundle on $\mathscr{H}$ extend to the ambient projective bundle, and tensoring its regular-section Koszul resolution gives terms

$$\Lambda^{\ell}\pi^{*}\mathcal{G}^{\vee}\otimes L^{\ell+e}\otimes\Lambda^{j}\overline{\mathcal{W}},\qquad\text{in degree }-\ell. \tag{60}$$

Put $b=a-1-j$. Exterior duality for the quotient bundle gives

$$\Lambda^{j}\overline{\mathcal{W}}\simeq\Lambda^{a-1-j}\overline{\mathcal{W}}^{\vee}\otimes\det\overline{\mathcal{W}}\simeq\Omega^{b}_{\pi}(b+1)\otimes\pi^{*}\det\mathcal{T}, \tag{61}$$

since $\overline{\mathcal{W}}^{\vee}=\Omega^{1}_{\pi}(1)$ and $\det\overline{\mathcal{W}}=\pi^{*}\det\mathcal{T}\otimes L^{-1}$. Thus the projective cohomology to be calculated is that of $\Omega^{b}_{\pi}(k)$ with $k=b+1-\ell-e=a-j-\ell-e$, up to a vector bundle pulled back from Y. Let us read (57) with $m=a-1$. The first condition $k>b$ reads $\ell+e<1$, forcing $(e,\ell)=(0,0)$ with relative degree $r=0$. The second, $k=0$, reads $\ell+e=a-j$, giving $\ell=a-j$ for $e=0$ and $\ell=a-j-1$ for $e=1$, with $r=b=a-j-1$. The third, $k<b-m$, reads $\ell+e\geq a+1$, giving $\ell=a+1$ for $e=0$ and $\ell\in\{a,a+1\}$ for $e=1$, with $r=a-1$. The complete list is

e	ℓ	relative cohomology degree r	$r-\ell$
0	0	0	0
0	$a-j$	$a-j-1$	-1
0	$a+1$	$a-1$	-2
1	$a-j-1$	$a-j-1$	0
1	a	$a-1$	-1
1	$a+1$	$a-1$	-2

(62)

For fixed (e,j) the three values of ℓ are distinct, including the case $a=1$. Each nonzero derived pushforward of a term (60), with its shift $-\ell$ included, is therefore a vector bundle in one of the degrees $-2,-1,0$ (locally free because relative cohomology of a homogeneous bundle on a projective bundle concentrated in one degree is locally free and commutes with base change [44, Thm. III.12.11]). Push the finite stupid filtration of the Koszul resolution through $R\pi_{*}$. It expresses the required object as a finite succession of extensions of these shifted vector bundles, and perfect complexes of Tor-amplitude $[-2,0]$ are closed under extensions (tensor the triangle with any quasicoherent module and use its cohomology sequence, see [73, Tag 0DJC]). Possible higher differentials in the associated spectral sequence change the maps in a locally free representative but cannot enlarge the amplitude interval. □

For reference, the three graded pieces in the pushed filtration are, as representations,

e	degree	bundle on Y
0	0	$\Lambda^{j}\mathcal{T}$
0	-1	$\Lambda^{a-j}\mathcal{G}^{\vee}\otimes\det\mathcal{T}$
0	-2	$\det\mathcal{G}^{\vee}\otimes\Lambda^{j+1}\mathcal{T}\otimes\det\mathcal{T}$
1	0	$\Lambda^{a-j-1}\mathcal{G}^{\vee}\otimes\det\mathcal{T}$
1	-1	$\Lambda^{a}\mathcal{G}^{\vee}\otimes\Lambda^{j+1}\mathcal{T}\otimes\det\mathcal{T}$
1	-2	$\det\mathcal{G}^{\vee}\otimes\mathbb{S}_{(2,1^{j})}\mathcal{T}\otimes\det\mathcal{T}$

(63)

(here $\mathbb{S}_{(2,1^{j})}$ is the hook Schur functor, and the top-cohomology factors follow from $\omega_{\pi}=L^{a}\otimes\pi^{*}(\det\mathcal{T})^{-1}$ and relative Serre duality). The table is not a claim of a canonical splitting, only the degrees and local freeness are used. The row $(e,j)=(0,0)$ has terms $\mathcal{T}\otimes\mathscr{D}^{-1},\mathcal{G}\otimes\mathscr{D}^{-1},\mathcal{O}_{Y}$ in degrees $-2,-1,0$, precisely the terms of the Hilbert–Burch resolution (43). This is a consistency check, not used below. The positions in (62) have been verified by an exact Bott-weight computation for all $a\leq 50$.

5.5.3. *The structure sheaf of the determinantal scheme.*

Lemma 5.19. *The determinantal scheme B is normal and irreducible.*

Proof. B is reduced and Cohen–Macaulay, and smooth wherever the even socle has rank one (Theorem 5.12). The complement is the locus with even socle rank at least two. If nonempty, (23) with $(a,b)=(2,0)$ gives its codimension in Y as $2\cdot 3=6$, hence codimension four in B. Thus B is regular in codimension one and satisfies S_2, and Serre's criterion [62, Thm. 23.8], [73, Tag 031S] gives normality. Irreducibility is [29, Thm. 1.3], and normality component by component would suffice below. □

Proposition 5.20. *Writing $p=i\bar{p}$ with $\bar{p}:\mathscr{H}\to B$,*

$$Rp_*\mathcal{O}_{\mathscr{H}}\simeq i_*\mathcal{O}_B,\qquad R\bar{p}_*\mathcal{O}_{\mathscr{H}}\simeq\mathcal{O}_B. \tag{64}$$

Proof. Theorem 5.18 with $(e,j)=(0,0)$ places $Rp_*\mathcal{O}_{\mathscr{H}}$ in cohomological degrees at most zero. As the derived pushforward of a sheaf it is also in degrees at least zero, hence all higher direct images vanish. The map $\bar{p}$ is projective, by the kernel incidence description, and birational. Indeed, it is an isomorphism over the dense exact-corank-one locus [29, Thm. 1.11], and every irreducible component of $\mathscr{H}$ dominates a component of B by a dimension count. Over the stratum $\{s_0=k\}\subset Y$, of codimension $k(k+1)$ by (23), the fiber of $\bar{p}$ is $\mathbb{P}(\mathrm{soc}_0)=\mathbb{P}^{k-1}$, hence the preimage of that stratum has dimension $2n-k(k+1)+(k-1)=2n-k^2-1<2n-2=\dim\mathscr{H}$ for $k\geq 2$. Since B is normal, Zariski's Main Theorem [44, Cor. III.11.4] gives $\bar{p}_*\mathcal{O}_{\mathscr{H}}=\mathcal{O}_B$ (alternatively, use Stein factorization [44, Cor. III.11.5], whose finite birational factor over a normal integral scheme is an isomorphism). As i_* is exact and faithful, the assertion about $R\bar{p}_*$ follows. □

5.6. **The negative window for the second projection.**

Proposition 5.21. *For the projection $q:\mathscr{H}\to Y'$,*

$$Rq_*\mathcal{L}^{-1}=Rq_*\mathcal{L}^{-2}=0, \tag{65}$$

and consequently $Rq_*(\mathcal{L}^{-t}\otimes q^*G)=0$ for every $G\in\mathrm{Perf}(Y')$ and $t=1,2$.

Proof. Use the regular-section realization of Theorem 5.11, with $\pi':\mathbb{P}_{\mathrm{quot},Y'}(\mathcal{V}')\to Y'$. The Koszul resolution of $\mathcal{O}_{\mathscr{H}}$ has terms $\Lambda^\ell\pi'^*\mathcal{R}_0'\otimes\mathcal{O}(-\ell)$, $0\leq\ell\leq n-1$, in degrees $-\ell$. After twisting by $\mathcal{L}^{-t}$ the line degrees are $-t-\ell$, and since the ambient bundle has fibers $\mathbb{P}^{n+1}$, every term is π'-acyclic for $t=1,2$, because $-(n+1)\leq -t-\ell\leq -1$ is the full negative acyclic range on $\mathbb{P}^{n+1}$. Pushing the finite resolution gives zero. The proof takes place on the ambient projective bundle and does not infer vanishing from the ordinary fibers of the nonflat map q. The projection formula for the perfect complex G [73, Tag 08EU] gives $Rq_*(\mathcal{L}^{-t}\otimes q^*G)\simeq(Rq_*\mathcal{L}^{-t})\overset{\mathbf{L}}{\otimes}G=0$. □

5.7. **Concentration and the sheaf-level Hecke realization.** Define, for $e=0,1$,

$$A_e=Rp_*(p^*\mathscr{D}\otimes\mathcal{L}^e\otimes q^*C'). \tag{66}$$

Proposition 5.22 (Concentration of both rows)**.** *$A_e\in\mathrm{Perf}^{[-n,0]}(Y)$, every $\mathcal{H}^i(A_e)$ is supported on $|F_n|$, and $\mathcal{H}^i(A_e)=0$ for $i\neq 0$. In other words, both A_0 and A_1 are coherent sheaves.*

Proof. The degree-$(-j)$ term of q^*C' is $\Lambda^j q^*\mathcal{T}'=\Lambda^j\mathcal{W}$, $0\leq j\leq n-2$. By Theorem 5.18 its contribution to (66), with its degree included, has Tor-amplitude $[-j-2,-j]$, and the finite filtration by the terms of C' and the projection formula for $\mathscr{D}$ give $A_e\in\mathrm{Perf}^{[-(n-2)-2,0]}(Y)=\mathrm{Perf}^{[-n,0]}(Y)$. This is a calculation with finite filtered perfect complexes, and no commutation of cohomology and pushforward is implicit. The complex q^*C' is contractible outside the zero set of its section, whose image under the proper map p is contained in F_n by (55), hence the cohomology of A_e is supported on F_n. Since $\dim Y=2n$ and $\dim F_n\leq n$, Theorem 3.8 with $m=c=n$ proves concentration in degree zero. □

Theorem 5.23 (Identification of the two rows)**.**

$$A_0\simeq i_*(\mathscr{D}|_B\otimes\mathcal{O}_Z),\qquad A_1\simeq i_*(\mathscr{D}|_B\otimes\mathcal{H}^{-1}(C_B))\simeq\mathcal{M}_n^+. \tag{67}$$

Proof. Tensor (56) with $p^*\mathscr{D}$ and apply Rp_*. Its middle term is

$$Rp_*(p^*\mathscr{D}\otimes\bar{p}^*C_B)\simeq i_*R\bar{p}_*(\bar{p}^*(\mathscr{D}|_B\otimes C_B))\simeq i_*((\mathscr{D}|_B\otimes C_B)\overset{\mathbf{L}}{\otimes}R\bar{p}_*\mathcal{O}_{\mathscr{H}})\simeq i_*(\mathscr{D}|_B\otimes C_B),$$

by Theorem 5.20 and the projection formula [73, Tag 08EU]. The projection formula applies because C_B is a bounded complex of vector bundles on B, hence perfect, and no flatness of $\bar{p}$ is required. We obtain the distinguished triangle

$$A_1[1]\longrightarrow i_*(\mathscr{D}|_B\otimes C_B)\longrightarrow A_0\longrightarrow A_1[2]. \tag{68}$$

Both A_e are sheaves by Theorem 5.22, and taking degrees -1 and 0 gives $A_1\simeq i_*(\mathscr{D}|_B\otimes\mathcal{H}^{-1}(C_B))$ and $A_0\simeq i_*(\mathscr{D}|_B\otimes\mathcal{H}^0(C_B))$, with $\mathcal{H}^0(C_B)=\mathcal{O}_Z$. The identification of the first sheaf with $\mathcal{M}_n^+$ is Theorem 5.15. □

Substituting (37) into the definition of A_1 gives

$$\mathcal{M}_n^+\simeq Rp_*(p^*\mathscr{D}\otimes\mathcal{L}\otimes q^*C')\simeq Rp_*(\mathcal{L}^{-1}\otimes q^*(\mathscr{D}'\otimes C')), \tag{69}$$

which is the realization assertion of Theorem 5.2, and the negative window (32) is Theorem 5.21. The amplitude-and-support argument proves that the entire pushforward is a sheaf before it is identified with $\mathcal{M}_n^+$, and we never take the zeroth cohomology of an object already known to lie in a null category.

5.8. **Proof of the vanishing theorem.**

5.8.1. *The common wall map.* The universal smaller algebra is a quotient of $S\otimes\mathcal{O}_{Y'}$, and its odd part is an S_0-linear quotient $S_1\otimes\mathcal{O}_{Y'}\twoheadrightarrow\mathcal{R}_1'$ of rank n, hence defines a morphism to the Quot scheme, which factors through its reduction Q_n since Y' is reduced (Theorem 2.1(a)). Denote it $h:Y'\to Q_n$. On $\mathscr{H}$ the odd quotients of the larger and the smaller cluster are the *same* quotient of S_1, not just isomorphic bundles. Hence the square

$$\begin{array}{ccc}\mathscr{H} & \xrightarrow{\ q\ } & Y'\\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle h}\\ Y & \xrightarrow[\ \rho\ ]{} & Q_n\end{array} \tag{70}$$

commutes as a square of moduli morphisms. It is not asserted to be Cartesian.

5.8.2. *Pushforward of the realized sheaf.* Put $G=\mathscr{D}'\otimes C'\in\mathrm{Perf}(Y')$. By (69), (70) and Theorem 5.21,

$$R\rho_*\mathcal{M}_n^+\simeq R(\rho p)_*(\mathcal{L}^{-1}\otimes q^*G)\simeq R(hq)_*(\mathcal{L}^{-1}\otimes q^*G)\simeq Rh_*(G\overset{\mathbf{L}}{\otimes}Rq_*\mathcal{L}^{-1})=0. \tag{71}$$

All compositions and projection formulas involve the indicated derived functors [73, Tag 08EU], the line bundle $\mathcal{L}$ is invertible, G is perfect, and none of p,q,ρ is assumed flat. This proves Theorem 5.1 for $n\geq 2$. For $n=1$, $\mathcal{R}_0$ is the unit line and $(-y,x):\mathcal{R}_0\to\mathcal{R}_1^{\oplus 2}$ is nowhere zero (a cyclic balanced algebra has dimension two). Its transpose is surjective, hence the even dual-socle sheaf, and therefore $\mathcal{M}_1^+$ by Theorem 3.14, is zero. □

Remark 5.24 (Which map ρ). The computation (71) uses nothing about the target of ρ beyond $\rho p=hq$. It proves $R\rho_*\mathcal{M}_n^+=0$ for every proper $\rho:Y\to Q$ that factors through the odd quotient $S_1\otimes\mathcal{O}_Y\twoheadrightarrow\mathcal{R}_1$, in particular for the VGIT contraction $Y_n=\mathfrak{M}_\theta((n,n),(1,0))\to\mathfrak{M}_{\theta_0}((n,n),(1,0))$, whose Y'-companion is the analogous contraction of $\mathfrak{M}_\theta((n-1,n),(1,0))$.

Remark 5.25 (What the proof does not use). The trace Koszul complex C_B is carried along rather than replaced by its zeroth cohomology, hence only the *support* of q^*C' enters, never its scheme structure, and no reducedness or Cohen–Macaulayness of F_n is used. The Grassmannian acyclicity of Theorem 4.5 reappears as the $(e,j)=(0,0)$ row of (59), but the vanishing is a consequence of amplitude and codimension, not of a single Borel–Weil–Bott group. There are four places where ordinary and derived operations must be kept separate. First, q^*C' is the termwise pullback, not $q^*\mathcal{O}_{Z'}$. Second, the projection formula for $\bar{p}$ uses only perfectness of C_B and $R\bar{p}_*\mathcal{O}_{\mathscr{H}}=\mathcal{O}_B$. Third, the two vanishings for q come from a regular-section resolution in a genuine projective bundle. Finally, the last computation uses only the equality of composed morphisms $\rho p = hq$.

5.8.3. *Cohen–Macaulay consequences.*

Corollary 5.26. *The trace-defined scheme $Z\subset B$ is Cohen–Macaulay of pure dimension n, and for $n\geq 2$ the sheaf $\mathcal{M}_n^+$ is nonzero, Cohen–Macaulay of pure dimension n, with set-theoretic support $|Z|$.*

Proof. By Theorems 5.22 and 5.23, both $i_*\mathcal{O}_Z$ (up to an invertible twist) and $\mathcal{M}_n^+$ are sheaves with local projective dimension at most n over the regular ambient Y and support of dimension at most n. At a closed point of the support the Auslander–Buchsbaum formula $\operatorname{depth} M+\operatorname{pd} M=\operatorname{depth} R$ [18, Thm. 1.3.3] gives depth at least $2n-n=n$, hence equality holds. Cohen–Macaulayness localizes and the same argument excludes components of smaller dimension. The scheme Z is nonempty, since the complete intersection $S/(xy,x^2+y^{2n-2})$ of Theorem 3.16 is a point of it. At a closed point of Z the trace ideal in the Cohen–Macaulay local ring of B has grade $(2n-2)-n=n-2$, and its Koszul complex has $n-1$ generators, hence grade sensitivity gives nonzero first homology. By Theorem 5.15, $\mathcal{M}_n^+$ is nonzero there, and its support equals $|Z|$. □

Warning 5.27. The corollary concerns the trace scheme Z inside the first even Brill–Noether locus. It does not prove that the entire scheme-theoretic punctual fiber F_n is Cohen–Macaulay or reduced, and the vanishing theorem does not by itself identify $R\rho_*\mathcal{I}_n$ with the structure sheaf of the symbolic central fiber.

Corollary 5.28 (Strong excess vanishing and the perfect reduction)**.** *For any perfect complex V on Q_n, $R\rho_*(\mathcal{M}_n^+ \overset{\mathbf{L}}{\otimes} L\rho^*V)=0$. In particular, with $\mathcal{L}_n=\rho^*\mathcal{O}_{Q_n}(1)$,*

$$R\rho_*(\mathcal{L}_n^{\otimes d}\otimes\mathcal{M}_n^+)=0 \qquad (d\in\mathbb{Z}),$$

and the positive-parity triangle (14) yields canonical isomorphisms

$$R\rho_*(\mathcal{L}_n^{\otimes d}\otimes\mathcal{K}_n^+)\xrightarrow{\sim} R\rho_*(\mathcal{L}_n^{\otimes d}\otimes\mathcal{I}_n) \qquad (d\in\mathbb{Z}). \tag{72}$$

Proof. Use the projection formula and Theorem 5.1, and then push forward the twisted triangle $\mathcal{M}_n^+[1]\to\mathcal{K}_n^+\to\mathcal{I}_n\to$. □

Part (3) of Theorem E is Theorem 5.1. The isomorphism (72) replaces the incidence sheaf $\mathcal{I}_n$, which can have infinite projective dimension over the singular wall variety, by the pushforward of the bounded complex of vector bundles $\mathcal{K}_n^+$, whose terms will be shown in Section 6 to be acyclic for the pushforward over the completion at every wall point.

5.9. **Rank two and rank six.** At $n=2$ the smaller moduli space is the single algebra $S/(x,y)^2$, hence Y' is a point and $\mathscr{H}=B\simeq\mathbb{P}^2$. The universal sequence has even part $0\to\mathcal{L}\to\mathcal{R}_0|_B\to\mathcal{O}_B\to 0$ with $\mathcal{L}=\mathcal{O}_{\mathbb{P}^2}(1)$, while $\mathcal{R}_1|_B$ is constant. Hence $\mathscr{D}|_B=\mathcal{L}^{-2}$ and $\mathcal{T}|_B=\mathcal{L}$. Every cluster on B is punctual and the augmentation vanishes on $\mathcal{L}$, hence $C_B\simeq\mathcal{O}_B\oplus\mathcal{L}[1]$ and $C'=\mathcal{O}_{Y'}$. The realization formula gives

$$\mathcal{M}_2^+\simeq i_*\mathcal{O}_{\mathbb{P}^2}(-1), \qquad R\rho_*\mathcal{M}_2^+=R\Gamma(\mathbb{P}^2,\mathcal{O}(-1))=0,$$

in agreement with a direct local computation of the module $\mathcal{M}_2^+$ on charts.

At $n = 6$, the first rank in which even socle rank two occurs (Theorem 4.9), we have $a = 5$, $\operatorname{rk}\mathcal{G} = 6$, $\operatorname{rk}\mathcal{T}' = 4$. The first ambient space has fibers $\mathbb{P}^4$ and the second $\mathbb{P}^7$, the two regular sections have ranks 6 and 5, the four smaller-trace exterior degrees extend the interval $[-2, 0]$ to $[-6, 0]$, and punctual support has codimension at least six. The $\mathrm{Gr}(3, 5)$ wall fiber and the rank-one strata meeting it are treated simultaneously, because the Hecke scheme contains all choices of one even socle line, including those inside a higher-dimensional even socle, and no separate rank-two induction is needed. The cancellation $-(n-2)-2 = -n$ is the reason the argument is uniform in n.

6. Symbolic descent

In this section we prove Theorem E(4):

$$R\rho_*(\mathcal{L}_n^{\otimes d} \otimes \mathcal{I}_n) \;\simeq\; \mathcal{O}_{\mathcal{Z}_n^{\mathrm{symb}}}(d), \qquad d \in \mathbb{Z}. \tag{73}$$

The inputs are the two-term homology of Theorem 3.13, the even-excess vanishing of Theorem 5.1, and, at each wall point, the vector-bundle part of the Bellamy–Craw normal form. The latter says that on the completed neighbourhood of a wall fiber the even cluster bundle is a trivial summand plus the tautological quotient bundle of a cotangent Grassmannian, and the odd cluster bundle is trivial.

The main idea is to compute the derived pushforward of the positive Haiman complex $\mathcal{K}_n^+$ instead of the derived pushforward of $\mathcal{O}_{F_n}$. The two agree by the even-excess vanishing, and $\mathcal{K}_n^+$ is a complex of vector bundles, to which Borel–Weil–Bott applies. Borel–Weil–Bott does two things. It makes every term of $\mathcal{K}_n^+$ acyclic for the completed pushforward, hence the pushforward is computed termwise. A second vanishing, with one subbundle factor, makes multiplication of pushed-forward sections surjective, and this identifies the image of the first differential. The pushed cluster algebra is then shown to be generated by the original polynomials, and the elementary normalized-trace lemma of Section 6.3 computes its punctual quotient, nilpotents included. Finally the even-excess vanishing forces the derived pushforward to be concentrated in degree zero. Nothing is assumed about a product decomposition of the cluster algebra, about flatness of ρ, or about reducedness or Cohen–Macaulayness of the punctual schemes. The scalar trace functions retain their full dependence on the cotangent and core coordinates throughout.

The section is organized as follows. In Sections 6.1 and 6.2 we define the canonical map u_n and explain why the statement is not a formal consequence of the vanishing theorems. Section 6.3 contains the normalized-trace lemma. In Sections 6.4 and 6.5 we collect the vanishing and surjectivity results on the cotangent Grassmannian, and in Sections 6.6 to 6.8 we transfer them to the formal neighbourhood of a wall fiber. The local computation and the proof of the theorem are in Sections 6.9 and 6.10. The case $n = 2$ is worked out in closed form in Section 6.11.

6.1. **The canonical map and the perfect reduction.** Write $\mathfrak{a} := \mathfrak{m}_0\mathcal{O}_{Q_n}$, hence $\mathcal{Z}_n^{\mathrm{symb}} = V(\mathfrak{a})$ and $F_n = V(\mathfrak{a}\mathcal{O}_{Y_n}) = Y_n \times_{Q_n} \mathcal{Z}_n^{\mathrm{symb}}$. These are ordinary scheme-theoretic fibers, $\mathcal{O}_{F_n} = \mathcal{O}_{Y_n}/\mathfrak{a}\mathcal{O}_{Y_n}$, and $\mathcal{I}_n \cong \mathcal{O}_{F_n}$ by Theorem 3.4. Let $j : \mathcal{Z}_n^{\mathrm{symb}} \hookrightarrow Q_n$ and $f = \rho|_{F_n} : F_n \to \mathcal{Z}_n^{\mathrm{symb}}$, hence the square

$$\begin{array}{ccc} F_n & \hookrightarrow & Y_n \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle \rho} \\ \mathcal{Z}_n^{\mathrm{symb}} & \xrightarrow[j]{} & Q_n \end{array}$$

is Cartesian. The adjunction unit $\mathcal{O}_{\mathcal{Z}} \to Rf_*\mathcal{O}_{F_n}$ for the proper morphism f, followed by the exact functor j_*, is a canonical T-equivariant morphism

$$u_n : \mathcal{O}_{\mathcal{Z}_n^{\mathrm{symb}}} \longrightarrow R\rho_*\mathcal{O}_{F_n} = R\rho_*\mathcal{I}_n. \tag{74}$$

It exists as a unit for the actual morphism f, and nothing is inferred from annihilation of cohomology. Since $\mathcal{L}_n = \rho^*\mathcal{O}_{Q_n}(1)$, the projection formula reduces (73) to the untwisted statement

$$u_n : \ \mathcal{O}_{\mathcal{Z}_n^{\mathrm{symb}}} \xrightarrow{\sim} R\rho_*\mathcal{O}_{F_n}, \tag{75}$$

which is what we shall prove.

Proposition 6.1 (Perfect reduction)**.** *The augmentation $\mathcal{K}_n^+ \to \mathcal{I}_n$ of (14) induces a canonical isomorphism $R\rho_*\mathcal{K}_n^+ \xrightarrow{\sim} R\rho_*\mathcal{I}_n$.*

Proof. Apply $R\rho_*$ to (14). Its first and last terms vanish by Theorem 5.1. □

The sheaf $\mathcal{M}_n^+$ does not vanish on Y_n (Theorem 5.26), only its derived image does. The proposition does not by itself identify $R\rho_*\mathcal{K}_n^+$ with a sheaf, let alone with $\mathcal{O}_{\mathcal{Z}}$, and that is the task of the rest of the section. Its advantage is that $\mathcal{K}_n^+$ is a bounded complex of vector bundles on the smooth Y_n, whereas $\mathcal{O}_{F_n}$ can have infinite projective dimension over the singular wall variety (Theorem 6.2).

6.2. **Descent is not Tor-independent base change.** Let $j : \mathcal{Z}_n^{\mathrm{symb}} \hookrightarrow Q_n$. Since Q_n has rational singularities (Theorem 2.1(a)), $R\rho_*\mathcal{O}_{Y_n} = \mathcal{O}_{Q_n}$, and the projection formula [73, Tag 08EU] gives

$$R\rho_*L\rho^*\mathcal{O}_{\mathcal{Z}_n^{\mathrm{symb}}} \simeq R\rho_*\mathcal{O}_{Y_n} \overset{\mathbf{L}}{\otimes}_{\mathcal{O}_{Q_n}} \mathcal{O}_{\mathcal{Z}_n^{\mathrm{symb}}} \simeq \mathcal{O}_{\mathcal{Z}_n^{\mathrm{symb}}}. \tag{76}$$

Since $F_n = \rho^{-1}(\mathcal{Z}_n^{\mathrm{symb}})$ scheme-theoretically (both are cut out by $\mathfrak{m}_0$), $\mathcal{H}^0(L\rho^*\mathcal{O}_{\mathcal{Z}_n^{\mathrm{symb}}}) = \mathcal{O}_{F_n} = \mathcal{I}_n$ by Theorem 3.4. Thus (75) is exactly the assertion that $\mathcal{I}_n$ may be substituted for $L\rho^*\mathcal{O}_{\mathcal{Z}_n^{\mathrm{symb}}}$, and it would follow at once if the higher Tor sheaves $\operatorname{Tor}_i^{\mathcal{O}_{Q_n}}(\mathcal{O}_{Y_n}, \mathcal{O}_{\mathcal{Z}_n^{\mathrm{symb}}})$, $i > 0$, vanished, for instance if $\mathcal{Z}_n^{\mathrm{symb}}$ were a local complete intersection of codimension n in the Cohen–Macaulay Q_n. This is the case for $n = 1$, where $Q_1 \cong T^*\mathbb{P}^1$ is smooth and $\mathcal{Z}_1^{\mathrm{symb}}$ is a divisor. It fails already at $n = 2$.

Proposition 6.2. *Let $z_J \in \mathcal{Z}_2^{\mathrm{symb}}$ be the heart lying under the wall fiber $F_J \cong \mathbb{P}^2$. Then*

$$\mu(\mathfrak{m}_0\mathcal{O}_{Q_2,z_J}) = 3 \ > \ 2 = \operatorname{ht}(\mathfrak{m}_0\mathcal{O}_{Q_2,z_J}),$$

hence $\mathcal{Z}_2^{\mathrm{symb}}$ is not a local complete intersection in Q_2 at z_J. Moreover $\operatorname{pd}_{\mathcal{O}_{Q_2,z_J}} \mathcal{O}_{\mathcal{Z}_2^{\mathrm{symb}},z_J} = \infty$, although $\mathcal{Z}_2^{\mathrm{symb}}$ itself is reduced, irreducible and Cohen–Macaulay of dimension 2.

Proof. Write $R_0 = \mathbb{C}[u,v,w]/(uv - w^2)$, hence $R_1 = R_0X + R_0Y$ with relations $vX = wY$ and $uY = wX$. The locus of $q \in \operatorname{Quot}_X^2(R_1)$ for which $(q(X), q(Y))$ is a basis of the length-two quotient is open, contains z_J, and is rigidified by that basis. A point of it is a triple of pairwise commuting 2×2 matrices U, V, W, the actions of u, v, w, with $UV = W^2$, $Ve_1 = We_2$, $Ue_2 = We_1$. With $W = (w_{ij})$ the two column conditions force $U = \left(\begin{smallmatrix} u_1 & w_{11} \\ u_2 & w_{21} \end{smallmatrix}\right)$, $V = \left(\begin{smallmatrix} w_{12} & v_1 \\ w_{22} & v_2 \end{smallmatrix}\right)$, hence the chart is the closed subscheme of $\mathbb{A}^8$ cut out by the entries of $[U,V]$, $[U,W]$, $[V,W]$ and $UV - W^2$, all homogeneous quadrics, and z_J is the origin. Hence $I(Q_2)_{z_J} \subseteq \mathfrak{m}_{z_J}^2$ and $\dim \mathfrak{m}_{z_J}/\mathfrak{m}_{z_J}^2 = 8$. The support morphism $Q_2 \to \operatorname{Sym}^2 X$ is given by polarized power sums, hence $\mathfrak{m}_0\mathcal{O}_{Q_2}$ contains $\operatorname{Tr} U = u_1 + w_{21}$, $\operatorname{Tr} V = w_{12} + v_2$, $\operatorname{Tr} W = w_{11} + w_{22}$, three linearly independent linear forms whose images in $\mathfrak{m}_{z_J}/\mathfrak{m}_{z_J}^2$ remain independent since $I(Q_2)$ contains no linear form. Thus $\mu(\mathfrak{m}_0\mathcal{O}_{Q_2,z_J}) \geq 3$, while its height is $\dim Q_2 - \dim \mathcal{Z}_2^{\mathrm{symb}} = 4 - 2 = 2$. The remaining assertions are Gröbner computations in this chart. The chart is reduced, irreducible, Gorenstein of codimension four (Betti numbers $1, 9, 16, 9, 1$ over $\mathbb{C}[\mathbb{A}^8]$), in agreement with the local model of Section 4.5, whose transverse singularity at $r = 1$ is $\mathfrak{N}_{1,3} = \overline{\mathcal{O}}_{\min} \subset \mathfrak{sl}_3$. The three linear traces generate $\mathfrak{m}_0\mathcal{O}_{Q_2,z_J}$, hence $\mu = 3$. The ideal $I(\mathcal{Z}_2^{\mathrm{symb}})$ has projective dimension $6 = \operatorname{codim}$ over $\mathbb{C}[\mathbb{A}^8]$, hence $\mathcal{Z}_2^{\mathrm{symb}}$ is Cohen–Macaulay, reduced and irreducible. Finally, the minimal free resolution of $\mathcal{O}_{\mathcal{Z}_2^{\mathrm{symb}},z_J}$ over $\mathcal{O}_{Q_2,z_J}$ begins $1, 3, 6, 18, 66, 246, 918$. Since $\mathcal{O}_{Q_2,z_J}$ is Cohen–Macaulay of depth 4, Auslander–Buchsbaum forces $\operatorname{pd} \leq 4$ whenever it is finite, and a nonzero fifth syzygy proves $\operatorname{pd} = \infty$. □

The failure is one of depth over a singular base, not of $\mathcal{Z}_n^{\mathrm{symb}}$ itself. Note that $\mu = 3 = n+1$ exceeds the height by one, the same deviation as $\operatorname{rk}\mathcal{E}_n = n+1$ against $\operatorname{codim} F_n = n$ upstairs. The higher-Tor defect must therefore be controlled by additional structure, and the structure used below is the positivity of the tautological quotient bundles on the wall fibers, together with the even-excess vanishing.

6.3. **The normalized-trace quotient.** In this subsection we prove an elementary lemma of commutative algebra. It computes the quotient of a parity-graded algebra by the ideal generated by the trace-zero elements and by the odd elements times x and y, and it identifies this quotient with the quotient of the base ring by an ideal of traces. In Section 6.9 the lemma is applied to the pushed-forward cluster algebra at a wall point, which is not locally free, and this application dictates the generality of the statement. Let us first fix the notation on Y_n. For $n>0$ put

$$\epsilon := \tfrac{1}{n}\operatorname{Tr}_{\mathcal{R}_0} : \mathcal{R}_0 \longrightarrow \mathcal{O}_{Y_n}, \qquad \epsilon(1)=1, \qquad \mathcal{R}_0 = \mathcal{O}_{Y_n}\cdot 1 \oplus \mathcal{T}_n, \quad \mathcal{T}_n = \ker\epsilon. \tag{77}$$

For an even polynomial $f \in R_0 = \mathbb{C}[x,y]^\Gamma$ we write $\bar f \in \mathcal{R}_0$ for its image in the universal cluster algebra, as in the proof of Theorem 3.4, and we put $\tau_f := \operatorname{Tr}_{\mathcal{R}_0}(m_{\bar f}) = n\,\epsilon(\bar f) \in \mathcal{O}_{Y_n}$. By Theorem 3.3 and Haiman's trace identity (proof of Theorem 3.4),

$$(\tau_f :\ f\in R_0,\ f(0)=0) = \mathfrak{m}_0\mathcal{O}_{Y_n}. \tag{78}$$

Here $\operatorname{Tr}_{\mathcal{R}_0}$ is the trace of an endomorphism of the rank-n vector bundle $\mathcal{R}_0$. We shall push ϵ forward, but we never take the trace of a module which is not locally free.

In the lemma below and in Sections 6.8 and 6.9 the letter R denotes an arbitrary base ring, which in the application is the completed local ring of Q_n at a wall point. It should not be confused with the polynomial ring $R=\mathbb{C}[x,y]$ of Section 2.1. We write x,y also for the images of the variables in B_1.

Lemma 6.3 (Normalized-trace quotient)**.** *Let R be a commutative $\mathbb{C}$-algebra and let $B = B_0 \oplus B_1$ be a commutative $\mathbb{Z}/2$-graded R-algebra. Assume that the parity-preserving homomorphism of R-algebras $R[x,y]\to B$, with x,y odd, is surjective, and for a polynomial $f\in R[x,y]$ write $\bar f\in B$ for its image. Let $\epsilon : B_0 \to R$ be an R-linear map with $\epsilon(1)=1$ and put $T=\ker\epsilon$, hence $B_0 = R\cdot 1\oplus T$. Fix a nonzero integer n, and for an even polynomial $f\in R[x,y]$ put $\tau_f := n\,\epsilon(\bar f)\in R$. Let*

$$I = (\tau_f :\ f\in R[x,y] \text{ even},\ f(0,0)=0) \subset R$$

be the ideal generated by the elements τ_f for the even polynomials f without constant term. Since ϵ is R-linear, I is generated by the elements $\tau_{x^ay^b} = n\,\epsilon(\overline{x^ay^b})$ with $a+b$ even and positive, and the factor n is immaterial for I. Then the unit $R\to B_0$ induces an isomorphism of R-algebras

$$R/I \xrightarrow{\ \sim\ } B_0/(B_0T + xB_1 + yB_1), \tag{79}$$

whose inverse is induced by ϵ. No multiplicativity of ϵ, no local freeness of B and no reducedness of either ring is assumed.

Proof. Step 1, generators of T. Since $R[x,y]\to B$ is surjective and preserves parity, B_0 is generated as an R-module by the images $\bar f$ of the even monomials f, including $f=1$, and B_1 by the images of the odd monomials. For an even monomial f put $g_f := \bar f - \epsilon(\bar f)1 = \bar f - \frac{\tau_f}{n}1 \in T$. Then $g_1=0$, and applying $b\mapsto b-\epsilon(b)1$ to an expression of $b\in T$ in the generators $\bar f$ shows that T is generated over R by the g_f with f an even monomial of positive degree.

Step 2, the quotient is generated by $[1]$. Each of B_0T, xB_1 and yB_1 is stable under multiplication by B_0, hence $B_0T+xB_1+yB_1$ is an ideal of B_0. Since T is killed and $B_0 = R\cdot 1\oplus T$, every element of the quotient is an R-multiple of $[1]$, that is, the unit map $R\to B_0/(B_0T+xB_1+yB_1)$ is surjective.

Step 3, I maps to zero. A positive-degree even monomial f is x or y times an odd monomial, hence $\bar{f} \in xB_1 + yB_1$. Thus both $\bar{f}$ and g_f vanish in the quotient, and therefore $\frac{\tau_f}{n}[1] = [\bar{f}] - [g_f] = 0$. The unit factors through R/I.

Step 4, ϵ kills the relations modulo I. For an odd monomial a the monomials xa and ya are even of positive degree, hence $\epsilon(x\bar{a}) = \epsilon(\overline{xa}) = \tau_{xa}/n \in I$ and likewise $\epsilon(y\bar{a}) \in I$. For an even monomial f and a positive-degree even monomial g, R-linearity gives

$$\epsilon(\bar{f}g_g) = \epsilon(\overline{fg}) - \tfrac{\tau_g}{n}\epsilon(\bar{f}) = \tfrac{\tau_{fg}}{n} - \tfrac{\tau_f\tau_g}{n^2} \in I.$$

Indeed, for $f = 1$ the expression vanishes because $\tau_1 = n\,\epsilon(1) = n$, and for $f \neq 1$ the monomial fg has positive degree. Since B_0T is spanned over R by the products $\bar{f}g_g$, and xB_1, yB_1 are spanned by the elements $x\bar{a}$, $y\bar{a}$, we conclude that $\epsilon(B_0T + xB_1 + yB_1) \subset I$. Hence ϵ induces an R-linear map $\bar{\epsilon}$ from the quotient to R/I.

Step 5, conclusion. The composite $R/I \to B_0/(B_0T + xB_1 + yB_1) \xrightarrow{\bar{\epsilon}} R/I$ is $r \mapsto \epsilon(r1) = r$. A surjection with a left inverse is an isomorphism, and the inverse of an isomorphism of algebras is an algebra map. □

Remark 6.4 (How the lemma is used). The lemma is used twice. First, on affine opens of Y_n we take $B = \mathcal{P}_n$ and ϵ the normalized trace of (77), hence $T = \mathcal{T}_n$ and $I = \mathfrak{m}_0\mathcal{O}_{Y_n}$ by (78). The lemma gives $\mathcal{I}_n = \mathcal{P}_n/(\mathcal{T}_n, x, y)\mathcal{P}_n \cong \mathcal{O}_{Y_n}/\mathfrak{m}_0\mathcal{O}_{Y_n}$, which is a second proof of Theorem 3.4. This proof exhibits the inverse map and therefore controls the scheme structure without a Nakayama argument on a kernel.

The essential application is in Section 6.9. There $R = \widehat{\mathcal{O}}_{Q_n,z_J}$ is the completed local ring of the wall variety at a fixed point z_J, the algebra B is the completion of the pushed-forward cluster algebra $\rho_*\mathcal{P}_n$, and ϵ is the pushforward of the normalized trace. At a wall point with a positive-dimensional fiber B_0 is not locally free over R, its fiber at z_J has dimension larger than n, and ϵ is not the trace of the R-module B_0. This is why the lemma assumes neither multiplicativity of ϵ nor local freeness of B. The hypotheses of the lemma are verified in Section 6.8. The surjectivity of $R[x,y] \to B$ is Theorem 6.20, the splitting $B_0 = R \cdot 1 \oplus T$ is (102), and the equality $I = \mathfrak{m}_0 R$ is (103), which comes from Haiman's trace identity (78). The conclusion of the lemma identifies the cokernel of the first differential of the pushed-forward Haiman complex, which is $B_0/(B_0T + xB_1 + yB_1)$ by (106), with $R/\mathfrak{m}_0R = \widehat{\mathcal{O}}_{\mathcal{Z}_n^{\mathrm{symb}},z_J}$, and it shows that the identification is induced by the unit, that is, by the descent map u_n. The lemma asserts an equality of ordinary quotients, not of radicals, and this is what makes the nilpotent structure of $\mathcal{Z}_n^{\mathrm{symb}}$ visible in the computation.

6.4. **Two Grassmannian vanishing theorems.** In this subsection $G = \mathrm{Gr}(k, V)$ with $\dim V = N = k + r$, $k, r \geq 1$, parametrizes k-dimensional subspaces, with tautological sequence

$$0 \longrightarrow \mathcal{U} \longrightarrow V \otimes \mathcal{O}_G \longrightarrow \mathcal{Q} \longrightarrow 0, \qquad \mathrm{rk}\,\mathcal{U} = k, \quad \mathrm{rk}\,\mathcal{Q} = r \tag{80}$$

($\mathcal{U}$ is the bundle written $\mathcal{S}$ in Section 4.2). Thus $T_G = \mathcal{U}^\vee \otimes \mathcal{Q}$, and for $X = T^*G$ with projection $p : X \to G$,

$$p_*\mathcal{O}_X = \mathrm{Sym}(T_G) = \bigoplus_{d\geq 0} \mathrm{Sym}^d(\mathcal{U}^\vee \otimes \mathcal{Q}). \tag{81}$$

We give fiber-polynomial functions degree 1 (the *auxiliary grading*, which is unrelated to the q,t-bigrading and is used only on the model). Let $\mu : X \to \mathfrak{N}$ be the Springer contraction onto the reduced square-zero orbit closure and $A = \Gamma(X, \mathcal{O}_X) = \bigoplus_{d\geq 0} A_d$, $A_0 = \mathbb{C}$, with $A_+ = \bigoplus_{d>0} A_d$.

Borel–Weil–Bott convention. For a dominant weight $\beta \in \mathbb{Z}^k$ we interpret $\mathbb{S}^\beta E = (\det E)^{\beta_k} \otimes \mathbb{S}^{\beta-\beta_k(1,\dots,1)}E$, which allows a last entry $\beta_k = -1$. For dominant $\alpha \in \mathbb{Z}^r$, $\beta \in \mathbb{Z}^k$ the homogeneous bundle $\mathbb{S}^\alpha\mathcal{Q} \otimes \mathbb{S}^\beta\mathcal{U}^\vee$ has, in the quotient-first convention, Bott weight

$$\gamma = (\alpha_1, \dots, \alpha_r, -\beta_k, \dots, -\beta_1) \in \mathbb{Z}^N. \tag{82}$$

Put $\varrho = (N-1, \dots, 1, 0)$. If $\gamma + \varrho$ has a repeated entry the bundle is acyclic, and otherwise its cohomology is concentrated in the degree equal to the length of the permutation sorting $\gamma + \varrho$, in particular in degree 0 when γ is dominant [17]. The signs are calibrated by $\mathcal{Q}$ (weight $(1, 0, \dots, 0)$, $H^0 = V$) and $\mathcal{U}$ (block boundary $0, 1$, acyclic).

Theorem 6.5 (Positive and subbundle-tensor vanishing)**.** *For every partition λ with at most r parts,*

$$H^i(X, p^*\mathbb{S}^\lambda \mathcal{Q}) = 0 \qquad (i > 0), \tag{83}$$

$$H^i(X, p^*(\mathcal{U} \otimes \mathbb{S}^\lambda \mathcal{Q})) = 0 \qquad (i > 0), \tag{84}$$

in each fiber-polynomial degree separately.

Proof. Step 1. Since p is affine, for a bundle $\mathcal{F}$ on G

$$H^i(X, p^*\mathcal{F}) = \bigoplus_{d \geq 0} H^i(G, \mathcal{F} \otimes \operatorname{Sym}^d(\mathcal{U}^\vee \otimes \mathcal{Q})), \tag{85}$$

cohomology commuting with the direct sum (compute with a finite affine Čech cover of G).

Step 2: the positive case. By the Cauchy formula

$$\operatorname{Sym}^d(\mathcal{U}^\vee \otimes \mathcal{Q}) = \bigoplus_{\nu \vdash d,\ \ell(\nu) \leq \min(k,r)} \mathbb{S}^\nu \mathcal{U}^\vee \otimes \mathbb{S}^\nu \mathcal{Q}. \tag{86}$$

Decomposing $\mathbb{S}^\lambda \mathcal{Q} \otimes \mathbb{S}^\nu \mathcal{Q}$ by the Littlewood–Richardson rule into bundles $\mathbb{S}^\alpha \mathcal{Q}$, α a partition, each summand $\mathbb{S}^\alpha \mathcal{Q} \otimes \mathbb{S}^\nu \mathcal{U}^\vee$ has Bott weight $(\alpha_1, \dots, \alpha_r, -\nu_k, \dots, -\nu_1)$, which is dominant since $\alpha_r \geq 0 \geq -\nu_k$. This proves (83).

Step 3: the subbundle factor. Write $E = \mathcal{U}^\vee$. Since $E^\vee \cong \Lambda^{k-1} E \otimes (\det E)^{-1}$, the Pieri rule for exterior powers shows that the weights occurring in $E^\vee \otimes \mathbb{S}^\nu E$ are exactly the dominant weights

$$\beta = \nu - \epsilon_j \qquad (1 \leq j \leq k), \tag{87}$$

because tensoring with $\Lambda^{k-1} E$ adds a vertical strip in $k-1$ distinct rows and $(\det E)^{-1}$ subtracts one from every row. Every entry of β is nonnegative except possibly $\beta_k = -1$.

Step 4: the three cases. A summand of $\mathcal{U} \otimes \mathbb{S}^\lambda \mathcal{Q} \otimes \operatorname{Sym}^d(\mathcal{U}^\vee \otimes \mathcal{Q})$ has weight

$$\gamma = (\alpha_1, \dots, \alpha_r, -\beta_k, \dots, -\beta_1),$$

with α a partition and β as in (87). If $\beta_k \geq 0$, γ is dominant. If $\beta_k = -1$ and $\alpha_r \geq 1$, then $\alpha_r \geq -\beta_k$ and γ is again dominant. If $\beta_k = -1$ and $\alpha_r = 0$, the entries of $\gamma + \varrho$ in positions r and $r+1$ are $0 + (N - r) = k$ and $1 + (N - r - 1) = k$, hence the shifted weight is singular and all cohomology of the summand vanishes. No regular non-dominant weight occurs, which proves (84). □

Corollary 6.6. *Both vanishings hold with $\mathbb{S}^\lambda \mathcal{Q}$ replaced by a finite direct sum of tensor powers of $\mathcal{Q}$ or by a direct summand of such a bundle. Since the target of μ is affine, $R^i\mu_*$ of the corresponding pulled-back bundles vanishes for $i > 0$.*

Proof. In characteristic zero tensor powers of $\mathcal{Q}$ are finite sums of polynomial Schur bundles, and cohomology is additive and respects split idempotents. □

The following variant, with one dual quotient factor, is used only in the proof of Theorem 6.13.

Lemma 6.7 (Vanishing with one dual quotient factor)**.** *For every partition λ with at most r parts, $H^i(X, p^*(\mathcal{Q}^\vee \otimes \mathbb{S}^\lambda \mathcal{Q})) = 0$ for $i > 0$. Consequently, for $E_0 = \mathcal{O}_G^{\oplus m} \oplus \mathcal{Q}$ with $m \geq 0$,*

$$H^i(G, \mathcal{E}nd(E_0) \otimes \operatorname{Sym}^d T_G) = 0 \qquad (i > 0,\ d \geq 0). \tag{88}$$

Proof. By (85) and (86) it suffices to treat $\mathcal{Q}^\vee \otimes \mathbb{S}^\lambda\mathcal{Q} \otimes \mathbb{S}^\nu\mathcal{Q} \otimes \mathbb{S}^\nu\mathcal{U}^\vee$. Decompose $\mathbb{S}^\lambda\mathcal{Q} \otimes \mathbb{S}^\nu\mathcal{Q}$ into $\mathbb{S}^\mu\mathcal{Q}$, μ a partition, and then $\mathcal{Q}^\vee \otimes \mathbb{S}^\mu\mathcal{Q}$ by (87) applied to the quotient block. The summands are $\mathbb{S}^\alpha\mathcal{Q}$ with $\alpha = \mu - \epsilon_j$ dominant, hence every entry of α is nonnegative except possibly $\alpha_r = -1$. The Bott weight is $\gamma = (\alpha_1, \dots, \alpha_r, -\nu_k, \dots, -\nu_1)$. If $\alpha_r \geq 0$, or if $\alpha_r = -1$ and $\nu_k \geq 1$, it is dominant. If $\alpha_r = -1$ and $\nu_k = 0$, the entries of $\gamma + \varrho$ in positions $r, r+1$ are both $k-1$, and the summand is acyclic. For (88) write $\mathcal{E}nd(E_0) = \mathcal{O}^{\oplus m^2} \oplus \mathcal{Q}^{\oplus m} \oplus (\mathcal{Q}^\vee)^{\oplus m} \oplus (\mathcal{Q} \otimes \mathcal{Q}^\vee)$ and use (83) with $\lambda = \varnothing, (1)$ for the first two summands and the vanishing just proved with $\lambda = \varnothing, (1)$ for the last two. Finally, $H^i(G, \mathcal{F} \otimes \operatorname{Sym}^d T_G)$ is the degree-d part of $H^i(X, p^*\mathcal{F})$ by (85). □

6.5. **Tensor surjectivity and the degree-zero restriction.** For a bundle $\mathcal{F}$ on G put $M_\mathcal{F} = \Gamma(X, p^*\mathcal{F})$, a finite A-module since μ is proper. There is a natural tensor-of-sections map

$$m_{\mathcal{F},\mathcal{G}} : M_\mathcal{F} \otimes_A M_\mathcal{G} \longrightarrow M_{\mathcal{F}\otimes\mathcal{G}}, \tag{89}$$

which in general need not be surjective. It is not the multiplication of a cluster algebra, the latter will be composed with it.

Proposition 6.8 (Surjectivity with one quotient factor)**.** *For $\mathcal{F} = \mathbb{S}^\lambda\mathcal{Q}$ the natural maps*

$$V \otimes_\mathbb{C} M_\mathcal{F} \twoheadrightarrow M_{\mathcal{Q}\otimes\mathcal{F}}, \qquad M_\mathcal{Q} \otimes_A M_\mathcal{F} \twoheadrightarrow M_{\mathcal{Q}\otimes\mathcal{F}} \tag{90}$$

are surjective.

Proof. Tensor (80) with $\mathcal{F}$, pull back to X (exact, as $\mathcal{F}$ is locally free and p flat) and take sections. This gives the exact sequence $V \otimes M_\mathcal{F} \to M_{\mathcal{Q}\otimes\mathcal{F}} \to H^1(X, p^*(\mathcal{U} \otimes \mathcal{F})) = 0$, where the last vanishing is (84). The first map factors through the second via the degree-zero sections $V \to M_\mathcal{Q}$. □

Proposition 6.9 (Tensor powers, sums and retracts)**.** *The map (89) is surjective for every pair of bundles on X which are direct summands of finite direct sums of the bundles $(p^*\mathcal{Q})^{\otimes a}$, $a \geq 0$, and the splitting idempotents need not be homogeneous for the auxiliary grading.*

Proof. Theorem 6.8 holds for $\mathcal{F} = \mathcal{Q}^{\otimes b}$ by its Schur decomposition. Applying it a times gives a surjection $V^{\otimes a} \otimes M_{\mathcal{Q}^{\otimes b}} \twoheadrightarrow M_{\mathcal{Q}^{\otimes(a+b)}}$, which factors through $M_{\mathcal{Q}^{\otimes a}} \otimes_A M_{\mathcal{Q}^{\otimes b}}$, hence the tensor map is surjective, and direct sums preserve this. If $\mathcal{F}, \mathcal{G}$ are retracts of $\mathcal{F}', \mathcal{G}'$ with inclusions i_F, i_G and retractions p_F, p_G, then for a section s of $\mathcal{F} \otimes \mathcal{G}$ include it into $\mathcal{F}' \otimes \mathcal{G}'$, lift to $\sum_a u_a \otimes v_a$, and apply p_F, p_G. Naturality of (89) gives $m(\sum_a p_F u_a \otimes p_G v_a) = (p_F \otimes p_G)(i_F \otimes i_G)s = s$. The argument is ungraded. □

The first map in (90) with $\mathcal{F} = \mathcal{O}_G$ is a surjection

$$A \otimes_\mathbb{C} V \twoheadrightarrow M_\mathcal{Q} \tag{91}$$

by generators of degree zero.

Lemma 6.10 (Restriction of the quotient bundle)**.** *Restriction to the zero section induces an isomorphism*

$$M_\mathcal{Q}/A_+M_\mathcal{Q} \xrightarrow{\sim} H^0(G, \mathcal{Q}) = V. \tag{92}$$

Proof. $M_\mathcal{Q} = \bigoplus_d (M_\mathcal{Q})_d$ with $(M_\mathcal{Q})_0 = H^0(G, \mathcal{Q})$ by (85). The surjection (91) is graded with V in degree 0, hence $(M_\mathcal{Q})_{>0} = A_+M_\mathcal{Q}$ and $M_\mathcal{Q}/A_+M_\mathcal{Q} = (M_\mathcal{Q})_0$, and restriction to the zero section kills positive fiber degree and is the identity in degree zero. Finally $V \to H^0(G, \mathcal{Q})$ is an isomorphism (Borel–Weil for $\mathcal{Q}$, or (80) with $\mathcal{U}$ acyclic). □

This is an ordinary residue-field quotient of a specified finite module, not a base change assertion for the nonflat μ, and nothing is claimed about $\operatorname{Tor}_i^A(M_\mathcal{Q}, \mathbb{C})$ or about derived pullback of $\mathcal{O}_\mathfrak{N}$.

6.6. **The formal bundle model at a wall point.** Fix a T-fixed point $z_J \in Q_n^T$ with $k = k_J \geq 2$ and put $r = k-1$, $V = V_J$, $N = \dim V = 2r+1$ (Theorem 4.1), $F_J = \rho^{-1}(z_J)$. From now on $G = \mathrm{Gr}(k, V)$, $X = T^*G$ and $\mathcal{U}, \mathcal{Q}$ refer to this Grassmannian, and $\mathcal{Q}_J = \mathcal{Q}$. Recall $J' = R_1 J$, $J'' = (J : R_1)$, $V_J = J''/J'$.

Lemma 6.11 (The scheme-theoretic fiber)**.** *$F_J \cong \mathrm{Gr}(k, V_J)$ as schemes, and on it there are exact sequences*

$$0 \longrightarrow \mathcal{U}_J \longrightarrow V_J \otimes \mathcal{O}_{F_J} \longrightarrow \mathcal{Q}_J \longrightarrow 0, \tag{93}$$

$$0 \longrightarrow \mathcal{U}_J \longrightarrow (R_0/J') \otimes \mathcal{O}_{F_J} \longrightarrow \mathcal{R}_0|_{F_J} \longrightarrow 0, \tag{94}$$

and $\mathcal{R}_1|_{F_J} = (R_1/J) \otimes \mathcal{O}_{F_J}$.

Proof. Q_n is the reduced Quot scheme of odd quotients (Theorem 2.1). The odd quotient of the universal cluster is flat, hence Y_n maps to the Quot scheme, the map factors through Q_n, and the fiber over the closed point z_J is the same whether computed in the Quot scheme or in Q_n. A T_0-point of the fiber is thus a T_0-flat family of balanced clusters $I \subset R \otimes \mathcal{O}_{T_0}$ whose odd part is $J \otimes \mathcal{O}_{T_0}$. Its even part I^+ contains $R_1 J \otimes \mathcal{O}_{T_0} = J' \otimes \mathcal{O}_{T_0}$, and $I^+/J' \otimes \mathcal{O}_{T_0} \subset (R_0/J') \otimes \mathcal{O}_{T_0}$ is the kernel of a surjection onto the locally free $\mathcal{R}_0$ of rank n, hence a subbundle of rank $k = \dim(R_0/J') - n$. The condition $R_1 I^+ \subset J \otimes \mathcal{O}_{T_0}$ says $I^+ \subset (J \otimes \mathcal{O}_{T_0} : R_1) = J'' \otimes \mathcal{O}_{T_0}$, the colon commuting with the flat extension of scalars from $\mathbb{C}$, hence the subbundle lies in $V_J \otimes \mathcal{O}_{T_0}$. Conversely every rank-$k$ subbundle of $V_J \otimes \mathcal{O}_{T_0}$ gives an ideal, by $\mathfrak{m}_X V_J = 0$, with flat balanced quotient. These constructions are inverse and functorial, including on nonreduced T_0, hence the fiber functor is the Grassmannian functor. The universal subbundle gives (93) and (94), and the odd quotient is constant by definition. □

By (94) and a vector-space splitting $R_0/J' = (R_0/J'') \oplus V_J$,

$$\mathcal{R}_0|_{F_J} \cong \mathcal{O}^{\oplus(n-r)} \oplus \mathcal{Q}_J, \qquad \mathcal{R}_1|_{F_J} \cong \mathcal{O}^{\oplus n}, \tag{95}$$

in agreement with Theorem 4.3(a) ($\dim R_0/J'' = n - r$ by Theorem 4.1(c)).

The formal normal form. By Section 4.5, the local model of ρ at z_J is the Springer map $\mu : T^* \mathrm{Gr}(r+1, V) \to \mathfrak{N}_{r,2r+1}$ times a smooth core of dimension 2ℓ, $\ell = n - r(r+1)$. We use the relative form of the Bellamy–Craw normal form [8, Thm. 3.2, eq. (3.5), Thm. 3.7] (Theorem 4.6(a),(b)). Completing along the scheme-theoretic fibers, there is a commutative diagram of formal schemes

$$\begin{array}{ccc} (\widehat{Y_n})_{F_J} & \xrightarrow{\ \sim\ } & (\widehat{T^*G \times \mathbb{A}^{2\ell}})_{G\times\{0\}} \\ \Big\downarrow{\scriptstyle \widehat{\rho}} & & \Big\downarrow{\scriptstyle \widehat{\mu\times \mathrm{id}}} \\ (\widehat{Q_n})_{z_J} & \xrightarrow{\ \sim\ } & (\widehat{\mathfrak{N}_{r,2r+1} \times \mathbb{A}^{2\ell}})_{(0,0)}. \end{array} \tag{96}$$

It is obtained from the étale-local statement by completing at z_J, the étale maps of Theorem 4.6(b) inducing isomorphisms of completions and the Cartesian squares giving the top row. Only the formal smoothness of the core is used.

Remark 6.12 (The citation, checked against its hypotheses). Theorem 3.2 of [8] is stated for the variation-of-GIT morphism $f : \mathfrak{M}_\theta(\mathbf{v}, \mathbf{w}) \to \mathfrak{M}_{\theta_0}(\mathbf{v}, \mathbf{w})$ with θ_0 in the boundary of the closure of the chamber of θ, at a closed point x with θ_0-polystable representative $M_\infty \oplus M_0^{\oplus m_0} \oplus \cdots \oplus M_k^{\oplus m_k}$. Its conclusion is the diagram of Theorem 4.6(a),(b), with Cartesian squares and étale horizontal maps, and (3.5) of *loc. cit.* is its completion. In our case θ is the cyclic chamber, $\theta_0 = (0, 1)$ lies on its wall, and at z_J the polystable representative is $E_{\mathrm{fr}} \oplus S_0^{\oplus r}$ by (17), hence $k = 0$, $M_\infty = E_{\mathrm{fr}}$ with $\beta^{(\infty)} = (1, n-r, n)$ on the vertices $(\infty, 0, 1)$, $M_0 = S_0$ with $\beta^{(0)} = (0, 1, 0)$ and $m_0 = r$. For

the framed doubled affine A_1 quiver with one framing edge at the vertex 0 the Cartan form is $(\alpha,\alpha) = 2(a^2+b^2+c^2) - 2ab - 4bc$ for $\alpha = (a,b,c)$, hence

$$\ell = p(\beta^{(\infty)}) = 1 - \tfrac{1}{2}(\beta^{(\infty)},\beta^{(\infty)}) = n - r(r+1), \qquad \mathbf{n}_0 = -(\beta^{(\infty)},\beta^{(0)}) = 2r+1, \qquad p(\beta^{(0)}) = 0.$$

Thus the Ext-graph is one vertex without loops, framed by $2r+1$, the core has dimension 2ℓ, and the local quiver varieties are $\mathfrak{M}_\varrho(r,2r+1) \to \mathfrak{M}_0(r,2r+1)$, in agreement with Theorem 4.1(c), Theorem 4.9 and (18).

Two points of the citation deserve comment. First, [8] takes $\mathfrak{M}_{\theta_0}(\mathbf{v},\mathbf{w})$ to be the GIT quotient of the scheme-theoretic moment-map fiber, whereas Q_n was given its reduced structure in Section 2.2, and the two agree. Indeed, for the local model the dimension vector $(1,r)$ of the deframed quiver (two vertices joined by $2r+1$ edges) lies in Crawley-Boevey's set Σ_0 because $p(1,r) = r(r+1) > s(2r+1-s) = p(1,s)$ for $0 \le s < r$, hence the moment-map fiber and its quotient $\mathfrak{M}_0(r,2r+1)$ carry their reduced structure and the quotient is normal ([10, Thm. 1.1], the reducedness of the natural scheme structures for dimension vectors in Σ_0 is proved in [9] and recalled in §1 of [10]). The étale map q of Theorem 4.6(b) then makes its source reduced, and the étale map p makes $\mathfrak{M}_{\theta_0}(\mathbf{v},\mathbf{w})$ reduced, and in fact normal, at every closed point (reducedness and normality ascend along flat local homomorphisms with reduced, respectively normal, fibers and descend along all of them [62, Thm. 23.9 and its corollary]). This also reproves the normality of Q_n used in Theorem 2.1. Second, the identification of the fibers in [8, Cor. 3.4] is scheme-theoretic, since the fiber of an étale map over a $\mathbb{C}$-point is a reduced point. On the model the fiber over 0 is the zero section, because $BA = 0$ with A surjective forces $B = 0$ (Section 4.5), hence (96) carries F_J to $G \times \{0\}$, in agreement with Theorem 6.11. The tautological-bundle statement [8, Prop. 3.8], whose displayed formula omits the trivial summands coming from M_∞, is not used, since the underlying bundles are determined below by their restriction to F_J.

Proposition 6.13 (Underlying tautological bundles)**.** *On* (96)*, as vector bundles,*

$$\widehat{\mathcal{R}_0} \cong \mathcal{O}^{\oplus(n-r)} \oplus \widehat{p^*\mathcal{Q}_J}, \qquad \widehat{\mathcal{R}_1} \cong \mathcal{O}^{\oplus n}, \tag{97}$$

where $p : T^*G \times \mathbb{A}^{2\ell} \to G$ is the projection and the completion is along $G \times \{0\}$. The quotient bundle here is the positive $\mathcal{Q}_J$, not $\mathcal{Q}_J^\vee$, and the formula includes the trivial summands contributed by the framed stable factor. No constancy of the multiplication of $\mathcal{P}_n$ is asserted.

Proof. Write $\widehat{Y} = (\widehat{Y_n})_{F_J}$, let $\mathcal{I}$ be the ideal of F_J and $Y_m = V(\mathcal{I}^{m+1})$ its thickenings. The isomorphism of formal schemes in (96) identifies $\mathcal{I}/\mathcal{I}^2$ with the conormal bundle of $G \times \{0\}$ in $T^*G \times \mathbb{A}^{2\ell}$, and $F_J \subset Y_n$ is a regular embedding, hence

$$\mathcal{I}^m/\mathcal{I}^{m+1} \cong \operatorname{Sym}^m(\mathcal{I}/\mathcal{I}^2) \cong \bigoplus_{d+e=m} \operatorname{Sym}^d T_G \otimes \operatorname{Sym}^e \mathbb{C}^{2\ell} \qquad (m \ge 1). \tag{98}$$

Lifting principle. Let $\mathcal{E},\mathcal{E}'$ be vector bundles on $\widehat{Y}$ and $\phi_0 : \mathcal{E}|_{F_J} \xrightarrow{\sim} \mathcal{E}'|_{F_J}$, and suppose

$$H^1(F_J, \mathcal{H}om(\mathcal{E}|_{F_J}, \mathcal{E}'|_{F_J}) \otimes \mathcal{I}^m/\mathcal{I}^{m+1}) = 0 \qquad (m \ge 1). \tag{99}$$

Then ϕ_0 lifts to an isomorphism $\mathcal{E} \cong \mathcal{E}'$ on $\widehat{Y}$. Indeed, given an isomorphism ϕ_m over Y_m, the inclusion $Y_m \subset Y_{m+1}$ is a square-zero thickening with ideal $\mathcal{I}^{m+1}/\mathcal{I}^{m+2}$ and both sheaves are locally free, hence ϕ_m lifts locally, two local lifts differ by a section of $\mathcal{H}om(\mathcal{E}|_{F_J}, \mathcal{E}'|_{F_J}) \otimes \mathcal{I}^{m+1}/\mathcal{I}^{m+2}$, and the differences on overlaps form a Čech 1-cocycle whose class is the obstruction to a global lift, and it vanishes by (99). The global lift ϕ_{m+1} is an isomorphism because ϕ_0 is and the thickening is nilpotent, and the compatible system (ϕ_m) is an isomorphism of locally free sheaves on the formal scheme.

Application. Take $\mathcal{E} = \widehat{\mathcal{R}_0}$ and $\mathcal{E}' = \widehat{p^*E_0}$ with $E_0 = \mathcal{O}^{\oplus(n-r)} \oplus \mathcal{Q}_J$. Their restrictions to F_J agree by (95), and (99) is (88) tensored with the constant factors $\operatorname{Sym}^e \mathbb{C}^{2\ell}$. For $\widehat{\mathcal{R}_1}$ take $E_0 = \mathcal{O}^{\oplus n}$ and use (83) with $\lambda = \varnothing$. □

Remark 6.14. Theorem 4.6(c) yields the same isomorphism directly through the étale slice. The pullback of $\mathcal{R}_i$ to the slice is the bundle associated with the restriction to $H = GL(U)$ of the vertex representation, which is $(E_{\mathrm{fr}})_0 \oplus U$ and $(E_{\mathrm{fr}})_1$ by (17), and the bundle associated with the standard representation U is the positive quotient $\mathcal{Q}_J$ because $A : V \to U$ is surjective in the cluster chamber. The proof above shows that this clause is not needed, since the isomorphism type of the underlying bundles is forced by their restriction to the fiber and Borel–Weil–Bott.

Warning 6.15 (Bundles, not algebras). The isomorphisms (97) are isomorphisms of vector bundles. The multiplication of $\mathcal{P}_n$, the trace-zero projector and the differential of $\mathcal{K}_n$ are transported to bundle maps on the formal model and may depend on every formal coordinate. On the big cell of (21), completion retains all powers of the cotangent block D and of the core variables, and a scalar trace τ_f remains a nonconstant formal function of (u, BA). Nothing below replaces it by its value on the zero section.

6.7. **Passage to the completed neighbourhood.** Let $A = \Gamma(T^*G, \mathcal{O})$ as in Section 6.4, $A' = A[u_1, \dots, u_{2\ell}]$, $\mathfrak{n} = (A_+, u_1, \dots, u_{2\ell})$ and $R_{\mathrm{mod}} = \widehat{A'_{\mathfrak{n}}}$. The maps $A \to A' \to A'_{\mathfrak{n}} \to R_{\mathrm{mod}}$ are flat, the last faithfully so [73, Tags 00MB, 00MC]. For a model bundle $\mathcal{F}_X = p^*\mathcal{F}$, flat base change gives

$$H^i\big(X \times_{\operatorname{Spec} A} \operatorname{Spec} R_{\mathrm{mod}},\ \mathcal{F}_X \otimes_A R_{\mathrm{mod}}\big) \cong H^i(X, \mathcal{F}_X) \otimes_A R_{\mathrm{mod}} \tag{100}$$

[73, Tag 02KH] (tensor a finite affine Čech complex with the flat algebra). For a proper scheme over a complete noetherian local ring and a coherent sheaf, the theorem on formal functions [73, Tags 02OC, 0A0H] identifies the cohomology, which is finite and hence complete, with the cohomology of the completed sheaf on the formal completion along the closed fiber. Applied to the model over R_{mod} and to $Y_n \times_{Q_n} \operatorname{Spec} \widehat{\mathcal{O}}_{Q_n, z_J}$ over $\widehat{\mathcal{O}}_{Q_n, z_J}$, and combined with the isomorphism of formal schemes (96), this identifies the completed higher direct images $(R^i\rho_*\mathcal{F})^{\wedge}_{z_J}$ of a bundle $\mathcal{F}$ on Y_n with the cohomology of the corresponding completed model bundle, together with all natural maps between such groups.

Lemma 6.16 (The completed positive-bundle package)**.** *After adding the core and completing at the origin, the following remain valid.* (i) *Higher direct images of positive tensor constructions in $p^*\mathcal{Q}$ vanish.* (ii) *The tensor-of-sections maps for those bundles are surjective.* (iii) *The restriction isomorphism* (92) *holds with A_+ replaced by the maximal ideal of the completed model base. Moreover,* (i) *and* (ii) *hold for arbitrary formal direct summands of these completed bundles, even when the splitting maps are not homogeneous.*

Proof. (i) is (100) applied to Theorem 6.5 and Theorem 6.6, transferred to completed bundles by formal functions. (ii) A surjection of finite A-modules remains surjective after $\otimes_A R_{\mathrm{mod}}$. By associativity of tensor products its source is the tensor product over R_{mod} of the two base-changed section modules, and naturality of flat base change and formal functions identifies the base-changed map with the formal tensor-of-sections map. The modules are finite over the complete R_{mod}, hence their ordinary tensor product is complete and equals the completed one. (iii) $(M_{\mathcal{Q}} \otimes_A R_{\mathrm{mod}}) \otimes_{R_{\mathrm{mod}}} \mathbb{C} = M_{\mathcal{Q}} \otimes_A \mathbb{C} \cong H^0(G, \mathcal{Q})$ by Theorem 6.10. The retract argument of Theorem 6.9 is natural and applies to formal bundle maps. □

6.8. **The pushed cluster algebra at a wall point.** Keep z_J with $r \geq 1$ and put

$$R = \widehat{\mathcal{O}}_{Q_n, z_J}, \qquad B_i = (\rho_*\mathcal{R}_i)^{\wedge}_{z_J}, \qquad B = B_0 \oplus B_1, \qquad T = (\rho_*\mathcal{T}_n)^{\wedge}_{z_J},$$

finite R-modules, and B is a finite commutative parity-graded R-algebra whose multiplication is induced by that of $\mathcal{P}_n$.

Proposition 6.17 (Termwise acyclicity)**.** $\big(R^i\rho_*(\mathcal{K}_n^{-a})^+\big)^{\wedge}_{z_J} = 0$ *for* $i > 0$ *and* $0 \leq a \leq n+1$.

Proof. The projector $\mathrm{id} - 1 \cdot \epsilon$ of (77) exhibits $\mathcal{T}_n$ as a retract of $\mathcal{R}_0$, hence $\mathcal{T}_n^{\otimes a}$ is a retract of $\mathcal{R}_0^{\otimes a}$, and the alternating idempotent, which commutes with it, cuts out $\Lambda^a \mathcal{T}_n$ as a direct summand of $\mathcal{R}_0^{\otimes a}$. The even terms of $\mathcal{K}_n$ are

$$(\mathcal{K}_n^{-a})^+ \cong \mathcal{R}_0 \otimes \Lambda^a \mathcal{T}_n \ \oplus \ \mathcal{R}_1 \otimes \Lambda^{a-1} \mathcal{T}_n \otimes (\mathcal{O}_q \oplus \mathcal{O}_t) \ \oplus \ \mathcal{R}_0 \otimes \Lambda^{a-2} \mathcal{T}_n \otimes \mathcal{O}_{qt}, \tag{101}$$

and by (97) tensor powers of $\widehat{\mathcal{R}_0}$ are sums of positive tensor powers of $p^*\mathcal{Q}_J$ and trivial bundles while $\widehat{\mathcal{R}_1}$ is trivial, and character lines are trivial as ordinary bundles. Hence every completed term is a retract of a finite sum of positive tensor constructions, and Theorem 6.16(i) applies. The trace projector may mix auxiliary degrees, but the retract argument is ungraded. □

Proposition 6.18 (The tensor map for the first differential)**.** *The natural map $B_0 \otimes_R T \to (\rho_*(\mathcal{R}_0 \otimes \mathcal{T}_n))^\wedge_{z_J}$ is surjective.*

Proof. The completed $\mathcal{R}_0$ is $\mathcal{O}^{n-r} \oplus p^*\mathcal{Q}_J$, hence the tensor-of-sections map for $\mathcal{R}_0, \mathcal{R}_0$ is surjective by Theorems 6.9 and 6.16. Since $\mathcal{T}_n$ is a retract of $\mathcal{R}_0$ via the normalized trace, the lift-and-retract argument replaces the second factor by $\mathcal{T}_n$. □

Lemma 6.19 (Residue-field restriction)**.** *For $i = 0, 1$ the restriction map $B_i/\mathfrak{m}_R B_i \to H^0(F_J, \mathcal{R}_i|_{F_J})$ is an isomorphism.*

Proof. Transport to the formal model by Theorem 6.13. For a trivial summand the map is $R/\mathfrak{m}_R = \mathbb{C} \to H^0(F_J, \mathcal{O})$, and for the summand $\widehat{p^*\mathcal{Q}_J}$ it is the base-changed map of Theorems 6.10 and 6.16(iii). The identifications are functorial for the bundle isomorphisms, hence the result is the original restriction map. □

Proposition 6.20 (Polynomial generation after pushforward)**.** *The evaluation maps $R \otimes_{\mathbb{C}} R_i \to B_i$ are surjective for $i = 0, 1$. Equivalently, $R[x, y] \to B$ is surjective, and finitely many polynomials generate each B_i over R.*

Proof. On the fiber, $H^0(F_J, \mathcal{R}_1|_{F_J}) = R_1/J$ is spanned by odd polynomials. Taking cohomology in (94) and using $H^0(F_J, \mathcal{U}_J) = H^1(F_J, \mathcal{U}_J) = 0$ (the boundary-singular case $d = 0$, $\lambda = \varnothing$ of Theorem 6.5, or (93) with $H^0(\mathcal{Q}_J) = V_J$) gives $R_0/J' \xrightarrow{\sim} H^0(F_J, \mathcal{R}_0|_{F_J})$, induced by the even polynomial sections. Thus polynomial sections span the residue-field quotients of Theorem 6.19. Choose monomial bases of R_0/J' and R_1/J. They define maps from finite free R-modules to B_0, B_1 whose cokernels C_i are finite with $C_i/\mathfrak{m}_R C_i = 0$, hence $C_i = 0$ by Nakayama. Products are preserved because the multiplication of B is induced from $\mathcal{P}_n$. □

At a positive-rank heart $\dim_{\mathbb{C}}(B_0/\mathfrak{m}_R B_0) = \dim R_0/J' = n + r + 1$ exceeds the generic rank n. Thus B_0 is not locally free, and no trace of its regular representation is ever taken. Instead, applying ρ_* to (77), using $\rho_*\mathcal{O}_{Y_n} = \mathcal{O}_{Q_n}$ (Theorem 2.1(a)) and completing,

$$B_0 = R \cdot 1 \oplus T, \qquad \epsilon : B_0 \to R, \quad \epsilon(1) = 1, \quad T = \ker \epsilon, \tag{102}$$

and for an even polynomial f

$$\epsilon(\bar{f}) = \tau_f/n, \qquad (\tau_f : \ f(0) = 0) = \mathfrak{m}_0 R \tag{103}$$

by (78). All hypotheses of Theorem 6.3 therefore hold for the actual finite R-algebra B, with the base ring R of the lemma equal to the complete local ring $\widehat{\mathcal{O}}_{Q_n, z_J}$ and with $I = \mathfrak{m}_0 R$.

6.9. **The local computation.** Let $(R\rho_*\mathcal{K}_n^+)^\wedge_{z_J}$ denote derived pushforward followed by the flat extension of scalars to R. The hypercohomology spectral sequence $E_1^{a,b} = (R^b\rho_*(\mathcal{K}_n^a)^+)^\wedge_{z_J} \Rightarrow H^{a+b}$ of the bounded complex has, by Theorem 6.17, only the row $b = 0$, whose differential is the pushforward of the Haiman differential. Equivalently, a bounded complex of ρ_*-acyclic sheaves computes $R\rho_*$, and

$$(R\rho_*\mathcal{K}_n^+)^\wedge_{z_J} \simeq (\rho_*\mathcal{K}_n^{\bullet,+})^\wedge_{z_J} \in D^{\leq 0}(R). \tag{104}$$

Concentration. By Theorem 6.1, $(R\rho_*\mathcal{K}_n^+)^\wedge_{z_J} \simeq (R\rho_*\mathcal{I}_n)^\wedge_{z_J} \in D^{\geq 0}(R)$, the derived pushforward of a sheaf followed by a flat extension of scalars. Hence

$$(R\rho_*\mathcal{K}_n^+)^\wedge_{z_J} \in D^{\leq 0}(R) \cap D^{\geq 0}(R) = \mathrm{Coh}(R), \tag{105}$$

that is, the completed derived pushforward is a single module, namely the cokernel of the first differential of (104). This is the one place where the even-excess vanishing turns a cokernel computation into a computation of the entire derived pushforward, and in particular it kills all higher direct images of $\mathcal{I}_n$ at z_J.

The first differential. By (101), $(\mathcal{K}_n^0)^+ = \mathcal{R}_0$ and $(\mathcal{K}_n^{-1})^+ = \mathcal{R}_0 \otimes \mathcal{T}_n \oplus \mathcal{R}_1 \otimes \mathcal{O}_q \oplus \mathcal{R}_1 \otimes \mathcal{O}_t$, with differential $(b \otimes t, a, c) \mapsto bt + xa + yc$. The pushforward of the first summand maps to B_0 through the commutative diagram

$$\begin{array}{ccc} B_0 \otimes_R T & \twoheadrightarrow & \big(\rho_*(\mathcal{R}_0 \otimes \mathcal{T}_n)\big)^\wedge_{z_J} \\ & \searrow^{\mathrm{mult}} & \downarrow{\scriptstyle \rho_*(\mathrm{mult})} \\ & & B_0, \end{array}$$

whose top arrow is surjective by Theorem 6.18. Hence its image is exactly B_0T, and not a possibly larger submodule. The two coordinate summands have images xB_1 and yB_1. Therefore

$$H^0\big((R\rho_*\mathcal{K}_n^+)^\wedge_{z_J}\big) = B_0/(B_0T + xB_1 + yB_1). \tag{106}$$

Acyclicity of the terms alone would not give this, the tensor surjectivity is what identifies the image.

The quotient. By Theorem 6.20 the map $R[x,y] \to B$ is surjective, and by (102)–(103) the pushed-forward normalized trace ϵ satisfies $\epsilon(1) = 1$ and has trace ideal $I = \mathfrak{m}_0 R$. Hence Theorem 6.3 applies to B and gives

$$R/\mathfrak{m}_0 R \xrightarrow{\sim} B_0/(B_0T + xB_1 + yB_1), \qquad 1 \longmapsto [1], \tag{107}$$

with inverse induced by the normalized trace. The kernel of the unit is exactly $\mathfrak{m}_0 R$, not a saturation or radical. Combining,

$$(R\rho_*\mathcal{O}_{F_n})^\wedge_{z_J} \cong R/\mathfrak{m}_0 R = \widehat{\mathcal{O}}_{\mathcal{Z},z_J}. \tag{108}$$

The map. The unit $\mathcal{O}_{Y_n} \to \mathcal{P}_n \to \mathcal{I}_n$ sends 1 to [1], and so does the augmentation $\mathcal{K}_n^+ \to \mathcal{I}_n$ on the degree-zero representative $1 \in \mathcal{R}_0$. By naturality of the adjunction unit for the Cartesian square of Section 6.1, the map induced by u_n on H^0 of the completions is the map (107). Both sides are concentrated in degree zero, hence $(u_n)^\wedge_{z_J}$ is an isomorphism. There is no undetermined scalar, the inverse sends [1] to 1.

Rank zero. At a fixed point with $k_J = 1$ the fiber is a single reduced point (Theorem 6.11), hence ρ is quasi-finite over an open neighbourhood of z_J. Being proper and birational onto the normal Q_n, it is an isomorphism there by Zariski's Main Theorem [44, Cor. III.11.4]. Hence $F_n \to \mathcal{Z}$ is an isomorphism near z_J and u_n is the identity. (Equivalently, the Bellamy–Craw local contraction is an isomorphism when the local Ext-quiver has no multiplicity vertex.)

6.10. **Proof of symbolic descent.**

Lemma 6.21 (Detection at the fixed hearts)**.** *Let $D_n = \mathrm{Cone}\big(u_n : \mathcal{O}_{\mathcal{Z}_n^{\mathrm{symb}}} \to R\rho_*\mathcal{O}_{F_n}\big)$. If $D_n \neq 0$, its completion at some T-fixed closed point of $\mathcal{Z}_n^{\mathrm{symb}}$ is nonzero. Consequently u_n is an isomorphism as soon as its completion is an isomorphism at every $z_J \in Q_n^T$.*

Proof. D_n is a bounded T-equivariant complex with coherent cohomology supported on the projective scheme $\mathcal{Z}_n^{\mathrm{symb}} = \pi_Q^{-1}(0)$. Let W be the reduced support of a nonzero cohomology sheaf, a nonempty closed T-stable subvariety. It contains a T-fixed point. Indeed, embed $\mathcal{Z}$ equivariantly in $\mathbb{P}(V)$ by a power of the T-linearized ample $\mathcal{O}(1)$, decompose a lift of a point of W into finitely many weight vectors, and take the limit under a one-parameter subgroup separating those weights. The limit lies

in W and is fixed by T, being a projective point in a single weight space. At that point the stalk of the sheaf is nonzero, and completion of a noetherian local ring is faithfully flat and commutes with cohomology [73, Tag 00MC], hence the completed complex is nonzero. □

Theorem 6.22 (Symbolic descent)**.** *For every $n \geq 0$ the canonical map u_n of (74) is an isomorphism, and for every $d \in \mathbb{Z}$*

$$R\rho_*(\mathcal{L}_n^{\otimes d} \otimes \mathcal{I}_n) \;\simeq\; \mathcal{O}_{\mathcal{Z}_n^{\mathrm{symb}}}(d),$$

canonically and T-equivariantly.

Proof. For $n = 0$ everything is a point. For $n \geq 1$, at every fixed point z_J with $k_J \geq 2$ the completion of u_n is an isomorphism by (108) and the identification of the map, and at every fixed point with $k_J = 1$ by the rank-zero argument. Hence D_n has zero completion at every fixed point, and $D_n = 0$ by Theorem 6.21. The local computations forget the torus grading, since the auxiliary grading of the model need not be compatible with it. This suffices, because u_n is equivariant and forgetting equivariance is conservative. Finally $\mathcal{L}_n^{\otimes d} = \rho^*\mathcal{O}_{Q_n}(d)$ with $\mathcal{O}_{Q_n}(d)$ invertible, hence the projection formula [73, Tag 01E6] gives $R\rho_*(\mathcal{L}_n^{\otimes d} \otimes \mathcal{O}_{F_n}) \simeq \mathcal{O}_{Q_n}(d) \otimes R\rho_*\mathcal{O}_{F_n} \simeq \mathcal{O}_{Q_n}(d) \otimes \mathcal{O}_{\mathcal{Z}} = \mathcal{O}_{\mathcal{Z}}(d)$, for negative d as well. □

This completes the proof of Theorem E.

Corollary 6.23. *$\rho_*\mathcal{O}_{F_n} = \mathcal{O}_{\mathcal{Z}_n^{\mathrm{symb}}}$, $R^i\rho_*\mathcal{O}_{F_n} = 0$ for $i > 0$, and*

$$R\Gamma(F_n, \mathcal{L}_n^{\otimes d}|_{F_n}) \cong R\Gamma(\mathcal{Z}_n^{\mathrm{symb}}, \mathcal{O}(d)) \qquad (d \in \mathbb{Z}).$$

In particular $H^{>0}(F_n, \mathcal{L}_n^{\otimes d}) = 0$ for $d \gg 0$ by Serre vanishing on the projective $\mathcal{Z}_n^{\mathrm{symb}}$.

The two statements of the corollary have different sources. The trace quotient identifies the degree-zero image, and the intersection of the two cohomological bounds in (105) kills the other degrees. The first alone would not imply the second.

Remark 6.24 (What the proof does not assume). The proof does not require ρ to be flat, the punctual scheme $\mathcal{Z}_n^{\mathrm{symb}}$ to be a local complete intersection in Q_n or its structure sheaf to be perfect over $\mathcal{O}_{Q_n}$ (both fail at $n = 2$, Theorem 6.2), either punctual scheme to be reduced or Cohen–Macaulay, or ordinary restriction to a wall fiber to agree with derived pullback. The only product description used is the formal description (96) of the ambient contraction with its underlying bundles. The multiplication, the trace-zero projector and the first Haiman differential remain their actual maps, and the scalar traces remain in the ordinary ideal with their full dependence on all formal coordinates.

6.11. **The case $n = 2$ in closed form.** As an illustration of Theorem 6.22 in the first nontrivial rank, and as an independent check, we describe the geometry of ρ over the unique wall point of Q_2 with positive-dimensional fiber completely. Every ingredient of the general proof is visible here in closed form.

Theorem 6.25 (The geometry at $n = 2$)**.** *Let $z_J \in Q_2$ be the wall point with $F_J \cong \mathbb{P}^2$, hence $J = R_1 \cap \mathfrak{m}^3$ and $V := V_J = \langle x^2, xy, y^2\rangle = \mathrm{Sym}^2\mathbb{C}^2$, and let $U \subset Q_2$ be the open set of quotients J for which x, y is a basis of R_1/J. Then the following hold.*

- (a) *$\rho^{-1}(U) \cong T^*\mathbb{P}^2$, $\mathbb{P}^2 = \mathrm{Gr}(2, V)$, equivariantly for T and for the group GL_2 acting on $\langle x, y\rangle$, and $\rho|_{\rho^{-1}(U)}$ is the affinization map. Thus $U \cong \overline{\mathcal{O}}_{\min} \subset \mathfrak{sl}(V)$ and ρ is the Springer map.*
- (b) *$\mathfrak{m}_0\mathcal{O}_{Q_2}|_U$ is generated by the three trace functions $\mathrm{Tr}\,U, \mathrm{Tr}\,V, \mathrm{Tr}\,W$ of Theorem 6.2, which under* (a) *are the components of the moment map of the SL_2-action on $T^*\mathbb{P}^2$, and $\mathcal{Z}_2^{\mathrm{symb}} \cap U$ is the cone over the rational normal quartic, $\mathcal{O}(\mathcal{Z}_2^{\mathrm{symb}} \cap U) = \bigoplus_{d\geq 0}\mathrm{Sym}^{4d}\mathbb{C}^2$, i.e. the quotient singularity $\mathbb{C}^2/\mu_4$.*
- (c) *$F_2 \cap \rho^{-1}(U)$ is a reduced local complete intersection, equal to $\mathbb{P}^2 \cup N_C^*$, where $C \subset \mathbb{P}^2$ is the conic of squares and $N_C^* \cong \mathrm{Tot}\,\mathcal{O}_{\mathbb{P}^1}(-4)$ is its conormal bundle.*

Consequently $R\rho_*\mathcal{O}_{F_2} \simeq \mathcal{O}_{\mathcal{Z}_2^{\mathrm{symb}}}$ *via* u_2*, in accordance with Theorem 6.22.*

Proof. By Theorem 4.9 and Section A the only fixed point of Q_2 with $k \geq 2$ is z_J, and by Theorem 6.21 together with the rank-zero case it suffices to prove (75) on a neighbourhood of z_J. The group GL_2 acting on $\langle x, y\rangle$ commutes with Γ, hence acts on Y_2, Q_2, S_2 compatibly with ρ, π_Q, the polarized power sums, and the torus $T \subset GL_2$. The symplectic form of Y_2 transforms by det, hence it is SL_2-invariant.

Step 1: U is the attracting chart of z_J. U is open, GL_2-stable, and contains z_J. Its coordinate ring is generated by the entries of the matrices of u, v, w acting on R_1/J in the basis x, y (the chart of Theorem 6.2), and their T-weights are $q^2,\ q^3t^{-1},\ t^2,\ t^3q^{-1},\ qt,\ t^2,\ q^2,\ qt$, all of weight s^2 under the central one-parameter subgroup $(q, t) = (s, s)$. Hence $\mathbb{C}[U]$ is positively graded by this subgroup with $\mathbb{C}[U]_0 = \mathbb{C}$. Thus every point of U flows to z_J, and every point of Q_2 flowing to z_J lies in U, U being open and T-stable.

Step 2: $\rho^{-1}(U) \cong T^*\mathbb{P}^2$. Every point of $\rho^{-1}(U)$ has a limit under the central subgroup, by properness of ρ, and the limit lies in $\rho^{-1}(z_J) = \mathbb{P}^2$. The fixed locus of the central subgroup in $\rho^{-1}(U)$ is exactly $\mathbb{P}^2$, since it maps to the unique fixed point z_J of U, and the subgroup acts trivially on $\mathrm{Gr}(2, V)$ because it acts on V by the scalar s^2. By the Białynicki-Birula decomposition [7, Thm. 4.1] the smooth variety $\rho^{-1}(U)$ is the attracting set of $\mathbb{P}^2$, a Zariski locally trivial fibration in affine spaces on which the subgroup acts with the weights of the normal bundle. At the three fixed points $(2, 2)$, $(3, 1)$, $(2, 1, 1)$ of $\mathbb{P}^2$ the tangent weights of Y_2 (Theorem 7.1) are $q^{\pm1}t^{\mp1}$ along $\mathbb{P}^2$ and q^2, t^2 in the normal directions. The normal weight is s^2 throughout, hence the transition functions of the fibration commute with a single scaling and are linear, and $\rho^{-1}(U) \cong N_{\mathbb{P}^2/Y_2}$ equivariantly. Since $\mathbb{P}^2$ is Lagrangian and the symplectic form is SL_2-invariant, $N_{\mathbb{P}^2/Y_2} \cong T^*\mathbb{P}^2$ SL_2-equivariantly. Finally $U = \operatorname{Spec}\Gamma(\rho^{-1}(U), \mathcal{O})$, because $\rho_*\mathcal{O}_{Y_2} = \mathcal{O}_{Q_2}$ and U is affine, and $\Gamma(T^*\mathbb{P}^2, \mathcal{O}) = \bigoplus_d H^0(\mathbb{P}^2, \mathrm{Sym}^d T_{\mathbb{P}^2})$ is the coordinate ring of the minimal nilpotent orbit closure $\overline{\mathcal{O}}_{\min} \subset \mathfrak{sl}(V)$, which is normal with ideal generated by the 2×2 minors [52], ρ being its Springer resolution. This is (a).

Step 3: the trace ideal. By Theorem 3.4, $\mathfrak{m}_0\mathcal{O}_{Q_2}|_U$ is generated by the functions $\mathrm{Tr}(m_f \,|\, R_1/J)$, f a Γ-invariant monomial of degree $2d > 0$. These are homogeneous of degree d in the matrix entries, and $f \mapsto \mathrm{Tr}(m_f)$ is GL_2-equivariant. Under (a), $\mathbb{C}[U]_1 = H^0(\mathbb{P}^2, T_{\mathbb{P}^2}) = \mathfrak{sl}(V) \cong \mathrm{Sym}^2 \oplus \mathrm{Sym}^4$ as an SL_2-module, and the traces of u, v, w span a nonzero submodule isomorphic to Sym^2 (nonzero since $\mathrm{Tr}\, U \neq 0$), hence exactly the image of $\mathfrak{sl}_2 \to \mathfrak{sl}(V)$. In other words, they are the vector fields ξ_E, ξ_F, ξ_H of the SL_2-action, viewed as fiber-linear functions on $T^*\mathbb{P}^2$, i.e. the components of the moment map of the SL_2-action. The ideal of $\overline{\mathcal{O}}_{\min}$ is generated by quadrics spanning $\mathfrak{sl}(V) \oplus \mathbb{C}$, namely $X \mapsto \mathrm{Tr}(MX^2)$ for $M \in \mathfrak{sl}(V)$ and $\mathrm{Tr}(X^2)$. Restricted to the linear subspace $\mathrm{Sym}^4 = \mathfrak{sl}_2^{\perp}$ they vanish on the cone $\{\ell^4\}$ over the rational normal quartic (the SL_2-orbit closure of the highest root vector $E_\theta \in \mathrm{Sym}^4$), hence they lie in the six-dimensional space $\mathrm{Sym}^4 \oplus \mathrm{Sym}^0 \subset \mathrm{Sym}^2(\mathrm{Sym}^4)^*$ of quadrics vanishing on that cone. Both isotypic components occur, the invariant one because the trace form is nondegenerate on Sym^4, the other because $\mathrm{Tr}(E_\theta X^2) = 2$ for $X = E_\theta + E_{-\theta} + \mathrm{diag}(1, -2, 1) \in \mathrm{Sym}^4$. Since the ideal of the rational normal quartic is generated by quadrics,

$$\mathbb{C}[U]/(\xi_E, \xi_F, \xi_H) = \mathbb{C}[\mathrm{Sym}^4]/(\text{quadrics of the cone}) = \bigoplus_{d\geq 0} \mathrm{Sym}^{4d}\,\mathbb{C}^2,$$

the coordinate ring of the cone over the quartic, which is the quotient $\mathbb{C}^2/\mu_4$. For $d \geq 1$ every SL_2-equivariant map $\mathrm{Sym}^{2d} \to \mathrm{Sym}^{4d}$ is zero, hence all higher traces lie in (ξ_E, ξ_F, ξ_H). This is (b).

Step 4: the punctual fiber. Thus $F_2 \cap \rho^{-1}(U) = V(\xi_E, \xi_F, \xi_H) \subset T^*\mathbb{P}^2$, with $\pi_*\mathcal{O}_{F_2} = \mathrm{Sym}^\bullet T_{\mathbb{P}^2}/(\xi)$. The SL_2-orbits on $\mathbb{P}^2 = \mathbb{P}(\mathrm{Sym}^2)$ are the conic $C = \{[\ell^2]\}$ and its complement. The three fields generate the subsheaf $T_{\mathbb{P}^2}(-\log C) \subset T_{\mathbb{P}^2}$ of fields tangent to C. Indeed, they lie in it, and they generate its fibers, off C because the orbit is open, at $w \in C$ because the stabilizer $\mathfrak{b}$ of w acts on the normal line $N_{C/\mathbb{P}^2,w}$ with a nonzero weight. In local coordinates (c, s) with $C = \{c = 0\}$ and

dual fiber coordinates (p_c, p_s) this ideal is $(p_s,\, cp_c)$, hence $F_2 \cap \rho^{-1}(U)$ is the reduced local complete intersection $\mathbb{P}^2 \cup N_C^*$, and $N_C^* = \mathrm{Tot}\ \mathcal{O}_{\mathbb{P}^1}(-4)$ since $N_{C/\mathbb{P}^2} = \mathcal{O}_{\mathbb{P}^2}(2)|_C$. This is (c).

Step 5: conclusion. Since $(p_s, p_c) \cap (p_s, c) = (p_s, cp_c)$ there is an exact sequence $0 \to \mathcal{O}_{F_2} \to \mathcal{O}_{\mathbb{P}^2} \oplus \mathcal{O}_{N_C^*} \to \mathcal{O}_C \to 0$ on $\rho^{-1}(U)$. Now $R\rho_*\mathcal{O}_{\mathbb{P}^2} = R\Gamma(\mathbb{P}^2, \mathcal{O}) = \mathbb{C}_{z_J}$ and $R\rho_*\mathcal{O}_C = R\Gamma(\mathbb{P}^1, \mathcal{O}) = \mathbb{C}_{z_J}$. The complement $N_C^* \setminus C$ is a single SL_2-orbit, with stabilizer $U \rtimes \mu_4$ (U the unipotent radical of $\mathfrak{b}$, the torus acting on the conormal line by a character of order four), mapping equivariantly onto the orbit $SL_2 \cdot E_\theta$, whose stabilizer is the same group. Hence $\rho|_{N_C^*}$ is birational onto $\mathcal{Z}_2^{\mathrm{symb}} \cap U$. Moreover $\Gamma(N_C^*, \mathcal{O}) = \bigoplus_d H^0(\mathbb{P}^1, \mathcal{O}(4d)) = \bigoplus_d \mathrm{Sym}^{4d}$, hence the injective pullback $\mathcal{O}(\mathcal{Z}_2^{\mathrm{symb}} \cap U) \to \Gamma(N_C^*, \mathcal{O})$ is an isomorphism degree by degree, and $H^1(N_C^*, \mathcal{O}) = \bigoplus_d H^1(\mathbb{P}^1, \mathcal{O}(4d)) = 0$. Therefore $R\rho_*\mathcal{O}_{N_C^*} = \mathcal{O}_{\mathcal{Z}_2^{\mathrm{symb}}}$ on U. In the pushed-forward sequence the map $\mathbb{C} \oplus \mathcal{O}_{\mathcal{Z}} \to \mathbb{C}$ is $(a, f) \mapsto a - f(z_J)$, surjective with kernel $\{(f(z_J), f)\} \cong \mathcal{O}_{\mathcal{Z}}$. Therefore $R\rho_*\mathcal{O}_{F_2} \simeq \mathcal{O}_{\mathcal{Z}_2^{\mathrm{symb}}}$ on U, and the isomorphism is the descent map u_2, which sends f to $(f(z_J), f|_{N_C^*})$. Twists follow by the projection formula. □

Remark 6.26 (Exact check). With $\mathcal{F}_d := (\mathrm{Sym}^\bullet T_{\mathbb{P}^2}/(\xi_E, \xi_F, \xi_H))_d$, a direct computation in Macaulay2 gives $H^1(\mathbb{P}^2, \mathcal{F}_d) = H^2(\mathbb{P}^2, \mathcal{F}_d) = 0$ and $h^0(\mathbb{P}^2, \mathcal{F}_d) = 4d + 1$ for $0 \le d \le 9$, and the ideal of $\mathfrak{sl}_3$ generated by the 2×2 minors, the trace, and the three pairings with $\mathfrak{sl}_2$ is prime of dimension two and degree four with Hilbert function $4d + 1$, whereas a generic triple of linear vector fields gives different numbers. This confirms (b) and (c) and the vanishing of $R^{>0}\rho_*\mathcal{O}_{F_2}$ independently of Steps 3–5, and it is the $n = 2$ instance of Theorem 6.23.

Remark 6.27. The germ $(\mathcal{Z}_2^{\mathrm{symb}}, z_J) \cong \mathbb{C}^2/\mu_4$ is Cohen–Macaulay and not Gorenstein, in agreement with Theorem 6.2, and it is resolved by the non-contracted component N_C^* of F_2 while the contracted component $\mathbb{P}^2$ contributes only the constants. At a wall point of rank r and dimension $n > r(r+1)$ the transverse core is positive-dimensional, the chart is no longer an attracting cone, and the trace functions depend on the core coordinates as well as on the cotangent block. The general proof works on the formal neighbourhood instead, where this dependence is retained (Theorem 6.15).

7. The all-twists localization polynomial

In this section we compute the q,t-characters. Equivariant localization on Y_n expresses the Euler characteristic of $\mathcal{L}_n^{\otimes m} \otimes \mathcal{K}_n$ as the fixed-point sum $F_{n,m}$ of (5), and the determinant-parity projection $P_{n,m}$ is the Euler characteristic of $\mathcal{L}_n^{\otimes m} \otimes \mathcal{I}_n$, by the two-term homology and the even-excess vanishing. Symbolic descent identifies $P_{n,m}$ with the Euler characteristic of $\mathcal{O}(m)$ on $\mathcal{Z}_n^{\mathrm{symb}}$, and Serre vanishing turns it into the Hilbert series of $\mathfrak{S}_{n,m}^{\mathrm{symb}}$ for $m \gg 0$ (Theorem 7.5).

The main idea of the principal specialization $t = q^{-1}$ (Theorem 7.2) is that only the hook partitions survive the specialization, because every partition which is not a hook contains the box $(1,1)$, whose factor $1 - qt$ vanishes. The hook contributions pair up, and the finite q-binomial theorem gives the closed form. This proves Theorem C and, together with Theorem 7.5, the dimension count $\binom{n(m+1)}{n}$ used in Sections 8 and 9.

7.1. The fixed-point formula.

Lemma 7.1. *The T-fixed points of Y_n are the monomial balanced ideals I_λ, $\lambda \in \mathrm{Bal}(2n)$, the tangent weights at I_λ are the $2n$ hook weights $q^{a(s)+1}t^{-l(s)}$, $q^{-a(s)}t^{l(s)+1}$ over the boxes $s \in \lambda$ with $a(s) + l(s)$ odd, and*

$$[\mathcal{P}_n]_\lambda = B_\lambda, \qquad [\mathcal{L}_n]_\lambda = \ell_\lambda, \qquad \sum_{a \ge 0} (-1)^a [\textstyle\bigwedge^a \mathcal{E}_n]_\lambda = \prod_{w \,\in\, \mathrm{wt}(\mathcal{E}_n|_\lambda)} (1 - w) = (1-q)(1-t)\,\Pi_\lambda^{(0)},$$

in the notation (4). Consequently, for every $m \in \mathbb{Z}$,

$$F_{n,m}(q,t) := \chi^T(Y_n, \mathcal{L}_n^{\otimes m} \otimes \mathcal{K}_n) = \sum_{\lambda \in \mathrm{Bal}(2n)} \frac{\ell_\lambda^m B_\lambda (1-q)(1-t)\Pi_\lambda^{(0)}}{D_\lambda}, \tag{109}$$

agreeing with (5)*, and the determinant-parity projection* $P_{n,m}$ *of* (5) *satisfies*

$$P_{n,m} = \chi^T(Y_n, \mathcal{L}_n^m \otimes \mathcal{I}_n) - \chi^T(Y_n, \mathcal{L}_n^m \otimes \mathcal{M}_n^+) = \chi^T(Y_n, \mathcal{L}_n^m \otimes \mathcal{I}_n), \tag{110}$$

the last equality by Theorem 5.28.

Proof. The fixed points of $\mathrm{Hilb}^{2n}(\mathbb{A}^2)^T$ are monomial ideals, and the Γ-fixed component Y_n selects the balanced ones. The tangent space of the Hilbert scheme at I_λ has the standard hook-weight basis, and the Γ-invariant part consists of the weights of even total degree, i.e. $a+l$ odd. The fiber of $\mathcal{P}_n$ is spanned by the standard monomials (B_λ), the line bundle $\mathcal{L}_n = \det \mathcal{R}_1$ gives the product of odd-box weights, and $[\mathcal{E}_n]_\lambda = (B_\lambda^{\mathrm{even}} - 1) + q + t$, hence the Koszul factor $\sum_a (-1)^a [\bigwedge^a \mathcal{E}_n]_\lambda = (1-q)(1-t)\Pi_\lambda^{(0)}$. Equivariant localization (Thomason) gives (109). The class $[\mathcal{K}_n] = [\mathcal{I}_n] - [\mathcal{M}_n]$ (Theorem 3.13) splits under the determinant-parity projector $\Pi^{(m)} f = \frac{1}{2}(f(q,t) + (-1)^{nm} f(-q,-t))$ into the even part $[\mathcal{I}_n] - [\mathcal{M}_n^+]$ (note $\ell_\lambda(-q,-t) = (-1)^n \ell_\lambda$ and $\mathcal{I}_n$ is even), giving the first equality of (110), and the second is Theorem 5.28. □

7.2. **Hook reduction and the principal specialization.**

Theorem 7.2. *Theorem C holds. For all* $n \geq 1$, $m \geq 0$, *with* $M = n(m+1)$, $Q = q^2$, *we have*

$$F_{n,m}(q, q^{-1}) = q^{-mn^2} \begin{bmatrix} M \\ n \end{bmatrix}_Q - q^{1-mn^2} [m]_{Q^n} \begin{bmatrix} M \\ n-1 \end{bmatrix}_Q, \tag{111}$$

with the two summands of opposite exponent parity. Consequently $P_{n,m}(q,q^{-1}) = q^{-mn^2} \left[\begin{smallmatrix} M \\ n \end{smallmatrix}\right]_Q$, $N_{n,m}(q,q^{-1}) = q^{1-mn^2} [m]_{q^{2n}} \left[\begin{smallmatrix} M \\ n-1 \end{smallmatrix}\right]_{q^2}$, *and the scalar values of Theorem C follow at* $q \to 1$.

Proof. Step 1: only hooks survive. If λ is not a hook it contains the box $(1,1)$, of even parity, contributing the factor $1 - qt$ to $\Pi_\lambda^{(0)}$, which vanishes at $t = q^{-1}$, while $D_\lambda(q, q^{-1}) = \prod (1 - q^{\pm h}) \neq 0$ in $\mathbb{Q}(q)$. Every hook of size $2n$ is balanced, and the hooks are

$$\lambda_j^+ = (2j+1, 1^{2n-2j-1}),\ 0 \leq j \leq n-1; \qquad \lambda_j^- = (2j, 1^{2n-2j}),\ 1 \leq j \leq n.$$

Step 2: a general hook. For $\lambda = (a+1, 1^b)$, $a + b = 2n - 1$, put $u = \lfloor a/2 \rfloor$, $v = \lfloor b/2 \rfloor$. Then

$$B_\lambda(q, q^{-1}) = q^{-b} \frac{1 - q^{2n}}{1 - q}, \qquad \Pi_\lambda^{(0)}(q, q^{-1}) = \prod_{r=1}^{u} (1 - Q^r) \prod_{s=1}^{v} (1 - Q^{-s}),$$

and, since the even hook lengths are $2n$ (origin) and $2, 4, \ldots, 2u$ and $2, 4, \ldots, 2v$ along the two arms,

$$D_\lambda(q, q^{-1}) = (1 - Q^n)(1 - Q^{-n}) \prod_{r=1}^{u} (1 - Q^r)(1 - Q^{-r}) \prod_{s=1}^{v} (1 - Q^s)(1 - Q^{-s}).$$

Using $\prod_{r \leq u} (1 - Q^{-r}) = (-1)^u Q^{-u(u+1)/2} (Q;Q)_u$ and $\frac{(1-q^{2n})(1-q^{-1})}{(1-Q^n)(1-Q^{-n})} = \frac{q^{2n-1}(1-q)}{1-Q^n}$, the contribution is

$$C_{a,b}^{(m)} = (-1)^u\, \ell_\lambda(q, q^{-1})^m\, q^{-b+2n-1+u(u+1)}\, \frac{1-q}{(1-Q^n)(Q;Q)_u (Q;Q)_v}. \tag{112}$$

Step 3: the two families. For both families the odd boxes have total exponent $j^2 - (n-j)^2$, hence $\ell_{\lambda_j^\pm}(q, q^{-1}) = q^{2nj - n^2}$. Substituting (a, b, u, v) and using $\frac{1}{(1-Q^n)(Q;Q)_j (Q;Q)_{n-j-1}} = \frac{1}{(Q;Q)_n} \left[\begin{smallmatrix} n-1 \\ j \end{smallmatrix}\right]_Q$,

$$C^{(m)}(\lambda_j^+) = \frac{1-q}{(Q;Q)_n} (-1)^j q^{-mn^2 + j^2 + (2nm+3)j} \begin{bmatrix} n-1 \\ j \end{bmatrix}_Q,$$

$$C^{(m)}(\lambda_j^-) = \frac{1-q}{(Q;Q)_n} (-1)^{j+1} q^{-mn^2 - 1 + j^2 + (2nm+1)j} \begin{bmatrix} n-1 \\ j-1 \end{bmatrix}_Q.$$

Step 4: pairing and the q-binomial theorem. Pair λ_k^+ with λ_{k+1}^- $(0 \le k \le n-1)$. Signs and Gaussian coefficients match, the second exponent exceeding the first by $2nm+1$, and we obtain

$$F_{n,m}(q,q^{-1}) = \frac{(1-q)(1+q^{2nm+1})q^{-mn^2}}{(Q;Q)_n} \sum_{k=0}^{n-1} (-1)^k q^{k^2+(2nm+3)k} \begin{bmatrix} n-1 \\ k \end{bmatrix}_Q .$$

With $q^{k^2+(2nm+3)k} = Q^{\binom{k}{2}}(Q^{nm+2})^k$ and the finite q-binomial theorem $\sum_k (-1)^k Q^{\binom{k}{2}} \begin{bmatrix} n-1 \\ k \end{bmatrix} z^k = (z;Q)_{n-1}$ at $z = Q^{nm+2}$,

$$F_{n,m}(q,q^{-1}) = q^{-mn^2}(1-q)(1+q^{2nm+1}) \frac{(Q;Q)_{n(m+1)}}{(Q;Q)_n (Q;Q)_{nm+1}} = q^{-mn^2} \begin{bmatrix} M \\ n \end{bmatrix}_Q \frac{(1-q)(1+q^{2nm+1})}{1-Q^{nm+1}}.$$

Step 5: parity separation. The identity $\frac{(1-q)(1+q^{2nm+1})}{1-Q^{nm+1}} = 1 - q\,\frac{1-Q^{nm}}{1-Q^{nm+1}}$ and $\begin{bmatrix} M \\ n \end{bmatrix}_Q \frac{1-Q^{nm}}{1-Q^{nm+1}} = [m]_{Q^n} \begin{bmatrix} M \\ n-1 \end{bmatrix}_Q$ (from $\begin{bmatrix} M \\ n-1 \end{bmatrix}_Q / \begin{bmatrix} M \\ n \end{bmatrix}_Q = \frac{1-Q^n}{1-Q^{nm+1}}$) yield (111). Exponents in the first summand are $\equiv nm$, in the second $\equiv nm+1 \pmod 2$. Under $t = q^{-1}$ total bidegree parity is preserved ($a+b \equiv a-b$), hence $\Pi^{(m)}$ selects the first summand. Let $q \to 1$ for the scalar identities, and note that $m\binom{M}{n-1} = \frac{nm}{nm+1}\binom{M}{n}$ gives $F_{n,m}(1,1) = \frac{1}{nm+1}\binom{M}{n}$. □

Remark 7.3 (Independent verification). The contents of this section are elementary and finite, hence machine-checkable. Implementing (5) verbatim in exact rational arithmetic, we have confirmed for all $n \le 4$ and $0 \le m \le 3$ that the fixed-point sum (109) simplifies to a Laurent polynomial, the full identity (111), the Gaussian specialization of $P_{n,m}$, and the three scalar identities $P_{n,m}(1,1) = \binom{n(m+1)}{n}$, $N_{n,m}(1,1) = m\binom{n(m+1)}{n-1}$, $F_{n,m}(1,1) = \frac{1}{nm+1}\binom{n(m+1)}{n}$. Likewise all statements of Theorem 7.4, and every assertion of Section A (length duality $|C|+|D| = 2n+1$ and $N = 2k-1$ at *every* heart, the fiber counts $|\rho^{-1}(z_J)^T| = \binom{N}{k}$, the census profile, and the thresholds $n = m(m-1)$ for $\mathrm{Gr}(m, 2m-1)$) have been verified for all $n \le 8$.

Example 7.4. For $n=2$ we have $P_{2,1} = [5]_{q,t} + qt$ and $N_{2,1} = qt[4]_{q,t}$. For $n=3$ we have $P_{3,1} = [10]+qt[6]+qt[4]$ and $N_{3,1} = qt[9]+q^2t^2[5]+q^4t^4$. For $n=4$ we have $P_{4,1} = [17]+qt[13]+qt[11]+q^2t^2[9]+qt[9]+q^3t^3[5]+q^2t^2[5]+q^4t^4$ and $N_{4,1} = qt[16]+q^2t^2[12]+q^2t^2[10]+q^3t^3[8]+q^4t^4[6]+q^4t^4[4]$. Here $[k]_{q,t} = \sum_{i+j=k-1} q^i t^j$. The positive parts agree with Stump's tables of $\mathrm{Cat}^{(1)}(B_n;q,t)$ [74, Cor. 3.18 and App. A], and $(P_{n,1}(1,1), F_{n,1}(1,1)) = (\binom{2n}{n}, \mathrm{Cat}_n)$ for $n = 2,3,4$.

Theorem 7.5. *For each n there is $m_0(n)$ such that for all $m \ge m_0(n)$ we have $H^i(\mathcal{Z}_n^{\mathrm{symb}}, \mathcal{O}(m)) = 0$ for $i > 0$, the canonical map $\widetilde{\mathfrak{S}}_{n,m}^{\mathrm{symb}} \to H^0(\mathcal{Z}_n^{\mathrm{symb}}, \mathcal{O}(m))$ is an isomorphism, and*

$$\mathrm{Hilb}_{q,t}\, \widetilde{\mathfrak{S}}_{n,m}^{\mathrm{symb}} = q^{mn^2} P_{n,m}(q,t), \qquad \mathrm{ch}_{q,t}\, \mathfrak{S}_{n,m}^{\mathrm{symb}} = P_{n,m}(q,t).$$

In particular $\dim \mathfrak{S}_{n,m}^{\mathrm{symb}} = \binom{n(m+1)}{n}$ *for* $m \ge m_0(n)$.

Proof. By (110) and Theorem 6.22, $P_{n,m} = \chi^T(\mathcal{Z}_n^{\mathrm{symb}}, \mathcal{O}(m))$ for every m. The morphism π_Q is projective with $\mathcal{O}(1)$ relatively ample (Theorem 2.1). Serre vanishing on the projective central fiber gives the cohomology vanishing, and relative Serre vanishing $R^{>0}\pi_{Q*}\mathcal{O}(m) = 0$ $(m \gg 0)$ with cohomology-and-base-change at $0 \in S_n$ identifies $\pi_{Q*}\mathcal{O}(m) \otimes k(0)$, which is $\widetilde{\mathfrak{S}}_{n,m}^{\mathrm{symb}}$ by (8), with H^0 of the central fiber. Hence $P_{n,m} = \mathrm{ch}_T\, H^0(\mathcal{Z}, \mathcal{O}(m)) = q^{-mn^2}\, \mathrm{Hilb}_{q,t}\, \widetilde{\mathfrak{S}}_{n,m}$, where the normalization shift is (7) (the equivariant structure of $\mathcal{O}(1)$ normalizes the generator Δ to degree zero). The dimension is Theorem 7.2 at $q = t = 1$. The bound $m_0(n)$ supplied by Serre vanishing is not effective. □

8. The Cherednik identification

In this section we prove Theorem D. The comparison between the symbolic model and the spherical Cherednik module $eL_{\frac{1}{2n}+m}(\mathrm{triv})$ is made through the shift bimodules of the rational Cherednik algebra. Kivinen's associated-graded theorem identifies the associated graded of the iterated shift bimodule with $e(\Delta^m I^{(m)})$, and the Procesi bundle theory of Losev and of Boixeda Alvarez–Losev, transported to the wall variety Q_n by Theorem 2.7, produces a canonical bigraded surjection from $\mathfrak{S}^{\mathrm{symb}}_{n,m}$ onto $\mathrm{gr}_F\, eL_{\frac{1}{2n}+m}(\mathrm{triv})$.

The main idea is a dimension count. The target has dimension $\binom{n(m+1)}{n}$ by the theorem of Berest, Etingof and Ginzburg on the finite-dimensional representations, which we make explicit in type B_n in Theorem 8.11, and the source has the same dimension for $m \gg 0$ by Theorem 7.5. A surjection between spaces of equal dimension is an isomorphism. The first subsection fixes the bigrading and checks that the parameters are spherical, and the remaining subsections assemble the surjection.

8.1. **Recollections and the bigrading.** Let $H_c = H_c(W(B_n))$ be the rational Cherednik algebra at constant parameter c (equal parameters on the two conjugacy classes of reflections), $U_c = eH_ce$. Its Euler element $\mathbf{h}$ satisfies $[\mathbf{h}, x] = x$, $[\mathbf{h}, y] = -y$, $[\mathbf{h}, w] = 0$ for $x \in \mathfrak{h}^*$, $y \in \mathfrak{h}$, $w \in W$, giving the Euler grading $\deg_{\mathbf{h}}(x) = 1$, $\deg_{\mathbf{h}}(y) = -1$, $\deg_{\mathbf{h}}(w) = 0$. Two filtrations occur. The PBW total-degree filtration has $\deg_{\mathrm{tot}}(x) = \deg_{\mathrm{tot}}(y) = 1$, $\deg_{\mathrm{tot}}(w) = 0$, with $\mathrm{gr}^{\mathrm{tot}} H_c \cong S\#W$, and the order filtration has $\mathrm{ord}(x) = \mathrm{ord}(w) = 0$, $\mathrm{ord}(y) = 1$, and is the ordinary order filtration after the Dunkl embedding, with $\mathrm{gr}^{\mathrm{ord}} U_c \cong A$. For a PBW monomial $x^\alpha w y^\beta$ one has $\deg_{\mathrm{tot}} = |\alpha| + |\beta|$, $\deg_{\mathbf{h}} = |\alpha| - |\beta|$, $\mathrm{ord} = |\beta|$, hence

$$\deg_{\mathrm{tot}} = \deg_{\mathbf{h}} + 2\,\mathrm{ord}, \qquad F^{\mathrm{ord}}_b H_c[\lambda] = F^{\mathrm{tot}}_{\lambda+2b} H_c[\lambda] \tag{113}$$

on the Euler eigenspace of eigenvalue λ (the PBW basis gives equality of the two filtered subspaces).

Definition 8.1 (Gordon–Stump bidegree)**.** Let M be a filtered H_c-module whose filtration is stable under $\mathbf{h}$. A class in $\mathrm{gr}\, M$ has bidegree (a, b) if it has filtration degree $a + b$ and Euler degree $a - b$, equivalently $a = \frac{\deg_{\mathrm{tot}} + \deg_{\mathbf{h}}}{2}$, $b = \frac{\deg_{\mathrm{tot}} - \deg_{\mathbf{h}}}{2}$. Under the PBW isomorphism, $\deg x = (1, 0)$, $\deg y = (0, 1)$, $\deg w = (0, 0)$, which is the convention of [74, §4.2.2]. For $L^{(m)} := L_{\frac{1}{h}+m}(\mathrm{triv})$, the filtration F is the *iterated tensor-product filtration*, which is defined as follows. Writing $H^{(j)} = H_{\frac{1}{h}+j}$, the shift isomorphisms give

$$eL^{(m)} \cong eH^{(m)}e_- \otimes_{eH^{(m-1)}e} eH^{(m-1)}e_- \otimes \cdots \otimes eH^{(1)}e_- \otimes_{eH^{(0)}e} \mathbb{C}, \tag{114}$$

and F is defined recursively by $F_s(A \otimes_R B) = \sum_{i+j=s} F_iA \otimes_R F_jB$, starting from the PBW filtrations on the shift factors and placing $\mathbb{C}$ in degree zero. The bidegree-(a, b) piece of $\mathrm{gr}^G eL^{(m)}$ is $\mathrm{gr}^F_{a+b}(eL^{(m)}[a - b])$. This is the bigrading for which Stump proves the surjection $DR^{(m)}(W) \otimes \varepsilon \twoheadrightarrow \mathrm{gr}^G L^{(m)}$ [74, Thm. 4.5].

By (113), on a single filtered Cherednik factor the order and total-degree filtrations determine each other in each Euler eigenspace. This is the bridge between Kivinen's associated-graded theorem, stated for the order filtration, and the Gordon–Stump bidegree. The bidegree obtained in this way is intrinsic. By [16, Rem. 3.6] the shift bimodules carry a $(\mathbb{C}^\times)^2$-action, the two factors scaling $\mathfrak{h}$ and $\mathfrak{h}^*$, and in the associated graded the Dunkl operator D_y becomes the coordinate y [51, proof of Thm. 4.3]. Hence the symbol of $x^\alpha D^\beta \Delta^m$ is $x^\alpha y^\beta \Delta^m$, of bidegree $(|\alpha| + mn^2, |\beta|)$, and the Gordon normalization places Δ^m in bidegree $(0, 0)$, which is the shift q^{mn^2} of (7).

Set $c_0 = \frac{1}{h} = \frac{1}{2n}$. The module $L_{c_0}(\mathrm{triv})$ is one-dimensional [6, Thm. 1.4, Prop. 1.7], and $L_0 := eL_{c_0}(\mathrm{triv})$ has $\mathrm{gr}\, L_0 \cong A/\mathfrak{m}$.

Proposition 8.2 (Sphericity at the parameters used)**.** *For $c = m + \frac{1}{h}$ the parameter is e-spherical for $m \geq 0$ and e_--spherical for $m > 0$.*

Proof. [16, Prop. 2.12], quoted there from [31, Lemma 4.5]. Independently, for $W(B_n) = G(2,1,n)$ the classification of aspherical parameters [23, Thm. 3.4], applied with $(c_0, d_0, d_1) = (c, c, -c)$, shows that $\frac{1}{2n} + j$ avoids the aspherical locus for all $j \geq 0$. "Constant" is a real hypothesis in type B_n, since a parameter is a pair $(c_{\mathrm{short}}, c_{\mathrm{long}})$, and we are on the diagonal. □

8.2. **Kivinen's associated-graded theorem.** Following Kivinen [51, §4], define the shift bimodules $Q_c^{c+1} = eH_{c+1}e_-\Delta$ and their iterates $B_{m0} = Q_{c_0+m-1}^{c_0+m} \cdots Q_{c_0}^{c_0+1}$, a (U_{c_0+m}, U_{c_0})-bimodule, with the order filtration.

Theorem 8.3 (Kivinen [51, Thm. 4.3(2), Cor. 4.5(2)])**.** *Let c be constant and spherical. Then, for $N_k = B_{k0}eH_c$, $\operatorname{ogr} N_k = I^{(k)}\Delta^k$ as graded submodules of S, and $\operatorname{gr} B_c$ is the $\mathbb{Z}$-algebra of $\bigoplus_d e(\Delta^d I^{(d)})$. In particular $\operatorname{gr}_F B_{m0} \cong e(\Delta^m I^{(m)})$.*

Both hypotheses hold at our parameters $c = m + \frac{1}{2n}$ by Theorem 8.2. In type A, where $I^{(m)} = J^m$, this recovers [33, Prop. 6.5].

8.3. **The Procesi package on the wall.** The shift bimodules of Boixeda Alvarez–Losev live on a Procesi sheaf over the wall variety, and we assemble the one we need from Losev's classification.

Write $W_n = W(B_n)$, $H = S\#W_n$ for the skew group algebra, $e = \frac{1}{|W_n|}\sum_{w\in W_n} w$ for the spherical idempotent and $f = \frac{1}{|W_{n-1}|}\sum_{w\in W_{n-1}} w$ for the parabolic averaging idempotent of $W_{n-1} = \mu_2 \wr S_{n-1} \subset W_n$.

Convention 8.4 (Side of the idempotents)**.** H acts on Procesi sheaves on the *left*, so that taking invariants under a subgroup is left multiplication by the averaging idempotent, that is, $e\mathscr{P} = \mathscr{P}^{W_n}$ and $f\mathscr{P} = \mathscr{P}^{W_{n-1}}$. This is the convention of [16], who write $\varepsilon\widetilde{\mathscr{P}} \cong \mathcal{O}_{\widetilde{X}}$.

Losev [60, Thm. 1.1] classifies the normalized Procesi bundles $\widetilde{\mathscr{P}}$ on a smooth Hamiltonian-reduction resolution, that is, those with $\operatorname{End}(\widetilde{\mathscr{P}}) \cong H$, $\operatorname{Ext}^{>0}(\widetilde{\mathscr{P}}, \widetilde{\mathscr{P}}) = 0$ and $e\widetilde{\mathscr{P}} \cong \mathcal{O}$, and proves the following theorem.

Theorem 8.5 (Losev [60, Thm. 1.2])**.** *For every generic stability chamber there is a normalized Procesi bundle $\widetilde{\mathscr{P}}_n$ on Y_n with $\widetilde{\mathscr{P}}_n^{W_{n-1}} \cong \mathcal{T}^\theta$, the tautological bundle of the wreath-product quiver variety.*

For the two-vertex McKay quiver of μ_2 the tautological bundle is the cluster bundle $\mathcal{P}_n = \mathcal{R}_0 \oplus \mathcal{R}_1$, hence in Theorem 8.4

$$f\widetilde{\mathscr{P}}_n \cong \mathcal{P}_n. \tag{115}$$

This fixes the choice between the tautological and dual-tautological normalizations.

Boixeda Alvarez–Losev [16, §2.2] call a maximal Cohen–Macaulay sheaf $\widetilde{\mathscr{P}}$ on a symplectic variety Z over $S_n = \operatorname{Spec} A$ a *Procesi sheaf* if (i) $\operatorname{End}(\widetilde{\mathscr{P}}) \cong H$ as graded algebras, (ii) $\operatorname{End}(\widetilde{\mathscr{P}})$ is maximal Cohen–Macaulay, (iii) $H^i(Z, \operatorname{End}(\widetilde{\mathscr{P}})) = 0$ for $i > 0$, and (iv) $e\widetilde{\mathscr{P}} \cong \mathcal{O}_Z$, $\mathbb{C}^\times$-equivariantly. All four can be taken $(\mathbb{C}^\times)^2$-equivariant, which is what makes the package compatible with the q,t-bigrading. Their Lemma 2.3 states that for their partial resolution $\bar\rho : \widetilde{X} \to X$ and a Procesi sheaf $\widetilde{\mathscr{P}}$ on $\widetilde{X}$, the pushforward $\bar\rho_*\widetilde{\mathscr{P}}$ is a Procesi sheaf on X, maximal Cohen–Macaulay, with the isomorphism $\operatorname{End}(\widetilde{\mathscr{P}}) \to \operatorname{End}(\bar\rho_*\widetilde{\mathscr{P}})$ produced in the proof. By Theorem 2.7, in type B_n their X is Q_n and $\bar\rho = \rho$.

Theorem 8.6 (Procesi package on the type-B_n wall)**.** *Let $\widetilde{\mathscr{P}}_n$ be the normalized Procesi bundle of Theorem 8.5, and set*

$$\mathscr{P}_n := \rho_*\widetilde{\mathscr{P}}_n, \qquad \mathscr{A}_n := \mathcal{E}nd_{Q_n}(\mathscr{P}_n).$$

Then the following hold.

(P1) *$\mathscr{P}_n$ and $\mathscr{A}_n$ are $(\mathbb{C}^\times)^2$-equivariant maximal Cohen–Macaulay sheaves, and $\Gamma(Q_n,\mathscr{A}_n)\cong S\#W_n$ as bigraded A-algebras, compatibly with the central map $A\to\Gamma(Q_n,\mathscr{A}_n)$ induced by π_Q;*
(P2) *$e\mathscr{P}_n\cong\mathcal{O}_{Q_n}$;*
(P3) *$f\mathscr{P}_n\cong\rho_*\mathcal{P}_n$;*
(P4) *$H^i(Q_n,\mathscr{A}_n)=0$ for every $i>0$.*

Every idempotent summand of $\mathscr{P}_n$ is maximal Cohen–Macaulay.

Proof. By Theorem 2.7, $\rho: Y_n\to Q_n$ is the morphism $\bar\rho:\widetilde{X}\to X$ of [16] after choosing the adjacent generic chamber whose smooth quiver variety is Y_n. Their Lemma 2.3 therefore applies to $\widetilde{\mathscr{P}}_n$ and gives (P1), (P2), (P4). For (P3), note that f is an idempotent acting on $\widetilde{\mathscr{P}}_n$, hence $\widetilde{\mathscr{P}}_n=f\widetilde{\mathscr{P}}_n\oplus(1-f)\widetilde{\mathscr{P}}_n$. The direct-image functor is additive, hence commutes with the image of the idempotent, and (115) gives $f\mathscr{P}_n=f\rho_*\widetilde{\mathscr{P}}_n\cong\rho_*(f\widetilde{\mathscr{P}}_n)\cong\rho_*\mathcal{P}_n$. No vanishing of higher direct images is used in this identity. A direct summand of a maximal Cohen–Macaulay sheaf is maximal Cohen–Macaulay. □

8.4. **The shift bimodule and the canonical quotient.** Let B_m be the classical Procesi shift bimodule of [16, §3],

$$B_m=\Gamma\big(Q_n^{\mathrm{reg}},(\mathscr{P}_n^{\mathrm{reg}})^\vee\otimes\mathcal{O}^{\mathrm{reg}}(m)\otimes\mathscr{P}_n^{\mathrm{reg}}\big),$$

an (H,H)-bimodule, which by [16, Prop. 3.1] (applicable since $\mathcal{O}_{Q_n}(1)$ is relatively ample and $\operatorname{codim}_{Q_n}Q_n^{\mathrm{sing}}=2k(k-1)\geq 4$ on the $\mathrm{Gr}(k,2k-1)$ strata, by semismallness) extends across the singular locus as a maximal Cohen–Macaulay sheaf with no higher cohomology. Set $M_m:=eB_m$ and $P_m:=eB_me$. By (P2) both corners are computed by $e\mathscr{P}_n^{\mathrm{reg}}\cong\mathcal{O}$, hence $P_m\cong\Gamma(Q_n^{\mathrm{reg}},\mathcal{O}(m))$, and (8) with normality gives

$$P_m\cong e\big(\Delta^m I^{(m)}\big), \tag{116}$$

the e–e corner of Theorem 8.3. Boixeda Alvarez–Losev construct a $\mathbb{C}[\hbar]$-flat Rees deformation $B_{\hbar,c_0+m\leftarrow c_0}$ of B_m whose $\hbar=1$ fiber $B_{c_0+m\leftarrow c_0}$ is an (H_{c_0+m},H_{c_0})-bimodule with $\operatorname{gr}B_{c_0+m\leftarrow c_0}=B_m$. The bimodule $B_{c_0+m\leftarrow c_0}$ is built from the shift bimodules $eH_{c+1}e_-$ [16, §3, Lemma 3.7], which in the Dunkl realization are Kivinen's Q_c^{c+1} up to multiplication by Δ. Hence $eB_{c_0+m\leftarrow c_0}e$ is B_{m0} up to the Vandermonde normalization, and its order-associated graded is P_m.

Let $\mathfrak{n}=S_+$ and $\mathbb{C}_{\mathrm{triv}}$ the one-dimensional H-module on which S acts through $S/\mathfrak{n}$ and W trivially, hence $H/H\mathfrak{n}\cong\mathbb{C}W$.

Definition 8.7. The *central symbolic specialization* is $\mathfrak{S}^{\mathrm{cent}}_{n,m}:=P_m/\mathfrak{m}_0P_m\cong\widetilde{\mathfrak{S}}^{\mathrm{symb}}_{n,m}$, and the *augmentation specialization* is $\mathfrak{S}^{\mathrm{aug}}_{n,m}:=M_m\otimes_H\mathbb{C}_{\mathrm{triv}}$.

Proposition 8.8 (Canonical quotient)**.** *There is a canonical bigraded surjection*

$$\pi_{n,m}:\mathfrak{S}^{\mathrm{cent}}_{n,m}\twoheadrightarrow\mathfrak{S}^{\mathrm{aug}}_{n,m},\qquad \mathfrak{S}^{\mathrm{aug}}_{n,m}\cong\frac{P_m}{(M_m\mathfrak{n})e},\qquad \ker\pi_{n,m}=\frac{(M_m\mathfrak{n})e}{\mathfrak{m}_0P_m}. \tag{117}$$

Proof. $M_m\otimes_H\mathbb{C}_{\mathrm{triv}}\cong(M_m/M_m\mathfrak{n})\otimes_{\mathbb{C}W}\mathbb{C}_{\mathrm{triv}}$, and for every right $\mathbb{C}W$-module N averaging gives $N\otimes_{\mathbb{C}W}\mathbb{C}_{\mathrm{triv}}\xrightarrow{\sim}Ne$, $v\otimes1\mapsto ve$. Since $M_m\mathfrak{n}$ is W-stable, $(M_m/M_m\mathfrak{n})e\cong P_m/(M_m\mathfrak{n})e$. Finally $\mathfrak{m}_0\subset\mathfrak{n}$ is central, hence $\mathfrak{m}_0P_m\subseteq(M_m\mathfrak{n})e$. □

Theorem 8.9 (Boixeda Alvarez–Losev [16, Thm. 1.2, Prop. 6.6, Prop. 2.11])**.** *Let $\mathfrak{g}$ be simple with Weyl group W and Coxeter number h, and $m\geq0$. The tensor-product filtration gives a surjection*

$$B_m\otimes_H\mathbb{C}_{\mathrm{triv}}\twoheadrightarrow\operatorname{gr}_F L_{m+1/h}(\mathrm{triv}), \tag{118}$$

and $L_{m+1/h}(\mathrm{triv})$ is the unique finite-dimensional simple $H_{m+1/h}$-module, of dimension $(mh+1)^{\dim\mathfrak{h}}$ and isomorphic as a W-module to $\mathbb{C}[\Lambda_0/(mh+1)\Lambda_0]$, Λ_0 the coroot lattice.

Source. The dimension and W-module structure are [16, Prop. 2.11], quoted from [6, Thm. 1.4, Prop. 1.7]. The surjection is the proof of [16, Prop. 6.6]. Flatness gives $\operatorname{gr} B_{m+1/h\leftarrow 1/h} = B_m$, hence a surjection $B_m \otimes_H \mathbb{C}_{\mathrm{triv}} \twoheadrightarrow \operatorname{gr}(B_{m+1/h\leftarrow 1/h} \otimes_{H_{1/h}} L_{1/h})$, and $B_{m+1/h\leftarrow 1/h}$ is a Morita equivalence bimodule (by induction on m using Theorem 8.2 and $e_- B_{\hbar,c+m+1\leftarrow c} \cong eB_{\hbar,c+m\leftarrow c}$ [16, Lem. 3.7(2)]), hence $B_{m+1/h\leftarrow 1/h} \otimes_{H_{1/h}} L_{1/h} \cong L_{m+1/h}$ by uniqueness. (Their Theorem 1.2 upgrades (118) to an isomorphism using affine Springer fibers, but we do not need this, since the dimension count below forces it.) □

Lemma 8.10 (Spherical tensor identity)**.** *Let $c = \frac{1}{h}$ and $m \geq 0$. Then, compatibly with the tensor-product filtrations,*

$$eL_{c+m} \cong (eB_{c+m\leftarrow c}e) \otimes_{U_c} (eL_c), \qquad \dim eL_c = 1.$$

Consequently $F_b(eL_{c+m})$ is the image of $F_b(eB_{c+m\leftarrow c}e)$.

Proof. By the proof of [16, Prop. 6.6], $B \otimes_{H_c} L_c \cong L_{c+m}$ for $B = B_{c+m\leftarrow c}$. The module L_c is the trivial one-dimensional representation of W on which $\mathfrak{h}, \mathfrak{h}^*$ act by zero, hence $eL_c = L_c$, and c is e-spherical (Theorem 8.2), hence $L_c \cong H_c e \otimes_{U_c} eL_c$. Therefore $eL_{c+m} = e(B \otimes_{H_c} L_c) = (eB) \otimes_{H_c} (H_c e \otimes_{U_c} eL_c) = (eBe) \otimes_{U_c} (eL_c)$, all steps filtered, the filtration on eL_c being trivial. □

Taking the left spherical part of (118) and composing with Theorem 8.8 gives a canonical bigraded surjection

$$\Phi_{n,m}:\ \mathfrak{S}^{\mathrm{cent}}_{n,m} = \frac{e(\Delta^m I^{(m)})}{\mathfrak{m}_0\, e(\Delta^m I^{(m)})} \twoheadrightarrow \mathfrak{S}^{\mathrm{aug}}_{n,m} \cong \operatorname{gr}_F eL_{m+1/h}(\mathrm{triv}), \tag{119}$$

where the last isomorphism is (118) restricted to spherical parts and Theorem 8.10, and bigradedness is (113) together with the intrinsic bigrading of the shift bimodule. In the normalization (7) this is a bigraded surjection $\mathfrak{S}^{\mathrm{symb}}_{n,m} \twoheadrightarrow \operatorname{gr}_F eL_{\frac{1}{2n}+m}(\mathrm{triv})$.

8.5. **The type-B_n dimension.** For $W(B_n)$, $h = 2n$ and $N_m := mh + 1 = 2nm + 1$ is odd. The coroot lattice is $\Lambda_0 = \{a \in \mathbb{Z}^n : \sum a_i \text{ even}\}$ (coroots $\pm e_i \pm e_j$ and $2e_i$), and inclusion into $\mathbb{Z}^n$ induces a $W(B_n)$-equivariant isomorphism $\Lambda_0/N_m\Lambda_0 \xrightarrow{\sim} (\mathbb{Z}/N_m)^n$. Indeed, the index $[\mathbb{Z}^n : \Lambda_0] = 2$ is invertible modulo the odd N_m, hence one of b, $b + N_m e_1$ lies in Λ_0 for every $b \in \mathbb{Z}^n$, and $N_m b \in \Lambda_0$ forces $\sum b_i$ even. (Whether one reads Λ_0 as the root or the coroot lattice, for $\mathfrak{so}_{2n+1}$ or $\mathfrak{sp}_{2n}$, all readings give the same W-module because N_m is odd.)

Theorem 8.11 (Type-B_n target dimension)**.** *For every $m \geq 0$,*

$$\dim \operatorname{gr}_F eL_{\frac{1}{2n}+m}(\mathrm{triv}) = \binom{n(m+1)}{n} = \prod_{i=1}^{n} \frac{2i + 2nm}{2i},$$

with an ungraded basis indexed by weakly increasing sequences $0 \leq a_1 \leq \cdots \leq a_n \leq nm$, i.e. by partitions in an $n \times nm$ rectangle.

Proof. By Theorem 8.9 the spherical dimension is the number of $W(B_n)$-orbits on $(\mathbb{Z}/N_m)^n$. The sign orbits in one coordinate are $\{0\}, \{\pm 1\}, \ldots, \{\pm nm\}$, hence there are $nm + 1$ coordinate types, and a signed-permutation orbit is a multiset of size n chosen from these, which gives $\binom{(nm+1)+n-1}{n} = \binom{n(m+1)}{n}$. Orbit sums give the basis. The product form is [74, Thm. 4.3]. □

8.6. **The asymptotic isomorphism.**

Theorem 8.12. *Theorem D holds. For $m \geq m_0(n)$, the canonical surjection (119) is an isomorphism of bigraded vector spaces for the iterated tensor-product filtration of Theorem 8.1,*

$$\mathfrak{S}^{\mathrm{symb}}_{n,m} \xrightarrow{\sim} \operatorname{gr}_F eL_{\frac{1}{2n}+m}(\mathrm{triv}), \qquad P_{n,m}(q,t) = \mathrm{ch}^G_{q,t} \operatorname{gr}_F eL_{\frac{1}{2n}+m}(\mathrm{triv}),$$

and $(M_m\mathfrak{n})e = \mathfrak{m}_0 P_m$, that is, taking the spherical corner commutes with specialization at the origin in the Serre range.

Proof. By Theorem 7.5 the source has dimension $\binom{n(m+1)}{n}$, and by Theorem 8.11 the target has the same dimension. A surjection of finite-dimensional spaces of equal dimension is an isomorphism, and it is bigraded since all constructions preserve the tensor-product filtration and the Euler grading. The character identity is Theorem 7.5, and the last assertion is the vanishing of the kernel in (117). □

Remark 8.13 (The two filtrations at higher rungs). At $m = 1$ there is a single shift factor in (114), hence the tensor-product filtration coincides with the ambient order filtration of the already-multiplied Dunkl operators, and the bidegree-(a, b) piece of $\mathrm{gr}^G\, eL^{(1)}$ is $\mathcal{F}_b(n^2+a-b)/\mathcal{F}_{b-1}(n^2+a-b)$ with $\mathcal{F}_b(r) = \left(e\,\mathrm{Span}\{x^\alpha D^\beta \Delta : |\beta| \le b\}\right)_{\deg_{\mathrm{pol}}=r}$. The parity $\deg_{\mathrm{tot}} \equiv \deg_{\mathbf{h}} \pmod 2$ of (113) is what makes the quotient by F_{b-1} the correct total-degree piece. For $m \ge 2$ multiplication of shift factors is filtered but need not be strict ($\mathrm{ord}(PQ) \le \mathrm{ord}\, P + \mathrm{ord}\, Q$), hence the ambient order of a product may be smaller than its tensor degree. The intrinsic bigrading is the tensor-product one of Theorem 8.1, and a reconstruction from the ambient order of $x^\alpha D^\beta \Delta^m$ is not part of the convention. This is visible already at $n = 2$, $m = 2$.

9. The spherical collapse and Stump's conjecture

In this section we prove Theorems A and B. The collapse $e_m J^m = e_m I^m = e_m I^{(m)}$ for $m \gg 0$ is deduced from Theorem D by a Nakayama argument. Stump's generalized Gordon surjection, from the minimal generating space of J^m onto $\mathrm{gr}_F\, L_{\frac{1}{\hbar}+m}(\mathrm{triv})$, is compatible with the surjection of Section 8, since both are induced by the same filtered multiplication of shift bimodules. Once the second surjection is known to be an isomorphism, the first forces $e_m J^m$ to generate $e_m I^{(m)}$ modulo $\mathfrak{m}$, and graded Nakayama gives the equality. Theorem B then follows by comparing the two punctual quotients of Section 2.3. We close the section with a list of open problems.

9.1. The collapse.

Theorem 9.1 (Stump [74, Thm. 4.5 with §4.2.1])**.** *The filtered iterated-shift construction* (114) *induces a bigraded surjection from the determinant component of the generalized diagonal coinvariants (equivalently, by* [74, Thm. 3.5]*, from the minimal generating space of J^m, up to the twist $\varepsilon^{\otimes(1-m)}$) onto $\mathrm{gr}_F\, L_{\frac{1}{\hbar}+m}(\mathrm{triv})$. Its spherical part is a surjection $\overline{A}_m \twoheadrightarrow T_m$, where $\overline{A}_m = e_m J^m/\mathfrak{m}\, e_m J^m$ and $T_m = \mathrm{gr}_F\, eL_{c_0+m}(\mathrm{triv})$.*

Theorem 9.2. *Theorem A holds. For $m \ge m_1(n)$ we have $e_m J^m = e_m I^m = e_m I^{(m)}$, and the three spherical models* (3) *coincide in type B_n/C_n.*

Proof. Set $A_m = e_m J^m \subseteq D_m = e_m I^m \subseteq C_m = e_m I^{(m)}$, finite graded A-modules (S is finite over A), and $\overline{A}_m = A_m/\mathfrak{m}A_m$, $\overline{C}_m = C_m/\mathfrak{m}C_m$, $T_m = \mathrm{gr}_F\, eL_{c_0+m}(\mathrm{triv})$.

Two maps. The map $\phi_m^{\mathrm{symb}} : \overline{C}_m \twoheadrightarrow T_m$ is (119). The map $\phi_m^J : \overline{A}_m \twoheadrightarrow T_m$ is Stump's generalized Gordon surjection, Theorem 9.1, that is, the same filtered iterated-shift construction applied to the subalgebra generated by products of alternants. Its spherical part is $\overline{A}_m$, since products of m alternating polynomials span $e_m J^m$ over A ($e_m(f a_1 \cdots a_m) = e(f)\, a_1 \cdots a_m$).

Compatibility. The diagram

$$\begin{array}{ccc} \overline{A}_m & \xrightarrow{\bar{\iota}_m} & \overline{C}_m \\ & \searrow^{\phi_m^J} & \downarrow{\scriptstyle \phi_m^{\mathrm{symb}}} \\ & & T_m \end{array}$$

commutes, because both maps are induced by the associated graded of one and the same filtered multiplication $Q_{c_0+m-1}^{c_0+m} \otimes \cdots \otimes Q_{c_0}^{c_0+1} \otimes eL_{c_0} \to eL_{c_0+m}$. Under Theorem 8.3, $A_m \subseteq C_m = \mathrm{gr}\, B_{m0}$ is the submodule generated by the products of the distinguished alternating symbols, and ϕ_m^J is by construction the restriction of ϕ_m^{symb}.

Nakayama. For $m \geq m_0(n)$, ϕ_m^{symb} is an isomorphism (Theorem 8.12). Surjectivity of ϕ_m^J then forces $\bar{\iota}_m$ surjective, i.e. $C_m = A_m + \mathfrak{m}C_m$, and graded Nakayama on the finite A-module C_m/A_m gives $A_m = C_m$, hence also $= D_m$. For the Proj statement compare r-th Veronese subalgebras for any $r \geq m_1(n)$, noting $(\Delta S^\varepsilon)^m = e(\Delta^m J^m)$ and the sandwich $e(\Delta^m J^m) \subseteq (e(\Delta I))^m \subseteq e(\Delta^m I^m) \subseteq e(\Delta^m I^{(m)})$, all equal in the range, and relative Proj is insensitive to Veronese. □

Remark 9.3. The collapse is a statement about the *spherical* pieces only. Kivinen's type-B_3 computation [51, Ex. B.2] shows $J \subsetneq I$ while $eJ = eI$, and the isospectral models remain distinct.

9.2. **Stump's quotient versus the invariant quotient, and the proof of Theorem B.**

Theorem 9.4. *Theorem B holds, and moreover* $\pi_m : \mathsf{Q}_m^{\mathrm{inv}}(J^m) \twoheadrightarrow \mathsf{Q}_m^{\mathrm{full}}(J^m)$ *of* (9) *is an isomorphism for* $m \gg 0$. *In particular Stump's conjecture* $\dim e_\varepsilon DR^{(m)}(B_n) = \binom{n(m+1)}{n}$ *holds for all* $m \gg 0$.

Proof. Let m be in the ranges of Theorems 7.5 and 9.2. Then $\dim \mathsf{Q}_m^{\mathrm{inv}}(J^m) = \dim \mathfrak{S}_{n,m}^{\mathrm{symb}} = \binom{n(m+1)}{n}$ by the collapse and Theorem 7.5, hence $\dim \mathsf{Q}_m^{\mathrm{full}}(J^m) \leq \binom{n(m+1)}{n}$. On the other hand Stump's surjection (Theorem 9.1, with target of dimension $\binom{n(m+1)}{n}$ by Theorem 8.11) factors through $\mathsf{Q}_m^{\mathrm{full}}(J^m)$, giving $\dim \mathsf{Q}_m^{\mathrm{full}}(J^m) \geq \binom{n(m+1)}{n}$. Hence equality holds, π_m is an isomorphism, and

$$\mathrm{Cat}^{(m)}(B_n; q, t) = \mathrm{Hilb}_{q,t}\, \mathsf{Q}_m^{\mathrm{full}}(J^m) = \mathrm{Hilb}_{q,t}\, \mathsf{Q}_m^{\mathrm{inv}}(J^m) = \mathrm{ch}_{q,t}\, \mathfrak{S}_{n,m}^{\mathrm{symb}} = P_{n,m}(q,t)$$

by Theorem 7.5. (The determinant twist $\varepsilon^{\otimes(1-m)}$ in (2) does not affect the bigrading.) □

Remark 9.5 (Type A versus type B). The binomial $\binom{n(m+1)}{n} = \prod_{i=1}^n \frac{2i+2nm}{2i}$ is the Fuss–Catalan number of $W(B_n)$. For the literal type-A_{n-1} diagonal ideal $\bigcap_{i<j}(x_i - x_j, y_i - y_j)^m$ the corresponding count is $\frac{1}{mn+1}\binom{(m+1)n}{n}$, and the coordinate-root and sum-diagonal factors of (6) are responsible for the difference. Compare $F_{n,m}(1,1) = \frac{1}{nm+1}\binom{n(m+1)}{n}$ in Theorem C, the *virtual* (unprojected) count, which is precisely the type-A Fuss–Catalan number.

9.3. **Open problems.** Let us put the results of the paper in the context of the conjectures of Haiman on diagonal coinvariants. For $W = S_n$ Haiman [39] conjectured, and later proved [41, 42], that the diagonal coinvariant ring $DR(W) = S/\mathfrak{m}S$ has dimension $(n+1)^{n-1}$ and that its determinantal part $e_\varepsilon DR(W)$ has dimension the Catalan number $\frac{1}{n+1}\binom{2n}{n}$, with bigraded Hilbert series the q,t-Catalan number of [39, 27].

The generalized diagonal coinvariants $DR^{(m)}(W)$ of Garsia and Haiman have dimension $(mn+1)^{n-1}$, and their determinantal part has dimension the Fuss–Catalan number $\frac{1}{mn+1}\binom{(m+1)n}{n}$ and Hilbert series the q,t-Fuss–Catalan number [42]. For a general Weyl group W of rank ℓ and Coxeter number h, Haiman [39, §7] computed the dimension of $DR(W)$ for B_4, B_5 and D_4, found it different from $(h+1)^\ell$ (it is $9^4 + 1$ for B_4). Haiman conjectured that $DR(W)$ has a natural quotient of dimension $(h+1)^\ell$ [39, Conj. 7.1.2] whose twist by the sign character is the permutation representation of W on $Q/(h+1)Q$, Q the root lattice.

The number of W-orbits on $Q/(h+1)Q$ is the Catalan number $\mathrm{Cat}(W) = \prod_{i=1}^\ell \frac{h+1+e_i}{1+e_i}$ of W [39, §7], which is $\binom{2n}{n}$ for $W(B_n)$ (compare Theorem 8.11), hence the conjecture predicts that the determinantal part of the quotient has dimension $\mathrm{Cat}(W)$. Gordon [32] constructed the quotient as $\mathrm{gr}\, L_{1+\frac{1}{h}}(\mathrm{triv})$, which is the case $m = 1$ of the surjection of Theorem 9.1. Haiman's conjecture on the determinantal part of the diagonal coinvariants [43, Conj. 7.2.5], recorded for all m as [74, Conj. 4.7], is that the kernel of the surjection $DR^{(m)}(W) \otimes \varepsilon \twoheadrightarrow \mathrm{gr}\, L_{m+\frac{1}{h}}(\mathrm{triv})$ contains no copy of the trivial representation, that is, that the determinantal part of the generalized diagonal coinvariants maps isomorphically onto $e\ \mathrm{gr}\, L_{m+\frac{1}{h}}(\mathrm{triv})$.

For $W = W(B_n)$ this is the statement that the spherical part of the surjection of Theorem 9.1 is an isomorphism, and Theorem 9.4 proves it for all $m \gg 0$, together with Conjectures 3.4 and 3.9 of

[74] on the dimension and on the principal specialization of $\mathrm{Cat}^{(m)}(W;q,t)$. The dimension of the full diagonal coinvariant ring of $W(B_n)$ exceeds $(2n+1)^n$ for every $n \geq 4$, by a quantity which grows at least quadratically in n [1], hence it is only the determinantal part which retains the Catalan behaviour.

The polynomials $P_{n,m}$ of Theorem B contain more information than the Fuss–Catalan numbers, and the questions below concern the low twists, the effectivity of the bounds, and the combinatorics of $P_{n,m}$ and of the opposite-parity part $N_{n,m} = P_{n,m} - F_{n,m}$ of Theorem C.

The conjectures and theorems of this paper lead to a collection of natural questions which is list below.

Question 9.6. (All twists). Do Theorems B and D hold for every $m \geq 1$? By Theorem 7.5 this reduces to $H^{>0}(\mathcal{Z}_n^{\mathrm{symb}}, \mathcal{O}(m)) = 0$ and base change in the finitely many low degrees. A Cohen–Macaulay property of the punctual section ring $\bigoplus_m \widetilde{\mathfrak{S}}_{n,m}$ with a-invariant ≤ 0 would give all $m \geq 1$ at once.

Question 9.7. (Effectivity.) Give an effective bound for $m_0(n)$ (a Castelnuovo–Mumford regularity bound for $\mathcal{Z}_n^{\mathrm{symb}}$).

Question 9.8. (The first rung.) At $m = 1$, (119) still provides the surjection $\mathfrak{S}_{n,1}^{\mathrm{symb}} \twoheadrightarrow \mathrm{gr}\, eL_{\frac{1}{2n}+1}(\mathrm{triv})$, whose target has dimension $\binom{2n}{n} = \mathrm{Cat}(B_n)$. Its bijectivity for all n, equivalently $H^{>0}$-vanishing at $m = 1$, would give the exact q,t-Fuss–Catalan character at the classical rung, and it would prove Haiman's conjecture [43, Conj. 7.2.5] on the determinantal part of the diagonal coinvariants of $W(B_n)$. It is verified for $n \leq 3$ by exact computation of the contravariant form on $L_{\frac{1}{2n}+1}(\mathrm{triv})$, in the bigrading of Theorem 8.13.

Question 9.9. (Combinatorics of $P_{n,m}$ and $N_{n,m}$.) All polynomials in Theorem 7.4 are positive linear combinations of the strings $(qt)^a[k]_{q,t}$. Stump [74, Conj. 3.14] conjectured that $\mathrm{Cat}^{(m)}(W;q,1)$ is the coheight generating function of the positive regions of the extended Shi arrangement of W, and in type B_n the specialization $\mathrm{Cat}^{(1)}(B_n;1,t)$ is expected to be the area generating function of the shifted Dyck paths of length n, that is, of the lattice paths with unit north and east steps which start on the anti-diagonal, end at (n,n) and stay above the diagonal [61, Conj. 1.17].

Is there a pair of statistics on lattice paths which realizes $P_{n,m}(q,t)$, and is there a geometric or representation-theoretic meaning of the opposite-parity part $N_{n,m}$? A conjectural answer to the first question is given in Section 10. For $m = 1$ the polynomials $N_{n,1}$ have appeared, up to the factor qt, in the thesis of Lingxi Lu [61]. That thesis attaches to the shifted Dyck paths of length n with base k, which start on the anti-diagonal $y = 1 - k - x$, polynomials $B_{n,k}(q,t)$ written as positive combinations of the strings $(qt)^a[l]_{q,t}$ and normalized by $B_{n,1}(q,t) = \mathrm{Cat}^{(1)}(B_n;q,t)$ and by the two specializations $B_{n,k}(1,t) = \sum_\pi t^{\mathrm{area}(\pi)}$ and $q^{n(n+k-1)}B_{n,k}(q,q^{-1}) = \begin{bmatrix} 2n+k-1 \\ n \end{bmatrix}_{q^2}$, and it computes them for $n \leq 3$. One has

$$N_{n,1}(q,t) = qt\, B_{n-1,3}(q,t) \qquad (2 \leq n \leq 4),$$

compare Theorem 7.4 with [61, Ch. 3, Ch. 4 and App. C], and for every n the specializations $N_{n,1}(1,1) = \binom{2n}{n-1}$ and $N_{n,1}(q,q^{-1}) = q^{1-n^2}\begin{bmatrix} 2n \\ n-1 \end{bmatrix}_{q^2}$ of Theorem C agree with those of $qt\, B_{n-1,3}(q,t)$. It would be interesting to know whether the equality holds for all n and what the base of the paths means on Y_n.

10. Shuffle conjecture in type B

In this section we formulate a combinatorial model for the polynomials $P_{n,m}$ of Theorem B, which answers, conjecturally, the first question of Section 9.3. The model is a sum over the $\binom{(m+1)n}{n}$ lattice paths of the $n \times mn$ rectangle, and it uses three statistics of such a path, the higher diagonal

inversion number dinv_m of Loehr [57], in the form of Lee, Li and Loehr [56], and two area statistics $A^{(m)}_{\mp}$ which measure the excursions of the path below and above the line $y = x/m$. We prove that the resulting polynomial $\widehat{P}_{n,m}$ shares with $P_{n,m}$ the positivity, the $q \leftrightarrow t$ symmetry and the two specializations of Theorem C, we prove the equality $P_{n,m} = \widehat{P}_{n,m}$ in ranks one and two for every m, and we verify it by exact computation in a range of small ranks.

The main idea behind the name of the section is the following. Let us write the localization polynomial as a pairing in the ring of two-colour symmetric functions. The wreath Macdonald polynomials $\widetilde{H}_\lambda$, $\lambda \in \mathrm{Bal}(2n)$, introduced by Haiman [43] and constructed by Bezrukavnikov and Finkelberg [14], are the bigraded Frobenius characters of the fibers of the normalized Procesi bundle of Theorem 8.5 at the fixed points I_λ, and the fixed-point data of Theorem 7.1 are pairings of $\widetilde{H}_\lambda$ with explicit symmetric functions. The Cauchy identity for the $\widetilde{H}_\lambda$ then converts the fixed-point sum (5) into the pairing

$$P_{n,m} = (-1)^{n-1}\Big\langle \nabla_1^m \nabla_0\, p_n\Big[\tfrac{(1+qt)X_0+(q+t)X_1}{(1-q^2)(1-t^2)}\Big],\ h_{n-1}[X_0]\, p_1\big[(1+qt)X_0-(q+t)X_1\big]\Big\rangle,$$

where ∇_0, ∇_1 are the two-colour nabla operators (Theorem 10.12). In type A the pairing $\langle \nabla^m e_n, e_n\rangle$ is the sum over the m-Dyck paths of the $n \times mn$ rectangle of $q^{\mathrm{dinv}_m} t^{\mathrm{area}}$, and the pairing $(-1)^{n-1}\langle \nabla p_n, e_n\rangle$ is a sum over all lattice paths of the $n \times n$ square, by the shuffle theorem of Carlsson and Mellit [19], conjectured in [36], its rational version by Mellit [63] and the square paths theorem of Sergel [72]. Theorem 10.5 is the type-B analogue of these statements at the Catalan level, with a two-colour nabla applied to a power sum on the left and a sum over all paths of the rectangle on the right. One of the inputs of the pairing formula, the evaluation of $\widetilde{H}_\lambda$ at the colour-zero power sum, is not available in the literature for two colours, and we prove it in Theorem 10.11 from an exterior-compatibility property of the normalized Procesi bundle, which we deduce from Losev's formal restriction theorem.

Throughout the section $m \geq 1$, $D := mn^2$, $[x]_+ := \max(x, 0)$, and $[k]_{q,t} = \sum_{i+j=k-1} q^i t^j$ as in Theorem 7.4, hence $[k]_{q,t}$ has k terms and $[k]_{q,t}(q, q^{-1}) = q^{1-k}[k]_{q^2}$.

10.1. **Paths in the rectangle and three statistics.** A lattice path π from $(0,0)$ to (mn, n) with unit north and east steps is determined by the numbers $e_1 \leq e_2 \leq \cdots \leq e_n$ of east steps which precede its first, second, …, n-th north step, $0 \leq e_i \leq mn$. Respectively, π is determined by the partition

$$\alpha(\pi) = (e_n, e_{n-1}, \ldots, e_1) \subseteq (mn)^n,$$

the partition with n rows and parts at most mn which fills the region between the path and the y-axis, and $|\alpha(\pi)| = e_1 + \cdots + e_n$. We set

$$b_i := mi - e_i \qquad (1 \leq i \leq n), \tag{120}$$

the height of the path above the line $y = x/m$ after its i-th north step, measured in units of $1/m$. Let us define the kernel

$$s_m(d) = \begin{cases} m + d, & 1 - m \leq d \leq 0, \\ m + 1 - d, & 1 \leq d \leq m, \\ 0, & \text{otherwise}, \end{cases} \tag{121}$$

and the three statistics

$$\begin{aligned} \mathrm{dinv}_m(\pi) &= \sum_{i<j} s_m(b_i - b_j), \\ A^{(m)}_-(\pi) = \sum_{i=1}^n \big([m - b_i]_+ + [-b_i]_+\big), &\qquad A^{(m)}_+(\pi) = \sum_{i=1}^n \big([b_i]_+ + [b_i - m]_+\big). \end{aligned} \tag{122}$$

For $m=1$ the kernel is $s_1(d)=1$ for $d\in\{0,1\}$ and $s_1(d)=0$ otherwise, thus dinv_1 is the usual diagonal inversion number of the sequence $(b_1,\dots,b_n)$, and $A_-^{(1)}=\sum_{b_i\le 0}(1-2b_i)$, $A_+^{(1)}=\sum_{b_i>0}(2b_i-1)$. The kernel s_m is the one of the higher q,t-Catalan numbers [57], [56, §1(b)], where it is applied to the m-Dyck paths, that is to the paths with all $b_i\ge 0$. Here it is applied to all paths of the rectangle, and the two area statistics count a step at height b in the strip $0<b<m$ on both sides.

Definition 10.1. For $n,m\ge 1$ let

$$\widehat{P}_{n,m}(q,t)=\sum_{\pi}(qt)^{\mathrm{dinv}_m(\pi)}\,q^{A_-^{(m)}(\pi)}\,t^{A_+^{(m)}(\pi)}, \tag{123}$$

the sum over all $\binom{(m+1)n}{n}$ lattice paths from $(0,0)$ to (mn,n).

The statistics have a second form, on the partition $\alpha=\alpha(\pi)$. For a box $s=(i,j)$ of α in row i and column j, both counted from 1 in English notation, let $a_\alpha(s)=\alpha_i-j$ and $l_\alpha(s)=\alpha'_j-i$ be its arm and leg, and put

$$H_m(\alpha)=\#\{s\in\alpha:\ 0\le a_\alpha(s)-m\,l_\alpha(s)\le m\},\qquad c_{m,n}(\alpha)=\sum_{i=1}^{n}\big[\,2(\alpha_i-m(n+1-i))-1\,\big]_+. \tag{124}$$

The hook statistic H_m is the statistic c_m of [56, §1(a)], applied here to all partitions of the rectangle. It does not depend on n, and the boundary correction $c_{m,n}$ vanishes exactly on the partitions inside the dilated staircase $\alpha_i\le m(n+1-i)$.

Theorem 10.2 (Hook–path identity)**.** *For every path π with partition α and heights b as in (120),*

$$H_m(\alpha)+c_{m,n}(\alpha)=\mathrm{dinv}_m(\pi)+A_-^{(m)}(\pi),\quad D-2|\alpha|+H_m(\alpha)+c_{m,n}(\alpha)=\mathrm{dinv}_m(\pi)+A_+^{(m)}(\pi). \tag{125}$$

Consequently

$$\widehat{P}_{n,m}(q,t)=\sum_{\alpha\subseteq(mn)^n}q^{\,H_m(\alpha)+c_{m,n}(\alpha)}\,t^{\,D-2|\alpha|+H_m(\alpha)+c_{m,n}(\alpha)}. \tag{126}$$

Proof. Let us number the rows of α from the bottom, hence row i has length $L_i:=e_i=mi-b_i$ and $L_0:=0\le L_1\le\cdots\le L_n$. A box (i,j) of row i has arm L_i-j, and its leg is the number of rows below row i of length at least j, which equals $i-1-k$ for $L_k<j\le L_{k+1}$, $0\le k\le i-1$. For such a box the hook condition $0\le \mathrm{arm}-m\,\mathrm{leg}\le m$ reads

$$0\le L_i-j-m(i-1-k)\le m\quad\Longleftrightarrow\quad mk-b_i\le j\le m(k+1)-b_i.$$

For integers $A\le B+1$ and $C\le D'$ the number of integers j with $A\le j\le B$ and $C\le j\le D'$ is $[B-C+1]_+-[B-D']_+-[A-C]_++[A-1-D']_+$, since $\#\{j\in[C,D']:\,j\le X\}=[X-C+1]_+-[X-D']_+$ for every integer X, and one subtracts the values at $X=B$ and $X=A-1$. With $A=L_k+1$, $B=L_{k+1}$, $C=mk-b_i$ and $D'=m(k+1)-b_i$ the number of hook boxes of row i in the columns $L_k<j\le L_{k+1}$ is therefore

$$[m+1+b_i-b_{k+1}]_+-[b_i-b_{k+1}]_+-[b_i-b_k+1]_++[b_i-b_k-m]_+.$$

We sum over $k=0,\dots,i-1$ and reindex the first two terms by $j=k+1\in[1,i]$ and the last two by $j=k\in[0,i-1]$. The indices $1\le j\le i-1$ occur in both groups and contribute, with $d:=b_j-b_i$,

$$[m+1-d]_+-[-d]_+-[1-d]_++[-m-d]_+=s_m(d),$$

as one checks on the four ranges $d\ge m+1$, $1\le d\le m$, $1-m\le d\le 0$ and $d\le -m$, where the left side equals 0, $m+1-d$, $(m+1-d)+d-(1-d)=m+d$ and $(m+1-d)+d-(1-d)+(-m-d)=0$. The index $j=i$ of the first group contributes $[m+1]_+-[0]_+=m+1$, and the index $j=0$ of the

second group contributes $-[b_i+1]_+ + [b_i-m]_+$, since $b_0 = 0$. Hence the number of hook boxes in row i is

$$g_m(b_i) + \sum_{j<i} s_m(b_j - b_i), \qquad g_m(b) := m + 1 - [b+1]_+ + [b-m]_+ = \#\{r \in \{0,\dots,m\} :\ r > b\},$$

and the sum over i gives $H_m(\alpha) = \sum_i g_m(b_i) + \mathrm{dinv}_m(\pi)$.

The row of α with index $k = n+1-i$ from the top is row i from the bottom, and $\alpha_k - m(n+1-k) = L_i - mi = -b_i$, thus $c_{m,n}(\alpha) = \sum_i [-2b_i - 1]_+$. For every integer b we have $g_m(b) + [-2b-1]_+ = [m-b]_+ + [-b]_+$, the three cases being $b \ge m$, where both sides vanish, $0 \le b < m$, where both sides equal $m - b$, and $b < 0$, where the left side is $(m+1) + (-2b-1)$ and the right side is $(m-b) + (-b)$. The sum over i gives the first identity of (125). For the second identity we use $[x]_+ - [-x]_+ = x$, which gives $A_+^{(m)}(\pi) - A_-^{(m)}(\pi) = \sum_i (b_i + (b_i - m)) = 2\sum_i b_i - mn$, and $\sum_i b_i = m\binom{n+1}{2} - |\alpha|$, hence $A_+^{(m)} - A_-^{(m)} = mn^2 - 2|\alpha| = D - 2|\alpha|$. Adding this to the first identity gives the second, and substituting the two exponents into (123) gives (126). □

Corollary 10.3.

(i) $\widehat{P}_{n,m} \in \mathbb{Z}_{\ge 0}[q,t]$, *and every monomial of* $\widehat{P}_{n,m}$ *has total degree* $\equiv mn \pmod 2$.

(ii) $\widehat{P}_{n,m}(q,t) = \widehat{P}_{n,m}(t,q)$. *More precisely, the complementation* $\alpha \mapsto \alpha^c$, $\alpha^c_k = mn - \alpha_{n+1-k}$, *exchanges the two exponents in* (126).

(iii) $\widehat{P}_{n,m}(1,1) = \binom{(m+1)n}{n}$ *and* $q^D\, \widehat{P}_{n,m}(q,q^{-1}) = \begin{bmatrix}(m+1)n\\ n\end{bmatrix}_{q^2}$.

Proof. (i) The three exponents in (123) are nonnegative. The total degree of the term of π is $2\,\mathrm{dinv}_m + \sum_i (|b_i| + |m - b_i|) \equiv \sum_i m \pmod 2$, since $[m-b]_+ + [-b]_+ + [b]_+ + [b-m]_+ = |b| + |m-b|$ and $|b| + |m-b| \equiv m \pmod 2$.

(ii) Under complementation $b_i^c = mi - \alpha^c_{n+1-i} = mi - mn + \alpha_i = m - b_{n+1-i}$. The bijection $(i,j) \mapsto (n+1-j, n+1-i)$ of the pairs $i<j$ satisfies $b_i^c - b_j^c = b_{n+1-j} - b_{n+1-i}$, thus dinv_m is preserved, and $[m - b_i^c]_+ + [-b_i^c]_+ = [b_{n+1-i}]_+ + [b_{n+1-i} - m]_+$, thus $A_-^{(m)}(\pi^c) = A_+^{(m)}(\pi)$ and $A_+^{(m)}(\pi^c) = A_-^{(m)}(\pi)$.

(iii) At $(1,1)$ every path has weight 1. At (q,q^{-1}) the term of π is $q^{A_-^{(m)} - A_+^{(m)}} = q^{2|\alpha| - D}$ by the proof of Theorem 10.2, and $\sum_{\alpha \subseteq (mn)^n} q^{2|\alpha|} = \begin{bmatrix}(m+1)n\\ n\end{bmatrix}_{q^2}$. □

10.2. **The conjecture.** Let us first write the parity projection of (5) as a fixed-point sum of its own. For $\lambda \in \mathrm{Bal}(2n)$ let

$$B_\lambda^{(i)} = \sum_{\substack{(a,b)\in\lambda \\ a+b\equiv i \ (2)}} q^a t^b \quad (i=0,1), \qquad u_{n,m} = \sum_{\lambda\in\mathrm{Bal}(2n)} \frac{\ell_\lambda^m \Pi_\lambda^{(0)} B_\lambda^{(0)}}{D_\lambda}, \qquad v_{n,m} = \sum_{\lambda\in\mathrm{Bal}(2n)} \frac{\ell_\lambda^m \Pi_\lambda^{(0)} B_\lambda^{(1)}}{D_\lambda},$$

hence $B_\lambda = B_\lambda^{(0)} + B_\lambda^{(1)}$ and $F_{n,m} = (1-q)(1-t)(u_{n,m} + v_{n,m})$.

Lemma 10.4. *For all* $n, m \ge 1$,

$$P_{n,m} = (1+qt)\,u_{n,m} - (q+t)\,v_{n,m}, \qquad N_{n,m} = (q+t)\,u_{n,m} - (1+qt)\,v_{n,m},$$
$$P_{n,m} + N_{n,m} = (1+q)(1+t)(u_{n,m} - v_{n,m}).$$

Proof. Let σ be the substitution $(q,t) \mapsto (-q,-t)$, hence $\sigma(q^a t^b) = (-1)^{a+b} q^a t^b$. Then $\sigma \Pi_\lambda^{(0)} = \Pi_\lambda^{(0)}$, since every monomial of $\Pi_\lambda^{(0)}$ has even content, and $\sigma D_\lambda = D_\lambda$, since for $a_\lambda(s) + l_\lambda(s)$ odd the exponent sums $a+1-l$ and $-a+l+1$ of the two factors of D_λ are even. Moreover $\sigma \ell_\lambda^m = (-1)^{nm} \ell_\lambda^m$, because ℓ_λ is a product of n monomials of odd content, and $\sigma B_\lambda^{(0)} = B_\lambda^{(0)}$, $\sigma B_\lambda^{(1)} = -B_\lambda^{(1)}$. Hence $\sigma u_{n,m} = (-1)^{nm} u_{n,m}$, $\sigma v_{n,m} = (-1)^{nm+1} v_{n,m}$ and $\sigma F_{n,m} = (-1)^{nm}(1+q)(1+t)(u_{n,m} - v_{n,m})$. Substituting

into (5) and using $(1-q)(1-t)+(1+q)(1+t) = 2(1+qt)$ and $(1-q)(1-t)-(1+q)(1+t) = -2(q+t)$ we obtain the formula for $P_{n,m}$. The other two follow from $N_{n,m} = P_{n,m} - F_{n,m}$. □

Conjecture 10.5 (Type-B shuffle conjecture at the Catalan level)**.** *For all $n, m \geq 1$ we have $P_{n,m} = \widehat{P}_{n,m}$, that is*

$$\sum_{\lambda \in \mathrm{Bal}(2n)} \frac{\ell_\lambda^m \, \Pi_\lambda^{(0)} ((1+qt)B_\lambda^{(0)} - (q+t)B_\lambda^{(1)})}{D_\lambda} = \sum_{\pi:(0,0)\to(mn,n)} (qt)^{\mathrm{dinv}_m(\pi)} \, q^{A_-^{(m)}(\pi)} \, t^{A_+^{(m)}(\pi)}. \tag{127}$$

By Theorem 10.4 the left side of (127) is $P_{n,m}$, and by Theorem 10.2 the right side is also $\sum_{\alpha \subseteq (mn)^n} q^{H_m(\alpha)+c_{m,n}(\alpha)} t^{D-2|\alpha|+H_m(\alpha)+c_{m,n}(\alpha)}$. Both sides of (127) are symmetric in q and t, both have $\binom{(m+1)n}{n}$ terms and both specialize at $t = q^{-1}$ to $q^{-mn^2} \left[\begin{smallmatrix}(m+1)n\\ n\end{smallmatrix}\right]_{q^2}$, by Theorem C and Theorem 10.3. For $m \gg 0$ the conjecture is a statement about the q,t-Fuss–Catalan number of Theorem B, and for every m it is a statement about the Euler characteristic $\chi^T(\mathcal{Z}_n^{\mathrm{symb}}, \mathcal{O}(m))$ of Theorem 7.5.

The conjecture was tested numerically for $m = 1, n \leq 13$, $m = 2, n \leq 8$, $m = 3, n \leq 7$. Theorem below offers another piece of evidence supporting the conjecture.

Theorem 10.6. *Theorem 10.5 holds in the following cases.*

1. *$n = 1$ and every m, where $P_{1,m} = \widehat{P}_{1,m} = [m+1]_{q,t}$ and $N_{1,m} = qt\,[m]_{q,t}$.*
2. *$n = 2$ and every m, where $P_{2,m} = \widehat{P}_{2,m} = \sum_{j=0}^{m} (qt)^j [4m-4j+1]_{q,t}$ (Theorem 10.7).*
3. *$n \leq 4$ and $m \leq 3$, and $(n,m) \in \{(5,1),(5,2)\}$, as an identity of polynomials, by an exact computation in the sense of Theorem 7.3.*

Moreover both sides of (127) agree at the four points $(q,t) = (2,3), (2,5), (3,5), (\frac{7}{3}, -\frac{5}{2})$ for $(n,m) \in \{(6,1),(4,4)\}$.

Proof. (1) $\mathrm{Bal}(2) = \{(2),(1,1)\}$. For $\lambda = (2)$ we have $\ell_\lambda = q$, $\Pi_\lambda^{(0)} = 1$, $B_\lambda = 1+q$ and $D_\lambda = (1-q^2)(1-q^{-1}t)$, the box $(0,0)$ being the only one with odd hook, hence its term in (5) is $q^{m+1}(1-t)/(q-t)$, and the term of $(1,1)$ is the mirror image $t^{m+1}(1-q)/(t-q)$. Therefore

$$F_{1,m} = \frac{q^{m+1} - t^{m+1}}{q-t} - qt\,\frac{q^m - t^m}{q-t} = [m+1]_{q,t} - qt\,[m]_{q,t},$$

and the parity projection keeps the first summand, whose total degree is m. On the path side a path of the $1 \times m$ rectangle is $e_1 \in \{0,\dots,m\}$, with $b_1 = m - e_1 \in [0,m]$, $\mathrm{dinv}_m = 0$, $A_-^{(m)} = e_1$ and $A_+^{(m)} = m - e_1$.

(2) is Theorem 10.7 below.

(3) The two sides were computed with the definitions (5) and (123) implemented verbatim in exact rational arithmetic, the left side as a rational function which simplifies to a Laurent polynomial, and compared coefficient by coefficient; the path form and the partition form of $\widehat{P}_{n,m}$ were computed separately and compared as well, for $n \leq 5$ and $m \leq 4$. The four-point check is the same computation with q,t specialized before summation. □

We record the polynomials of Theorem 10.6(3) for $n = 3$ and $n = 4$, in the notation of Theorem 7.4 and with $w = qt$:

$$\begin{aligned}
P_{3,2} &= [19] + w([15]+[13]) + w^2([11]+[9]+[7]) + w^3[7] + w^4[3],\\
N_{3,2} &= w[18] + w^2([14]+[12]) + w^3([10]+[8]) + w^4[6] + w^5[4],\\
P_{3,3} &= [28] + w([24]+[22]) + w^2([20]+[18]+[16]) + w^3([16]+[14]+[12]+[10])\\
&\quad + w^4([12]+[10]) + w^5([8]+[6]) + w^6[4],\\
P_{4,2} &= [33] + w([29]+[27]+[25]) + w^2([25]+[23]+2[21]+[19]+[17])\\
&\quad + w^3([21]+[19]+2[17]+[15]+[13]) + w^4([17]+[15]+2[13]+[11]+[9])
\end{aligned}$$

$$+ w^5([13]+[11]+2[9]) + w^6([9]+[7]+2[5]) + w^7[5] + w^8 + w^9.$$

All of them are positive combinations of the strings $w^a[k]_{q,t}$, as are the polynomials of Theorem 7.4, and among the polynomials $P_{n,m}$ with $n \le 4$ and $m \le 3$ a coefficient 2 occurs first for $n = 4$, $m = 2$.

10.3. **Rank two.** In rank two both sides of (127) can be computed in closed form for every m. The main idea for the localization side is to sum the geometric series in ℓ_λ^m over the five balanced partitions of 4, which produces a rational generating function in an auxiliary variable z, and the main idea for the path side is that $\widehat{P}_{2,m}$ is multiplicity free, so that it suffices to identify the set of exponents.

Proposition 10.7. *For all $m \ge 1$,*

$$P_{2,m} = \widehat{P}_{2,m} = \sum_{j=0}^{m} (qt)^j [4m-4j+1]_{q,t}, \qquad N_{2,m} = qt\,[4]_{q,t} \sum_{j=0}^{m-1} (qt)^j\, h_{m-1-j}(q^4,t^4),$$

where $h_k(q^4,t^4) = \sum_{i=0}^{k} q^{4i}t^{4k-4i}$. Equivalently $P_{2,m} = [4m+1]_{q,t} + qt\,P_{2,m-1}$ with $P_{2,0} = 1$.

Proof. The localization side. The five balanced partitions of 4 have $\ell_{(4)} = q^4$, $\ell_{(1^4)} = t^4$ and $\ell_\lambda = qt$ for $\lambda = (3,1), (2,2), (2,1,1)$, and their terms $(1-q)(1-t)B_\lambda \Pi_\lambda^{(0)}/D_\lambda$ in (5) are

$$\frac{q^4(1-t)}{(q-t)(q^3-t)}, \qquad -\frac{q^3t(1-q)(1-t)(1+q+q^2+t)}{(q-t)^2(q+t)(q^3-t)}, \qquad -\frac{qt(1-qt)}{(q-t)^2},$$
$$\frac{qt^3(1-q)(1-t)(1+t+t^2+q)}{(q-t)^2(q+t)(q-t^3)}, \qquad \frac{t^4(1-q)}{(q-t)(q-t^3)},$$

in the order $(4), (3,1), (2,2), (2,1,1), (1^4)$. Multiplying the term of λ by $\ell_\lambda^m z^m$, summing over $m \ge 0$ and adding the five rational functions we obtain

$$\sum_{m\ge 0} F_{2,m}\, z^m = \frac{1 + qt\,[3]_{q,t}\, z - qt\,[4]_{q,t}\, z}{(1-q^4z)(1-t^4z)(1-qtz)}. \tag{128}$$

The denominator is invariant under $(q,t) \mapsto (-q,-t)$, and in the numerator $qt[3]_{q,t}$ has total degree 4 while $qt[4]_{q,t}$ has total degree 5. Since $(-1)^{2m} = 1$, the parity projection (5) keeps the even part of the numerator, hence

$$\sum_{m\ge 0} P_{2,m} z^m = \frac{1 + qt[3]_{q,t} z}{(1-q^4z)(1-t^4z)(1-qtz)}, \qquad \sum_{m\ge 0} N_{2,m} z^m = \frac{qt[4]_{q,t} z}{(1-q^4z)(1-t^4z)(1-qtz)}.$$

From $[4k+1]_{q,t} = (q^{4k+1} - t^{4k+1})/(q-t)$ we get

$$\sum_{k\ge 0} [4k+1]_{q,t} z^k = \frac{1}{q-t}\Big(\frac{q}{1-q^4z} - \frac{t}{1-t^4z}\Big) = \frac{1 + qt[3]_{q,t} z}{(1-q^4z)(1-t^4z)},$$

and the expansion of $1/(1-qtz)$ gives the formula for $P_{2,m}$. The formula for $N_{2,m}$ follows from $1/((1-q^4z)(1-t^4z)) = \sum_k h_k(q^4,t^4) z^k$.

The path side. A path of the $2 \times 2m$ rectangle is a pair (b_1, b_2) with $-m \le b_1 \le m$ and $0 \le b_2 \le m + b_1$, by (120) and $0 \le e_1 \le e_2 \le 2m$. Let us write $a(\pi) = \mathrm{dinv}_m + A_-^{(m)}$ and $b(\pi) = \mathrm{dinv}_m + A_+^{(m)}$ for its two exponents, and let

$$S_m = \{(a,b) \in \mathbb{Z}_{\ge 0}^2 :\ a + b \le 4m,\ a + b \equiv 0\ (2),\ 3\min(a,b) + \max(a,b) \ge 4m\}.$$

We claim that $\pi \mapsto (a(\pi), b(\pi))$ is a bijection from the paths onto S_m. Note that $s_m(d) = m + d$ for all $-m \le d \le 0$. There are three families of paths.

If $b_1 < 0$, then $b_2 \le m-1$ and $-m \le b_1 - b_2 \le -1$, hence $\mathrm{dinv}_m = m + b_1 - b_2$, $A_-^{(m)} = 2m - 2b_1 - b_2$, $A_+^{(m)} = b_2$, and $(a,b) = (4m - b - 2b_2,\ b)$ with $b := m + b_1 \in [0, m-1]$ and $b_2 \in [0, b]$. As b_2 varies,

a runs over $4m-b, 4m-b-2, \dots, 4m-3b$. These are exactly the points of S_m with $b \le m-1$, since such a point has $a \ge b$ (otherwise $3a+b \le 4b < 4m$), hence $\min(a,b) = b$ and the conditions read $4m-3b \le a \le 4m-b$ with $a \equiv b \pmod 2$.

If $b_1 \ge 0$ and $b_2 > m$, then $b_1 \ge 1$, $-m \le b_1 - b_2 \le -1$, hence $\mathrm{dinv}_m = m + b_1 - b_2$, $A_-^{(m)} = m - b_1$, $A_+^{(m)} = b_1 + 2b_2 - m$, and $(a,b) = (2m - b_2,\, 2b_1 + b_2)$. With $a = 2m - b_2 \in [0, m-1]$ and $b_1 \in [m-a, m]$ the exponent $b = 2b_1 + 2m - a$ runs over $4m-3a, \dots, 4m-a$ in steps of 2. These are exactly the points of S_m with $a \le m-1$, by the symmetry $a \leftrightarrow b$ of S_m.

If $0 \le b_1 \le m$ and $0 \le b_2 \le m$, then $A_-^{(m)} = 2m - b_1 - b_2$ and $A_+^{(m)} = b_1 + b_2$, hence $a + b = 2m + 2s_m(d)$ and $a - b = 2m - 2\sigma$ with $d = b_1 - b_2$ and $\sigma = b_1 + b_2$. For fixed d the sum σ runs over the integers of the parity of d in $[|d|, 2m - |d|]$. For $0 \le s \le m$ the equation $s_m(d) = s$ has the solutions $d = s - m$ and, when $s \ge 1$, $d = m + 1 - s$. The first gives $a - b \in \{2s, 2s-4, \dots, -2s\}$, the second $a - b \in \{2s-2, 2s-6, \dots, 2-2s\}$, and together all $a - b \in \{-2s, -2s+2, \dots, 2s\}$, each once. Hence the third family maps bijectively onto the points with $a + b = 2m + 2s$ and $a, b \ge m$, $0 \le s \le m$, which are exactly the points of S_m with $\min(a,b) \ge m$.

The three families exhaust the paths, and the three sets of points exhaust S_m and are disjoint, since a point with $a \le m-1$ and $b \le m-1$ has $3\min(a,b) + \max(a,b) < 4m$. This proves the claim. Finally S_m is the disjoint union over $0 \le j \le m$ of the sets $\{a + b = 4m - 2j,\ \min(a,b) \ge j\}$, because for $a + b = 4m - 2j$ the condition $3\min(a,b) + \max(a,b) \ge 4m$ reads $\min(a,b) \ge j$, and $\sum_{a+b=4m-2j,\ \min(a,b)\ge j} q^a t^b = (qt)^j [4m - 4j + 1]_{q,t}$. □

Remark 10.8. By Theorem 6.25(b) the germ of $\mathcal{Z}_2^{\mathrm{symb}}$ at the wall point is the cone over the rational normal quartic, that is $\mathbb{C}^2/\mu_4$. The polynomial $\sum_j (qt)^j [4m - 4j + 1]_{q,t}$ is the character of the monomials $x^a y^b z^c$ with $a + b + 4c = 4m$ and weights $q^a t^b (qt)^c$, that is of the degree-$4m$ part of the graded ring $\mathbb{C}[x,y,z]$ with $\deg x = \deg y = 1$ and $\deg z = 4$, the section ring of the weighted projective plane $\mathbb{P}(1,1,4)$, which is the projective cone over the rational normal quartic. Thus Theorem 10.7 identifies $P_{2,m}$ with the character of $H^0(\mathbb{P}(1,1,4), \mathcal{O}(4m))$ for every $m \ge 1$, in accordance with Theorem 7.5, which identifies $P_{2,m}$ with the character of $H^0(\mathcal{Z}_2^{\mathrm{symb}}, \mathcal{O}(m))$ for $m \gg 0$.

10.4. **The wreath Macdonald form.** In this subsection we rewrite the left side of (127) as a pairing of two explicit two-colour symmetric functions, one of them obtained from a power sum by the two-colour nabla operators. We follow the survey of Orr and Shimozono [67], to which we refer for the definitions, and we use their results in the identity chamber, which is the chamber of Y_n.

Let $\Lambda^{(2)} = \Lambda_{X_0} \otimes \Lambda_{X_1}$ be the ring of symmetric functions in two alphabets over $K = \mathbb{Q}(q,t)$, with the tensor Schur functions $s_{\mu^\bullet} = s_{\mu^{(0)}}[X_0] s_{\mu^{(1)}}[X_1]$ as an orthonormal basis for the Hall pairing $\langle\,,\rangle$, hence $\langle p_k[X_i], p_k[X_j]\rangle = k\delta_{ij}$. For a 2×2 matrix M over K and $f \in \Lambda^{(2)}$ we write $f[MX]$ for the image of f under the K-algebra endomorphism $p_k[X_i] \mapsto \sum_j M_{ij}(q^k, t^k) p_k[X_j]$, and for $c \in K$ we write $p_k[cX_i] = c(q^k, t^k) p_k[X_i]$ as usual. Let χ be the permutation matrix of the colour swap, and let inv be the involution of $\Lambda^{(2)}$ which inverts q and t in the coefficients. Let us identify $\mathrm{Bal}(2n)$ with the set of pairs of partitions of total size n by the 2-quotient in the convention of [67, §3.1], and let $\trianglerighteq$ be the dominance order on $\mathrm{Bal}(2n)$. The wreath Macdonald polynomial $\widetilde{H}_\lambda \in \Lambda^{(2)}$, $\lambda \in \mathrm{Bal}(2n)$, is the unique element with

$$\widetilde{H}_\lambda[(1-q\chi)X] \in \bigoplus_{\mu \trianglerighteq \lambda} K s_{\mu^\bullet}, \qquad \widetilde{H}_\lambda[(1-t^{-1}\chi)X] \in \bigoplus_{\mu \trianglelefteq \lambda} K s_{\mu^\bullet}, \qquad \langle \widetilde{H}_\lambda, h_n[X_0]\rangle = 1, \tag{129}$$

where $\mu^\bullet$ is the 2-quotient of μ [67, (3.7)–(3.9)]. For $\lambda \in \mathrm{Bal}(2n)$ put

$$T_0(\lambda) = \prod_{\substack{(a,b)\in\lambda \\ a+b \text{ even}}} q^a t^b, \qquad T_1(\lambda) = \ell_\lambda, \qquad A = \frac{1}{(1-q^2)(1-t^2)} \begin{pmatrix} 1+qt & q+t \\ q+t & 1+qt \end{pmatrix}, \tag{130}$$

hence A is the bigraded character of $\mathbb{C}[x,y]$ split by the Γ-parity, $A^{-1}=(1-t\chi)(1-q\chi)$, and $\langle f,g\rangle_{q,t}:=\langle f,g[A^{-1}X]\rangle$ is a symmetric bilinear form, since A is symmetric.

Theorem 10.9 (Bezrukavnikov–Finkelberg [14], Orr–Shimozono [67]). *Let $\lambda,\mu\in\mathrm{Bal}(2n)$.*

(a) *The $\widetilde{H}_\lambda$ exist, they form a basis of the degree-n part of $\Lambda^{(2)}$, and $\widetilde{H}_\lambda$ is the bigraded Frobenius character of the fiber at I_λ of the normalized Procesi bundle $\widetilde{\mathscr{P}}_n$ of Theorem 8.5, the trivial representation of W corresponding to $h_n[X_0]$* [67, §3.3, Thm. 4.6 and (4.3)].

(b) $\langle\widetilde{H}_\lambda, h_{n-1}[X_0]\,p_1[X_i]\rangle=B_\lambda^{(i)}$ *for* $i=0,1$ [67, Prop. 4.9 and (3.9)].

(c) $\langle\widetilde{H}_\lambda, e_n[X_i]\rangle=T_i(\lambda)$ *for* $i=0,1$ [67, Thm. 4.14].

(d) $\langle\widetilde{H}_\lambda, \mathrm{inv}\,\widetilde{H}_\mu\rangle_{q,t}=\delta_{\lambda\mu}D_\lambda$ [67, Prop. 3.30, 3.31 and Thm. 3.32], *with D_λ as in* (4).

In (b) we used the Pieri rule $h_{n-1}h_1=h_n+s_{(n-1,1)}$ and the normalization in (129) to rewrite the case $i=0$ of [67, Prop. 4.9]. In (d) the permutation neg of [67, §3.6] is the identity for $r=2$, and the product in [67, Thm. 3.32] runs over the boxes with even hook length, that is over the boxes with $a_\lambda(s)+l_\lambda(s)$ odd. By (a) and (b), the representation $1\oplus U\oplus V=\mathrm{Ind}_{W_{n-1}}^{W}\mathbf{1}$ of Theorem 8.5, where U is the reflection representation of S_n inflated to W and V is the reflection representation of W, corresponds to $h_{n-1}[X_0](p_1[X_0]+p_1[X_1])$, with U corresponding to $s_{(n-1,1)}[X_0]$ and V to $h_{n-1}[X_0]p_1[X_1]$, in accordance with (115) and $[\mathcal{R}_i]_\lambda=B_\lambda^{(i)}$. Following [67, §4.5] we define the two-colour nabla operators by

$$\nabla_i\,\widetilde{H}_\lambda=T_i(\lambda)\,\widetilde{H}_\lambda\qquad(i=0,1),\tag{131}$$

that is ∇_i multiplies $\widetilde{H}_\lambda$ by the product of the weights of the i-th colour component of the cluster bundle at I_λ, and by (c) this product is $\langle\widetilde{H}_\lambda,e_n[X_i]\rangle$.

The remaining input is the evaluation of $\widetilde{H}_\lambda$ at the colour-zero power sum. In type A one has $\langle\widetilde{H}_\mu,s_{(n-k,1^k)}\rangle=e_k[B_\mu-1]$ for all k, which is the coefficient extraction of the specialization $\widetilde{H}_\mu[1-u]=\prod_{(a,b)\in\mu}(1-uq^at^b)$ [27, 43], and the alternating sum over k gives $\langle\widetilde{H}_\mu,p_n\rangle=\Pi_\mu=\prod_{(a,b)\neq(0,0)}(1-q^at^b)$. For $r\geq 3$ colours the wreath analogue is proved by Romero and Wen by operator methods [70, Prop. 6.6, Cor. 6.7], and these methods are not yet available for two colours [71, footnote 1]. We give a geometric proof, valid for two colours, through an exterior-compatibility property of the normalized Procesi bundle.

For a representation L of W let

$$\mathsf{F}(L):=\mathrm{Hom}_W\big(L,\widetilde{\mathscr{P}}_n\big)=\big(L^*\otimes\widetilde{\mathscr{P}}_n\big)^W,\tag{132}$$

a T-equivariant vector bundle on Y_n, the invariants of a finite group being a direct summand. Let $\mathbb{C}^n=1\oplus U$ be the permutation representation of S_n inflated to W, with U the reflection representation of S_n, and let V be the reflection representation of W, as after Theorem 10.9. By the normalization $\widetilde{\mathscr{P}}_n^W=\mathcal{O}$, by (115) and by the decomposition $\mathrm{Ind}_{W_{n-1}}^{W}\mathbf{1}=1\oplus U\oplus V$, in which the residual sign change acts trivially on $1\oplus U$ and by -1 on V,

$$\mathsf{F}(1)=\mathcal{O},\qquad\mathsf{F}(1\oplus U)=\mathcal{R}_0,\qquad\mathsf{F}(V)=\mathcal{R}_1,\qquad\mathsf{F}(U)\cong\mathcal{R}_0/\mathcal{O}\cong\mathcal{T}_n,\tag{133}$$

where $\mathcal{T}_n=\ker(\mathrm{Tr}_{\mathcal{R}_0})$ is the trace-zero even part of Section 3. The fixed-point characters of these bundles at I_λ are 1, $B_\lambda^{(0)}$, $B_\lambda^{(1)}$ and $B_\lambda^{(0)}-1$.

Lemma 10.10 (Exterior compatibility). *For $L\in\{1\oplus U,\ U,\ V\}$ and every $j\geq 0$ there is a T-equivariant isomorphism*

$$\alpha_{j,L}:\ \textstyle\bigwedge^j\mathsf{F}(L)\xrightarrow{\sim}\mathsf{F}\big(\bigwedge^j L\big),\tag{134}$$

exterior powers above the rank being zero.

Proof. Write $\pi = \pi_Q \circ \rho : Y_n \to S_n = (\mathfrak{h} \oplus \mathfrak{h}^*)/W$, $S = \mathbb{C}[\mathfrak{h} \oplus \mathfrak{h}^*]$ and $H = S \# W$, hence $\mathrm{End}(\widetilde{\mathscr{P}}_n) = H$ by Theorem 8.5.

Step 1, the free locus. Let $(\mathfrak{h} \oplus \mathfrak{h}^*)^\circ$ be the open set of points with trivial stabilizer, S_n° its image and $Y_n^\circ = \pi^{-1}(S_n^\circ)$. The quotient map $\eta : (\mathfrak{h} \oplus \mathfrak{h}^*)^\circ \to S_n^\circ$ is a W-torsor, and π is an isomorphism over S_n°, since it is proper and birational, S_n° is smooth, and the fibre over n distinct free Γ-orbits is the single cluster supported on their union. Let $1 \in \widetilde{\mathscr{P}}_n^W = \mathcal{O}$ be the unit section. The action of $S \subset H$ on $\widetilde{\mathscr{P}}_n$ defines a T-equivariant morphism

$$\iota : \ \eta_* \mathcal{O}_{(\mathfrak{h} \oplus \mathfrak{h}^*)^\circ} \longrightarrow \widetilde{\mathscr{P}}_n|_{Y_n^\circ}, \qquad f \longmapsto f \cdot 1,$$

which intertwines the actions of H, where H acts on functions by multiplication and by the W-action. At a point $x \in S_n^\circ$ the fibre $H \otimes_{S^W} k(x)$ is the skew group algebra of the functions on the free orbit $\eta^{-1}(x)$, that is the full matrix algebra of the $|W|$-dimensional space of these functions, and the fibre of $\widetilde{\mathscr{P}}_n$ is its unique simple module, the regular representation of W [67, Thm. 4.6]. In the module of functions the W-invariants are the constants, thus 1_x corresponds to a nonzero constant and $f \mapsto f \cdot 1_x$ is injective between spaces of dimension $|W|$. Hence ι is an isomorphism. Consequently $\mathsf{F}(L)|_{Y_n^\circ} = \mathrm{Hom}_W(L, \eta_* \mathcal{O})$ is the bundle associated with the torsor η and the representation L^*, and the alternating products of functions

$$\alpha_{j,L}(\varphi_1 \wedge \cdots \wedge \varphi_j)(v_1 \wedge \cdots \wedge v_j) = \det\big(\varphi_a(v_b)\big)_{a,b}$$

define an isomorphism $\alpha_{j,L}$ over Y_n°, which fibrewise is the canonical map $\bigwedge^j L^* \to (\bigwedge^j L)^*$. Since Y_n° is dense and both sides of (134) are locally free, $\alpha_{j,L}$ is a T-equivariant rational map of vector bundles on Y_n, and it remains to prove that it and its inverse are regular in codimension one.

Step 2, the prime divisors outside Y_n°. The complement of $(\mathfrak{h} \oplus \mathfrak{h}^*)^\circ$ is the union of the fixed loci of the nontrivial elements of W, each of codimension at least two, hence $S_n \setminus S_n^\circ$ has codimension two and every prime divisor $D \subset Y_n \setminus Y_n^\circ$ is contracted by π. If c is the codimension of $\pi(D)$, the generic fibre of $D \to \pi(D)$ has dimension $c-1$, and the semismallness of the symplectic resolution π [49, Lem. 2.11] gives $2(c-1) \le c$, thus $c = 2$. Hence a generic point $y \in D$ lies over $\eta(b)$ for a point $b \in \mathfrak{h} \oplus \mathfrak{h}^*$ whose stabilizer $W_b = \langle s \rangle$ is generated by one reflection, a short reflection ε_i, where one Γ-orbit reaches the origin, or a long one, conjugate to $(i\,j)$, where two free orbits collide; these are the two leaves of Theorem 2.4.

Step 3, formal restriction. Let $\widehat{Y}_b$ be the formal neighbourhood of $\pi^{-1}(\eta(b))$ in Y_n. It is identified with the formal neighbourhood of the zero fibre in the symplectic resolution of the slice $(\mathfrak{h} \oplus \mathfrak{h}^*)^\wedge_0 / W_b$, the product of the minimal resolution $T^*\mathbb{P}^1$ of an A_1 singularity with a smooth factor. By Losev's restriction theorem [60, Prop. 4.1] there are a normalized Procesi bundle $\widetilde{\mathscr{P}}'$ of rank two on the slice resolution and an isomorphism

$$\Lambda : \ \widetilde{\mathscr{P}}_n|_{\widehat{Y}_b} \xrightarrow{\sim} \mathrm{Hom}_{W_b}(\mathbb{C}W, \widetilde{\mathscr{P}}')^\wedge$$

of modules over the completed skew group algebra $H^\wedge$, the latter being identified with the centralizer algebra of the completed skew group algebra of the slice by the construction of Bezrukavnikov and Etingof [13, §3.2], in which W acts on the coinduced module by translation and functions act by multiplication [13, Thm. 3.2]. In particular this identification takes the completed module $S^\wedge$ of functions to $\mathrm{Hom}_{W_b}(\mathbb{C}W, S_0^\wedge)$, the functions on the formal neighbourhood of the orbit Wb. Taking W-invariants, $\Lambda(1)$ is a nowhere vanishing section of $(\widetilde{\mathscr{P}}')^{W_b} = \mathcal{O}$, that is a unit u of the completed base ring, and we replace Λ by $u^{-1}\Lambda$, so that $\Lambda(1) = 1$. On the free part of $\widehat{Y}_b$ we now have two isomorphisms of $H^\wedge$-modules from $\widetilde{\mathscr{P}}_n$ onto the module of functions on the torsor, the completion of ι^{-1}, and Λ followed by the isomorphism of Step 1 for $\widetilde{\mathscr{P}}'$ and the group W_b. Both send 1 to the constant function 1. Over the free part the skew group algebra is the full endomorphism algebra of the module of functions, thus an automorphism of this module is the multiplication by an invertible function, and one which fixes 1 is the identity. Hence the two isomorphisms coincide, and in the

formal neighbourhood of y the map $\alpha_{j,L}$ is the alternating-product map $\alpha'_{j,L}$ of Step 1 formed with $\widetilde{\mathscr{P}'}$ and $L|_{W_b}$, under the identification $\mathsf{F}(L)^\wedge = \mathrm{Hom}_{W_b}(L|_{W_b}, \widetilde{\mathscr{P}'}^\wedge)$ of Frobenius reciprocity.

Step 4, the local computation. Let ε be the sign character of $W_b = \langle s\rangle$. Since $\widetilde{\mathscr{P}'}$ has rank two and $(\widetilde{\mathscr{P}'})^{W_b} = \mathcal{O}$, we have $\widetilde{\mathscr{P}'} = \mathcal{O} \oplus \mathcal{N} \otimes \varepsilon$ for a line bundle $\mathcal{N}$, which on the free part is the module of s-anti-invariant functions. For the three representations $L|_{W_b} = T \oplus \varepsilon^{\oplus m}$ with T trivial and $m \le 1$: for a long reflection $m = 1$ in all three cases, and for a short reflection $m = 0$ for $1 \oplus U$ and U, on which the sign changes act trivially, and $m = 1$ for V. Therefore

$$\mathrm{Hom}_{W_b}(L|_{W_b}, \widetilde{\mathscr{P}'}) = T^* \otimes \mathcal{O} \;\oplus\; \mathcal{N}^{\oplus m}, \qquad \textstyle\bigwedge^j (L|_{W_b}) = \bigwedge^j T \;\oplus\; \big(\bigwedge^{j-1} T \otimes \varepsilon\big)^{\oplus m},$$

and both $\bigwedge^j \mathrm{Hom}_{W_b}(L|_{W_b}, \widetilde{\mathscr{P}'})$ and $\mathrm{Hom}_{W_b}(\bigwedge^j L|_{W_b}, \widetilde{\mathscr{P}'})$ are $\bigwedge^j T^* \otimes \mathcal{O} \oplus (\bigwedge^{j-1} T^* \otimes \mathcal{N})^{\oplus m}$. An exterior product of sections contains at most one section of $\mathcal{N}$, hence the determinants which define $\alpha'_{j,L}$ involve only products of invariant functions and the $\mathcal{O}$-module structure of $\mathcal{N}$, never the product $\mathcal{N} \otimes \mathcal{N} \to \mathcal{O}$ of two anti-invariant functions. In the two decompositions $\alpha'_{j,L}$ is therefore the identity up to the normalization constants of the exterior algebra, and it is regular with a regular inverse on the whole formal neighbourhood.

Step 5, extension. Completion at y is faithfully flat, thus $\alpha_{j,L}$ and $\alpha_{j,L}^{-1}$ are regular at the generic point of every prime divisor of $Y_n \setminus Y_n^\circ$. They are then sections of locally free sheaves on the smooth variety Y_n defined outside a closed subset of codimension at least two, hence they extend. The two compositions are the identity on Y_n°, hence everywhere, and the extensions are T-equivariant by uniqueness. □

Theorem 10.11 (Exterior evaluations)**.** *For every $\lambda \in \mathrm{Bal}(2n)$ and $0 \le j \le n$,*

$$\langle \widetilde{H}_\lambda,\ h_{n-j}[X_0]\, e_j[X_0]\rangle = e_j[B_\lambda^{(0)}], \qquad \langle \widetilde{H}_\lambda,\ h_{n-j}[X_0]\, e_j[X_1]\rangle = e_j[B_\lambda^{(1)}], \tag{135}$$

equivalently

$$\widetilde{H}_\lambda[1-z,\,0] = \prod_{\substack{(a,b)\in\lambda\\ a+b \text{ even}}} (1 - zq^a t^b), \qquad \widetilde{H}_\lambda[1,\,-z] = \prod_{\substack{(a,b)\in\lambda\\ a+b \text{ odd}}} (1 - zq^a t^b). \tag{136}$$

For $0 \le k \le n-1$,

$$\langle \widetilde{H}_\lambda,\ s_{(n-k,1^k)}[X_0]\rangle = e_k[B_\lambda^{(0)} - 1], \tag{137}$$

the k-th elementary symmetric function of the $n-1$ monomials $q^a t^b$, $(a,b) \in \lambda \setminus \{(0,0)\}$ of even content, and consequently

$$\langle \widetilde{H}_\lambda,\ p_n[X_0]\rangle = \Pi_\lambda^{(0)}. \tag{138}$$

Proof. By Theorem 10.9(a), $\widetilde{H}_\lambda$ is the Frobenius character of $\widetilde{\mathscr{P}}_n|_\lambda$, hence for every representation L of W the fixed-point character of $\mathsf{F}(L)$ at I_λ is $\langle \widetilde{H}_\lambda, \mathrm{ch}\, L\rangle$, where $\mathrm{ch}\, L$ is the Frobenius character of L. A basis of $\bigwedge^j(1 \oplus U)$ is indexed by the j-subsets of $\{1,\dots,n\}$, which S_n permutes with the sign of the induced permutation and which the sign changes fix, hence $\bigwedge^j(1\oplus U) = \mathrm{Ind}_{W_{n-j}\times W_j}^W(\mathbf{1} \boxtimes \mathrm{sgn}_{S_j})$ and $\mathrm{ch}\bigwedge^j(1\oplus U) = h_{n-j}[X_0]e_j[X_0]$. The same basis of $\bigwedge^j V$ is permuted in the same way, and the sign change ε_i acts by -1 exactly when i lies in the subset, hence $\mathrm{ch}\bigwedge^j V = h_{n-j}[X_0]e_j[X_1]$, because the character of W_j in which every ε_i acts by -1 and S_j by the sign corresponds to $e_j[X_1]$. Finally $\mathrm{ch}\bigwedge^k U = s_{(n-k,1^k)}[X_0]$ for $0 \le k \le n-1$, the exterior powers of the reflection representation of S_n being the hook representations; this is compatible with the Pieri rule $h_{n-j}e_j = s_{(n-j,1^j)} + s_{(n-j+1,1^{j-1})}$ and $\bigwedge^j(1\oplus U) = \bigwedge^j U \oplus \bigwedge^{j-1} U$. Now Theorem 10.10 and (133) give (135) and (137), since the fixed-point characters of $\bigwedge^j \mathcal{R}_0$, $\bigwedge^j \mathcal{R}_1$ and $\bigwedge^k \mathcal{T}_n$ at I_λ are $e_j[B_\lambda^{(0)}]$, $e_j[B_\lambda^{(1)}]$ and $e_k[B_\lambda^{(0)} - 1]$. For (136), the degree-n part of $\Omega[(1-z)X_0]$ is $\sum_j (-z)^j h_{n-j}[X_0]e_j[X_0]$ and that of $\Omega[X_0 - zX_1]$ is $\sum_j (-z)^j h_{n-j}[X_0]e_j[X_1]$, hence the two specializations equal $\sum_j (-z)^j e_j[B_\lambda^{(i)}] = \prod(1 - zq^a t^b)$ over the

boxes of content parity i. For (138), $p_n = \sum_{k=0}^{n-1}(-1)^k s_{(n-k,1^k)}$ and $\sum_k (-1)^k e_k[B_\lambda^{(0)}-1] = \prod(1-q^a t^b)$ over the $n-1$ monomials of $B_\lambda^{(0)} - 1$. □

Now we can prove the plethystic interpretation of the localization formula from the main result of the paper:

Proposition 10.12 (Plethystic formula)**.** *Let $n \geq 1$ and put*

$$R_n := p_n\Big[\frac{(1+qt)X_0 + (q+t)X_1}{(1-q^2)(1-t^2)}\Big] = \frac{(1+(qt)^n)\,p_n[X_0] + (q^n+t^n)\,p_n[X_1]}{(1-q^{2n})(1-t^{2n})}, \tag{139}$$

$$\mathscr{K}_n^{(m)} := (-1)^{n-1}\nabla_1^m \nabla_0\, R_n, \qquad \Phi_n := h_{n-1}[X_0]\; p_1\big[(1+qt)X_0 - (q+t)X_1\big]. \tag{140}$$

Then for all $m \geq 1$

$$\mathscr{K}_n^{(m)} = \sum_{\lambda \in \mathrm{Bal}(2n)} \frac{\ell_\lambda^m\, \Pi_\lambda^{(0)}}{D_\lambda}\, \widetilde{H}_\lambda, \tag{141}$$

$$u_{n,m} = \langle \mathscr{K}_n^{(m)}, h_{n-1}[X_0]p_1[X_0]\rangle, \qquad v_{n,m} = \langle \mathscr{K}_n^{(m)}, h_{n-1}[X_0]p_1[X_1]\rangle,$$

and therefore

$$P_{n,m} = \langle \mathscr{K}_n^{(m)}, \Phi_n\rangle, \qquad F_{n,m} = (1-q)(1-t)\langle \mathscr{K}_n^{(m)}, h_{n-1}[X_0]p_1[X_0+X_1]\rangle. \tag{142}$$

Proof. Step 1, the expansion of R_n. By Theorem 10.9(d) the basis dual to $\{\widetilde{H}_\lambda\}$ for the form $\langle\,,\rangle_{q,t}$ is $\{D_\lambda^{-1}\,\mathrm{inv}\,\widetilde{H}_\lambda\}$, hence every f of degree n satisfies $f = \sum_\lambda \langle f, \mathrm{inv}\,\widetilde{H}_\lambda\rangle_{q,t} D_\lambda^{-1}\widetilde{H}_\lambda$. The substitution $X \mapsto A^{-1}X$ is self-adjoint for the Hall pairing, since A^{-1} is symmetric, and $R_n = p_n[X_0][AX]$ by (139), since $p_n[X_0][AX] = \sum_j A_{0j}(q^n,t^n)p_n[X_j]$. Substitutions compose as matrix products, therefore

$$\langle R_n, \mathrm{inv}\,\widetilde{H}_\lambda\rangle_{q,t} = \langle R_n[A^{-1}X], \mathrm{inv}\,\widetilde{H}_\lambda\rangle = \langle p_n[X_0], \mathrm{inv}\,\widetilde{H}_\lambda\rangle = \mathrm{inv}\langle p_n[X_0], \widetilde{H}_\lambda\rangle = \mathrm{inv}\,\Pi_\lambda^{(0)},$$

where we used that $p_n[X_0]$ is inv-invariant, that the Hall pairing is defined over $\mathbb{Q}$, and (138). The product $\Pi_\lambda^{(0)}$ has $n-1$ factors $1-q^at^b$, and $\mathrm{inv}(1-q^at^b) = -q^{-a}t^{-b}(1-q^at^b)$, hence $\mathrm{inv}\,\Pi_\lambda^{(0)} = (-1)^{n-1}T_0(\lambda)^{-1}\Pi_\lambda^{(0)}$ and

$$R_n = (-1)^{n-1}\sum_{\lambda\in\mathrm{Bal}(2n)} \frac{\Pi_\lambda^{(0)}}{T_0(\lambda)D_\lambda}\,\widetilde{H}_\lambda. \tag{143}$$

Step 2, the operators. We apply $\nabla_1^m\nabla_0$ to (143) using (131). The operator ∇_0 contributes $T_0(\lambda)$, which cancels the denominator, and ∇_1^m contributes ℓ_λ^m. This is the first formula of (141).

Step 3, the pairings. We pair the first formula of (141) with $h_{n-1}[X_0]p_1[X_i]$ and use Theorem 10.9(b). This gives the formulas for $u_{n,m}$ and $v_{n,m}$, and (142) follows from Theorem 10.4 by bilinearity, since $\Phi_n = (1+qt)h_{n-1}[X_0]p_1[X_0] - (q+t)h_{n-1}[X_0]p_1[X_1]$. □

Corollary 10.13. *Theorem 10.5 is equivalent to the identity*

$$(-1)^{n-1}\Big\langle \nabla_1^m\nabla_0\, p_n\Big[\tfrac{(1+qt)X_0+(q+t)X_1}{(1-q^2)(1-t^2)}\Big],\; h_{n-1}[X_0]\,p_1\big[(1+qt)X_0-(q+t)X_1\big]\Big\rangle = \sum_{\pi:(0,0)\to(mn,n)} (qt)^{\mathrm{dinv}_m(\pi)} q^{A_-^{(m)}(\pi)} t^{A_+^{(m)}(\pi)} \tag{144}$$

for all $m \geq 1$.

Let us compare (144) with type A. By [42] the higher q,t-Catalan number $\langle \nabla^m e_n, e_n\rangle$ is the Hilbert series of the determinantal part of $DR^{(m)}(S_n)$, and by the rational shuffle theorem of Mellit [63] it equals $\sum_\pi q^{\mathrm{dinv}_m(\pi)}t^{\mathrm{area}(\pi)}$ over the m-Dyck paths of the $n\times mn$ rectangle, with the same kernel s_m in dinv_m [57, 56, 35].

For $m=1$ this is the Catalan case of the shuffle theorem [36, 19]. The companion statement for $(-1)^{n-1}\nabla p_n$ is the square paths theorem, conjectured by Loehr and Warrington [59] and proved by Sergel [72], in which the paths of the square are allowed to go below the diagonal.

The left side of (144) is the two-colour version of the pairing of $(-1)^{n-1}\nabla^m p_n$ with e_n, and the right side is a sum over all paths of the rectangle, with $A_-^{(m)}$ counting the excursions below the line $y=x/m$ and $A_+^{(m)}$ those above. The two ingredients which have no counterpart in type A are the parity projection, which is responsible for the test function Φ_n in place of e_n, and the Koszul factor $\Pi_\lambda^{(0)}$ of the trace-zero even bundle, which enters through (138).

A labelled version of the conjecture and the matching of its statistics with those of type A are the subject of Section 11.

Example 10.14. For $n=2$ the wreath Macdonald polynomials, computed from (129), are

$$\begin{aligned}
\widetilde{H}_{(4)} &= s_2[X_0] + q^2 s_{11}[X_0] + (q+q^3)\, s_1[X_0]s_1[X_1] + q^2 s_2[X_1] + q^4 s_{11}[X_1],\\
\widetilde{H}_{(3,1)} &= s_2[X_0] + q^2 s_{11}[X_0] + (q+t)\, s_1[X_0]s_1[X_1] + \tfrac{t}{q}\, s_2[X_1] + qt\, s_{11}[X_1],\\
\widetilde{H}_{(2,2)} &= s_2[X_0] + qt\, s_{11}[X_0] + (q+t)\, s_1[X_0]s_1[X_1] + s_2[X_1] + qt\, s_{11}[X_1],\\
\widetilde{H}_{(2,1,1)} &= s_2[X_0] + t^2 s_{11}[X_0] + (q+t)\, s_1[X_0]s_1[X_1] + \tfrac{q}{t}\, s_2[X_1] + qt\, s_{11}[X_1],\\
\widetilde{H}_{(1^4)} &= s_2[X_0] + t^2 s_{11}[X_0] + (t+t^3)\, s_1[X_0]s_1[X_1] + t^2 s_2[X_1] + t^4 s_{11}[X_1].
\end{aligned}$$

The coefficients of $s_{11}[X_0]$ are $B_\lambda^{(0)}-1$, the coefficients of $s_1[X_0]s_1[X_1]$ are $B_\lambda^{(1)}$, the coefficients of $s_{11}[X_1]$ are ℓ_λ, and $\langle \widetilde{H}_\lambda, p_2[X_0]\rangle = 1-(B_\lambda^{(0)}-1)$ is $1-q^2, 1-q^2, 1-qt, 1-t^2, 1-t^2$, in accordance with Theorem 10.9 and Theorem 10.11. The Laurent monomials t/q and q/t reflect the fact that the fibers of the Procesi bundle are not quotients of $\mathbb{C}[x_1,x_2,y_1,y_2]$ [67, Rem. 4.7].

10.5. **Rank three.** Let us write out both sides of (127) for $n=3$ and $m=1,2$. The ten balanced partitions of 6 are all partitions of 6 except the 2-core $(3,2,1)$. Their fixed-point data are listed in Table 1, where w_1,w_2,w_3 are the tangent weights $q^{a(s)+1}t^{-l(s)}$ of Theorem 7.1 at the three boxes s with $a(s)+l(s)$ odd, the other three tangent weights being qt/w_i, hence $D_\lambda=\prod_{i=1}^3(1-w_i)(1-qt/w_i)$.

λ	ℓ_λ	$\Pi_\lambda^{(0)}$	$B_\lambda^{(0)}$	$B_\lambda^{(1)}$	w_1, w_2, w_3
(6)	q^9	$(1-q^2)(1-q^4)$	$1+q^2+q^4$	$q+q^3+q^5$	$q^6,\ q^4,\ q^2$
$(5,1)$	q^4t	$(1-q^2)(1-q^4)$	$1+q^2+q^4$	$q+q^3+t$	$q^5t^{-1},\ q^4,\ q^2$
$(4,2)$	q^4t	$(1-q^2)(1-qt)$	$1+q^2+qt$	$q+q^3+t$	$q^3t^{-1},\ q^2,\ q^2$
$(4,1,1)$	q^4t	$(1-q^2)(1-t^2)$	$1+q^2+t^2$	$q+q^3+t$	$q^4t^{-2},\ q^2,\ qt^{-1}$
$(3,3)$	q^3t^2	$(1-q^2)(1-qt)$	$1+q^2+qt$	$q+t+q^2t$	$q^3t^{-1},\ qt^{-1},\ q^2$
$(3,1,1,1)$	qt^4	$(1-q^2)(1-t^2)$	$1+q^2+t^2$	$q+t+t^3$	$q^3t^{-3},\ q^2,\ qt^{-1}$
$(2,2,2)$	q^2t^3	$(1-t^2)(1-qt)$	$1+qt+t^2$	$q+t+qt^2$	$q^2t^{-2},\ qt^{-1},\ q^2$
$(2,2,1,1)$	qt^4	$(1-t^2)(1-qt)$	$1+qt+t^2$	$q+t+t^3$	$qt^{-1},\ q^2t^{-2},\ qt^{-1}$
$(2,1^4)$	qt^4	$(1-t^2)(1-t^4)$	$1+t^2+t^4$	$q+t+t^3$	$q^2t^{-4},\ qt^{-3},\ qt^{-1}$
(1^6)	t^9	$(1-t^2)(1-t^4)$	$1+t^2+t^4$	$t+t^3+t^5$	$qt^{-5},\ qt^{-3},\ qt^{-1}$

TABLE 1. Fixed-point data of Y_3.

Summing the ten terms of (127) with these data we obtain, with $w=qt$,

(145)

$$\begin{aligned}
P_{3,1} &= [10] + w[6] + w[4], & N_{3,1} &= w[9] + w^2[5] + w^4,\\
P_{3,2} &= [19] + w([15]+[13]) + w^2([11]+[9]+[7]) + w^3[7] + w^4[3],\\
N_{3,2} &= w[18] + w^2([14]+[12]) + w^3([10]+[8]) + w^4[6] + w^5[4],
\end{aligned}$$

and $F_{3,m} = P_{3,m} - N_{3,m}$. The polynomials $P_{3,1}$, $N_{3,1}$ are those of Theorem 7.4, and $P_{3,1}(1,1) = 20$, $P_{3,2}(1,1) = 84$, $q^9 P_{3,1}(q,q^{-1}) = \begin{bmatrix}6\\3\end{bmatrix}_{q^2}$, $q^{18} P_{3,2}(q,q^{-1}) = \begin{bmatrix}9\\3\end{bmatrix}_{q^2}$, in accordance with Theorem C.

For $m = 1$ the right side of (127) is the sum over the 20 paths of the 3×3 square, listed in Table 2 with their partitions $\alpha = (e_3, e_2, e_1)$, their heights $b = (b_1, b_2, b_3)$, $b_i = i - e_i$, the three statistics and the resulting monomial. The sum of the last column is $[10] + qt[6] + qt[4] = P_{3,1}$.

α	e	b	dinv_1	$A_-^{(1)}$	$A_+^{(1)}$	term	α	e	b	dinv_1	$A_-^{(1)}$	$A_+^{(1)}$	term
$(0,0,0)$	$(0,0,0)$	$(1,2,3)$	0	0	9	t^9	$(1,1,1)$	$(1,1,1)$	$(0,1,2)$	0	1	4	qt^4
$(1,0,0)$	$(0,0,1)$	$(1,2,2)$	1	0	7	qt^8	$(2,1,1)$	$(1,1,2)$	$(0,1,1)$	1	1	2	q^2t^3
$(2,0,0)$	$(0,0,2)$	$(1,2,1)$	2	0	5	q^2t^7	$(3,1,1)$	$(1,1,3)$	$(0,1,0)$	2	2	1	q^4t^3
$(3,0,0)$	$(0,0,3)$	$(1,2,0)$	1	1	4	q^2t^5	$(2,2,1)$	$(1,2,2)$	$(0,0,1)$	1	2	1	q^3t^2
$(1,1,0)$	$(0,1,1)$	$(1,1,2)$	1	0	5	qt^6	$(3,2,1)$	$(1,2,3)$	$(0,0,0)$	3	3	0	q^6t^3
$(2,1,0)$	$(0,1,2)$	$(1,1,1)$	3	0	3	q^3t^6	$(3,3,1)$	$(1,3,3)$	$(0,-1,0)$	2	5	0	q^7t^2
$(3,1,0)$	$(0,1,3)$	$(1,1,0)$	3	1	2	q^4t^5	$(2,2,2)$	$(2,2,2)$	$(-1,0,1)$	0	4	1	q^4t
$(2,2,0)$	$(0,2,2)$	$(1,0,1)$	2	1	2	q^3t^4	$(3,2,2)$	$(2,2,3)$	$(-1,0,0)$	1	5	0	q^6t
$(3,2,0)$	$(0,2,3)$	$(1,0,0)$	3	2	1	q^5t^4	$(3,3,2)$	$(2,3,3)$	$(-1,-1,0)$	1	7	0	q^8t
$(3,3,0)$	$(0,3,3)$	$(1,-1,0)$	1	4	1	q^5t^2	$(3,3,3)$	$(3,3,3)$	$(-2,-1,0)$	0	9	0	q^9

TABLE 2. The 20 paths of the 3×3 square, $m = 1$.

For $m = 2$ the right side of (127) is a sum over the 84 paths of the 3×6 rectangle, with $b_i = 2i - e_i$ and the kernel $s_2(-1) = 1$, $s_2(0) = 2$, $s_2(1) = 2$, $s_2(2) = 1$. Table 3 lists, for each value d of dinv_2, the number of paths and the sum of $q^{A_-^{(2)}} t^{A_+^{(2)}}$ over them, hence $\widehat{P}_{3,2} = \sum_d (qt)^d r_d$ with r_d the polynomial in the last column. The result is the polynomial $P_{3,2}$ of (145). For instance the seven paths with $\mathrm{dinv}_2 = 0$ are the paths $e_1 = e_2 = e_3 = e$ with three consecutive north steps, and the seven paths with $\mathrm{dinv}_2 = 6$ are the paths with $b_1 \geq b_2 \geq b_3 \geq b_1 - 1$, whose three pairs contribute $s_2(0) = s_2(1) = 2$ each.

d	paths	$r_d = \sum_{\mathrm{dinv}_2 = d} q^{A_-^{(2)}} t^{A_+^{(2)}}$
0	7	$t^{18} + qt^{13} + q^2t^8 + q^5t^5 + q^8t^2 + q^{13}t + q^{18}$
1	14	$t^{16} + t^{14} + qt^{11} + qt^9 + q^2t^8 + q^2t^6 + q^3t^5 + q^5t^3 + q^6t^2 + q^8t^2 + q^9t + q^{11}t + q^{14} + q^{16}$
2	21	$t^{14} + t^{12} + t^{10} + 2qt^9 + qt^7 + 2q^2t^6 + q^2t^4 + q^3t^5 + q^3t^3 + q^4t^2 + q^5t^3 + 2q^6t^2 + q^7t + 2q^9t + q^{10} + q^{12} + q^{14}$
3	14	$t^{12} + t^{10} + 2qt^7 + q^2t^6 + q^2t^4 + 2q^3t^3 + q^4t^2 + q^6t^2 + 2q^7t + q^{10} + q^{12}$
4	14	$t^{10} + t^8 + qt^7 + qt^5 + 2q^2t^4 + 2q^3t^3 + 2q^4t^2 + q^5t + q^7t + q^8 + q^{10}$
5	7	$t^8 + qt^5 + q^2t^4 + q^3t^3 + q^4t^2 + q^5t + q^8$
6	7	$t^6 + qt^5 + q^2t^4 + q^3t^3 + q^4t^2 + q^5t + q^6$

TABLE 3. The 84 paths of the 3×6 rectangle, $m = 2$, grouped by dinv_2.

11. The labelled conjecture and a monomial basis

In this section we lift Theorem 10.5 from an identity of polynomials to an identity of symmetric functions. In type A the q,t-Fuss–Catalan identity $\langle \nabla^m e_n, e_n \rangle = \sum_\pi q^{\mathrm{dinv}_m} t^{\mathrm{area}}$ is the $\langle \cdot\, , e_n \rangle$-coefficient of the shuffle theorem of Carlsson and Mellit [19], conjectured in [36]. There the paths acquire labels, each labelled path contributes a fundamental quasisymmetric function, and the numerical statement is recovered from the coefficient of one Schur function. We formulate the type-B analogue of this passage.

The main idea is to let the labels interact with the kernel s_m of (121). The support of s_m splits into two halves, and an inversion of the labels cancels one unit of dinv_m on one half, a non-inversion

on the other. The resulting labelled series $\mathcal{H}^{+}_{n,m}$ is a sum of fundamental quasisymmetric functions, and nothing in its definition says that it is symmetric.

To prove that it is symmetric and Schur positive we follow the type-A strategy of Haglund, Haiman, Loehr, Remmel and Ulyanov [36, Thm. 3.1.3 and §5], see also [35, Ch. 6]. The contribution of one path is identified with an LLT polynomial of Lascoux, Leclerc and Thibon [54] (Theorem 11.9), the symmetry of LLT polynomials is a theorem, and their Schur positivity, which for the tuples we need rests on the parabolic Kazhdan–Lusztig polynomials [36, Prop. 5.3.1], [55, 50], gives the Schur positivity of $\mathcal{H}^{+}_{n,m}$ (Theorem 11.15).

The type-B features are that the heights of a path take both signs and that the kernel s_m is supported on $2m$ values, and the identification requires splitting every height into a content and a colour.

At $m=1$ we prove a type-C analogue of the schedule formula of Haglund and Loehr [37], [35, Thm. 5.3]. The series $\mathcal{H}^{+}_{n,1}$ is a sum over the permutations of $\{0,1,\dots,n\}$ of a monomial in q,t times a product of qt-integers (Theorem 11.23), and the two extreme sectors of the sum are the type-A formula itself (Theorem 11.26). In type A the schedule formula was the key to the monomial basis of the diagonal coinvariants found by Carlsson and Oblomkov [20, Thm. B], whose specialization $x_i \mapsto q$, $y_i \mapsto t$ is the schedule formula [38, §4]. This is the origin of the second conjecture below.

There are two conjectures. The first, Theorem 11.16, is a symmetric-function identity whose s_n-coefficient is exactly Theorem 10.5 (Theorem 11.19). Conjecturally it computes the bigraded Frobenius character of the G_n-invariant part of Gordon's module, of which the spherical part of Theorem 8.12 is the S_n-invariant part. The second, Theorem 11.30, is a monomial basis for that module at $m=1$, indexed by the permutations of $\{0,1,\dots,n\}$ and by a box of independent schedule choices attached to each of them. It is the type-B analogue of the basis of Carlsson and Oblomkov [20, Thm. B] for the type-A diagonal coinvariants, and it implies the Hilbert-series half of the first conjecture at $m=1$.

Neither conjecture implies the other, and neither is proved here. What we prove is the comparison with Section 10 and the elementary properties of the labelled model. The evidence is collected in Theorem 11.34 and Theorem 11.35.

11.1. **Labelled paths.** Let us fix the notation. We keep $m \geq 1$, $D = mn^2$, the heights (120) and the statistics (122). For a set $S \subseteq \{1,\dots,n-1\}$ let

$$F_{n,S}(X) = \sum_{\substack{i_1 \leq \cdots \leq i_n \\ i_j < i_{j+1} \text{ if } j \in S}} x_{i_1} \cdots x_{i_n}$$

be Gessel's fundamental quasisymmetric function, and for $w \in S_n$ let $\mathrm{ides}(w) = \{j : w^{-1}(j) > w^{-1}(j+1)\}$ be the inverse descent set of w. We use the expansion $s_\lambda = \sum_Q F_{n,\mathrm{Des}(Q)}$ of the Schur functions, the sum over the standard Young tableaux Q of shape λ [36, Prop. 2.4.1], and the linear independence of the $F_{n,S}$.

Definition 11.1. A *labelled path* of the $n \times mn$ rectangle is a pair $P = (e, \sigma)$ with $0 \leq e_1 \leq \cdots \leq e_n \leq mn$ and $\sigma \in S_n$, subject to

$$e_i = e_{i+1} \implies \sigma_i < \sigma_{i+1}.$$

We write $\mathsf{PF}^{+}_{n,m}$ for the set of labelled paths. It has $(mn+1)^n$ elements. The *reading word* $w(P)$ lists the labels σ_i in increasing order of (b_i, i). The *defect* and the *labelled* dinv are

$$\mathrm{def}_m(P) = \sum_{i<j} \Big(\mathbf{1}_{1-m \leq b_i - b_j \leq 0} \mathbf{1}_{\sigma_i > \sigma_j} + \mathbf{1}_{1 \leq b_i - b_j \leq m} \mathbf{1}_{\sigma_i < \sigma_j} \Big), \qquad \mathrm{dinv}^{\mathrm{lab}}_m = \mathrm{dinv}_m - \mathrm{def}_m, \tag{146}$$

and the *labelled path series* is

$$\mathcal{H}^{+}_{n,m}(X;q,t) = \sum_{P\in \mathsf{PF}^{+}_{n,m}} q^{A^{(m)}_{-}(P)+\mathrm{dinv}^{\mathrm{lab}}_{m}(P)}\, t^{A^{(m)}_{+}(P)+\mathrm{dinv}^{\mathrm{lab}}_{m}(P)}\, F_{n,\mathrm{ides}(w(P))}(X). \tag{147}$$

The three path statistics of (122) depend only on e, and the labels enter only through def_m and ides. The two summands of (146) are the two halves of the support of the kernel s_m of (121). Namely, a pair $i<j$ with $b_i-b_j\in[1-m,0]$ is an attack in which the later north step is weakly higher, and it is cancelled by an inversion of the labels, while a pair with $b_i-b_j\in[1,m]$ is an attack in which the earlier north step is higher, and it is cancelled by a non-inversion.

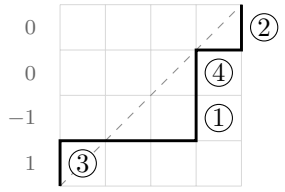
Since $s_m(d)=[m+d]_+$ for $d\le 0$ and $s_m(d)=[m+1-d]_+$ for $d\ge 1$, we have $s_m(d)\ge 1$ exactly on $1-m\le d\le m$, and the two summands of (146) partition this range. Hence $\mathrm{def}_m\le \mathrm{dinv}_m$ pair by pair, and $\mathrm{dinv}^{\mathrm{lab}}_m\ge 0$. At $m=1$ the kernel is an indicator function and (147) simplifies to

$$\mathcal{H}^{+}_{n,1} = \sum_{P} q^{A_-}t^{A_+}(qt)^{D(P)}F_{n,\mathrm{ides}(w(P))},$$

$$D(P) = \#\{i<j : b_i=b_j,\ \sigma_i<\sigma_j\} + \#\{i<j : b_i=b_j+1,\ \sigma_i>\sigma_j\}. \tag{148}$$

11.2. **Worked examples.** In this subsection we compute the statistics on three labelled paths, and we write out the whole of $\mathcal{H}^{+}_{2,1}$. We draw a labelled path as the lattice path from $(0,0)$ to (mn,n) whose i-th north step joins $(e_i,i-1)$ to (e_i,i). The label σ_i is circled in the cell immediately east of that step, the dashed line is $y=x/m$, and the column of small numbers at the left lists the heights $b_1,\dots,b_n$, read upwards. Thus the number in the i-th row is the height b_i of the i-th north step, which by (120) is the height of the path above the line after that step, measured in units of $1/m$. Unlike in type A the path is an arbitrary lattice path of the rectangle, hence the b_i take both signs and the line is crossed in both directions. This is what the statistics have to accommodate.

Example 11.2. Let $n=4$, $m=1$, $e=(0,3,3,4)$ and $\sigma=(3,1,4,2)$. The picture is



Here $b=(1,-1,0,0)$. The only equality among the e_i is $e_2=e_3$, and there $\sigma_2=1<4=\sigma_3$. Hence P is admissible, the labels increase up the vertical run.

Areas. The summands of $A^{(1)}_-$ are $[1-b_i]_+ + [-b_i]_+ = 0,3,1,1$ and those of $A^{(1)}_+$ are $[b_i]_+ + [b_i-1]_+ = 1,0,0,0$, hence $A^{(1)}_-=5$ and $A^{(1)}_+=1$.

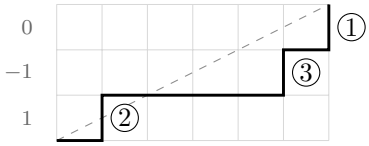
Kernel and dinv. At $m=1$ the kernel (121) is the indicator of $d\in\{0,1\}$, hence dinv_1 counts the pairs $i<j$ with $b_i-b_j\in\{0,1\}$. These are $(1,3)$ and $(1,4)$, with $d=1$, and $(3,4)$, with $d=0$. Thus $\mathrm{dinv}_1=3$. The pairs $(1,2)$, $(2,3)$, $(2,4)$, with $d=2,-1,-1$, do not attack.

Defect. A pair with $d=0$ is counted by (146) when its labels are in decreasing order, a pair with $d=1$ when they are in increasing order. Here $\sigma_1=3<4=\sigma_3$ (counted), $\sigma_1=3>2=\sigma_4$ (not counted), and $\sigma_3=4>2=\sigma_4$ (counted). Therefore $\mathrm{def}_1=2$ and $\mathrm{dinv}^{\mathrm{lab}}_1=1$.

Reading word. Sorting by increasing (b_i,i) visits the steps in the order $2,3,4,1$, hence $w(P)=1\,4\,2\,3$. In this word 4 precedes 3 and no other letter precedes its predecessor, hence $\mathrm{ides}(w(P))=\{3\}$. The contribution of P to (147) is therefore $q^6t^2F_{4,\{3\}}$.

The next example has almost the same heights but $m=2$, which widens the attacking window from two values of d to $2m=4$ and lets the kernel exceed 1.

Example 11.3. Let $n=3$, $m=2$, $e=(1,5,6)$ and $\sigma=(2,3,1)$. The picture is



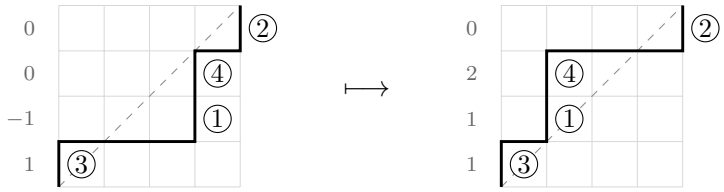
Here $b=(1,-1,0)$. Now $s_2(d)=2$ for $d\in\{0,1\}$, $s_2(d)=1$ for $d\in\{-1,2\}$ and $s_2(d)=0$ otherwise. The three pairs have $d=2,1,-1$, hence all three attack and $\mathrm{dinv}_2=1+2+1=4$. At $m=1$ the pairs with $d=2$ and $d=-1$ would not have attacked at all.

The pairs with $d=2$ and $d=1$ lie in the second range of (146) and are counted when the labels increase, the pair with $d=-1$ lies in the first range and is counted when they decrease. Here $\sigma_1=2<3=\sigma_2$ (counted), $\sigma_1=2>1=\sigma_3$ (not counted), and $\sigma_2=3>1=\sigma_3$ (counted). Therefore $\mathrm{def}_2=2$ and $\mathrm{dinv}_2^{\mathrm{lab}}=2$. Note that the pair $(1,3)$ contributes 2 to dinv_2 and 0 to def_2. The inequality $\mathrm{def}_m\le\mathrm{dinv}_m$ holds pair by pair, but it is far from tight.

The areas are $A_-^{(2)}=1+4+2=7$ and $A_+^{(2)}=1+0+0=1$, the reading word is $w(P)=3\,1\,2$ with $\mathrm{ides}=\{2\}$, and P contributes $q^9t^3F_{3,\{2\}}$.

The complementation of Theorem 11.14 is visible in the picture. It turns the path by a half-turn about the centre of the rectangle and replaces each label c by $n+1-c$. On Theorem 11.2 the two operations on the labels cancel.

Example 11.4. Let P be as in Theorem 11.2. Then $e_i^c=4-e_{5-i}$ gives $e^c=(0,1,1,4)$ and $\sigma_i^c=5-\sigma_{5-i}$ gives $\sigma^c=(3,1,4,2)=\sigma$. The two pictures are



and $b^c=(1,1,2,0)=(1-b_4,1-b_3,1-b_2,1-b_1)$. The attacking pairs are transported by $(i,j)\mapsto(5-j,5-i)$. The pair $(3,4)$ of P, of type $d=0$ and counted, becomes the pair $(1,2)$ of P^c, again of type $d=0$ and again counted, and the two pairs with $d=1$ become the two pairs $(1,4)$ and $(2,4)$ with $d=1$, of which again exactly one is counted. Thus $\mathrm{dinv}_1=3$ and $\mathrm{def}_1=2$ are unchanged, while $A_-^{(1)}$ and $A_+^{(1)}$ exchange the values 5 and 1 and the weight becomes q^2t^6. The reading word is $w(P^c)=2\,3\,1\,4$ with $\mathrm{ides}=\{1\}$, and indeed $1=n-3$.

Finally we write out the whole of $\mathcal{H}_{2,1}^+$, which the reader can compare with Theorem 7.4. The nine labelled paths of the 2×2 square, with $F_{2,\varnothing}=s_2$ and $F_{2,\{1\}}=s_{11}$, are

e	σ	b	$A_-^{(1)}$	$A_+^{(1)}$	dinv_1	def_1	$w(P)$	contribution
00	12	1, 2	0	4	0	0	12	t^4s_2
01	12	1, 1	0	2	1	0	12	qt^3s_2
01	21	1, 1	0	2	1	1	21	t^2s_{11}
02	12	1, 0	1	1	1	1	21	$qt\,s_{11}$
02	21	1, 0	1	1	1	0	12	$q^2t^2s_2$
11	12	0, 1	1	1	0	0	12	$qt\,s_2$
12	12	0, 0	2	0	1	0	12	$q^3t\,s_2$
12	21	0, 0	2	0	1	1	21	q^2s_{11}
22	12	−1, 0	4	0	0	0	12	q^4s_2

Summing the last column we obtain

$$\mathcal{H}_{2,1}^+=(q^4+q^3t+q^2t^2+qt^3+t^4+qt)s_2+(q^2+qt+t^2)s_{11}.$$

Its s_2-coefficient is $\widehat{P}_{2,1}$, in accordance with Theorem 11.5(i), and setting $q=t=1$ gives $6s_2+3s_{11}=h_2[3X]$, as in Theorem 11.5(iii).

Lemma 11.5. *Let $n,m\geq 1$.*

(i) *Every lattice path π of the rectangle carries exactly one labelling P with $\mathrm{ides}(w(P))=\varnothing$, namely the one whose reading word is the identity, and for it $\mathrm{def}_m(P)=0$. Consequently*

$$[s_n]\,\mathcal{H}^+_{n,m}=\widehat{P}_{n,m}.$$

(ii) $\langle \mathcal{H}^+_{n,m},h_1^n\rangle=\sum_P q^{A_-^{(m)}+\mathrm{dinv}_m^{\mathrm{lab}}}t^{A_+^{(m)}+\mathrm{dinv}_m^{\mathrm{lab}}}$, *the weight enumerator of all $(mn+1)^n$ labelled paths.*

(iii) $\mathcal{H}^+_{n,m}(X;1,1)=h_n[(mn+1)X]$, *the Frobenius character of the permutation representation of S_n on the functions $[n]\to\{0,1,\dots,mn\}$.*

Proof. (i) Fix π, hence e and b. The reading order is the linear order on $\{1,\dots,n\}$ given by $i\prec j$ iff $(b_i,i)<(b_j,j)$ lexicographically. The condition $\mathrm{ides}(w)=\varnothing$ holds iff w^{-1} has no descent, that is iff w is the identity, that is iff σ is increasing along $\prec$. This determines σ uniquely, and this σ is admissible. Indeed, if $e_i=e_{i+1}$ then $b_{i+1}=b_i+m>b_i$, hence $i\prec i+1$ and $\sigma_i<\sigma_{i+1}$.

For this σ and $i<j$ we have $\sigma_i>\sigma_j$ iff $j\prec i$ iff $b_j<b_i$, since ties in b are broken by the index. In the first sum of (146) the two conditions $b_i-b_j\leq 0$ and $b_i-b_j>0$ are incompatible, and in the second sum the two conditions $b_i-b_j\geq 1$ and $b_i-b_j\leq 0$ are incompatible. Hence $\mathrm{def}_m(P)=0$, and the weight of P is $(qt)^{\mathrm{dinv}_m(\pi)}q^{A_-^{(m)}(\pi)}t^{A_+^{(m)}(\pi)}$, the term of π in (123).

Finally, $s_\lambda=\sum_Q F_{n,\mathrm{Des}(Q)}$ over the standard Young tableaux Q of shape λ, and the only shape with a tableau of empty descent set is the single row. Therefore a symmetric $f=\sum_S c_S F_{n,S}$ has $c_\varnothing=[s_n]f$.

(ii) Each $F_{n,S}$ contains the monomial $x_1\cdots x_n$ exactly once, and $\langle\,\cdot\,,h_1^n\rangle$ is the coefficient of $x_1\cdots x_n$.

(iii) Fix a path e, let $R_1,\dots,R_k$ be its maximal vertical runs, the maximal blocks of consecutive indices i with equal e_i, and let $C_s\subseteq\{1,\dots,n\}$ be the set of positions in the reading order occupied by the steps of R_s. Along a run the heights increase, hence the steps of R_s appear in the reading order in the order of their indices, and a labelling σ is admissible iff its labels increase along each run. Therefore the words $w(e,\sigma)$, σ admissible for e, are exactly the permutations $u\in S_n$ which are increasing on each block C_s.

For a permutation u the function $F_{n,\mathrm{ides}(u)}$ is the sum of the monomials $x_{v_1}\cdots x_{v_n}$ over the sequences $v_1\leq\cdots\leq v_n$ with $v_c<v_{c+1}$ whenever $u^{-1}(c)>u^{-1}(c+1)$.

We claim that the pairs (u,v) correspond bijectively to the functions $g\colon\{1,\dots,n\}\to\{1,2,\dots\}$, by $g(p)=v_{u(p)}$, and that under this correspondence u is increasing on a block C exactly when g is weakly increasing on C.

Indeed, given g let u be its standardization with ties broken from left to right, that is $u(p)<u(p')$ iff $g(p)<g(p')$, or $g(p)=g(p')$ and $p<p'$, and let $v_c=g(u^{-1}(c))$. Then v is weakly increasing, and if $u^{-1}(c)>u^{-1}(c+1)$ then $v_c\neq v_{c+1}$, since an equality would be broken the other way, hence $v_c<v_{c+1}$.

Conversely, given (u,v) put $g(p)=v_{u(p)}$. If $u(p)<u(p')$ then $g(p)\leq g(p')$, and in case of equality all v_c with $u(p)\leq c\leq u(p')$ coincide, hence no c in $\{u(p),\dots,u(p')-1\}$ lies in $\mathrm{ides}(u)$, hence $p<p'$. Thus u is the standardization of g, and the two constructions are inverse to each other.

Finally, if g is weakly increasing on C then its standardization is increasing on C, and if u is increasing on C then $g=v\circ u$ is weakly increasing on C. This proves the claim, and it gives

$$\sum_\sigma F_{n,\mathrm{ides}(w(e,\sigma))}(X)=\sum_g\prod_{p=1}^n x_{g(p)}=\prod_{s=1}^k h_{|R_s|}(X),$$

the middle sum over the functions g which are weakly increasing on each block C_s.

A path e is a multiset of n values from $\{0,1,\dots,mn\}$, and the sizes of its runs are the multiplicities $a_0,\dots,a_{mn}$ of the values. Summing over the paths we obtain

$$\mathcal{H}^{+}_{n,m}(X;1,1)=\sum_{a_0+\cdots+a_{mn}=n}\prod_{j=0}^{mn}h_{a_j}(X)=h_n\big[(mn+1)X\big],$$

by the expansion $h_n[(y_0+\cdots+y_{mn})X]=\sum_a\prod_j y_j^{a_j}h_{a_j}[X]$ at $y_j=1$. The permutation representation of S_n on the functions $[n]\to\{0,\dots,mn\}$ is the direct sum over the multisets of the representations induced from the trivial representation of the Young subgroups $\prod_j S_{a_j}$, whose Frobenius character is the same product. □

Since $A^{(m)}_{\pm}$ and dinv_m depend only on e, the sum (147) groups by path,

$$\mathcal{H}^{+}_{n,m}(X;q,t)=\sum_e q^{A^{(m)}_{-}(e)}\,t^{A^{(m)}_{+}(e)}\,(qt)^{\mathrm{dinv}_m(e)}\;L_e(X;(qt)^{-1}),\tag{149}$$

where e runs over the $\binom{(m+1)n}{n}$ paths of the rectangle and

$$L_e(X;z)=\sum_{\sigma} z^{\mathrm{def}_m(e,\sigma)}\,F_{n,\mathrm{ides}(w(e,\sigma))}(X),\tag{150}$$

the inner sum over the labellings admissible for e. The polynomial L_e is the part of $\mathcal{H}^{+}_{n,m}$ contributed by the labels of one path. It is a quasisymmetric function of degree n with coefficients in $\mathbb{N}[z]$, and nothing in its definition says that it is symmetric.

The aim of the rest of this subsection is to identify L_e with an LLT polynomial. This is the type-B counterpart of the LLT expansion of the type-A shuffle formula [36, Thm. 3.1.3], [35, Ch. 6], and it is the tool by which the two properties of a Frobenius character, symmetry and Schur positivity, are proved for $\mathcal{H}^{+}_{n,m}$ in Theorem 11.15 below. The main idea is to split every height into its quotient and its remainder modulo m, $b_i=mq_i+r_i$. The quotient q_i becomes the content of a cell and the remainder r_i orders the components, and the $2m$ values of b_i-b_j on which the kernel s_m is supported collapse to the two values of the attacking relation of LLT theory.

We use the LLT polynomials in the form of the definition of Blasiak, Haiman, Morse, Pun and Seelinger [15, Def. 4.1.1]. Let $\boldsymbol{\nu}=(\nu^{(1)},\dots,\nu^{(k)})$ be a tuple of skew diagrams. A cell of $\boldsymbol{\nu}$ is a cell of one of the components, the components being kept apart, hence $\boldsymbol{\nu}$ has $|\nu^{(1)}|+\cdots+|\nu^{(k)}|$ cells. Let $c(u)$ be the content of a cell u, that is its column index minus its row index, and for $u\in\nu^{(s)}$ put

$$\tilde{c}(u)=c(u)+\varepsilon s,\qquad 0<\varepsilon<1/k,$$

the *adjusted content*.

The *reading order* lists all the cells of $\boldsymbol{\nu}$ by increasing $\tilde{c}$, that is by increasing content, the cells of one content in the order of their components, and cells of equal content in one component, which do not occur for the tuples of rows below, in any order.

Two cells u and v of $\boldsymbol{\nu}$, in the same component or in different ones, form an *attacking pair* (u,v) if $0<\tilde{c}(v)-\tilde{c}(u)<1$. The pair is ordered, u being the cell which comes first in the reading order, and the condition says that v follows u by less than one unit of content. Explicitly, (u,v) is attacking if either $c(v)=c(u)$ and v lies in a later component than u, or $c(v)=c(u)+1$ and v lies in an earlier component than u. Two cells of the same component never attack.

A *semistandard filling* T of $\boldsymbol{\nu}$ is a semistandard filling of each component, and an attacking pair (u,v) is an *inversion* of T if the earlier cell carries the larger entry, $T(u)>T(v)$. Writing $\mathrm{inv}(T)$ for the number of inversions of T and x^T for the product of the $x_{T(u)}$ over the cells, the LLT polynomial is $G_{\boldsymbol{\nu}}(X;z)=\sum_T z^{\mathrm{inv}(T)}x^T$, summed over the semistandard fillings, and it is a symmetric function [15, §4.1].

When every component is a single row, a semistandard filling is a weakly increasing filling of each row, and a standard filling, one with the entries $1,\dots,n$, is a filling which increases along each row.

Construction 11.6 (The tuple of rows of a path)**.** Let e be a path of the $n \times mn$ rectangle with heights $b_i = mi - e_i$. The tuple $\boldsymbol{\nu}(e)$ is built in three steps.

(a) Split every height as $b_i = mq_i + r_i$ with $q_i = \lfloor b_i/m \rfloor$ and $0 \le r_i < m$. We call q_i the *content* and r_i the *colour* of the i-th north step. At $m = 1$ every colour is 0 and $q_i = b_i$.

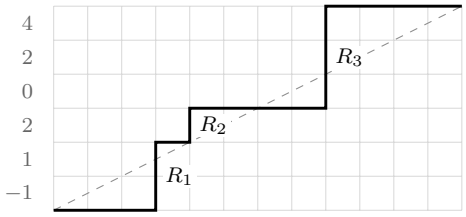
(b) Cut the sequence of north steps into the maximal vertical runs $R_1, \dots, R_k$ of the path, the maximal blocks of consecutive indices i with the same e_i. Along a run b increases by m at each step, hence q increases by 1 and the colour is constant. We write $r(R)$ for the colour of the run R.

(c) Replace the run $R = \{i_1 < \dots < i_l\}$ by a single row of l cells, the cell of the step i being placed at content q_i, hence the row occupies the consecutive contents $q_{i_1}, q_{i_1}+1, \dots, q_{i_1}+l-1$. The row has as many cells as the run has north steps. Order the rows by increasing colour, and the rows of one colour by increasing first index, that is in the order in which they occur along the path.

The resulting ordered tuple of rows is $\boldsymbol{\nu}(e) = (\nu^{(1)}, \dots, \nu^{(k)})$. It has one component for each vertical segment of the path, that is k is the number of distinct values among $e_1, \dots, e_n$, the sizes $|\nu^{(1)}|, \dots, |\nu^{(k)}|$ are the lengths of the segments, that is the multiplicities of these values, and they add up to n. Every cell is the cell of one north step i, and $c(i) = q_i$.

Every component of $\boldsymbol{\nu}(e)$ is a row, hence a horizontal strip. A standard filling of $\boldsymbol{\nu}(e)$ assigns the entries $1, \dots, n$ to the north steps so that they increase along each run, and by Theorem 11.1 this is exactly a labelling σ admissible for e, the label σ_i being placed in the cell of the step i. The tuple forgets the heights themselves, but it keeps what the kernel and the defect see, namely the contents q_i and the colours r_i, which together decide whether two steps attack and in which order.

Example 11.7. Let $n = 6$, $m = 2$ and $e = (3, 3, 4, 8, 8, 8)$, a path of the 6×12 rectangle with three vertical segments $R_1 = \{1, 2\}$, $R_2 = \{3\}$ and $R_3 = \{4, 5, 6\}$, at the abscissae 3, 4 and 8,

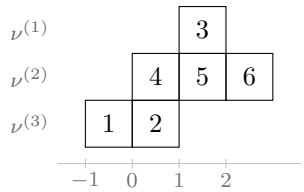


The heights $b_i = 2i - e_i$, listed at the left, and their quotients and remainders modulo 2 are

i	1	2	3	4	5	6
e_i	3	3	4	8	8	8
b_i	−1	1	2	0	2	4
q_i	−1	0	1	0	1	2
r_i	1	1	0	0	0	0

Along each segment the content rises by one and the colour is constant, as step (b) says.

The segments R_1, R_2, R_3 have the colours 1, 0, 0. The two segments of colour 0 come first, in the order of their first indices, and the segment of colour 1 comes last. Hence $\nu^{(1)}$ is the row of R_2, one cell at content 1, $\nu^{(2)}$ is the row of R_3, three cells at contents $0, 1, 2$, and $\nu^{(3)}$ is the row of R_1, two cells at contents $-1, 0$,



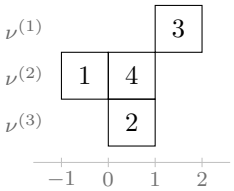
where each cell carries the index of its north step and the horizontal position of a cell is its content.

The sizes of the components are $1, 3, 2$, the lengths of the segments, and they add up to $n = 6$. A labelling σ writes σ_i in the cell of the step i, and it is admissible exactly when every row increases from left to right.

Note that the row of R_3 precedes the row of R_1 in the tuple although R_3 comes later along the path, because the colour is compared first.

At $m = 1$ all colours are 0 and the rows stay in their order along the path. We return to the running example, in which the attacking pairs can also be seen.

Example 11.8. We take the labelled path of Theorem 11.2, with $n = 4$, $m = 1$, $e = (0, 3, 3, 4)$, $\sigma = (3, 1, 4, 2)$ and $b = (1, -1, 0, 0)$. Since $m = 1$ every r_i is 0 and $q_i = b_i$. The maximal vertical runs are $\{1\}$, $\{2, 3\}$, $\{4\}$. They all have $r = 0$, hence they are ordered by their first index. Thus $\boldsymbol{\nu}(e)$ is the triple consisting of one cell of content 1, a row of two cells of contents $-1, 0$, and one cell of content 0, carrying the labels 3, $(1, 4)$ and 2,



where the horizontal position of a cell is its content. With ε small the adjusted contents are $\tilde{c} = 1+\varepsilon$ for the label 3, $-1 + 2\varepsilon$ and 2ε for the labels 1 and 4, and 3ε for the label 2.

An ordered pair attacks when the difference of the adjusted contents lies in $(0, 1)$, which here happens for three pairs, namely the labels 4 and 2, with difference ε, the labels 4 and 3, with difference $1 - \varepsilon$, and the labels 2 and 3, with difference $1 - 2\varepsilon$. In terms of the north steps these are the pairs $(3, 4)$, $(1, 3)$ and $(1, 4)$ of Theorem 11.2, which are exactly the three pairs counted there by dinv_1. The pair $(2, 3)$ of the same component and the pairs $(1, 2)$, $(2, 4)$ do not attack, again as there.

Of the three attacking pairs, the first two carry the larger entry first and are inversions, the third does not. Hence $\mathrm{inv} = 2 = \mathrm{def}_1(P)$, as Theorem 11.9 below asserts.

The next theorem says that the defect of an admissible labelling is the inversion number of the corresponding standard filling, that the reading orders of the path and of the tuple agree, and consequently that L_e is the LLT polynomial of $\boldsymbol{\nu}(e)$.

Theorem 11.9. *For all $n, m \geq 1$ and every path e of the $n \times mn$ rectangle, the polynomial L_e of (150) is the LLT polynomial of the tuple of rows $\boldsymbol{\nu}(e)$ of Theorem 11.6,*

$$L_e(X; z) = G_{\boldsymbol{\nu}(e)}(X; z).$$

In particular L_e is a symmetric function, and by (149) $\mathcal{H}^+_{n,m}$ is a sum of LLT polynomials, evaluated at $z = (qt)^{-1}$, with monomial coefficients.

Proof. The main points are in Steps 2 and 3, and we first describe them in words. The defect (146) sees a pair $i < j$ of north steps when $b_i - b_j$ lies in $[1 - m, m]$, a range of $2m$ integers, and it counts the pair when the labels form an inversion, $\sigma_i > \sigma_j$, on the half $[1 - m, 0]$ and a non-inversion, $\sigma_i < \sigma_j$, on the half $[1, m]$. The inversion number of a filling of $\boldsymbol{\nu}(e)$ sees a pair of cells when their adjusted contents differ by less than one, which leaves only two possibilities for their contents, equal or differing by one, and it counts the pair when the earlier cell in the reading order carries the larger entry.

Step 2 shows that the two descriptions agree. Writing $b_i - b_j = m\Delta + \rho$ with Δ the difference of the contents and ρ the difference of the colours, the range $[1 - m, m]$ corresponds to $\Delta \in \{0, 1\}$

when $\rho \le 0$ and to $\Delta \in \{-1, 0\}$ when $\rho \ge 1$, where the sign of ρ is the order of the two components by Step 1. In each case the cell read first is the cell of i on the half $[1-m, 0]$, where $b_i \le b_j$, and the cell of j on the half $[1, m]$, where $b_i > b_j$, so that an inversion of the labels on the first half and a non-inversion on the second half both say that the earlier cell carries the larger entry.

Step 3 shows that the reading order of $\boldsymbol{\nu}(e)$, by increasing adjusted content, is the reading order of the path, by increasing (b_i, i). Hence the reading words of a labelling and of the corresponding standard filling have the same inverse descent set, and the passage from standard to semistandard fillings is the usual standardization.

Write $\pi(i)$ for the index of the run containing i. Then the cell of the label i has $\tilde{c}(i) = q_i + \varepsilon\pi(i)$, and for $i < j$ we put

$$\Delta = q_i - q_j, \qquad \rho = r_i - r_j \in [1-m, m-1], \qquad b_i - b_j = m\Delta + \rho.$$

Step 1*, the component order.* Let $i < j$ lie in different runs. If $r_i < r_j$ then $\pi(i) < \pi(j)$ by the first coordinate of the ordering. If $r_i = r_j$ then the two runs are distinct blocks of consecutive indices, hence the one containing i lies entirely to the left of the one containing j and has the smaller minimum, whence again $\pi(i) < \pi(j)$. If $r_i > r_j$ then $\pi(i) > \pi(j)$. Thus for $i < j$ in different runs,

$$\pi(i) < \pi(j) \iff r_i \le r_j \iff \rho \le 0. \tag{151}$$

Step 2*, attacking pairs and inversions.* Suppose first that $i < j$ lie in the same run, say $j = i + s$ with $s \ge 1$. Then $q_j = q_i + s$ and $\pi(i) = \pi(j)$, hence $\tilde{c}(i) - \tilde{c}(j) = -s \le -1$ and $\tilde{c}(j) - \tilde{c}(i) = s \ge 1$, and neither ordered pair is attacking. On the other side $b_i - b_j = -ms \le -m < 1 - m$, hence the pair lies in neither range of (146) and def_m ignores it. Both statistics ignore the pair.

Now let $i < j$ lie in different runs and put $\eta = \pi(j) - \pi(i) \ne 0$, hence

$$\tilde{c}(j) - \tilde{c}(i) = -\Delta + \varepsilon\eta, \qquad |\varepsilon\eta| < 1.$$

Assume first $\rho \le 0$, hence $\eta > 0$ by (151). There are four cases. If $\Delta = 0$ then $\tilde{c}(j) - \tilde{c}(i) = \varepsilon\eta \in (0, 1)$, hence (i, j) is attacking and it is an inversion iff $\sigma_i > \sigma_j$. On the other side $b_i - b_j = \rho \in [1-m, 0]$, and the pair is counted by def_m iff $\sigma_i > \sigma_j$. If $\Delta = 1$ then $\tilde{c}(i) - \tilde{c}(j) = 1 - \varepsilon\eta \in (0, 1)$, hence (j, i) is attacking and it is an inversion iff $\sigma_j > \sigma_i$. On the other side $b_i - b_j = m + \rho$ lies in $[1, m]$ because $1 - m \le \rho \le 0$, and the pair is counted iff $\sigma_i < \sigma_j$.

If $\Delta \ge 2$ then $\tilde{c}(i) - \tilde{c}(j) = \Delta - \varepsilon\eta > 2 - 1 = 1$ and neither ordered pair is attacking, while $b_i - b_j = m\Delta + \rho \ge 2m + (1 - m) = m + 1$ lies outside both ranges. If $\Delta \le -1$ then $\tilde{c}(j) - \tilde{c}(i) = -\Delta + \varepsilon\eta > 1$ and neither ordered pair is attacking, while $b_i - b_j = m\Delta + \rho \le -m + 0 = -m$ lies outside both ranges.

Assume now $\rho \ge 1$, hence $\eta < 0$. If $\Delta = 0$ then $\tilde{c}(i) - \tilde{c}(j) = -\varepsilon\eta \in (0, 1)$, hence (j, i) is attacking and it is an inversion iff $\sigma_j > \sigma_i$. On the other side $b_i - b_j = \rho \in [1, m-1] \subseteq [1, m]$, and the pair is counted iff $\sigma_i < \sigma_j$. If $\Delta = -1$ then $\tilde{c}(j) - \tilde{c}(i) = 1 + \varepsilon\eta \in (0, 1)$, hence (i, j) is attacking and it is an inversion iff $\sigma_i > \sigma_j$. On the other side $b_i - b_j = -m + \rho \in [1-m, -1] \subseteq [1-m, 0]$, and the pair is counted iff $\sigma_i > \sigma_j$.

If $\Delta \ge 1$ then $\tilde{c}(i) - \tilde{c}(j) = \Delta - \varepsilon\eta > 1$, and $b_i - b_j = m\Delta + \rho \ge m + 1$ lies outside both ranges. If $\Delta \le -2$ then $\tilde{c}(j) - \tilde{c}(i) = -\Delta + \varepsilon\eta > 2 - 1 = 1$, and $b_i - b_j = m\Delta + \rho \le -2m + (m-1) = -m - 1$ lies outside both ranges.

In every case the ordered pair attacks exactly when def_m sees the pair, and the inversion condition is exactly the condition counted by def_m. Hence for a standard filling S of $\boldsymbol{\nu}(e)$, corresponding to the admissible labelling σ,

$$\mathrm{inv}(S) = \mathrm{def}_m(e, \sigma). \tag{152}$$

Step 3*, the reading order and the quasisymmetric expansion.* The reading order of $\boldsymbol{\nu}(e)$ sorts the labels by increasing $\tilde{c}(i) = q_i + \varepsilon\pi(i)$, that is by increasing q_i with ties broken by increasing $\pi(i)$. Two labels with equal q lie in different runs, since q is injective on a run, hence (151) applies to them and

breaks the tie by increasing (r_i, i). The reading order is therefore the order of increasing (q_i, r_i, i), which is the order of increasing (b_i, i), the reading order of the path. Thus $\mathrm{ides}(S) = \mathrm{ides}(w(e,\sigma))$.

Let T be a semistandard filling of $\boldsymbol{\nu}(e)$ and let $S = \mathrm{std}(T)$ be its standardization, obtained by replacing the entries by $1, \dots, n$ preserving their order and breaking ties by the reading order. Along a row the reading order increases, hence the weak increase of T along rows becomes a strict increase in S. Thus S is a standard filling, and every standard filling occurs.

If (u, v) is attacking then $T(u) > T(v)$ iff $S(u) > S(v)$. Indeed, the two inequalities agree when $T(u) \neq T(v)$, and when $T(u) = T(v)$ both fail, the second because the tie is broken by the reading order and $\tilde{c}(u) < \tilde{c}(v)$. Thus inv is constant on the fibres of std.

Finally, fix S and write v_c for the entry of T in the cell where S takes the value c. Then $\mathrm{std}(T) = S$ if and only if $v_1 \leq v_2 \leq \cdots \leq v_n$ with $v_c < v_{c+1}$ whenever the cell carrying $c+1$ precedes the cell carrying c in the reading order, that is whenever $c \in \mathrm{ides}(S)$. Such a T is automatically semistandard, because v is weakly increasing and S is increasing along each row.

Summing the monomials of one fibre gives $F_{n,\mathrm{ides}(S)}$ by the definition of the fundamental quasisymmetric function. Therefore

$$G_{\boldsymbol{\nu}(e)}(X; z) = \sum_S z^{\mathrm{inv}(S)} F_{n,\mathrm{ides}(S)}(X) = \sum_\sigma z^{\mathrm{def}_m(e,\sigma)} F_{n,\mathrm{ides}(w(e,\sigma))}(X) = L_e(X; z),$$

where the middle equality is (152) together with the identification of the reading orders. □

Schur positivity of $G_{\boldsymbol{\nu}}$ for an arbitrary tuple of skew diagrams is proved only in the unpublished manuscript of Grojnowski and Haiman [34, Cor. 6.9]. We do not use it, because the tuples of Theorem 11.6 lie in a smaller class for which the statement is published.

Lemma 11.10 (Rotation)**.** *Write $\nu^{(s)}[1]$ for $\nu^{(s)}$ translated so that every content increases by 1, and set $\boldsymbol{\nu}^{\circlearrowleft} = (\nu^{(k)}[1], \nu^{(1)}, \dots, \nu^{(k-1)})$. Then $G_{\boldsymbol{\nu}^{\circlearrowleft}} = G_{\boldsymbol{\nu}}$. Translating every component by the same amount also leaves $G_{\boldsymbol{\nu}}$ unchanged.*

Proof. Both operations leave the shape of each component, hence the semistandard condition, unchanged. Therefore it suffices to show that they preserve the attacking pairs together with the order of the two cells in each. A common translation adds a constant to every $\tilde{c}$, which changes no difference.

For the rotation write $\tilde{c}'$ for the adjusted content afterwards. If u and v both lie in $\nu^{(k)}$, or neither of them does, then their contents and their component indices are shifted by the same amounts, hence $\tilde{c}'(u) - \tilde{c}'(v) = \tilde{c}(u) - \tilde{c}(v)$.

Let $u \in \nu^{(k)}$ and $v \in \nu^{(s)}$ with $s < k$, and put $\delta = c(u) - c(v)$. Before the rotation $\tilde{c}(u) - \tilde{c}(v) = \delta + \varepsilon(k - s)$ with $0 < \varepsilon(k - s) < 1$, hence the ordered pair (v, u) is attacking exactly when $\delta = 0$, and (u, v) exactly when $\delta = -1$. Afterwards u carries the component index 1 and v the index $s + 1$, hence $\tilde{c}'(u) - \tilde{c}'(v) = \delta + 1 - \varepsilon s$ with $0 < \varepsilon s < 1$, and again (v, u) is attacking exactly when $\delta = 0$ and (u, v) exactly when $\delta = -1$. □

Lemma 11.11. *For every path e there is a partition μ such that a tuple obtained from $\boldsymbol{\nu}(e)$ by rotations and a common translation is $\mathrm{quot}_k(\mu)$ over the inner shape $\mathrm{core}_k(\mu)$, in the normalization of* [36, §5.1].

Proof. By [36, §5.1] a k-core ν is determined by its content sequence $(s_0, \dots, s_{k-1})$, $s_i \equiv i \pmod k$, the contents of the k ribbons which can be added to ν. The partitions μ with $\mathrm{core}_k(\mu) = \nu$ correspond to the k-tuples of partitions $(\mu^{(0)}, \dots, \mu^{(k-1)})$, and the adjusted content of $x \in \mu^{(i)}$ is $\tilde{c}(x) = k\, c(x) + s_i$ [36, (71)]. Writing $s_i = ka_i + i$, the i-th component of the resulting tuple is the partition $\mu^{(i)}$ with its contents shifted by a_i.

In the abacus description of cores the vectors $(a_0, \dots, a_{k-1})$ which arise from the k-cores are exactly the integer vectors of sum 0 [28, Bijection 2]. The components of $\boldsymbol{\nu}(e)$ are the one-row partitions $(l_1), \dots, (l_k)$ with contents shifted by $d_1, \dots, d_k$, hence all that is needed is $\sum_s d_s = 0$.

A common translation changes that sum by a multiple of k and one rotation raises it by 1. Thus at most $k-1$ rotations followed by a translation achieve it, and by Theorem 11.10 neither move changes G. □

Corollary 11.12. *$G_{\boldsymbol{\nu}(e)}(X;z) \in \bigoplus_\lambda \mathbb{N}[z]\, s_\lambda$ for every path e.*

Proof. Let μ and $\nu = \mathrm{core}_k(\mu)$ be as in Theorem 11.11. We recall the objects of [36, §5.1–5.2]. A k-ribbon is a connected skew shape of k cells which contains no 2×2 square, and a semistandard k-ribbon tableau T of μ/ν is a tiling of μ/ν by k-ribbons together with a filling of the ribbons which is weakly increasing along the rows and the columns of μ/ν and whose level sets are horizontal ribbon strips with the tiling prescribed there. The quotient quot_k identifies these tableaux with the semistandard fillings of $\mathrm{quot}_k(\mu)$.

The spin of a ribbon is one less than its number of rows, hence a horizontal ribbon has spin 0 and a vertical one spin $k-1$, and the spin $\mathrm{sp}(T)$ of the tableau is half the sum of the spins of its ribbons, normalized by subtracting the minimum over all tableaux of the shape, hence a nonnegative integer. It measures how far the tiling is from the flattest tiling of the shape. For instance the shape $(2,2)$ with $k=2$ has two tilings, by two horizontal dominoes, of spin 0, and by two vertical dominoes, of spin 1.

Let $G_{\mu/\nu}(z;u) = \sum_T u^{\mathrm{sp}(T)} z^T$ be the spin generating function of the semistandard k-ribbon tableaux of μ/ν, where z^T records the entries. By [36, Cor. 5.2.4] there is a constant c with $u^c G_{\mu/\nu}(z;u^{-1}) = G_{\mathrm{quot}_k(\mu)}(z;u)$. This is the semistandard form, [36, Lem. 5.2.3], of the identity $\mathrm{sp} = c - \mathrm{inv}\circ \mathrm{quot}_k$ proved for standard tableaux in [36, Lem. 5.2.2].

The two statistics are complementary, every unit of spin of the ribbon tableau being traded for an inversion of the corresponding filling of the quotient, as in the example, where the tiling of spin 0 corresponds to the filling with one inversion and the tiling of spin 1 to the filling with none.

By [36, Prop. 5.3.1], whose hypothesis is exactly that ν be $\mathrm{core}_k(\mu)$ and which Theorem 11.11 supplies, the Schur coefficients of $G_{\mu/\nu}$ are, up to a power of the variable and the substitution $u \mapsto u^2$, the parabolic Kazhdan–Lusztig polynomials $P^-_{\mu+\rho,\,\nu+k\lambda+\rho}$, and these have non-negative coefficients [50]. Inverting the variable permutes the degrees within each Schur coefficient and hence preserves non-negativity. □

Remark 11.13. The hypothesis of [36, Prop. 5.3.1] cannot be dropped. The same paper records the case of ν not a k-core as a conjecture, observing that the Kazhdan–Lusztig expressions then acquire negative terms. That case is [34, Cor. 6.9], where the authors note that empty ν is [55, Thm. 4.1] and ν a k-core is [36, Prop. 5.3.1]. Their manuscript is unpublished, and nothing here depends on it. The published case of empty ν alone would not suffice, because it forces all the a_i to be equal, whereas the offsets d_s of Theorem 11.6 are the values q_{i_1} at the first labels of the runs and are unequal for most paths. At $(n,m)=(5,1)$ only 47 of the 252 paths have all offsets equal, and at $(4,2)$ only 107 of 495.

Lemma 11.14 (Complementation)**.** *For $P=(e,\sigma)\in \mathsf{PF}^+_{n,m}$ put*

$$e^c_i = mn - e_{n+1-i}, \qquad \sigma^c_i = n+1-\sigma_{n+1-i} \qquad (1\le i\le n).$$

Then $P\mapsto P^c=(e^c,\sigma^c)$ is an involution of $\mathsf{PF}^+_{n,m}$ under which

$$\mathrm{dinv}_m(e^c) = \mathrm{dinv}_m(e), \quad \mathrm{def}_m(P^c) = \mathrm{def}_m(P), \quad A^{(m)}_{\mp}(e^c) = A^{(m)}_{\pm}(e),$$

and $\mathrm{ides}(w(P^c)) = \{\, n-c : c\in \mathrm{ides}(w(P))\,\}$.

Proof. Both formulas are visibly involutive. From $b_i = mi - e_i$ we get

$$b^c_i = mi - e^c_i = mi - mn + e_{n+1-i} = mi - mn + m(n+1-i) - b_{n+1-i} = m - b_{n+1-i},$$

hence for $i<j$, writing $i' = n+1-i$ and $j' = n+1-j$, we have $j'<i'$ and

$$b^c_i - b^c_j = b_{j'} - b_{i'}. \tag{153}$$

The map $(i,j)\mapsto(j',i')$ is a bijection of the pairs $i<j$. Hence (153) gives $\mathrm{dinv}_m(e^c)=\mathrm{dinv}_m(e)$ at once, and $[m-b_i^c]_+ + [-b_i^c]_+ = [b_{i'}]_+ + [b_{i'}-m]_+$ gives $A_-^{(m)}(e^c)=A_+^{(m)}(e)$ and $A_+^{(m)}(e^c)=A_-^{(m)}(e)$, which is Theorem 10.3(ii).

For the labels, $\sigma_i^c>\sigma_j^c$ means $n+1-\sigma_{i'}>n+1-\sigma_{j'}$, that is $\sigma_{j'}>\sigma_{i'}$. Thus the pair (i,j) of P^c is counted in the first sum of (146) exactly when the pair (j',i') of P is, and likewise for the second sum. Hence $\mathrm{def}_m(P^c)=\mathrm{def}_m(P)$. The pair P^c is admissible, since $e_i^c=e_{i+1}^c$ means $e_{n-i}=e_{n+1-i}$, whence $\sigma_{n-i}<\sigma_{n+1-i}$ and therefore $\sigma_i^c<\sigma_{i+1}^c$.

For the reading words, the order of increasing (b_i^c,i) is, by $b_i^c=m-b_{n+1-i}$, the order of decreasing $(b_{n+1-i},n+1-i)$. Hence $w(P^c)$ is the reverse of the word obtained from $w(P)$ by replacing each letter v by $n+1-v$. Writing $\mathrm{pos}_w(v)$ for the position of v in w, this says $\mathrm{pos}_{w(P^c)}(v)=n+1-\mathrm{pos}_{w(P)}(n+1-v)$. Hence $c\in\mathrm{ides}(w(P^c))$, that is $\mathrm{pos}_{w(P^c)}(c+1)<\mathrm{pos}_{w(P^c)}(c)$, holds if and only if $\mathrm{pos}_{w(P)}(n-c)>\mathrm{pos}_{w(P)}(n+1-c)$, that is if and only if $n-c\in\mathrm{ides}(w(P))$. □

Proposition 11.15. *$\mathcal{H}^+_{n,m}$ is a symmetric function, it is Schur positive, it satisfies $\mathcal{H}^+_{n,m}(X;q,t)=\mathcal{H}^+_{n,m}(X;t,q)$, and*

$$q^D\,\mathcal{H}^+_{n,m}(X;q,q^{-1})=h_n\big[(1+q^2+\cdots+q^{2mn})X\big].$$

Proof. By (149) and Theorem 11.9,

$$\mathcal{H}^+_{n,m}(X;q,t)=\sum_e q^{A_-^{(m)}(e)}t^{A_+^{(m)}(e)}(qt)^{\mathrm{dinv}_m(e)}\,G_{\boldsymbol{\nu}(e)}(X;(qt)^{-1}).$$

By Theorem 11.12 each $G_{\boldsymbol{\nu}(e)}$ is symmetric, with Schur coefficients in $\mathbb{N}[z]$. Its degree in z is $\max_\sigma \mathrm{def}_m(e,\sigma)$, at most $\mathrm{dinv}_m(e)$ because $\mathrm{def}_m\le\mathrm{dinv}_m$ pair by pair. Hence each summand

$$(qt)^{\mathrm{dinv}_m(e)}\,G_{\boldsymbol{\nu}(e)}(X;(qt)^{-1})$$

has coefficients in $\mathbb{N}[qt]$, and the sum is symmetric and Schur positive.

For the third assertion, we apply Theorem 11.14 and substitute $P\mapsto P^c$ in (147), which gives

$$\mathcal{H}^+_{n,m}(X;q,t)=\sum_P q^{A_+^{(m)}(P)+\mathrm{dinv}_m^{\mathrm{lab}}(P)}\,t^{A_-^{(m)}(P)+\mathrm{dinv}_m^{\mathrm{lab}}(P)}\,F_{n,\mathrm{ides}(w(P))^{\mathrm{rev}}}(X),$$

where $S^{\mathrm{rev}}=\{n-c : c\in S\}$. Its bidegree-$(a,b)$ part is therefore obtained from the bidegree-(a,b) part of $\mathcal{H}^+_{n,m}(X;t,q)$, which is the same sum with $F_{n,\mathrm{ides}(w(P))}$ in place of $F_{n,\mathrm{ides}(w(P))^{\mathrm{rev}}}$, by replacing every $F_{n,S}$ by $F_{n,S^{\mathrm{rev}}}$.

That operation fixes every symmetric function. Indeed, let us truncate to $N\ge n$ variables and substitute $x_i\mapsto x_{N+1-i}$. Writing $k_j=N+1-i_{n+1-j}$ turns a weakly increasing index sequence $i_1\le\cdots\le i_n$ into a weakly increasing $k_1\le\cdots\le k_n$, and the strictness required at $j\in S$ into strictness at $n-j$. Hence the substitution carries $F_{n,S}(x_1,\dots,x_N)$ to $F_{n,S^{\mathrm{rev}}}(x_1,\dots,x_N)$. A symmetric $f=\sum_S c_S F_{n,S}$ is fixed by the substitution, which is a permutation of the variables, hence $\sum_S c_S F_{n,S}=\sum_S c_S F_{n,S^{\mathrm{rev}}}$ in N variables.

For $N\ge n$ the $F_{n,S}(x_1,\dots,x_N)$, $S\subseteq\{1,\dots,n-1\}$, are linearly independent. Indeed, the coefficient of $x_1^{\alpha_1}x_2^{\alpha_2}\cdots$ in $F_{n,S}$ is 1 when S is contained in the set of partial sums of the composition α and 0 otherwise, hence the matrix of coefficients on the monomials indexed by compositions is unitriangular for inclusion. Therefore $c_S=c_{S^{\mathrm{rev}}}$ for every S. Each bidegree part of $\mathcal{H}^+_{n,m}(X;t,q)$ is symmetric by the first assertion, hence the two agree.

For the last assertion, $\mathrm{dinv}_m^{\mathrm{lab}}$ occurs with the same exponent in q and in t, hence it cancels at $t=q^{-1}$ and the term of P becomes $q^{A_-^{(m)}(e)-A_+^{(m)}(e)}$. Since $[m-b]_+-[b-m]_+=m-b$ and $[-b]_+-[b]_+=-b$ for every b,

$$A_-^{(m)}-A_+^{(m)}=\sum_{i=1}^n(m-2b_i)=mn-2m\tbinom{n+1}{2}+2\sum_i e_i=2\sum_i e_i-D.$$

Hence

$$q^D\,\mathcal{H}^+_{n,m}(X;q,q^{-1}) = \sum_P q^{2\sum_i e_i}\,F_{n,\mathrm{ides}(w(P))}(X).$$

For a fixed path e the sum over the admissible labellings is $\prod_s h_{|R_s|}(X)$, the product over the maximal vertical runs of e, by the proof of Theorem 11.5(iii), and $\sum_i e_i = \sum_j j a_j$, where a_j is the number of north steps of e at abscissa j. Therefore the sum equals $\sum_a \prod_j q^{2ja_j} h_{a_j}(X) = h_n\big[(1+q^2+\cdots+q^{2mn})X\big]$, by the expansion $h_n[(y_0+\cdots+y_{mn})X] = \sum_a \prod_j y_j^{a_j} h_{a_j}[X]$ at $y_j = q^{2j}$. □

The specialization at $t = q^{-1}$ refines Theorem 7.2. Taking $[s_n]$ and using Theorem 11.5(i) returns $q^D \widehat{P}_{n,m}(q,q^{-1}) = \begin{bmatrix}(m+1)n\\ n\end{bmatrix}_{q^2}$ of Theorem 10.3(iii), because $[s_n]h_n[(1+q^2+\cdots+q^{2mn})X]$ is the generating function of the partitions in the $n\times mn$ rectangle by twice the size.

11.3. **The conjecture.** In this subsection we state the labelled conjecture and we relate it to its operator form and to its representation-theoretic form. The main idea is to promote each fixed-point weight of (127) to the colour-zero pure part of the corresponding wreath Macdonald polynomial, with the coefficients inverted.

Recall ℓ_λ, $\Pi^{(0)}_\lambda$, D_λ from (4), $B^{(i)}_\lambda$ from Section 10.2, the wreath Macdonald polynomials $\widetilde{H}_\lambda$ of (129) and the involution inv of Section 10.4. Write $\widetilde{H}_\lambda[X,0] \in \Lambda_{X_0}$ for the colour-zero pure part. Put

$$C_\lambda = (1+qt)B^{(0)}_\lambda - (q+t)B^{(1)}_\lambda = \langle \widetilde{H}_\lambda, \Phi_n\rangle, \tag{154}$$

where the second equality holds by Theorem 10.9(b) and (140). Thus $P_{n,m} = \sum_\lambda \ell_\lambda^m \Pi^{(0)}_\lambda C_\lambda D_\lambda^{-1}$ is the left side of (127).

Conjecture 11.16 (Type-B shuffle conjecture, labelled form)**.** *For all $n,m\ge 1$,*

$$\sum_{\lambda\in \mathrm{Bal}(2n)} \frac{\ell_\lambda^m\,\Pi^{(0)}_\lambda\,C_\lambda}{D_\lambda}\ \mathrm{inv}\big(\widetilde{H}_\lambda[X,0]\big) \;=\; \mathcal{H}^+_{n,m}(X;q,t). \tag{155}$$

Here inv is the involution of Section 10.4, which inverts q and t in the coefficients and fixes the Schur functions. Applied to the one-alphabet symmetric function $\widetilde{H}_\lambda[X,0] = \sum_\mu c_\mu(q,t)s_\mu$ it gives $\sum_\mu c_\mu(q^{-1},t^{-1})s_\mu$. For instance Theorem 10.14 gives $\widetilde{H}_{(3,1)}[X,0] = s_2 + q^2 s_{11}$ and $\mathrm{inv}(\widetilde{H}_{(3,1)}[X,0]) = s_2 + q^{-2}s_{11}$.

The inversion is not a normalization. It is what converts the colour-one pure part, which is the one produced by the operator form below, into the colour-zero pure part, which is the one normalized by (129). The precise statement is the following.

Lemma 11.17. *For every $\lambda\in\mathrm{Bal}(2n)$,*

$$\mathrm{inv}\,\widetilde{H}_\lambda = \ell_\lambda^{-1}\,\Omega\,\widetilde{H}_\lambda, \qquad \Omega = \chi\,(\omega\otimes\omega), \tag{156}$$

where χ exchanges the two alphabets, and consequently

$$\widetilde{H}_\lambda[0,X] = \ell_\lambda\,\omega\,\mathrm{inv}\big(\widetilde{H}_\lambda[X,0]\big). \tag{157}$$

Proof. Orr and Shimozono give two symmetries of the wreath Macdonald basis, each only up to a scalar, which they write $\equiv$. The inversion symmetry is $\downarrow \widetilde{H}^w_{\mu^\bullet} \equiv \widetilde{H}^{w_0 w}_{w_0\mu^\bullet}$ [67, Prop. 3.23], where $\downarrow = \mathrm{inv}\circ\mathrm{neg}\circ\omega$ and ω is applied to both alphabets, and the rotational symmetry is $\chi\widetilde{H}^w_{\mu^\bullet} \equiv \widetilde{H}^{\chi w}_{\chi\mu^\bullet}$ [67, Thm. 3.26]. At $r=2$ the three permutations of $I=\mathbb{Z}/2\mathbb{Z}$ that occur degenerate, namely neg is the identity, and χ and w_0 are both the transposition.

Let us take $w=1$, which is the chamber attached to an empty 2-core. The first symmetry gives $\mathrm{inv}\,\omega\widetilde{H}_\lambda \equiv \widetilde{H}^{w_0}_{w_0\mu^\bullet}$, and the second, applied with $w=w_0$ to $\widetilde{H}^{w_0}_{w_0\mu^\bullet}$, gives $\chi\,\mathrm{inv}\,\omega\widetilde{H}_\lambda \equiv \widetilde{H}^{\chi w_0}_{\chi w_0\mu^\bullet} = \widetilde{H}_\lambda$,

because $\chi w_0 = 1$. Since ω commutes with the colour permutations and with inv, this says $\Omega\,\mathrm{inv}\,\widetilde{H}_\lambda = c\,\widetilde{H}_\lambda$ for a scalar c.

It remains to evaluate c. Orr and Shimozono say of the symmetries of their §3.5 that "the constant of proportionality is a Laurent monomial which may easily be deduced using the normalization condition (3.9)" [67, §3.5], and they carry this out for the inversion symmetry. Namely, [67, Rem. 3.25] records that the c with $\downarrow \widetilde{H}^{w}_{\mu\bullet} = c\,\widetilde{H}^{w_0 w}_{w_0\mu\bullet}$ is $\langle s_n[X^{(0)}], \downarrow \widetilde{H}^{w}_{\mu\bullet}\rangle = \mathrm{inv}\langle e_n[X^{(0)}], \widetilde{H}^{w}_{\mu\bullet}\rangle$.

The constant of [67, Thm. 3.26] is not written out there, and (156) needs the composite. What follows is their computation applied to $\Omega\,\mathrm{inv}$ at $r = 2$, where it is the colour-one rather than the colour-zero monomial that appears.

We pair both sides with $s_n[X_0]$ and use the normalization $\langle \widetilde{H}_\lambda, s_n[X_0]\rangle = 1$ of (129). On the right this gives c. On the left, χ exchanges the alphabets, ω is self-adjoint for the Hall pairing and carries $s_n[X_1]$ to $e_n[X_1]$, and inv acts on coefficients only. Hence

$$\langle \Omega\,\mathrm{inv}\,\widetilde{H}_\lambda,\ s_n[X_0]\rangle = \langle \mathrm{inv}\,\widetilde{H}_\lambda,\ e_n[X_1]\rangle = \mathrm{inv}\langle \widetilde{H}_\lambda, e_n[X_1]\rangle = \mathrm{inv}(\ell_\lambda) = \ell_\lambda^{-1},$$

where the third equality is [67, Thm. 4.14], attributed there to Haiman, which identifies $\langle \widetilde{H}_\lambda, e_n[X_i]\rangle$ with the product of the box weights $q^a t^b$ of λ over the boxes of residue i. At $r = 2$ the residue is the parity of $a + b$, hence the value at $i = 1$ is ℓ_λ.

Here residue 1 is the odd runner, which carries the alphabet X_1 in the convention of Section 10.4 fixed by Theorem 10.14. Orr and Shimozono weight x by t and y by q, and (156) was checked directly in the present conventions (Theorem 11.34(9)), hence the difference does not affect it.

Since Ω is an involution, (156) follows. Taking the colour-zero pure part of (156) and using $(\Omega f)[X, 0] = \omega(f[0, X])$ gives (157). □

Passing from $\omega\widetilde{H}_\lambda[0, X]$ to $\mathrm{inv}(\widetilde{H}_\lambda[X, 0])$ absorbs one factor ℓ_λ. This is why the exponent is ℓ_λ^m in (155) and ℓ_λ^{m-1} in (159) below. Dropping inv from (155) leaves Theorem 11.19 intact, since $[s_n]$ of the extra factor is 1 either way, but it destroys the conjecture in every other Schur coordinate. At $n = 2$, $m = 1$ it gives $[s_{11}] = (q^2 + qt + t^2)(q^4 + qt + t^4)$ in place of the correct $[s_{11}]\mathcal{H}^+_{2,1} = q^2 + qt + t^2$.

Remark 11.18 (A conjectural proportionality of the pure parts). Write $\ell_\lambda^{(i)} = \prod_{a+b\equiv i\ (2)} q^a t^b$ for the product of the box weights of residue i, hence $\ell_\lambda^{(1)} = \ell_\lambda$. Computation suggests that the two pure parts are proportional, namely that for every $\lambda \in \mathrm{Bal}(2n)$

$$\widetilde{H}_\lambda[0, X] = \frac{\ell_\lambda^{(1)}}{\ell_\lambda^{(0)}}\,\widetilde{H}_\lambda[X, 0]. \tag{158}$$

This was verified symbolically over $\mathbb{Q}(q, t)$ for $n \le 4$ and at rational points for every one of the 110 diagrams with $n \le 7$, but it is not proved.

It is not a consequence of Theorem 11.17, which combined with (158) would say that the colour-zero pure part has the type-A self-duality $\mathrm{inv}(\widetilde{H}_\lambda[X, 0]) = \ell_\lambda^{(0)\,-1}\,\omega\,\widetilde{H}_\lambda[X, 0]$.

Granting (158), (155) takes the inv-free form

$$\sum_{\lambda\in\mathrm{Bal}(2n)} \frac{\ell_\lambda^m\,\Pi_\lambda^{(0)}\,C_\lambda}{D_\lambda\,\ell_\lambda^{(0)}}\,\omega\,\widetilde{H}_\lambda[X, 0] \;=\; \mathcal{H}^+_{n,m}(X; q, t),$$

which is a statement about the G_n-invariant part of the Procesi bundle alone. Equivalently, with ∇_i the diagonal operators $\nabla_i \widetilde{H}_\lambda = \ell_\lambda^{(i)} \widetilde{H}_\lambda$, the identity (158) says that $f \mapsto f[0, X]$ agrees with $f \mapsto (\nabla_1 \nabla_0^{-1} f)[X, 0]$ on all of $\Lambda_n^{(2)}$.

The left side of (155) is the left side of (127) with each fixed-point weight promoted to a symmetric function. Namely, the scalar term of λ is multiplied by $\mathrm{inv}(\widetilde{H}_\lambda[X, 0])$, whose s_n-coefficient is 1. Hence we obtain the following.

Proposition 11.19. *The s_n-coefficient of the left side of (155) is $P_{n,m}$, and the s_n-coefficient of the right side is $\widehat{P}_{n,m}$. Consequently Theorem 11.16 implies Theorem 10.5 in the same rank and twist.*

Proof. For a one-alphabet symmetric function f of degree n we have $[s_n]f = \langle f, h_n\rangle$, and the third condition of (129) gives $[s_n]\widetilde{H}_\lambda[X,0] = \langle \widetilde{H}_\lambda, h_n[X_0]\rangle = 1$. The involution inv acts on coefficients only, hence $[s_n]\operatorname{inv}(\widetilde{H}_\lambda[X,0]) = \operatorname{inv}(1) = 1$. The first assertion is now Theorem 10.4, and the second is Theorem 11.5(i). □

Remark 11.20 (The operator form). Let $\operatorname{pr}_1 f = f[0,X]$ be the projection to the colour-one pure part. The expected operator form of Theorem 11.16 is

$$\omega\, \operatorname{pr}_1 \nabla_1^m e_n[X_1] \;=\; \mathcal{H}^+_{n,m}(X;q,t), \tag{159}$$

the two-colour analogue of $\nabla^m e_n = \sum_{\mathrm{PF}} q^{\mathrm{dinv}} t^{\mathrm{area}} F$.

The two left sides are related as follows. Expanding $e_n[X_1]$ in the basis $\{\widetilde{H}_\lambda\}$ by Theorem 10.9(d) gives $e_n[X_1] = \sum_\lambda E_\lambda \ell_\lambda^{-1} D_\lambda^{-1} \widetilde{H}_\lambda$ with $E_\lambda = \widetilde{H}_\lambda[1+qt, -(q+t)]$, and $\widetilde{H}_\lambda[0,X] = \ell_\lambda\, \omega \operatorname{inv}(\widetilde{H}_\lambda[X,0])$ by Theorem 11.17. Hence the left side of (159) is $\sum_\lambda \ell_\lambda^m E_\lambda D_\lambda^{-1} \operatorname{inv}(\widetilde{H}_\lambda[X,0])$. The two left sides therefore differ by

$$\sum_u u^m \sum_{\lambda:\, \ell_\lambda = u} \frac{E_\lambda - \Pi_\lambda^{(0)} C_\lambda}{D_\lambda} \operatorname{inv}(\widetilde{H}_\lambda[X,0]), \tag{160}$$

which vanishes for every m if and only if each inner sum vanishes.

The inner sums do not vanish termwise, since $E_\lambda \neq \Pi_\lambda^{(0)} C_\lambda$ for 6 of the 10 diagrams at $n=3$ and for 15 of the 20 at $n=4$. Their vanishing is a separate conjecture, verified symbolically for $n \le 6$ and at rational points for $n = 7, 8$. Under it, (155) and (159) are equivalent. Without it they are independent statements, and it is (155), not (159), which refines Theorem 10.5.

Remark 11.21 (The representation-theoretic form). Let $L^{(m)} = L_{\frac{1}{2n}+m}(\mathrm{triv})$ with the bigrading of Theorem 8.1 and let

$$DH^+_{n,m} := \big(\operatorname{gr}_F L^{(m)}\big)^{G_n}, \qquad DH^+_n := DH^+_{n,1},$$

an S_n-module by the residual action, with $e\, DH^+_{n,m} = \operatorname{gr}_F eL^{(m)}$. The representation-theoretic form of the conjecture is

$$\operatorname{Frob}^{q,t}_{S_n}(DH^+_{n,m}) = \mathcal{H}^+_{n,m}(X;q,t). \tag{161}$$

Applying $[s_n]$ to (161) gives $\operatorname{ch}\operatorname{gr}_F eL^{(m)} = \widehat{P}_{n,m}$ by Theorem 11.5(i), which is Theorem 8.12 combined with Theorem 10.5. For $m \gg 0$ the first half is a theorem.

At $q=t=1$, (161) is a theorem for all n, m. Indeed, $(L^{(m)} \otimes \varepsilon)^{G_n} \cong \mathbb{C}[(\mathbb{Z}/(2mn+1))^n]^{G_n}$ is the permutation module on the functions $[n] \to \{0, \dots, mn\}$ by [32], of character $h_n[(mn+1)X]$, which is Theorem 11.5(iii).

For $n=2$ and $n=3$ at $m=1$ the right side of (155) is, in the notation of Theorem 7.4 and with $w = qt$,

$$\begin{aligned}
\mathcal{H}^+_{2,1} &= \big([5]+w\big)s_2 + [3]\, s_{11},\\
\mathcal{H}^+_{3,1} &= \big([10]+w[6]+w[4]\big)s_3 + \big([8]+[6]+w[4]+w[2]\big)s_{21} + [4]\, s_{111},
\end{aligned}$$

whose s_n-coefficients are the polynomials $P_{2,1}$ and $P_{3,1}$ of Theorem 7.4, as Theorem 11.19 requires.

11.4. **Rooted schedules.** In this subsection we prove the schedule formula for $\mathcal{H}^{+}_{n,1}$. At $m=1$ the labelled paths can be reorganized into boxes of independent choices, in the way that Haglund and Loehr [37] reorganize the ordinary parking functions. The main idea is to build a labelled path by inserting its labels one at a time into a word with two sentinels, the labels above the diagonal from the right and the labels on or below it from the left. The index set is $S_{\{0,1,\dots,n\}}$, and the role of the extra symbol 0 is to mark the line $y=x$, which the paths of the square cross in both directions.

Definition 11.22. Let τ be a permutation of $\{0,1,\dots,n\}$ with maximal increasing runs $R_1|\dots|R_s$, and let $0\in R_r$. Since 0 is the smallest symbol it begins its run. For an ordinary label $c\in R_j$ put $h_\tau(c)=r-j$ and $B_h=\{c: h_\tau(c)=h\}$. The *schedule bounds* are

$$s_\tau(c)=\begin{cases}\#\{d\in B_h: d>c\}+\#\{d\in B_{h-1}: d<c\}, & h\ge 2,\\ 1+\#\{d\in B_1: d>c\}, & h=1,\\ 1+\#\{d\in B_0: d<c\}+\#\{d\in B_1: d>c\}, & h=0,\\ \#\{d\in B_h: d<c\}+\#\{d\in B_{h+1}: d>c\}, & h\le -1,\end{cases}\qquad (c\in B_h), \tag{162}$$

all of them positive, and the *schedule box* is $\mathcal{B}_\tau=\prod_{c=1}^{n}\{0,1,\dots,s_\tau(c)-1\}$. Put

$$a_-(\tau)=\sum_{h_\tau(c)\le 0}(1-2h_\tau(c)),\qquad a_+(\tau)=\sum_{h_\tau(c)\ge 1}(2h_\tau(c)-1),$$

and for $\mathbf{k}\in\mathcal{B}_\tau$ define the *rooted monomial*

$$M_{\tau,\mathbf{k}}=\prod_{h_\tau(c)\ge 1}x_c^{k_c}y_c^{2h_\tau(c)-1+k_c}\prod_{h_\tau(c)\le 0}x_c^{1-2h_\tau(c)+k_c}y_c^{k_c}=g^B_\tau\prod_{c=1}^{n}(x_cy_c)^{k_c}, \tag{163}$$

of bidegree $(a_-(\tau)+|\mathbf{k}|,\ a_+(\tau)+|\mathbf{k}|)$, where $g^B_\tau=\prod_{h\ge 1}y_c^{2h-1}\prod_{h\le 0}x_c^{1-2h}$.

Every variable pair of $M_{\tau,\mathbf{k}}$ has odd total exponent, and the exponents (a_c,b_c) recover $h_\tau(c)=(b_c-a_c+1)/2$ and $k_c=\min(a_c,b_c)$, hence they recover τ. Thus the $M_{\tau,\mathbf{k}}$ are pairwise distinct monomials, and they span a space of dimension $(n+1)^n$. The baseline g^B_τ is an odd root-centred Garsia–Stanton monomial, in the sense that $h_\tau(c)=d_\tau(c)-d_\tau(0)$, where $d_\tau(c)$ is the number of descents of τ at or after the position of c.

Theorem 11.23 (Rooted insertion)**.** *There is a bijection $\bigsqcup_\tau\mathcal{B}_\tau\to\mathsf{PF}^{+}_{n,1}$, given by a two-sentinel insertion algorithm, under which*

$$A_-(P)=a_-(\tau),\qquad A_+(P)=a_+(\tau),\qquad D(P)=|\mathbf{k}|,$$

and under which $\operatorname{ides}(w(P))=\mathcal{D}_\tau(\mathbf{k})$, *where $c\in\mathcal{D}_\tau(\mathbf{k})$ iff $h_\tau(c+1)<h_\tau(c)$, or $h_\tau(c+1)=h_\tau(c)\ge 1$ and $k_c\le k_{c+1}$, or $h_\tau(c+1)=h_\tau(c)\le 0$ and $k_c\ge k_{c+1}$. Consequently*

$$\mathcal{H}^{+}_{n,1}=\sum_{\tau\in S_{\{0,\dots,n\}}}q^{a_-(\tau)}t^{a_+(\tau)}\sum_{\mathbf{k}\in\mathcal{B}_\tau}(qt)^{|\mathbf{k}|}F_{n,\mathcal{D}_\tau(\mathbf{k})}(X),\qquad \langle\mathcal{H}^{+}_{n,1},h_1^n\rangle=\sum_\tau q^{a_-(\tau)}t^{a_+(\tau)}\prod_{c=1}^{n}[s_\tau(c)]_{qt}, \tag{164}$$

with $[r]_{qt}=1+qt+\cdots+(qt)^{r-1}$.

Remark 11.24 (Relation to the type-A schedule formula). Theorem 11.23 is a type-C form of a formula of Haglund and Loehr. In type A one takes $\tau\in S_n$ with descents at $i_1<\cdots<i_k$, divides it into the maximal increasing runs $R_1|\dots|R_{k+1}$, lets $\operatorname{cars}(\tau)$ be the set of parking functions whose cars in the rows of length $k+1-j$ are the elements of R_j, and for $\tau_i\in R_j$ sets

$$w_i(\tau)=\#\{d\in R_j: d>\tau_i\}+\#\{d\in R_{j+1}: d<\tau_i\}, \tag{165}$$

where, in Haglund's words, "it will prove convenient to call element 0 the $k+2$nd run of τ" [35, p. 79]. Then

(166)
$$\sum_{P\in\mathrm{cars}(\tau)} q^{\mathrm{dinv}(P)}t^{\mathrm{area}(P)} = t^{\mathrm{maj}(\tau)}\prod_{i=1}^{n}[w_i(\tau)]_q, \qquad \sum_{P\in\mathcal{P}_n} q^{\mathrm{dinv}(P)}t^{\mathrm{area}(P)} = \sum_{\tau\in S_n} t^{\mathrm{maj}(\tau)}\prod_{i=1}^{n}[w_i(\tau)]_q,$$

which are [35, Thm. 5.3 and Cor. 5.3.1].

The first of these is due to Haglund and Loehr [37]. It is Theorem 1 of the Melbourne 2002 preliminary version of their paper, cited as such in [36, (56)], where it is restated as $H(\sigma;q,t) = t^{\mathrm{comaj}(\sigma)}\prod_{i=2}^{n}[v(\sigma,i)+\chi(i\le r_1)]_q$ with a different bookkeeping of the runs.

The word *schedule* is later. It was introduced by Garsia and Hicks while Hicks's thesis [46] was being written [38, Historical Remark 3.2]. Both identities in (166) became theorems with the shuffle theorem [19], and a second proof is the Carlsson–Oblomkov basis [20] specialized at $x_i \mapsto q$, $y_i \mapsto t$ [38, §4].

Three features of (162) are visible already in (165), and one is not.

(i) *The bounds.* By construction $B_h = R_{r-h}$, hence B_{h-1} is the run immediately to the right of B_h, and the first line of (162) is (165) verbatim.

(ii) *The root is a sentinel.* Haglund's convention that element 0 is the $(k+2)$nd run contributes to a car c in the last run the single count $\#\{d\in\{0\} : d<c\} = 1$. That is the $+1$ in the second line of (162). The left sentinel $(0,0)$ of the insertion below is Haglund's element 0. In [38, §3.1] the same device is described as appending a 0 to τ, where "the appended 0 accounts for the extra point of insertion in the special case $k=0$".

(iii) *The insertion.* The proof of (166) in [35, pp. 80–81] builds $\mathrm{cars}(\tau)$ from right to left and identifies $w_i(\tau)$ with "the number of ways to insert a row containing car τ_i into a partial parking function containing cars $\tau_{i+1},\dots,\tau_n$, in rows of the appropriate length, and still obtain a partial parking function", the $w_i(\tau)$ choices raising dinv by $0,1,\dots,w_i(\tau)-1$. Steps 2 and 5 of the proof below are that argument carried out with a second sentinel.

(iv) *What is not there.* A type-A path stays on one side of $y=x$, hence all its heights have the same sign and the insertion has one direction. A path in $\mathsf{PF}^{+}_{n,1}$ crosses the diagonal. The index set is $S_{\{0,\dots,n\}}$ rather than S_n with a 0 appended, the insertion proceeds from the right for $h\ge 1$ and from the left for $h\le 0$, the third line of (162) counts in both directions at once and has no type-A counterpart, and a unit of D raises both degrees, hence the bracket is $[\,\cdot\,]_{qt}$ and the prefactor is a monomial in two variables instead of $t^{\mathrm{maj}(\tau)}$.

Chapter 5 of [35] (pp. 77–90) carries the type-A story in full. The dinv statistic is in Chapter 3 (pp. 48–52), the shuffle conjecture and the LLT expansion used in Theorem 11.9 above are in Chapter 6 (pp. 91–99), and the m-parameter of Theorem 11.29 below is in Chapter 6 (pp. 107–112).

Proof. The main points are in Steps 2, 5 and 6. They are the count of admissible slots, which must come out equal to (162) and must not depend on the earlier choices, the identification of D with $|\mathbf{k}|$, and the comparison of the anchor sets of two labels of equal height, which is what produces the descent rule. The other three steps are bookkeeping.

Let us write a labelled path $P=(e,\sigma)$ as the word

$$W(P) = (\sigma_0,b_0)(\sigma_1,b_1)\cdots(\sigma_n,b_n)(\sigma_{n+1},b_{n+1}), \qquad (\sigma_0,b_0)=(0,0), \quad (\sigma_{n+1},b_{n+1})=(n+1,1),$$

each entry being a pair (label, height), with $b_i = i-e_i$ for $1\le i\le n$. The two extreme entries are the *sentinels*, and they are the first and the last entry of every word considered below. We say that a word of pairs satisfies the *local rule* if for every two consecutive entries (u,β), (v,γ) we have

$$\gamma\le\beta+1, \qquad \text{and} \qquad \gamma=\beta+1 \implies u<v.$$

Step 0, the local rule is the definition. For $1\le i<n$ the inequality $b_{i+1}\le b_i+1$ is $e_i\le e_{i+1}$, and $b_{i+1}=b_i+1$ is $e_i=e_{i+1}$, in which case the local rule demands $\sigma_i<\sigma_{i+1}$. These are exactly the

conditions of Theorem 11.1 for the inner pairs. At the left sentinel, $b_1 \le b_0 + 1 = 1$ is $e_1 \ge 0$, and if $b_1 = 1$ the local rule demands $\sigma_0 = 0 < \sigma_1$, which holds because 0 is smaller than every label. At the right sentinel, $b_{n+1} = 1 \le b_n + 1$ is $e_n \le n$, and if $b_n = 0$ the local rule demands $\sigma_n < n + 1 = \sigma_{n+1}$, which holds because $n + 1$ is larger than every label. Thus $P \in \mathsf{PF}^+_{n,1}$ if and only if $W(P)$ satisfies the local rule at all $n + 1$ consecutive pairs.

Step 1*, the heights are a rooted permutation.* Let P be a labelled path and let $B_h = \{c : h(c) = h\}$ be the set of ordinary labels of height h, where $h(\sigma_i) = b_i$. If $B_h \neq \varnothing$ for some $h \ge 2$ then $B_{h-1} \neq \varnothing$. Indeed, the word starts at height $b_0 = 0$ and the heights increase by at most one from one entry to the next, hence some entry strictly between the left sentinel and an entry of height h has height $h - 1$. It is not the left sentinel, and it is not the right sentinel either, that being the last entry of the word. Moreover, at the first pair at which the height rises from $h - 1$ to h the local rule gives $d < c$ with $d \in B_{h-1}$ and $c \in B_h$, hence

$$\max B_h > \min B_{h-1} \qquad (h \ge 2). \tag{167}$$

Symmetrically, if $B_h \neq \varnothing$ for some $h \le -1$ then $B_{h+1} \neq \varnothing$. Indeed, the word ends at height $b_{n+1} = 1$ and the heights increase by at most one from one entry to the next, hence some entry strictly to the right of an entry of height h has height $h + 1 \le 0$. That entry is not the right sentinel, whose height is 1, and it is not the left sentinel, which is the first entry of the word and lies to the right of nothing. At the first pair at which the height rises from h to $h + 1$ the local rule gives

$$\max B_{h+1} > \min B_h \qquad (h \le -1). \tag{168}$$

Let $H = \max\{h : B_h \neq \varnothing\}$, let $L = \min\{h : B_h \neq \varnothing\}$ if $B_h \neq \varnothing$ for some $h \le 0$ and $L = 1$ otherwise, and set

$$\tau = (B_H \mid B_{H-1} \mid \cdots \mid B_1 \mid 0, B_0 \mid B_{-1} \mid \cdots \mid B_L),$$

each block written in increasing order.

Each block is increasing, and each junction is a descent. Namely, between B_h and B_{h-1} for $h \ge 2$ this is (167), between B_1 and the block $0, B_0$ it holds because $\max B_1 > 0$, between $0, B_0$ and B_{-1} it is (168) with $h = -1$, whose left side is $\max B_0$, and between B_h and B_{h-1} for $h \le -1$ it is (168) applied with $h - 1$ in place of h. The last two junctions occur only when $B_0 \neq \varnothing$, and if $B_0 = \varnothing$ then $B_{-1} = \varnothing$ as well, hence no junction is left unaccounted for.

Therefore the blocks are the maximal increasing runs of τ. The run containing 0 is the $(H + 1)$-st, and $c \in B_h$ lies in the $(H + 1 - h)$-th, hence $h_\tau(c) = h$ in the sense of Theorem 11.22. Conversely the blocks of any τ satisfy (167) and (168), because its runs are maximal. Thus $P \mapsto \tau$ is the map recording the heights, and the possible height functions are exactly the rooted permutations.

Step 2*, the insertion and the count of slots.* Fix τ. Starting from the word consisting of the two sentinels, we insert the ordinary labels in the following order. First the labels of height 1, then those of height 2, and so on up to H, with the labels of one height inserted in decreasing order. Then the labels of height 0, then those of height -1, and so on down to L, with the labels of one height inserted in increasing order. A label c of height $h \ge 1$ is inserted immediately to the right of an entry, a label of height $h \le 0$ immediately to the left of an entry. The sentinels are never displaced, hence the chosen entry is not the right sentinel in the first case and not the left sentinel in the second.

Let $c \in B_h$ with $h \ge 1$, and suppose that the labels of $B_{h'}$ for $h' < h$ and the labels of B_h greater than c have been inserted, and no others. All entries then present have heights in $\{0, 1, \ldots, h\}$, the value 0 occurring only at the left sentinel and the value 1 also at the right sentinel.

Inserting c between two consecutive entries (u, β) and (v, γ) destroys the pair (u, v) and creates the pairs (u, c) and (c, v), and the local rule at all other pairs is unaffected. For (c, v) the rule requires $\gamma \le h + 1$, which holds because $\gamma \le h$, and the implication is vacuous because $\gamma = h + 1$ never occurs. For (u, c) the rule requires $h \le \beta + 1$, that is $\beta \ge h - 1$, hence $\beta \in \{h - 1, h\}$, and when $\beta = h - 1$ it requires $u < c$.

Therefore the admissible positions are exactly those immediately to the right of an entry u with either $\beta = h$, or $\beta = h-1$ and $u < c$, the right sentinel excluded. The entries with $\beta = h$ are the labels of B_h greater than c, together with the right sentinel when $h = 1$, which is excluded. The entries with $\beta = h-1$ are the labels of B_{h-1}, all of which are present, when $h \geq 2$, together with the right sentinel when $h = 2$, whose label $n+1$ is not smaller than c, and when $h = 1$ they reduce to the left sentinel, whose label 0 is smaller than c, since B_0 has not yet been inserted. The number of admissible positions is therefore

$$\begin{aligned} &\#\{d \in B_h : d > c\} + \#\{d \in B_{h-1} : d < c\} \quad (h \geq 2), \\ &1 + \#\{d \in B_1 : d > c\} \quad (h = 1), \end{aligned}$$

which are the first two lines of (162).

Let now $c \in B_h$ with $h \leq 0$, and suppose that the labels of $B_{h'}$ for $h' > h$ and the labels of B_h smaller than c have been inserted, and no others. All entries present have heights in $\{h, h+1, \ldots, H\} \cup \{0, 1\}$, and in particular all are $\geq h$.

For the pair (u, c) created by inserting c immediately to the left of (v, γ), the rule requires $h \leq \beta + 1$, which holds because $\beta \geq h$, and the implication is vacuous because $\beta = h - 1$ never occurs. For the pair (c, v) the rule requires $\gamma \leq h + 1$, hence $\gamma \in \{h, h+1\}$, and when $\gamma = h + 1$ it requires $c < v$.

Therefore the admissible positions are exactly those immediately to the left of an entry v with either $\gamma = h$, or $\gamma = h + 1$ and $v > c$, the left sentinel excluded. The entries with $\gamma = h$ are the labels of B_h smaller than c, together with the left sentinel when $h = 0$, which is excluded. The entries with $\gamma = h + 1$ are the labels of B_{h+1}, all of which are present, together with the right sentinel when $h = 0$, whose label $n + 1$ is larger than c. The left sentinel is not among them, since its height is $0 = h + 1$ only for $h = -1$, and then its label 0 is not larger than c. The number of admissible positions is therefore

$$\begin{aligned} &1 + \#\{d \in B_0 : d < c\} + \#\{d \in B_1 : d > c\} \quad (h = 0), \\ &\#\{d \in B_h : d < c\} + \#\{d \in B_{h+1} : d > c\} \quad (h \leq -1), \end{aligned}$$

which are the last two lines of (162).

In all four cases the set of admissible entries, which we call the set $\mathcal{A}(c)$ of *anchors* of c, is given uniformly by

$$\mathcal{A}(c) = \begin{cases} \{d \in B_h : d > c\} \ \cup \ \{d \in \widehat{B}_{h-1} : d < c\}, & h \geq 1, \\ \{d \in B_h : d < c\} \ \cup \ \{d \in \widehat{B}_{h+1} : d > c\}, & h \leq 0, \end{cases} \tag{169}$$

where $\widehat{B}_{h-1} = B_{h-1}$ for $h \geq 2$ and $\widehat{B}_0 = \{0\}$ is the left sentinel for $h = 1$, and where $\widehat{B}_{h+1} = B_{h+1}$ for $h \leq -1$ and $\widehat{B}_1 = B_1 \cup \{n+1\}$ adjoins the right sentinel for $h = 0$.

Since the labels of a word are distinct we read $\mathcal{A}(c)$ as a set of labels. It depends only on τ and c, not on the positions chosen for the earlier insertions. Only the left-to-right order of its members in the current word depends on those choices. We number the anchors of c from right to left by $0, 1, 2, \ldots$ when $h \geq 1$ and from left to right when $h \leq 0$, and we let k_c be the number of the anchor used.

This identifies the possible insertion sequences for the given τ with $\mathcal{B}_\tau$, and each bound $s_\tau(c)$ is positive. Indeed, for $h \in \{0, 1\}$ the summand 1 is present. For $h \geq 2$ the first term is nonzero unless $c = \max B_h$, and for that c the second term is nonzero by (167). For $h \leq -1$ the first term is nonzero unless $c = \min B_h$, and for that c the second term is nonzero by (168). By Step 0 the word obtained at the end is $W(P)$ for a labelled path P, whose heights are those of τ by construction.

Step 3, the inverse. Let P be a labelled path and let τ be its rooted permutation from Step 1. We delete the ordinary labels in the reverse order. First the labels of height L in decreasing order,

then those of height $L+1$, and so on up to height 0. Then the labels of height H in increasing order, then those of height $H-1$, and so on down to height 1.

We check that each deletion is the inverse of an admissible insertion. Let $c \in B_h$ with $h \geq 1$ be the label to be deleted, hence the labels present are those of $B_{h'}$ with $h' < h$ and those of B_h greater than or equal to c, and let (u,β) and (v,γ) be the entries adjacent to c. The local rule at (u,c) gives $\beta \geq h-1$, and $\beta \leq h$ because all present heights are at most h. If $\beta = h-1$ the rule gives $u < c$. Thus u is an anchor of c, and c was inserted immediately to its right.

Deleting c restores the pair (u,v), which satisfies the local rule. Indeed $\gamma \leq h \leq \beta+1$, and if $\gamma = \beta+1$ then $\gamma = h$ and $\beta = h-1$, whence $u < c$ from the rule at (u,c) and $c < v$ because v is either the right sentinel or a label of B_h present at this moment, hence $u < v$. Counting the anchors of c lying to its right recovers k_c.

For $c \in B_h$ with $h \leq 0$ the argument is the mirror image. The present heights are all at least h, and the local rule at (c,v) gives $\gamma \leq h+1$ and $\gamma \geq h$, hence v is an anchor and c was inserted immediately to its left. The restored pair (u,v) satisfies $\gamma \leq h+1 \leq \beta+1$, and if $\gamma = \beta+1$ then $\beta = h$ and $\gamma = h+1$, whence $c < v$ from the rule at (c,v) and $u < c$ because u is either the left sentinel or a label of B_h present at this moment, hence $u < v$. Counting the anchors of c lying to its left recovers k_c. Deletion is therefore inverse to insertion, and the map of Step 2 is a bijection.

Step 4, the two area statistics. At $m=1$ the kernel $[1-b]_+ + [-b]_+$ equals $1-2b$ for $b \leq 0$ and 0 for $b \geq 1$, and $[b]_+ + [b-1]_+$ equals 0 for $b \leq 0$ and $2b-1$ for $b \geq 1$. Summing over the ordinary labels and using $h(\sigma_i) = b_i$ gives $A_-(P) = \sum_{h(c)\leq 0}(1-2h(c)) = a_-(\tau)$ and $A_+(P) = \sum_{h(c)\geq 1}(2h(c)-1) = a_+(\tau)$.

Step 5, $D = |\mathbf{k}|$. Recall from Theorem 11.1 that at $m=1$

$$D(P) = \#\{i<j:\ b_i = b_j,\ \sigma_i < \sigma_j\} + \#\{i<j:\ b_i = b_j+1,\ \sigma_i > \sigma_j\},$$

a sum over the unordered pairs of ordinary labels which depends only on their heights, their labels and their left-to-right order. An insertion does not change the relative order, the heights or the labels of the entries already present, hence it does not change the contribution of any pair already present. It suffices to show that inserting c at the anchor number k_c creates exactly k_c new contributions.

Let $h = h(c) \geq 1$ and let d be an ordinary label already present, hence $h(d) \in \{1,\dots,h\}$ and, if $h(d) = h$, then $d > c$. If $h(d) \leq h-2$ the pair $\{c,d\}$ contributes to neither sum, since the two heights are neither equal nor consecutive. If $h(d) = h$, hence $d > c$, the pair contributes to the first sum if and only if c precedes d, because the earlier of the two must carry the smaller label, and it never contributes to the second sum, the heights being equal. If $h(d) = h-1$ the pair contributes to the second sum if and only if c precedes d and $c > d$, since the earlier of the two must have the larger height, which forces it to be c, and it never contributes to the first sum.

Thus a new contribution arises exactly from the labels d lying to the right of the insertion point with $d \in B_h$, $d > c$, or with $d \in B_{h-1}$, $d < c$, that is, exactly from the anchors of c lying to the right of the insertion point.

The two counts agree even though D counts only ordinary labels while $\mathcal{A}(c)$ may contain a sentinel. The only sentinel it can contain is the left one, at $h=1$, and that is the first entry of the word, hence never to the right of the insertion point. Since c is inserted immediately to the right of the anchor number k_c and the anchors are numbered from the right, there are exactly k_c of them.

Let now $h = h(c) \leq 0$ and let d be an ordinary label already present, hence $h(d) \geq h$ and, if $h(d) = h$, then $d < c$. If $h(d) \geq h+2$ the pair contributes to neither sum. If $h(d) = h$, hence $d < c$, the pair contributes to the first sum if and only if d precedes c, the earlier of the two carrying the smaller label. If $h(d) = h+1$ the pair contributes to the second sum if and only if d precedes c and $d > c$, the earlier of the two having the larger height.

Thus a new contribution arises exactly from the anchors of c lying to the left of the insertion point. Again the only sentinel that $\mathcal{A}(c)$ can contain is the right one, at $h=0$, and that is the last entry of the word, hence never to the left of the insertion point. Since c is inserted immediately to

the left of the anchor number k_c and the anchors are numbered from the left, there are exactly k_c of them. Summing over the insertions, $D(P) = \sum_c k_c = |\mathbf{k}|$.

Step 6, the inverse descents. The reading word lists the ordinary labels by increasing (b_i, i), that is by increasing height and, within one height, from left to right. Hence $c \in \mathrm{ides}(w(P))$, which means that $c+1$ precedes c in the reading word, holds if and only if $h(c+1) < h(c)$, or $h(c+1) = h(c)$ and $c+1$ lies to the left of c in $W(P)$. It remains to treat the case $h(c+1) = h(c) = h$.

We isolate the mechanism as a separate observation. Let V be a word, let X be a set of entries of V, and let V' be obtained from V by inserting a new entry e immediately to the right of the member of X whose right-to-left index in V is j. Then in V' the members of X retain their relative order, and the right-to-left index of e inside $X \cup \{e\}$ is again j. Indeed, nothing separates e from its anchor, hence the members of X lying to the right of e in V' are exactly those lying strictly to the right of the anchor in V, namely the j members of index $0, \dots, j-1$. Reading V' from the right one meets those j first, then e, then the rest of X. The mirror statement holds with "right" and "left" exchanged throughout. We refer to this as the *index lemma*.

Let $h \geq 1$. Since the labels of B_h are inserted in decreasing order and $c, c+1 \in B_h$ are consecutive integers, $c+1$ is the immediate predecessor of c in the insertion order, hence no label at all is inserted between them. We claim that

$$\mathcal{A}(c) = \mathcal{A}(c+1) \sqcup \{c+1\}. \tag{170}$$

Indeed, by (169) the first constituent of $\mathcal{A}(c)$ is $\{d \in B_h : d > c\} = \{d \in B_h : d > c+1\} \cup \{c+1\}$, the union being disjoint because $c+1 \in B_h$. The second constituents agree, $\{d \in \widehat{B}_{h-1} : d < c\} = \{d \in \widehat{B}_{h-1} : d < c+1\}$, because the two sets can differ only in the element c, and $c \notin \widehat{B}_{h-1}$. Namely, for $h \geq 2$ the height function is single-valued and $c \in B_h$, while for $h = 1$ one has $\widehat{B}_0 = \{0\}$ and $c \geq 1$. Finally $c+1 \notin \mathcal{A}(c+1)$, because the first constituent of $\mathcal{A}(c+1)$ has all its elements strictly greater than $c+1$, and $c+1 \notin \widehat{B}_{h-1}$ by the argument just given with $c+1$ in place of c. This proves (170).

We apply the index lemma with V the word just before $c+1$ is inserted, $X = \mathcal{A}(c+1)$, $e = c+1$ and $j = k_{c+1}$. Since no label is inserted between $c+1$ and c, the word V' it produces is exactly the word just before c is inserted, and by (170) the set $X \cup \{e\}$ is exactly $\mathcal{A}(c)$. Hence at that moment the right-to-left index of the entry $c+1$ inside $\mathcal{A}(c)$ is k_{c+1}.

Write $\mathcal{A}(c) = \{b_0, b_1, \dots, b_r\}$ in that right-to-left order, hence $b_{k_{c+1}} = c+1$. The label c is inserted immediately to the right of b_{k_c}, hence it lies to the right of $b_{k_c}, b_{k_c+1}, \dots, b_r$ and to the left of $b_{k_c-1}, \dots, b_0$, and no later insertion changes the relative order of the entries already present. Therefore $c+1$ lies to the left of c in $W(P)$ if and only if $k_{c+1} \geq k_c$, which is the second clause of the rule.

Let $h \leq 0$. Now the labels of B_h are inserted in increasing order, hence c is the immediate predecessor of $c+1$ in the insertion order and no label is inserted between them. This time

$$\mathcal{A}(c+1) = \mathcal{A}(c) \sqcup \{c\}, \tag{171}$$

by the mirrored computation. Namely, $\{d \in B_h : d < c+1\} = \{d \in B_h : d < c\} \cup \{c\}$ disjointly because $c \in B_h$, while $\{d \in \widehat{B}_{h+1} : d > c+1\} = \{d \in \widehat{B}_{h+1} : d > c\}$ because the two sets can differ only in the element $c+1$, and $c+1 \notin \widehat{B}_{h+1}$. Indeed, for $h \leq -1$ the height function is single-valued and $c+1 \in B_h$, while for $h = 0$ one has $\widehat{B}_1 = B_1 \cup \{n+1\}$, and $c+1 \notin B_1$ for the same reason and $c+1 \leq n < n+1$.

Also $c \notin \mathcal{A}(c)$, both constituents of $\mathcal{A}(c)$ being defined by inequalities strict at c.

By the mirrored index lemma, applied with V the word just before c is inserted, $X = \mathcal{A}(c)$, $e = c$ and $j = k_c$, the left-to-right index of the entry c inside $\mathcal{A}(c+1)$, in the word just before $c+1$ is inserted, is k_c.

Writing $\mathcal{A}(c+1) = \{b'_0, \dots, b'_{r'}\}$ in left-to-right order, hence $b'_{k_c} = c$, the label $c+1$ is inserted immediately to the left of $b'_{k_{c+1}}$, hence it lies to the left of $b'_{k_{c+1}}, \dots, b'_{r'}$ and to the right of

$b'_{k_{c+1}-1},\dots,b'_0$. Therefore $c+1$ lies to the left of c if and only if $k_c \geq k_{c+1}$, which is the third clause of the rule.

In the remaining case $h(c+1) > h(c)$ the label $c+1$ follows c in the reading word, hence $c \notin \mathrm{ides}(w(P))$, in agreement with the rule.

Steps 2 and 3 together give the bijection. Step 2 produces from each $(\tau,\mathbf{k})$ a labelled path whose rooted permutation is τ, Step 3 produces from each labelled path a pair $(\tau,\mathbf{k})$, and each construction undoes the other insertion by insertion. The first formula of (164) now follows by summing (147) over the fibres of the bijection, using Steps 4, 5 and 6 for the three statistics and the descent set. The second formula follows from the first by Theorem 11.5(ii), which identifies $\langle \mathcal{H}^{+}_{n,1}, h^n_1\rangle$ with the plain weight enumerator, together with the independence of the choices k_c, each contributing $[s_\tau(c)]_{qt}$. □

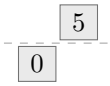
Example 11.25. Let $n=4$ and $\tau = 3\,0\,2\,4\,1$, with maximal increasing runs $3 \mid 0,2,4 \mid 1$. The run containing 0 is the second one, hence by Theorem 11.22 $h_\tau(3)=1$, $h_\tau(2)=h_\tau(4)=0$ and $h_\tau(1)=-1$. Thus $B_1=\{3\}$, $B_0=\{2,4\}$, $B_{-1}=\{1\}$, and $a_-(\tau)=3+1+1=5$, $a_+(\tau)=1$. The bounds (162) are

$$\begin{aligned} s_\tau(3) &= 1 + \#\{d\in B_1 : d>3\} = 1,\\ s_\tau(2) &= 1 + \#\{d\in B_0 : d<2\} + \#\{d\in B_1 : d>2\} = 2,\\ s_\tau(4) &= 1 + \#\{d\in B_0 : d<4\} + \#\{d\in B_1 : d>4\} = 2,\\ s_\tau(1) &= \quad\ \#\{d\in B_{-1} : d<1\} + \#\{d\in B_0 : d>1\} = 2, \end{aligned}$$

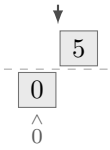
hence $\mathcal{B}_\tau$ has $1\cdot 2\cdot 2\cdot 2 = 8$ elements. We take $\mathbf{k}=(k_1,k_2,k_3,k_4)=(0,1,0,0)$ and run the insertion of Theorem 11.23. The labels enter in the order $3,2,4,1$, that is first height 1, then height 0 in increasing order of the label, then height -1.

The insertion is easiest to follow with the word drawn as a skyline. The j-th entry is a box at horizontal position j, at height equal to its own height, the two sentinels are shaded, and the dashed line separates the entries of height ≥ 1 from those of height ≤ 0, that is the part of the path strictly above the line $y=x$ from the part on or below it.

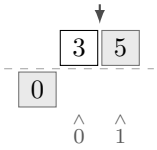
Below each picture the carets mark the anchors $\mathcal{A}(c)$ of the label about to be inserted, numbered in the direction in which Theorem 11.23 counts them, from the right for a label of height ≥ 1 and from the left for a label of height ≤ 0, and the arrow marks the slot chosen by k_c. We start from the two sentinels,



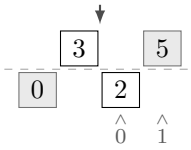
The label 3, of height 1, has the single anchor 0, hence $k_3=0$ and it goes immediately to its right,



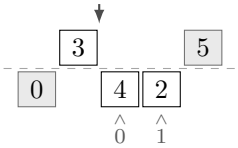
The label 2, of height 0, has the two anchors 3 and 5 counted from the left, and $k_2=1$ puts it immediately left of 5,



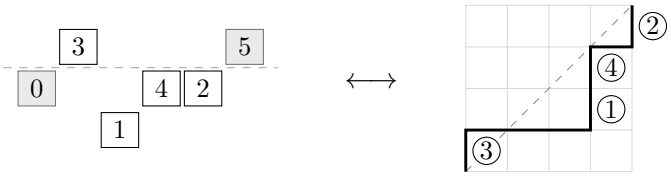
The label 4, also of height 0, has the anchors 2 and 5, and $k_4 = 0$ puts it immediately left of 2,



Finally the label 1, of height -1, has the anchors 4 and 2, and $k_1 = 0$ puts it immediately left of 4,



Deleting the sentinels from the completed word leaves $\sigma = (3, 1, 4, 2)$ with heights $b = (1, -1, 0, 0)$, and the skyline is the sequence of heights of the labelled path of Theorem 11.2 read from bottom to top,



The four assertions of Theorem 11.23 can be read off against that example. Namely $a_-(\tau) = 5 = A_-^{(1)}$, $a_+(\tau) = 1 = A_+^{(1)}$, $|\mathbf{k}| = 1 = D(P)$, and $\mathcal{D}_\tau(\mathbf{k}) = \{3\} = \mathrm{ides}(w(P))$, the last because $h_\tau(4) = 0 < 1 = h_\tau(3)$ while the pairs $(1, 2)$ and $(2, 3)$ have $h_\tau(2) > h_\tau(1)$ and $h_\tau(3) > h_\tau(2)$.

The two extreme sectors of Theorem 11.23 are the type-A formula of Theorem 11.24, and the dictionary is exact.

Proposition 11.26. *Let $n \geq 1$.*

(1) Let $\tau \in S_{\{0,\dots,n\}}$ have $\tau_{n+1} = 0$, and let $\tau' \in S_n$ be τ with the 0 deleted. Then $h_\tau(c) \geq 1$ for every c, and

$$s_\tau(\tau'_i) = w_i(\tau') \quad (1 \leq i \leq n), \qquad a_-(\tau) = 0, \qquad a_+(\tau) = 2\,\mathrm{maj}(\tau') + n,$$

with w_i as in (165). Conversely $a_-(\tau) = 0$ forces $\tau_{n+1} = 0$.

(2) The bijection of Theorem 11.23 carries $\bigsqcup_{\tau_{n+1}=0} \mathcal{B}_\tau$ onto $\mathcal{P}_n := \{P \in \mathsf{PF}^+_{n,1} : b_i(P) \geq 1 \text{ for all } i\}$, which is the set of type-A parking functions. On $\mathcal{P}_n$ one has $D = \mathrm{dinv}$, $A_-^{(1)} = 0$ and $A_+^{(1)} = 2\,\mathrm{area} + n$, and $\mathrm{ides}(w(P))$ is the complement in $\{1, \dots, n-1\}$ of the inverse descent set of Haglund's reading word $\mathrm{read}(P)$.

(3) Hence

$$\sum_{\substack{\tau \in S_{\{0,\dots,n\}} \\ \tau_{n+1}=0}} q^{a_-(\tau)} t^{a_+(\tau)} \prod_{c=1}^{n} [s_\tau(c)]_{qt} = t^n \Big(\sum_{P \in \mathcal{P}_n} q^{\mathrm{dinv}(P)} t^{\mathrm{area}(P)} \Big) \Big|_{q \mapsto qt,\ t \mapsto t^2},$$

and the sector $\tau_1 = 0$ contributes the image of this under $q \leftrightarrow t$.

Proof. (1) If $\tau_{n+1} = 0$ then $\tau_n > 0$, hence 0 is a maximal run by itself, $s = r$ and $R_s = \{0\}$. Deleting 0 leaves the runs $R_1 | \dots | R_{s-1}$ of τ', and $h_\tau(c) = s - j \geq 1$ for $c \in R_j$, $j \leq s - 1$. For $h \geq 2$ we have $B_h = R_{s-h}$ and $B_{h-1} = R_{s-h+1}$, the run of τ' immediately to the right of B_h, hence the first line of (162) is (165).

For $h = 1$ we have $B_1 = R_{s-1}$, the last run of τ', whose successor in Haglund's convention is the $(k+2)$nd run $\{0\}$. Since $0 < c$ for every c, his second count equals 1, which is the constant in the second line of (162). There is no c with $h_\tau(c) \leq 0$, hence $a_-(\tau) = 0$ and $a_+(\tau) = \sum_c (2h_\tau(c) - 1) =$

$2\sum_c h_\tau(c) - n$. For $c \in R_j$ the number $h_\tau(c) - 1 = (s-1) - j$ is exactly the length of the row of c in cars(τ'), whose total is maj(τ') [35, p. 81]. Hence $a_+(\tau) = 2(\mathrm{maj}(\tau') + n) - n$.

Conversely $a_-(\tau) = 0$ means $B_h = \emptyset$ for every $h \le 0$, that is $R_j = \emptyset$ for $j > r$ and $R_r = \{0\}$. Runs are nonempty, hence $r = s$ and $\tau_{n+1} = 0$.

(2) The statistic $A_-^{(1)}(P) = \sum_i([1-b_i]_+ + [-b_i]_+)$ vanishes iff $b_i \ge 1$ for all i, and $A_-^{(1)}(P) = a_-(\tau)$, hence the sector is carried onto the set of P with all $b_i \ge 1$. Such a P is a path from $(0,0)$ to (n,n) weakly above $y = x$ together with a labelling increasing up the columns, that is a type-A parking function, and its i-th row has length $a_i = b_i - 1$. Haglund defines

$$\mathrm{dinv}(P) = \#\{i < j : a_i = a_j,\ \mathrm{occ}(i) < \mathrm{occ}(j)\} + \#\{i < j : a_i = a_j + 1,\ \mathrm{occ}(i) > \mathrm{occ}(j)\}$$

[35, pp. 78–79], the rows being indexed so that a larger index means a higher row, and his gloss is "pairs of rows of P of the same length, with the row above containing the larger car, or which differ by one in length, with the longer row below the shorter, and the longer row containing the larger car".

Substituting $a_i = b_i - 1$ and $\mathrm{occ}(i) = \sigma_i$ turns this into (148), term for term. Next, $A_+^{(1)}(P) = \sum_i(b_i + [b_i - 1]_+) = \sum_i(2b_i - 1) = 2\sum_i(b_i - 1) + n$ and $\mathrm{area}(P) = \sum_i(b_i - 1)$.

Finally, read(P) is obtained by reading the cars "along diagonals in a SE direction, starting with the diagonal farthest from the line $y = x$, then working inwards" [35, p. 79], that is by decreasing (b_i, i), whereas $w(P)$ reads by increasing (b_i, i). The two words are reverses of each other. For any word u and any c, exactly one of "$c+1$ lies left of c in u" and "$c+1$ lies left of c in the reverse of u" holds, hence the two inverse descent sets are complementary in $\{1, \dots, n-1\}$.

(3) We substitute (1) into the second identity of (164) and use (2) with $\langle F_{n,S}, h_1^n \rangle = 1$. For the last sentence, the map which reverses τ and then replaces each nonzero c by $n+1-c$ is an involution of $S_{\{0,\dots,n\}}$ exchanging the two sectors, and by Theorem 11.14 the corresponding involution $P \mapsto P^c$ of $\mathsf{PF}_{n,1}^+$ fixes D and exchanges $A_-^{(1)}$ with $A_+^{(1)}$. □

Because Theorem 11.23 tracks ides and not only the bidegree, the same computation refines (166) to the quasisymmetric level. For $\tau \in S_n$ with runs $R_1 | \dots | R_{k+1}$ write $a_\tau(c) = k + 1 - j$ for $c \in R_j$, the length of the row of c in cars(τ), and $w_\tau(c) = w_i(\tau)$ for $\tau_i = c$.

Corollary 11.27. *For every $n \ge 1$,*

$$\sum_{P \in \mathcal{P}_n} q^{\mathrm{dinv}(P)} t^{\mathrm{area}(P)} F_{n,\mathrm{Des}(\mathrm{read}(P)^{-1})}(X) = \sum_{\tau \in S_n} t^{\mathrm{maj}(\tau)} \sum_{\mathbf{k} \in \mathcal{B}_\tau^A} q^{|\mathbf{k}|} F_{n,E_\tau(\mathbf{k})}(X), \tag{172}$$

where $\mathcal{B}_\tau^A = \prod_{c=1}^n \{0, 1, \dots, w_\tau(c) - 1\}$ and $c \in E_\tau(\mathbf{k})$ iff $a_\tau(c+1) > a_\tau(c)$, or $a_\tau(c+1) = a_\tau(c)$ and $k_c > k_{c+1}$. By the shuffle theorem [19] both sides equal ∇e_n.

Proof. We apply Theorem 11.26 to the first identity of (164) restricted to the sector $\tau_{n+1} = 0$, and we complement both descent sets. The substitution $h_\tau(c) = a_{\tau'}(c) + 1$ turns the three clauses defining $\mathcal{D}_\tau(\mathbf{k})$ in Theorem 11.23 into two, the third being vacuous on this sector, and complementing gives E_τ. □

We did not find (172) in the literature. Every schedule identity in [35, 37, 38] is a statement about $q^{\mathrm{dinv}} t^{\mathrm{area}}$ alone, and a search for a quasisymmetric or Gessel-F refinement of [35, Thm. 5.3] returned only the Hilbert-series form. The descent rule $E_\tau(\mathbf{k})$ is what the monomial $M_{\tau,\mathbf{k}}$ contributes, hence (172) is the quasisymmetric shadow of the statement that the Carlsson–Oblomkov basis [20] carries the S_n-character and not only the bigraded dimension. The type-C statement is Theorem 11.30.

The intermediate sectors are governed by the third line of (162), which couples the labels on the two sides of the root. They all carry the same (D, ides) distribution, by the circular-street argument which Haglund uses to count parking functions [35, p. 77].

Proposition 11.28. *Identify* $\mathsf{PF}^{+}_{n,1}$ *with the set of functions* $f\colon\{1,\dots,n\}\to\{0,1,\dots,n\}$ *by sorting the pairs* $(f(c),c)$ *lexicographically into* (e_i,σ_i)*, and let* $\varsigma(f)(c)=f(c)+1 \bmod (n+1)$*. Then* ς *fixes* $w(P)$ *and* $D(P)$*, and each* ς*-orbit has* $n+1$ *elements, one in each sector* $\mathcal{S}_p=\{P:\#\{i:b_i\geq 1\}=p\}$*,* $0\leq p\leq n$*. Consequently*

$$\sum_{P\in\mathcal{S}_p}(qt)^{D(P)}F_{n,\mathrm{ides}(w(P))}=\sum_{\tau\in S_n}\sum_{\mathbf{k}\in\mathcal{B}^A_\tau}(qt)^{|\mathbf{k}|}F_{n,\{1,\dots,n-1\}\setminus E_\tau(\mathbf{k})}(X)$$

for every p*, independently of* p*, and by Theorem 11.27 this is* $(\omega\nabla e_n)\big|_{t=1,\ q\mapsto qt}$*.*

Proof. Write $a=\#f^{-1}(n)$, hence $e_{n-a+1}=\cdots=e_n=n$ and $\sigma_{n-a+1}<\cdots<\sigma_n$. Applying ς moves these a cars to the value 0 and adds 1 to the remaining $n-a$ values, hence

$$e(\varsigma f)=(\underbrace{0,\dots,0}_{a},e_1+1,\dots,e_{n-a}+1),\qquad \sigma(\varsigma f)=(\sigma_{n-a+1},\dots,\sigma_n,\sigma_1,\dots,\sigma_{n-a}),$$

which is again a labelled path. Its heights are $b'_i=i=b_{n-a+i}+a$ for $i\leq a$ and $b'_i=b_{i-a}+a-1$ for $i>a$. Thus every height rises by $a-1$, and the a cars that wrapped around rise by one more.

A pair of cars both inside the wrapped block, or both outside it, therefore keeps its height difference and its relative order, and it contributes the same to (148). For a pair consisting of a wrapped car, with old index $i'>n-a$, and an unwrapped one, with old index $j'\leq n-a$, the order reverses and the difference increases by one, $b'_{i'}-b'_{j'}=b_{i'}-b_{j'}+1$ with i' now preceding j'.

The old pair (j',i') is counted by the first clause of (148) iff $b_{j'}=b_{i'}$ and $\sigma_{j'}<\sigma_{i'}$, which is exactly the condition for the new pair (i',j') to be counted by the second clause, namely $b'_{i'}=b'_{j'}+1$ and $\sigma_{i'}>\sigma_{j'}$, and symmetrically the second clause for (j',i') becomes the first for (i',j'). Thus D is preserved pair by pair.

The reading word is unchanged. Two wrapped cars, or two unwrapped ones, keep their relative position in the order by increasing (b_i,i).

For a wrapped car with new index $i\leq a$ and an unwrapped one with new index $l+a$, the comparison is between $(i-a,n-a+i)$ and (b_l,l) before the shift and between (i,i) and $(b_l+a-1,l+a)$ after it. Before the shift the wrapped car precedes the other iff $i-a<b_l$, the tie $i-a=b_l$ going the other way because $l\leq n-a<n-a+i$. After the shift it precedes iff $i<b_l+a-1$, or $i=b_l+a-1$, the tie now going its way because $i\leq a<l+a$. That is, it precedes iff $i\leq b_l+a-1$, which is again $i-a<b_l$.

For the sectors, let $m_j=\#f^{-1}(j)$ and $W(k)=\sum_{l<k}(m_l-1)$ for $0\leq k\leq n+1$, hence $W(0)=0$ and $W(n+1)=-1$, and extend W to $\mathbb{Z}$ by $W(k+n+1)=W(k)-1$. Since $e_i\leq i-1$ iff at least i of the values of f are $\leq i-1$, that is iff $m_0+\cdots+m_{i-1}\geq i$, that is iff $W(i)\geq 0$, we have

$$p(f)=\#\{i:b_i\geq 1\}=\#\{i\in\{1,\dots,n\}:W(i)\geq 0\}=\#\{k\in\{1,\dots,n+1\}:W(k)\geq 0\},$$

the last step because $W(n+1)=-1$.

Now ς rotates $(m_0,\dots,m_n)$, and the translates of f are the functions whose W-walks are $i\mapsto W(k+i)-W(k)$ for $0\leq k\leq n$, the translate $\varsigma^{-k}f$ corresponding to k. Hence

$$\begin{aligned}p(\varsigma^{-k}f)&=\#\{i\in\{k+1,\dots,k+n+1\}:W(i)\geq W(k)\}\\&=\#\{i>k:W(i)\geq W(k)\}+\#\{i<k:W(i)>W(k)\},\end{aligned}$$

where in the second expression i runs over $\{0,\dots,n\}$. The terms with $i\leq k$ have been reduced modulo $n+1$, which lowers W by one, and the term $i=k$ contributes $W(k+n+1)=W(k)-1<W(k)$, that is nothing.

The right-hand side is the number of indices strictly greater than k in the total order on $\{0,\dots,n\}$ in which $i\prec k$ iff $W(i)<W(k)$, or $W(i)=W(k)$ and $i<k$. As k runs over $\{0,\dots,n\}$ that number runs over $\{0,\dots,n\}$ bijectively, which is the assertion about the orbits. In particular all orbits are free.

For the displayed identity, we apply what has just been proved to the sector $p = n$, which is $\mathcal{P}_n$, and we use Theorem 11.26(2) and Theorem 11.27 at $t = 1$, the reading word $w(P)$ being the reverse of read(P). The identification with $\omega\nabla e_n$ uses $\omega F_{n,S} = F_{n,\{1,\dots,n-1\}\setminus S}$ for the F-expansion of a symmetric function, which follows from $s_\lambda = \sum_Q F_{n,\mathrm{Des}(Q)}$ over the standard Young tableaux of shape λ, already used in the proof of Theorem 11.5, together with $\omega s_\lambda = s_{\lambda'}$ and $\mathrm{Des}(Q^{\mathsf{t}}) = \{1,\dots,n-1\} \setminus \mathrm{Des}(Q)$. □

Remark 11.29 (Why $m = 1$). Theorem 11.23 is stated at $m = 1$ only, and the obstruction at $m \geq 2$ is the one which Loehr and Remmel record. Indexing by a permutation works in type A because the word $S_n|S_{n-1}|\dots|S_1|S_0$ formed from the diagonals, increasing inside each block, has two properties. Namely, $S_j = \varnothing$ forces $S_l = \varnothing$ for all $l > j$, and the largest element of S_j exceeds the smallest element of S_{j-1} whenever both are nonempty. Together these say that the descents of the word fall exactly at the bars, hence the bars are recoverable from the permutation.

Loehr and Remmel observe that "unfortunately, the two properties above are no longer guaranteed in the case where $k > 0$ or $m > 1$" [58, p. 32], they replace the permutation by an arbitrary function $f\colon \{1,\dots,n\} \to \{0,\dots,k+m(n-1)\}$, and they arrive at a formula [58, Thm. 55] whose summand $q^{\mathrm{maj}(f)}t^{x_0(f)}\prod_j t^{x_j(f)}[\mathrm{count}(f,j)]_t$ carries correction exponents $x_j(f) \leq 0$ and is not a monomial times a product of brackets.

Their two properties are exactly the two halves of Step 1 above. The implication $B_h \neq \varnothing \Rightarrow B_{h-1} \neq \varnothing$ for $h \geq 2$ and its mirror say together that $\{b_i\} \cup \{0\}$ is an interval of consecutive integers, and (167), (168) say that the junctions are descents. Both are proved from $b_{i+1} \leq b_i + 1$, that is from $m = 1$.

At $m \geq 2$ only $b_{i+1} \leq b_i + m$ is available and the first property already fails. The smallest example is $(n,m) = (2,2)$ with $e = (0,0)$, whose heights are $b = (2,4)$, and no permutation of $\{0,1,2\}$ has nonempty runs at the levels 2 and 4 and nothing between.

For the m-analogue in type A see [35, Ch. 6, pp. 107–112], in particular Conjectures 6.30 and 6.34 there, and Haiman's observation $\mathrm{dinv}_m(P) = \mathrm{dinv}_1(P^{(m)})$ [35, (6.70)], which realizes dinv_m by magnifying each north step m times. The magnified object is not an arbitrary path, hence it does not transport the schedule formula.

A second point of contact is [35, Open Problem 5.4], which asks for a proof, ideally bijective, that $\sum_{P\in\mathcal{P}_n} q^{\mathrm{dinv}(P)}t^{\mathrm{area}(P)}$ is symmetric in q,t. Haglund notes in [35, Remark 5.5] that, by (166), setting either variable to 0 already reduces the symmetry to the fact that inv and maj are both Mahonian, hence a bijective proof may have to generalize Foata's map.

In type C the corresponding symmetry does have an involutive proof, Theorem 11.14, since $P \mapsto P^c$ fixes $\mathrm{dinv}_m^{\mathrm{lab}}$ and exchanges $A_{\mp}^{(m)}$. This says nothing about the type-A problem, because the involution does not preserve the sectors. It exchanges $\mathcal{S}_p$ with $\mathcal{S}_{n-p}$, and no single sector is symmetric, since at $n = 2$ the sector $p = 0$ contributes $q^4 + q^3t + q^2$. The complement of a path in the square is again a path in the square, while the complement of a Dyck path is not a Dyck path, and that is the whole of the difference.

11.5. **The basis conjecture.** In this subsection we formulate the basis conjecture. The main idea is that the rooted monomials of Theorem 11.22 are the type-C counterpart of the monomials of Carlsson and Oblomkov, with the root in the place of the diagonal.

Let $\mathcal{P}_n = \mathbb{Q}[x_1,\dots,x_n,y_1,\dots,y_n]$ with $W_n = W(B_n)$ acting diagonally, $\mathfrak{b}_n = \langle(\mathcal{P}_n^{W_n})_+\rangle$, $DR(B_n) = \mathcal{P}_n/\mathfrak{b}_n$, and let $DR(B_n)^\chi$ be the isotypic component of the character $\chi(\varepsilon) = \prod_i \varepsilon_i$ of G_n, which is spanned by the classes of the monomials with all $a_i + b_i$ odd. Stump's surjection Theorem 9.1 restricts on this component to a bigraded surjection

$$DR(B_n)^\chi \twoheadrightarrow DH_n^+ = \bigl(\mathrm{gr}_F L^{(1)}\bigr)^{G_n}, \qquad x^\alpha y^\beta \longmapsto [x^\alpha D^\beta \bar{\Delta}], \tag{173}$$

where D_i are the Dunkl operators and $\bar{\Delta}$ is the image of the type-B Vandermonde. By (163) every $M_{\tau,\mathbf{k}}$ lies in the odd component.

Conjecture 11.30 (Rooted basis)**.** *For every $n \geq 1$ the $(n+1)^n$ classes $[x^\alpha D^\beta \bar{\Delta}]$, where $x^\alpha y^\beta$ runs over the rooted monomials (163), form a basis of DH_n^+, the class of $M_{\tau,\mathbf{k}}$ lying in bidegree $(a_-(\tau)+|\mathbf{k}|,\, a_+(\tau)+|\mathbf{k}|)$.*

Conjecture 11.31 (Rooted subbasis)**.** *For every $n \geq 1$ the classes of the $M_{\tau,\mathbf{k}}$ in $DR(B_n)^\chi$ are linearly independent, that is* $\mathrm{Span}\{M_{\tau,\mathbf{k}}\} \cap \mathfrak{b}_n = 0$.

Theorem 11.30 implies Theorem 11.31, since a relation among the classes in $DR(B_n)^\chi$ maps to a relation in DH_n^+ under (173). The converse fails to be automatic, and the two differ already at $n = 4$. The dimension of $DR(B_4)^\chi$ is $626 = 5^4 + 1$, in accordance with the excess $\dim DR(B_4) = 9^4 + 1$ of [39] and [1], hence the rooted classes cannot span there, while they do span DH_4^+.

Proposition 11.32. *Theorem 11.30 implies that the bigraded Hilbert series of DH_n^+ is the weight enumerator $\langle \mathcal{H}_{n,1}^+, h_1^n \rangle$ of (164), which is the Hilbert series predicted by (161). If in addition the filtration of DH_n^+ by the rooted index is compatible with the residual S_n-action in the sense that the associated graded pieces have Frobenius character $F_{n,\mathcal{D}_\tau(\mathbf{k})}$, then (161) holds at $m = 1$.*

Proof. The first assertion is Theorem 11.23 together with the bidegree in (163). The second is the first form of (164). □

For $n \leq 4$ the Frobenius character of DH_n^+ was computed directly (Theorem 11.34(8)), hence the conclusion of the second assertion holds there without any knowledge of the filtration. The compatibility of the rooted filtration with the S_n-action remains open in every rank.

Remark 11.33 (Relation to the type-A basis, and what a proof would need). Carlsson and Oblomkov [20, Thm. B] prove that the monomials $N_{\pi,\mathbf{k}} = g_\pi^A(\mathbf{v}) \prod_c u_c^{k_c}$, $\pi \in S_n$ and $0 \leq k_c < s_\pi^A(c)$, form a basis of the type-A diagonal coinvariants $\mathbb{Q}[\mathbf{u},\mathbf{v}]/\langle (\mathbb{Q}[\mathbf{u},\mathbf{v}]^{S_n})_+ \rangle$, where g_π^A is the Garsia–Stanton descent monomial and s_π^A the type-A schedule. The two extreme rooted sectors are that basis, before any quotient, namely

$$M_{(\pi,0),\mathbf{k}} = (y_1 \cdots y_n)\, N_{\pi,\mathbf{k}}(\mathbf{u} = \mathbf{xy},\ \mathbf{v} = \mathbf{y}^2), \qquad M_{(0,\pi),\mathbf{k}} = (x_1 \cdots x_n)\, \mathcal{R}\big[N_{\pi^*,\mathbf{k}^*}(\mathbf{u} = \mathbf{xy},\ \mathbf{v} = \mathbf{x}^2)\big],$$

with $\mathcal{R}$ the reversal $c \mapsto n+1-c$ of the variable labels and $\pi^*, \mathbf{k}^*$ its effect on the index.

The substitutions descend to S_n-equivariant linear maps $DR_n^A \to DR(B_n)^\chi$. The step degree is $(1,1)$ here and $(1,0)$ there, which is why qt and not q appears in (164). The one-body boundary module $M_1/(xy)M_1 = x\mathbb{C}[x^2] \oplus y\mathbb{C}[y^2]$ has one free direction on each side of the root.

The proof in [20, §7.3] is not a count of schedules. It filters the quotient by the descent order, identifies each associated graded piece with a principal ideal in the coinvariant algebra through an affine Springer fibre, and proves the required non-vanishing of fixed-point coefficients by a Hessenberg evaluation.

That mechanism does not transfer unchanged, because the span of the rooted classes in $DR(B_4)^\chi$ is not a module over $\mathbb{Q}[z_1, \ldots, z_n]$, $z_c = x_c y_c$. Explicitly, for $\tau = (1,2,0,3,4)$ and $\mathbf{k} = 0$ one has $M_{\tau,0} = y_1 y_2 x_3 x_4$, and in bidegree $(3,3)$, where the odd component has dimension 18 and the rooted classes number 17, the product $z_3 M_{\tau,0} = x_3^2 x_4 y_1 y_2 y_3$ is the one class outside their span.

A proof of Theorem 11.30 therefore requires either a different filtration of DH_n^+ or an independently defined quotient in which the rooted classes are visibly a basis.

11.6. **Evidence.** In this subsection we record the computational evidence for the statements of this section, in the sense of Theorem 7.3. Every finite assertion was checked by an implementation of the definitions in exact arithmetic, independently of the proofs.

Theorem 11.34. *The following hold, in the sense of Theorem 7.3.*

(1) *Theorem 11.5 and Theorem 11.19 hold for all n, m.*

(2) *Theorem 11.15 holds for all n, m. Its two harder assertions were also checked directly for (n, m) with $n \le 5$, $m \le 3$ and $mn \le 8$, together with $[s_n]\mathcal{H}^+_{n,m} = \widehat{P}_{n,m}$ and $\mathcal{H}^+_{n,m}(X; 1, 1) = h_n[(mn+1)X]$. Every clause of Theorem 11.14 was verified for every labelled path at (n, m) with $n \le 5$, $m \le 3$ and $mn \le 9$, as was the invariance of the multiset $\{\mathrm{Des}(Q) : Q \in \mathrm{SYT}(\lambda)\}$ under $S \mapsto S^{\mathrm{rev}}$ for all $\lambda \vdash n \le 7$.*

(3) *Theorem 11.9 was checked independently of its proof, by expanding both sides of $L_e = G_{\boldsymbol{\nu}(e)}$ into monomials in n variables and comparing coefficient by coefficient, for every path e of the rectangle at $(n, m) \in \{(2, m)_{m\le 4}, (3, m)_{m\le 3}, (4, m)_{m\le 3}, (5, m)_{m\le 2}\}$. The comparison was run in the two mirror conventions for the inversion rule and with $c = \pm q_i$. Only the normalization of Theorem 11.6 and its mirror image reproduce L_e, and they do so for every path. Separately, each z-coefficient of each L_e is symmetric and Schur positive in the same range, as Theorem 11.15 requires.*

(4) *Theorem 11.10 was checked on monomial expansions for every path at (n, m) with $n \le 4$, $m \le 3$ and $mn \le 6$, and Theorem 11.11 for every path at (n, m) with $n \le 5$, $m \le 3$ and $mn \le 9$, by constructing the abacus of μ from the charges and recovering $\mathrm{core}_k(\mu)$ and $\mathrm{quot}_k(\mu)$ from it.*

The shape μ/ν itself was then tested against ribbons. For every path at (n, m) with $n \le 4$, $m \le 3$ and $mn \le 6$ the standard k-ribbon tableaux of μ/ν were enumerated directly on the shapes, without using the quotient, and in every case the ribbon additions agree with the bead moves $p \mapsto p + k$, each ribbon's height equals 1 plus the number of beads strictly between p and $p + k$, the content of each ribbon equals the adjusted content $k\,c(x) + s_i$ of the corresponding cell of $\mathrm{quot}_k(\mu)$, the number of such tableaux is the multinomial $\binom{n}{l_1,\dots,l_k}$, and $\mathrm{sp}(T) + \mathrm{inv}(\mathrm{quot}_k(T))$ is constant, as [36, Lem. 5.2.2] requires.

Summed over paths this produced 625 standard ribbon tableaux at $(4, 1)$ and 343 at $(3, 2)$, in each case $(mn+1)^n$ in total.

The constraint $\sum_i a_i = 0$ used in Theorem 11.11 was itself checked in both directions, without imposing it. The charge vector of the k-core of every partition of size at most 16 has sum 0 for $k = 2, 3, 4$, and conversely every integer vector of sum 0 with entries in $[-2, 2]$ is the charge vector of a k-core for $k \le 5$. The rotation is not cosmetic. The offsets already satisfy $\sum_s d_s \equiv 0 \pmod k$ for only 93 of the 252 paths at $(n, m) = (5, 1)$ and 183 of the 495 at $(4, 2)$.

(5) *The identity (164), including the descent rule of Theorem 11.23, was verified coefficient by coefficient in the fundamental basis for $n \le 5$.*

Independently, every step of the proof of Theorem 11.23 was checked mechanically for $n \le 6$. For each of the $(n+1)^n$ pairs $(\tau, \mathbf{k})$, the number of admissible slots at each insertion equals the corresponding value of (162), the resulting word satisfies the local rule and has the heights prescribed by τ, its statistics are $a_-(\tau)$, $a_+(\tau)$ and $|\mathbf{k}|$, its inverse descent set is $\mathcal{D}_\tau(\mathbf{k})$, the deletion procedure of Step 3 returns $(\tau, \mathbf{k})$, and the $(n+1)^n$ words produced are pairwise distinct and exhaust $\mathsf{PF}^+_{n,1}$.

The intermediate assertions of Step 6 were checked in the same range and not merely its conclusion. For each of the 197,860 pairs $(c, c+1)$ of equal height arising at $n = 6$, the identity (170) or (171) holds, the two labels are consecutive in the insertion order, and the index supplied by the index lemma is the one claimed.

(6) *The comparisons with type A were checked mechanically in the same way, for $n \le 6$. These are the identities $s_\tau(\tau'_i) = w_i(\tau')$, $a_-(\tau) = 0$ and $a_+(\tau) = 2\,\mathrm{maj}(\tau') + n$ of Theorem 11.26(1) on the sector $\tau_{n+1} = 0$, and their mirrors on $\tau_1 = 0$, the identities $D = \mathrm{dinv}$, $A^{(1)}_+ = 2\,\mathrm{area} + n$ and the complementarity of $\mathrm{ides}(w(P))$ with the inverse descent set of $\mathrm{read}(P)$ of Theorem 11.26(2), on all $(n+1)^{n-1}$ type-A parking functions, 16,807 of them at $n = 6$, and the two assertions of Theorem 11.28, that ς fixes the reading word and D and that every*

orbit is free and meets each sector exactly once, on all $(n+1)^n$ *functions,* 117,649 *of them at* $n = 6$.

The sector identity of Theorem 11.26(3) and (172) *were verified coefficient by coefficient for* $n \le 5$, *the latter in the fundamental basis.*

(7) *Theorem 11.16 was verified symbolically at* $n = 2$ *for* $m \le 3$, *directly from the wreath basis of Theorem 10.14. The involution* inv *is needed there. Omitting it leaves the* s_2*-coefficient, which is* $P_{n,m}$ *and hence insensitive to* inv, *but gives* $[s_{11}] = (q^2+qt+t^2)(q^4+qt+t^4)$ *at* $m = 1$ *in place of* q^2+qt+t^2.

For $n \le 6$ *and* every $m \ge 1$ *the conjecture follows from two fixed-rank certificates, namely the operator form* (159), *which is proved for all* $m \ge 1$ *and* $n \le 6$ *by an exact residue certificate over* $\mathbb{Q}(q,t)$ *for the face expansion of the path side, and the cancellation* (160), *verified symbolically for* $n \le 6$, *which by Theorem 11.20 makes the two forms equivalent in a fixed rank. The cancellation was also checked at rational points for* $n = 7, 8$. *Identity* (159) *was in addition reproduced symbolically, by an independent implementation, for* $(n,m) \in \{(2,m)_{m\le 3}, (3,m)_{m\le 2}, (4,m)_{m\le 2}\}$.

(8) (161) *at* $m = 1$ *and Theorem 11.30 were verified for* $n \le 4$, *in every bidegree* (a,b) *with* $a \le b$. *The computation realizes* $L^{(1)} = \mathbb{C}[x]/J$ *with* J *the ideal generated by* $J_{2n+1} = \{f \in \mathbb{C}[x]_{2n+1} : D_i f = 0 \ \forall i\}$, *which is the radical of* $M(\mathrm{triv})$ *in degree* $2n+1$ *because the radical vanishes in degrees* $\le 2n$.

Since J *is contained in the radical,* J *coincides with the radical in the degrees* $\le n^2$ *used by the verification if and only if* $\dim L^{(1)}_d = [q^d]\,[2n+1]_q^n$ *for* $d \le n^2$, *the graded form of* $\dim L^{(1)} = (2n+1)^n$ [32]. *The quotient* $\mathbb{C}[x]/J$ *was checked to have exactly these dimensions in every degree* $\le n^2$, *and for* $n \le 2$ *the graded form itself was confirmed in all degrees by the ranks of the contravariant form. The verification therefore assumes the graded form for* $n = 3, 4$.

For $n \le 3$ *the linear algebra is exact over* $\mathbb{Q}(q,t)$, *and for* $n = 4$ *it is exact modulo the prime* $2^{31}-1$ *with the* S_4*-characters lifted to integers, which suffices because the Schur coefficients being compared are polynomials with small integer coefficients.*

The bidegrees with $a > b$ *follow from the Fourier automorphism* $x_i \mapsto y_i$, $y_i \mapsto -x_i$ *of* H_c, *which exists because* $\mathfrak{h} \cong \mathfrak{h}^*$ *as* $W(B_n)$*-modules and the parameters are equal. It carries* $L(\mathrm{triv})$ *to a finite-dimensional simple module whose lowest weight space is the top degree* $2n^2$ *of* $L(\mathrm{triv})$, *one-dimensional and trivial, hence to* $L(\mathrm{triv})$ *itself.*

It fixes $\bar{\Delta}$ *up to a scalar and exchanges the two PBW degrees, and since reordering* $y^\beta x^\alpha$ *into PBW form only produces terms of lower filtration degree it acts on* $\operatorname{gr} L^{(1)}$ *by* $[x^\alpha D^\beta \bar{\Delta}] \mapsto \pm c\,[x^\beta D^\alpha \bar{\Delta}]$. *Finally the rooted family is stable under* $x \leftrightarrow y$ *composed with* $c \mapsto n+1-c$ *(Theorem 11.33).*

At $n = 4$ *the* 625 *rooted classes are independent in* $DR(B_4)^\chi$, *and the single class they miss is* $q^3t^3 s_{1111}$, *which is the kernel of* (173).

(9) *Theorem 11.17 was verified independently of its proof by computing the wreath Macdonald basis from the triangularities of* (129), *symbolically over* $\mathbb{Q}(q,t)$ *for* $n \le 3$, *and at the rational points* $(q,t) = (5,7)$ *and* $(3/2, 7/5)$ *for* $n = 4$. *In each case* (156) *was checked on every one of the* $\dim \Lambda^{(2)}_n$ *coefficients of every* $\widetilde{H}_\lambda$, *not only on the pure parts.*

The computation reproduces Theorem 10.14 coefficient by coefficient, which is what fixes the two conventions that matter, namely which component of quot_2 *carries the alphabet* X_0 *and the direction of the colour rotation. The one ingredient of the proof taken from the literature was checked against the same computation. For all* 17 *diagrams* $\lambda \in \mathrm{Bal}(2n)$ *with* $n \le 3$, *both pairings* $\langle \widetilde{H}_\lambda, e_n[X_i] \rangle$ *equal the product of the box weights* $q^a t^b$ *over the cells of* λ *of residue* i, *which is* [67, Thm. 4.14] *at* $r = 2$ *in these conventions.*

(10) *Theorem 11.31 holds for* $n \le 3$ *by direct computation of* $DR(B_n)^\chi$, *and for* $n = 4$ *by* (8).

Remark 11.35 (Geometric evidence). The type-A proof of [20, Thm. B] computes the associated graded pieces of the descent filtration as cohomology of Hessenberg strata of the affine Springer fibre of $\mathfrak{gl}_n$ at slope $\frac{n+1}{n}$. The corresponding type-C object is the affine Springer fibre Fl_γ of Sp_{2n} at slope $m+\frac{1}{2n}$, for γ homogeneous and elliptic, whose $\mathbb{G}_m(\nu)$-fixed points are the alcoves of a dilated alcove. Namely, the affine roots of weight $2mn+1$ are exactly $\alpha_i+m\delta$, $0\le i\le n$, hence the fixed-point set is the set of alcoves of $\{x:\alpha_i(x)+m>0\}$, a translate of $(2mn+1)A_0$, of cardinality $(2mn+1)^n$.

Granting the affine paving of an equivalued element, in which the attracting cell at a fixed point is an affine space of the dimension of its Zariski tangent space, the following were computed. The number of cells of dimension $D-i$ equals $\dim DR(B_n)_{\deg_x=i}$. For $n=2$ this is $(8,8,5,3,1)$ against the cell profile $1,3,5,8,8$, and for $n=3$ the four computable slices are $(48,72,66,56)$ against $\ldots,56,66,72,48$. The number of cells of top dimension D is $\dim DR(B_n)_{\deg_x=0}=|W(B_n)|=2^n n!$, and there is a unique cell of dimension zero.

Moreover the terms of (147) with $\mathrm{ides}(w(P))=\varnothing$, which by Theorem 11.5(i) are the terms of $\widehat{P}_{n,m}$, have cell-dimension profile equal to that of the affine Springer fibre Gr_γ in the affine Grassmannian, at $(n,m)=(2,1),(3,1),(2,2),(3,2)$. These are the type-$C$ shadows of Hikita's theorem [47] in type A, and they support the geometric route to Theorem 11.30.

They also delimit it. For $(n,m)=(2,1)$ and $(3,1)$ there is no closed union of cells of Fl_γ consisting of $(mn+1)^n$ cells with the dimensions $D-A_-^{(m)}-\mathrm{dinv}_m^{\mathrm{lab}}$ read off from (147). Those dimensions require $n!$ cells of top dimension D, whereas the closure of a single top-dimensional cell already contains $2^n n!$ fixed points and the union of the closures of two contains more than $(mn+1)^n$, namely at least $12>9$ and $88>64$.

Thus the G_n-invariant sector is not the homology of a subvariety of Fl_γ, and a geometric proof must use Fl_γ together with its Springer $W(B_n)$-action, or a parahoric quotient Fl_γ^J. The latter can have $(mn+1)^n$ fixed points and $n!$ top cells only if W_J is generated by n commuting reflections, hence only if the affine Dynkin diagram is a star, hence only for $n\le 3$. For $n=3$ the unique candidate, $J=\{0,1,3\}$ for SO_7, has the predicted profile at $m=1$ and fails it at $m=2$ by one cell.

Appendix A. Census of fixed wall fibers

In this appendix we record the census of the fixed wall fibers which is used in Theorem 4.2 and in the proof of Theorem 6.25. Exact enumeration of hearts gives the following profiles of $\rho^{-1}(z_J)$, $z_J\in Q_n^T$ (verified through $n=16$, shown to $n=12$).

n	$\lvert Q_n^T\rvert$	pos.-dim. fibers	profile
1	2	0	$2\,\mathrm{Gr}(1,1)$
2	3	1	$2\,\mathrm{Gr}(1,1)+\mathrm{Gr}(2,3)$
3	6	2	$4\,\mathrm{Gr}(1,1)+2\,\mathrm{Gr}(2,3)$
4	10	5	$5\,\mathrm{Gr}(1,1)+5\,\mathrm{Gr}(2,3)$
5	16	10	$6\,\mathrm{Gr}(1,1)+10\,\mathrm{Gr}(2,3)$
6	26	16	$10\,\mathrm{Gr}(1,1)+15\,\mathrm{Gr}(2,3)+\mathrm{Gr}(3,5)$
7	40	28	$12\,\mathrm{Gr}(1,1)+26\,\mathrm{Gr}(2,3)+2\,\mathrm{Gr}(3,5)$
8	60	45	$15\,\mathrm{Gr}(1,1)+40\,\mathrm{Gr}(2,3)+5\,\mathrm{Gr}(3,5)$
9	90	70	$20\,\mathrm{Gr}(1,1)+60\,\mathrm{Gr}(2,3)+10\,\mathrm{Gr}(3,5)$
10	131	105	$26\,\mathrm{Gr}(1,1)+85\,\mathrm{Gr}(2,3)+20\,\mathrm{Gr}(3,5)$
11	188	156	$32\,\mathrm{Gr}(1,1)+120\,\mathrm{Gr}(2,3)+36\,\mathrm{Gr}(3,5)$
12	269	229	$40\,\mathrm{Gr}(1,1)+170\,\mathrm{Gr}(2,3)+58\,\mathrm{Gr}(3,5)+\mathrm{Gr}(4,7)$

$\mathrm{Gr}(m,2m-1)$ appears first at $n=m(m-1)$, where its dimension is n (Theorem 4.2). For each positive-dimensional fixed fiber with $n\le 4$ (all of type $\mathbb{P}^2$), the fixed-point classes $[\mathcal{M}_n^+]_\lambda$ of the excess sheaf have been computed exactly on charts, and the fiberwise localized cancellation $\sum_{\lambda\in F_J^T}\ell_J^d[\mathcal{M}_n^+]_\lambda/D_\lambda=0$ holds for all twists d and all wall fixed points with $n\le 4$, which is the

K-theoretic shadow of Theorem 5.1. The normalized classes $[\mathcal{M}_n^+]_\lambda D_\lambda^F/(D_\lambda^Y e_W)$, with e_W the fixed-point character of $\mathcal{Q}\otimes\mathcal{O}_{\mathrm{Pl}}(1-N)$, are constant along each fiber, as Theorem 4.5 predicts.

Appendix B. Two lemmas on regular sections and the ADHM dictionary

In this appendix we collect two elementary facts which are used in Section 5. The first is the standard fact that a section with a smooth zero scheme of the expected codimension is regular, and the second is the rank-one ADHM dictionary over an arbitrary base, which identifies cyclic framed representations with commutative cluster algebras.

B.1. **Regular sections.** Locally the ambient projective bundles of Section 5.2 are regular, hence Cohen–Macaulay. If a section of a rank-r vector bundle has a smooth zero scheme of codimension r, its ideal is generated by r functions and has height r. In a Cohen–Macaulay local ring such generators form a regular sequence [62, Thm. 17.4], and the Koszul complex is a resolution of the ordinary structure sheaf of the zero scheme [24, Cor. 17.5]. This is used twice in Section 5.2, and no derived enhancement is needed.

B.2. **The rank-one framed ADHM dictionary.** The cyclic chamber is important because it makes the universal object a commutative cluster algebra, not merely a representation with the correct ranks.

Lemma B.1. *Let A be a complex algebra, V a finite locally free A-module, and $X,Y\in\mathrm{End}_A(V)$, $i:A\to V$, $j:V\to A$ with $[X,Y]+ij=0$. Suppose the images of all words in X,Y applied to $i(1)$ generate V. Then $j=0$ and $[X,Y]=0$.*

Proof. Trace of the relation gives $ji=0$. We prove by induction on word length that $jwi=0$ for every word w. Suppose this holds for words of length less than d. Swapping an adjacent YX in a length-d word acting on $i(1)$ changes the result by a term $uijvi$ with v of length at most $d-2$, which is zero by induction. Thus every word of length d has the same value on $i(1)$ as its ordered form $X^aY^bi(1)$, $a+b=d$. Trace the commutator $[X,X^aY^{b+1}]$ and expand with $[X,Y]=-ij$. Cyclicity of trace for a rank-one product gives $0=-\sum_{r=0}^{b} jY^{b-r}X^aY^ri$, and each summand reorders on $i(1)$, using only the lower-length identities, to jX^aY^bi. Hence $(b+1)jX^aY^bi=0$, and $b+1$ is invertible in A. Since the vectors $wi(1)$ generate V, $j=0$, and the moment-map relation gives $[X,Y]=0$. □

For the McKay quiver, X and Y exchange the even and odd summands and the framing lies in the even summand, and the two vertex moment-map equations combine to the displayed ADHM equation. The lemma therefore identifies the cyclic framed representation with a quotient of $A[x,y]$ generated by the unit, with its Γ-grading, over arbitrary complex base algebras, and this is the family-level dictionary used in Section 5.2.

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